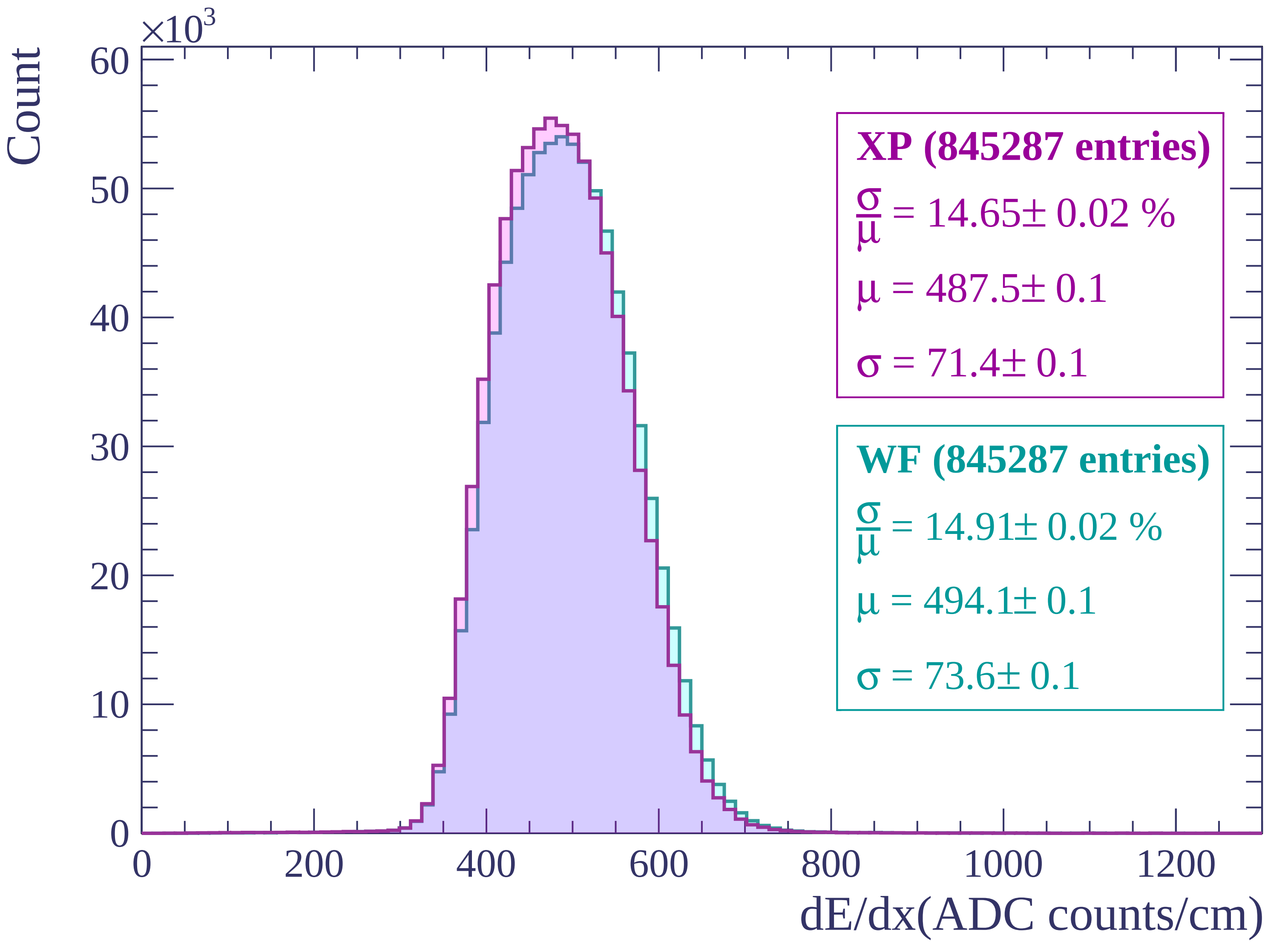
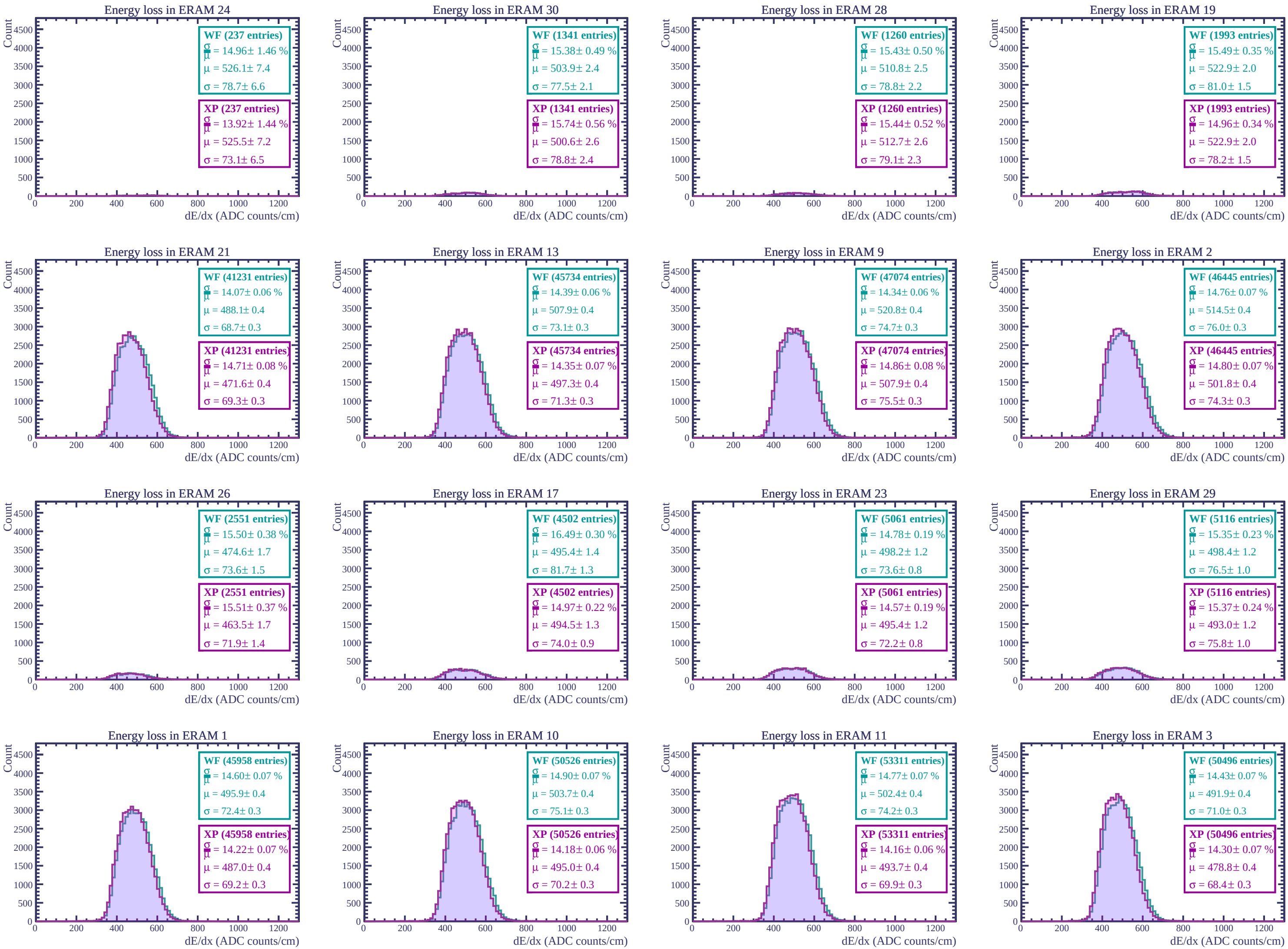
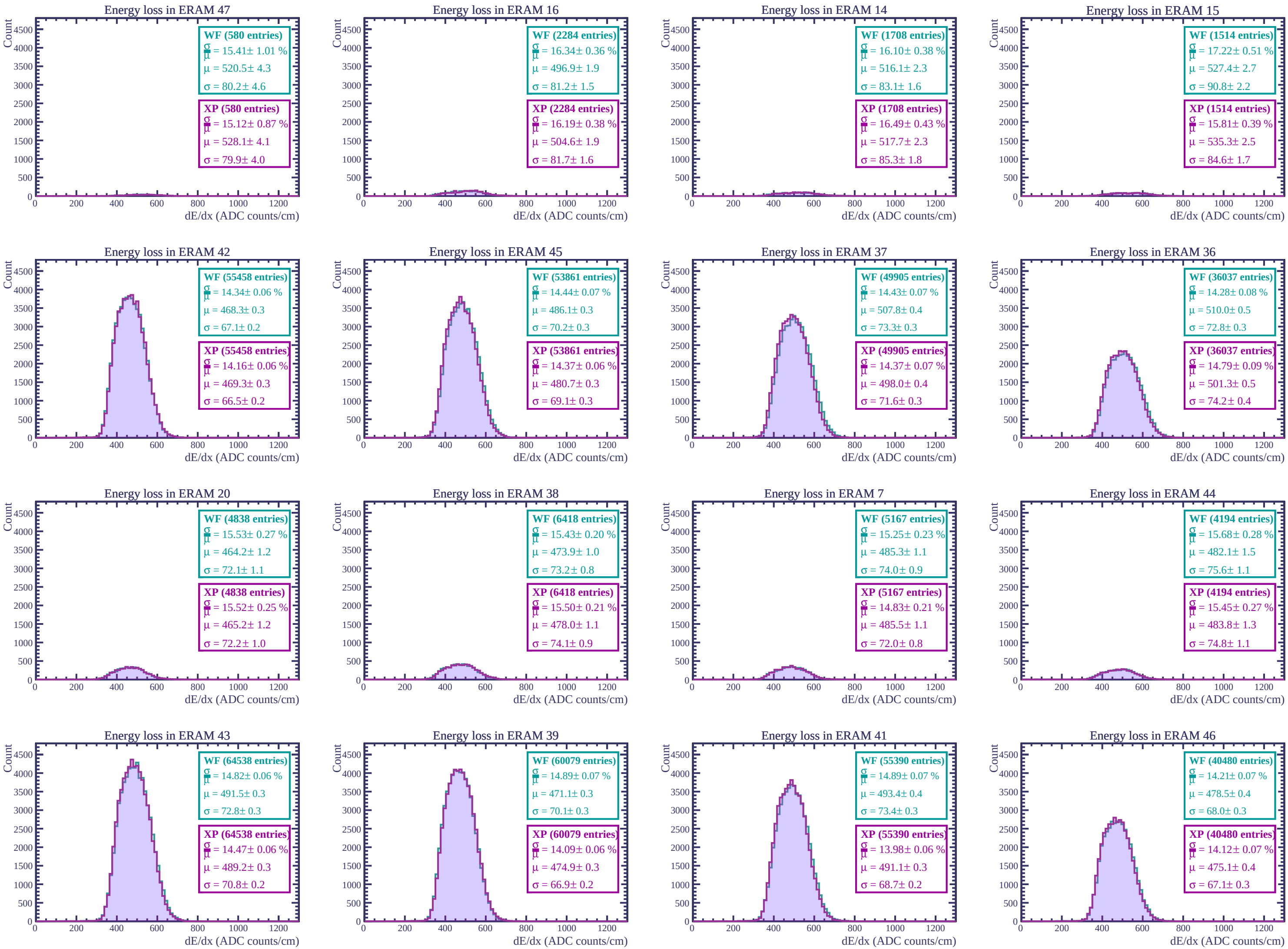


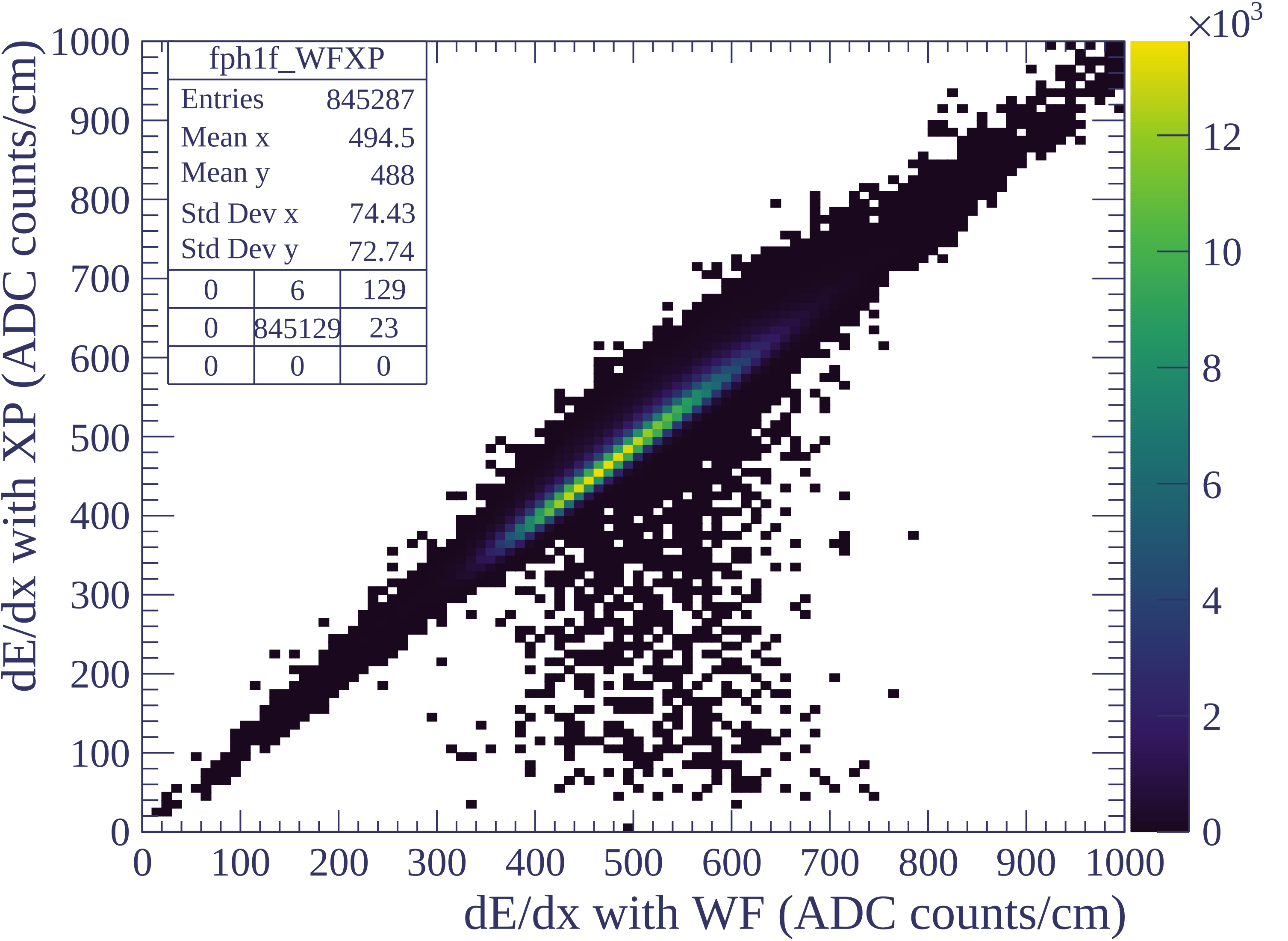
Count
$\times 10^3$
60
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx(ADC counts/cm)
XP (845287 entries)
$\frac{\sigma}{\mu}$ = 14.65± 0.02 %
μ = 487.5± 0.1
σ = 71.4± 0.1
WF (845287 entries)
$\frac{\sigma}{\mu}$ = 14.91± 0.02 %
μ = 494.1± 0.1
σ = 73.6± 0.1

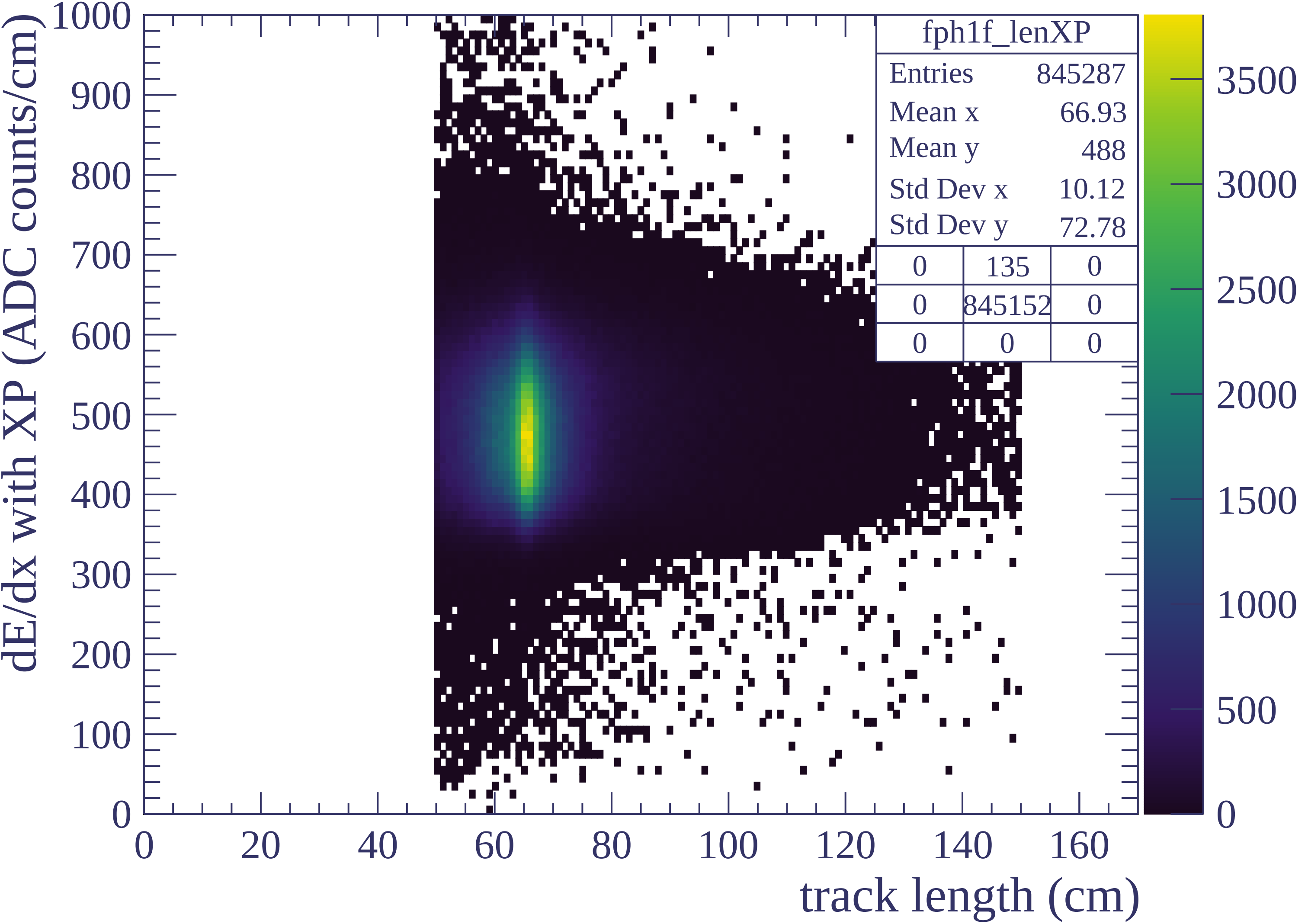


Energy loss in ERAM 24
WF (237 entries)
σ/μ = 14.96± 1.46 %
μ = 526.1± 7.4
σ = 78.7± 6.6
XP (237 entries)
σ/μ = 13.92± 1.44 %
μ = 525.5± 7.2
σ = 73.1± 6.5
Energy loss in ERAM 30
WF (1341 entries)
σ/μ = 15.38± 0.49 %
μ = 503.9± 2.4
σ = 77.5± 2.1
XP (1341 entries)
σ/μ = 15.74± 0.56 %
μ = 500.6± 2.6
σ = 78.8± 2.4
Energy loss in ERAM 28
WF (1260 entries)
σ/μ = 15.43± 0.50 %
μ = 510.8± 2.5
σ = 78.8± 2.2
XP (1260 entries)
σ/μ = 15.44± 0.52 %
μ = 512.7± 2.6
σ = 79.1± 2.3
Energy loss in ERAM 19
WF (1993 entries)
σ/μ = 15.49± 0.35 %
μ = 522.9± 2.0
σ = 81.0± 1.5
XP (1993 entries)
σ/μ = 14.96± 0.34 %
μ = 522.9± 2.0
σ = 78.2± 1.5
Energy loss in ERAM 21
WF (41231 entries)
σ/μ = 14.07± 0.06 %
μ = 488.1± 0.4
σ = 68.7± 0.3
XP (41231 entries)
σ/μ = 14.71± 0.08 %
μ = 471.6± 0.4
σ = 69.3± 0.3
Energy loss in ERAM 13
WF (45734 entries)
σ/μ = 14.39± 0.06 %
μ = 507.9± 0.4
σ = 73.1± 0.3
XP (45734 entries)
σ/μ = 14.35± 0.07 %
μ = 497.3± 0.4
σ = 71.3± 0.3
Energy loss in ERAM 9
WF (47074 entries)
σ/μ = 14.34± 0.06 %
μ = 520.8± 0.4
σ = 74.7± 0.3
XP (47074 entries)
σ/μ = 14.86± 0.08 %
μ = 507.9± 0.4
σ = 75.5± 0.3
Energy loss in ERAM 2
WF (46445 entries)
σ/μ = 14.76± 0.07 %
μ = 514.5± 0.4
σ = 76.0± 0.3
XP (46445 entries)
σ/μ = 14.80± 0.07 %
μ = 501.8± 0.4
σ = 74.3± 0.3
Energy loss in ERAM 26
WF (2551 entries)
σ/μ = 15.50± 0.38 %
μ = 474.6± 1.7
σ = 73.6± 1.5
XP (2551 entries)
σ/μ = 15.51± 0.37 %
μ = 463.5± 1.7
σ = 71.9± 1.4
Energy loss in ERAM 17
WF (4502 entries)
σ/μ = 16.49± 0.30 %
μ = 495.4± 1.4
σ = 81.7± 1.3
XP (4502 entries)
σ/μ = 14.97± 0.22 %
μ = 494.5± 1.3
σ = 74.0± 0.9
Energy loss in ERAM 23
WF (5061 entries)
σ/μ = 14.78± 0.19 %
μ = 498.2± 1.2
σ = 73.6± 0.8
XP (5061 entries)
σ/μ = 14.57± 0.19 %
μ = 495.4± 1.2
σ = 72.2± 0.8
Energy loss in ERAM 29
WF (5116 entries)
σ/μ = 15.35± 0.23 %
μ = 498.4± 1.2
σ = 76.5± 1.0
XP (5116 entries)
σ/μ = 15.37± 0.24 %
μ = 493.0± 1.2
σ = 75.8± 1.0
Energy loss in ERAM 1
WF (45958 entries)
σ/μ = 14.60± 0.07 %
μ = 495.9± 0.4
σ = 72.4± 0.3
XP (45958 entries)
σ/μ = 14.22± 0.07 %
μ = 487.0± 0.4
σ = 69.2± 0.3
Energy loss in ERAM 10
WF (50526 entries)
σ/μ = 14.90± 0.07 %
μ = 503.7± 0.4
σ = 75.1± 0.3
XP (50526 entries)
σ/μ = 14.18± 0.06 %
μ = 495.0± 0.4
σ = 70.2± 0.3
Energy loss in ERAM 11
WF (53311 entries)
σ/μ = 14.77± 0.07 %
μ = 502.4± 0.4
σ = 74.2± 0.3
XP (53311 entries)
σ/μ = 14.16± 0.06 %
μ = 493.7± 0.4
σ = 69.9± 0.3
Energy loss in ERAM 3
WF (50496 entries)
σ/μ = 14.43± 0.07 %
μ = 491.9± 0.4
σ = 71.0± 0.3
XP (50496 entries)
σ/μ = 14.30± 0.07 %
μ = 478.8± 0.4
σ = 68.4± 0.3
Count
dE/dx (ADC counts/cm)

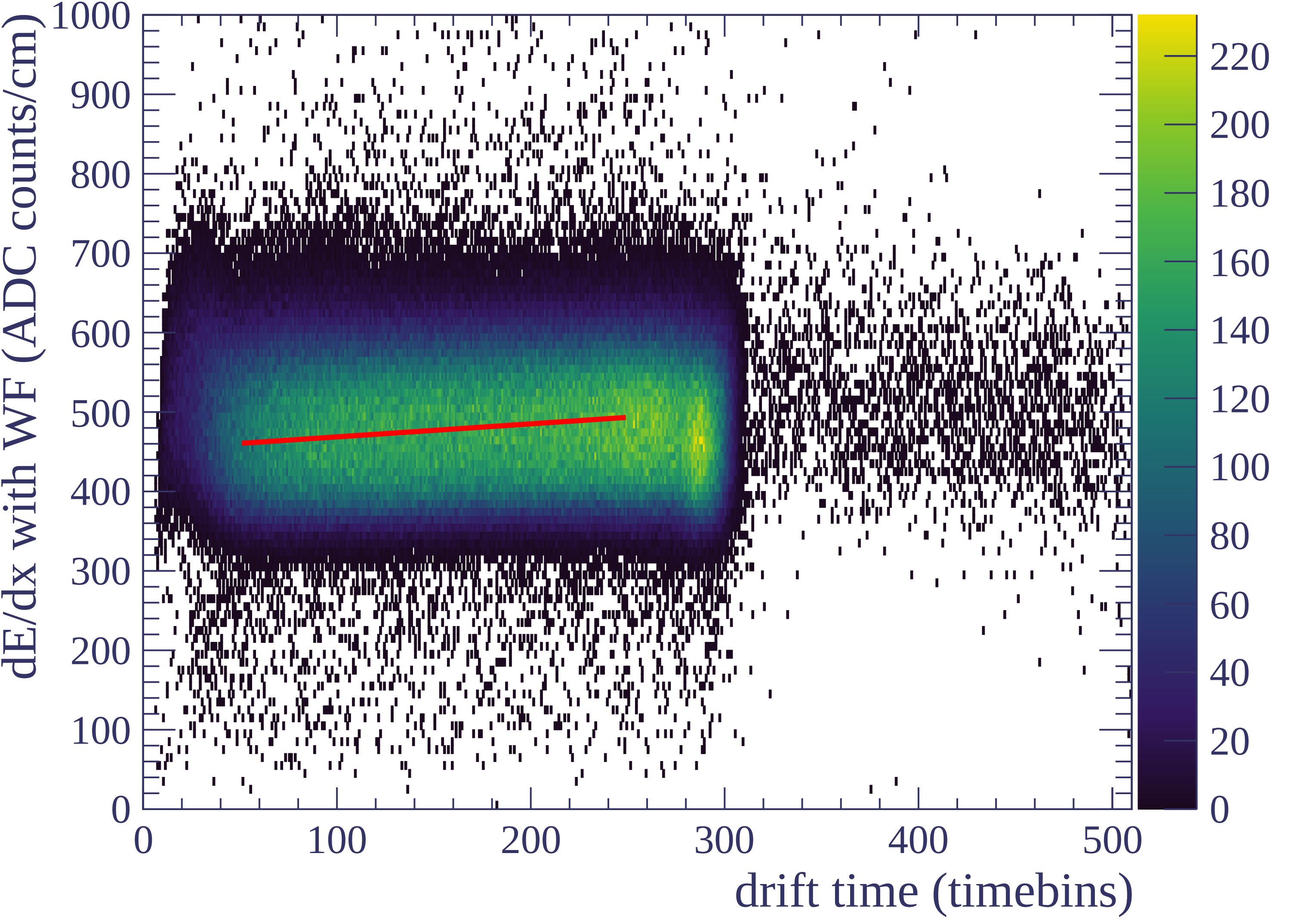


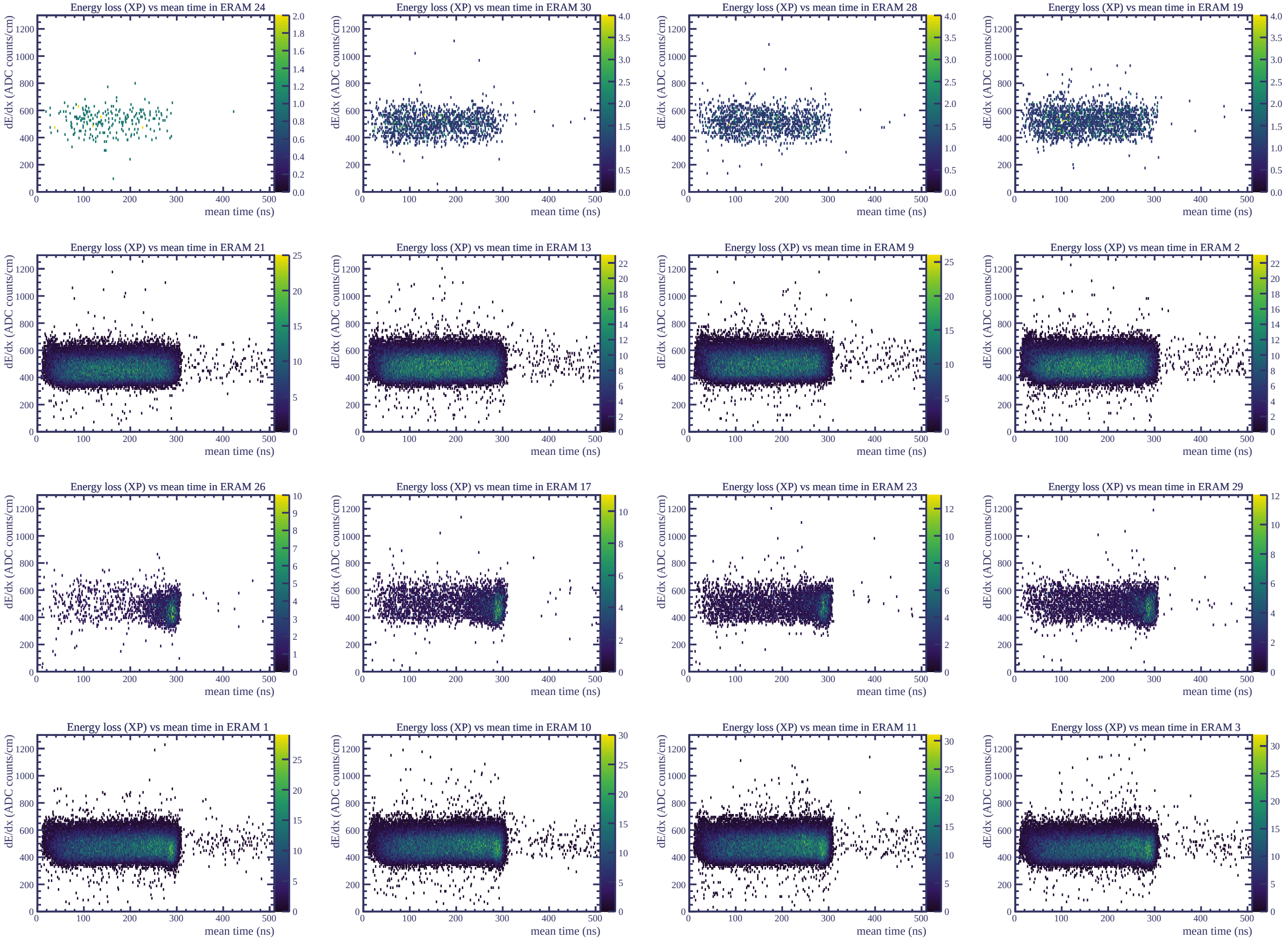
Energy loss in ERAM 47
WF (580 entries) σ/μ = 15.41± 1.01 % μ = 520.5± 4.3 σ = 80.2± 4.6
XP (580 entries) σ/μ = 15.12± 0.87 % μ = 528.1± 4.1 σ = 79.9± 4.0
Energy loss in ERAM 16
WF (2284 entries) σ/μ = 16.34± 0.36 % μ = 496.9± 1.9 σ = 81.2± 1.5
XP (2284 entries) σ/μ = 16.19± 0.38 % μ = 504.6± 1.9 σ = 81.7± 1.6
Energy loss in ERAM 14
WF (1708 entries) σ/μ = 16.10± 0.38 % μ = 516.1± 2.3 σ = 83.1± 1.6
XP (1708 entries) σ/μ = 16.49± 0.43 % μ = 517.7± 2.3 σ = 85.3± 1.8
Energy loss in ERAM 15
WF (1514 entries) σ/μ = 17.22± 0.51 % μ = 527.4± 2.7 σ = 90.8± 2.2
XP (1514 entries) σ/μ = 15.81± 0.39 % μ = 535.3± 2.5 σ = 84.6± 1.7
Energy loss in ERAM 42
WF (55458 entries) σ/μ = 14.34± 0.06 % μ = 468.3± 0.3 σ = 67.1± 0.2
XP (55458 entries) σ/μ = 14.16± 0.06 % μ = 469.3± 0.3 σ = 66.5± 0.2
Energy loss in ERAM 45
WF (53861 entries) σ/μ = 14.44± 0.07 % μ = 486.1± 0.3 σ = 70.2± 0.3
XP (53861 entries) σ/μ = 14.37± 0.06 % μ = 480.7± 0.3 σ = 69.1± 0.3
Energy loss in ERAM 37
WF (49905 entries) σ/μ = 14.43± 0.07 % μ = 507.8± 0.4 σ = 73.3± 0.3
XP (49905 entries) σ/μ = 14.37± 0.07 % μ = 498.0± 0.4 σ = 71.6± 0.3
Energy loss in ERAM 36
WF (36037 entries) σ/μ = 14.28± 0.08 % μ = 510.0± 0.5 σ = 72.8± 0.3
XP (36037 entries) σ/μ = 14.79± 0.09 % μ = 501.3± 0.5 σ = 74.2± 0.4
Energy loss in ERAM 20
WF (4838 entries) σ/μ = 15.53± 0.27 % μ = 464.2± 1.2 σ = 72.1± 1.1
XP (4838 entries) σ/μ = 15.52± 0.25 % μ = 465.2± 1.2 σ = 72.2± 1.0
Energy loss in ERAM 38
WF (6418 entries) σ/μ = 15.43± 0.20 % μ = 473.9± 1.0 σ = 73.2± 0.8
XP (6418 entries) σ/μ = 15.50± 0.21 % μ = 478.0± 1.1 σ = 74.1± 0.9
Energy loss in ERAM 7
WF (5167 entries) σ/μ = 15.25± 0.23 % μ = 485.3± 1.1 σ = 74.0± 0.9
XP (5167 entries) σ/μ = 14.83± 0.21 % μ = 485.5± 1.1 σ = 72.0± 0.8
Energy loss in ERAM 44
WF (4194 entries) σ/μ = 15.68± 0.28 % μ = 482.1± 1.5 σ = 75.6± 1.1
XP (4194 entries) σ/μ = 15.45± 0.27 % μ = 483.8± 1.3 σ = 74.8± 1.1
Energy loss in ERAM 43
WF (64538 entries) σ/μ = 14.82± 0.06 % μ = 491.5± 0.3 σ = 72.8± 0.3
XP (64538 entries) σ/μ = 14.47± 0.06 % μ = 489.2± 0.3 σ = 70.8± 0.2
Energy loss in ERAM 39
WF (60079 entries) σ/μ = 14.89± 0.07 % μ = 471.1± 0.3 σ = 70.1± 0.3
XP (60079 entries) σ/μ = 14.09± 0.06 % μ = 474.9± 0.3 σ = 66.9± 0.2
Energy loss in ERAM 41
WF (55390 entries) σ/μ = 14.89± 0.07 % μ = 493.4± 0.4 σ = 73.4± 0.3
XP (55390 entries) σ/μ = 13.98± 0.06 % μ = 491.1± 0.3 σ = 68.7± 0.2
Energy loss in ERAM 46
WF (40480 entries) σ/μ = 14.21± 0.07 % μ = 478.5± 0.4 σ = 68.0± 0.3
XP (40480 entries) σ/μ = 14.12± 0.07 % μ = 475.1± 0.4 σ = 67.1± 0.3
Count
dE/dx (ADC counts/cm)

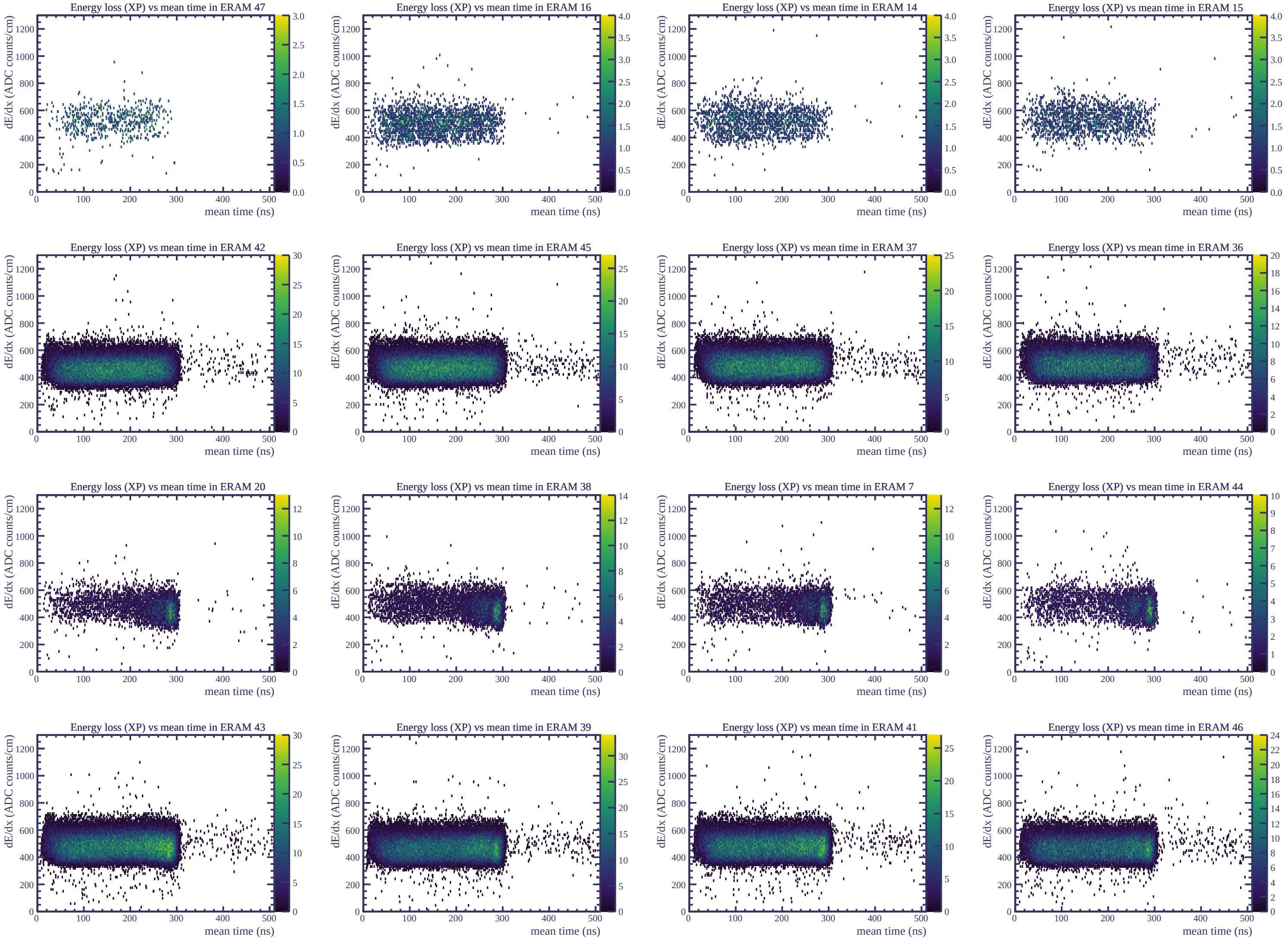

fph1f_WFXP
Entries 845287
Mean x 494.5
Mean y 488
Std Dev x 74.43
Std Dev y 72.74
0 6 129
0 845129 23
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
×10^3
0 2 4 6 8 10 12
0 100 200 300 400 500 600 700 800 900 1000

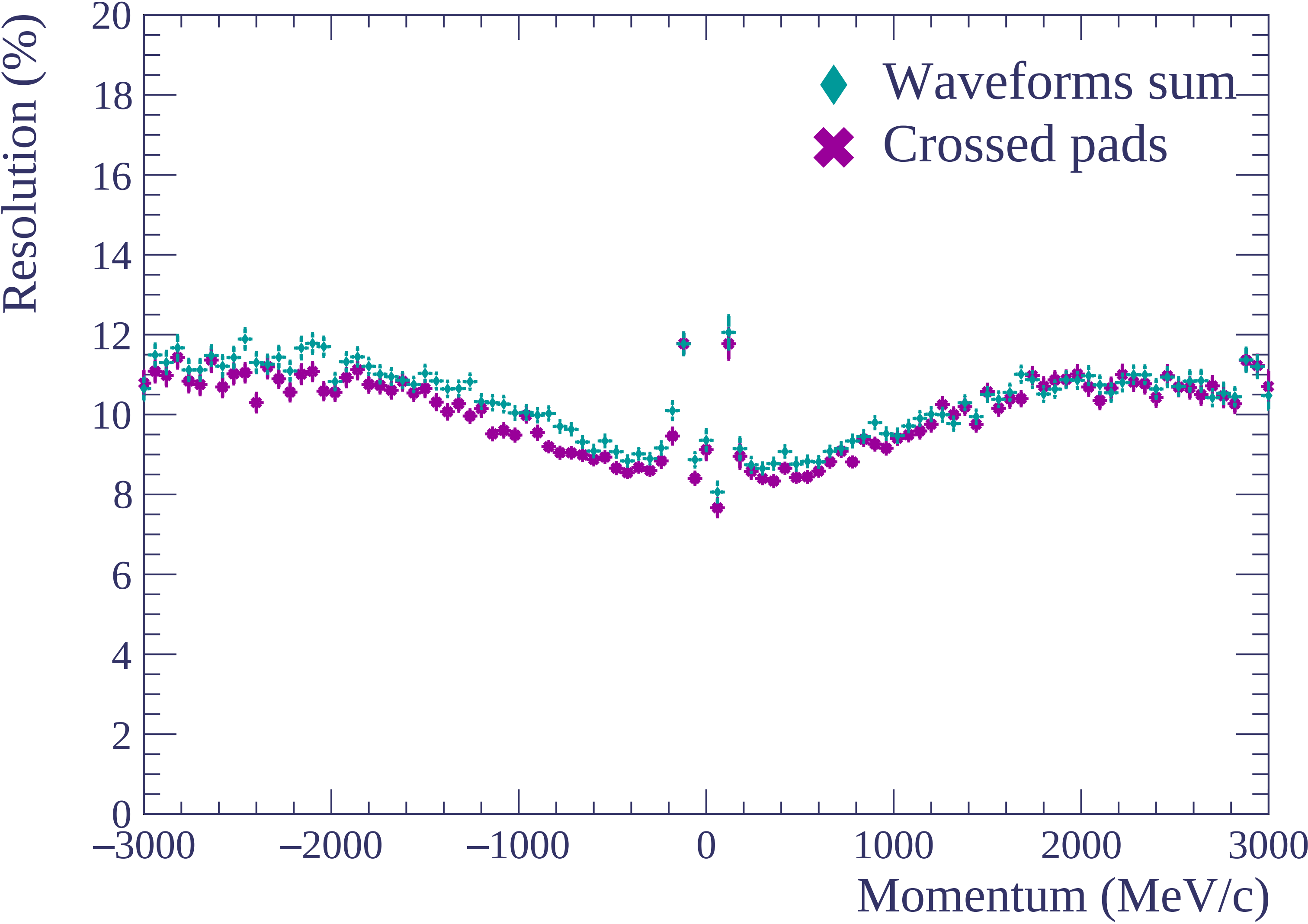


fph1f_lenXP
Entries 845287
Mean x 66.93
Mean y 488
Std Dev x 10.12
Std Dev y 72.78
0 135 0
0 845152 0
0 0 0
dE/dx with XP (ADC counts/cm)
track length (cm)

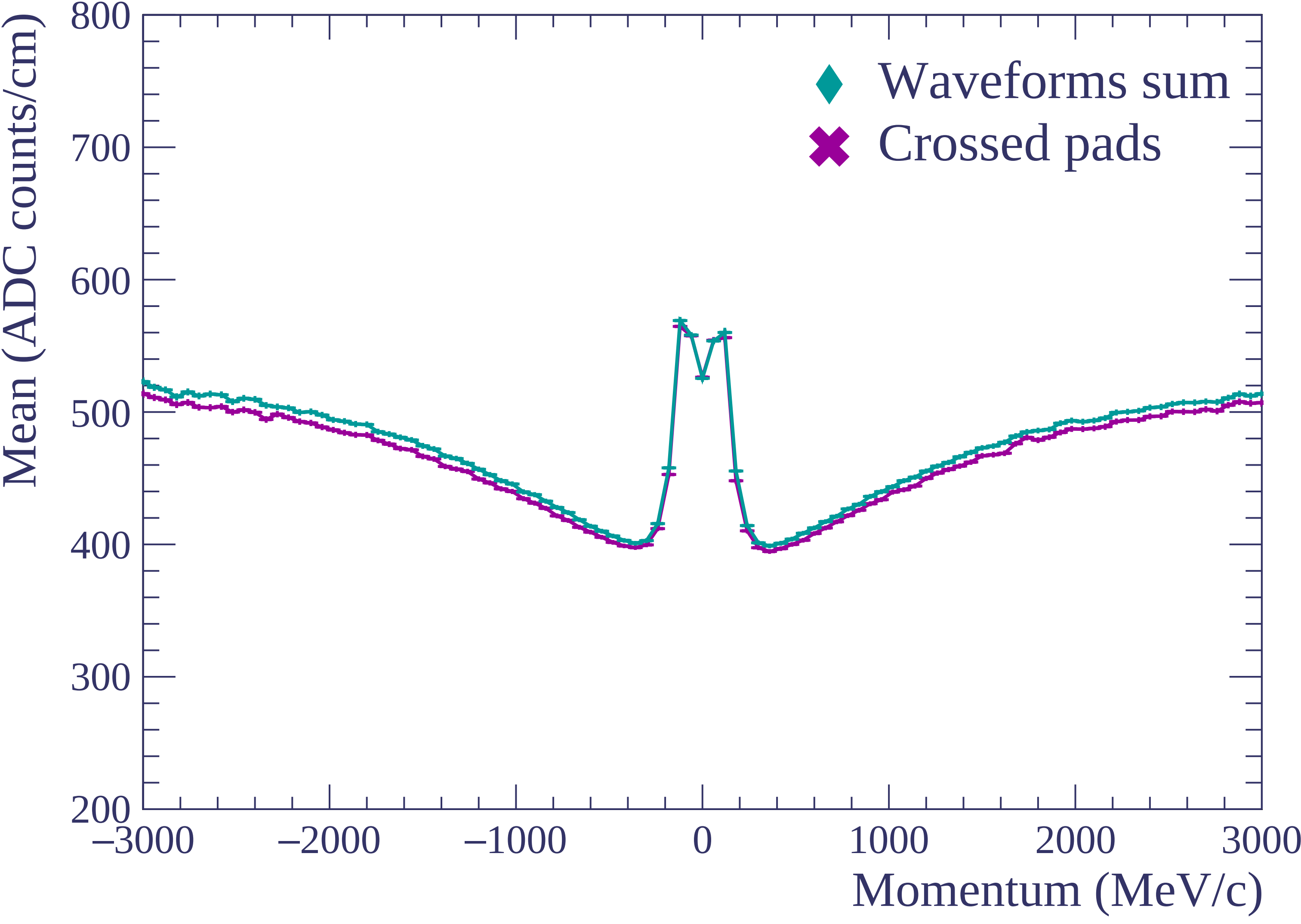

dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
drift time (timebins)
0
100
200
300
400
500
0
20
40
60
80
100
120
140
160
180
200
220


Energy loss (XP) vs mean time in ERAM 24
Energy loss (XP) vs mean time in ERAM 30
Energy loss (XP) vs mean time in ERAM 28
Energy loss (XP) vs mean time in ERAM 19
Energy loss (XP) vs mean time in ERAM 21
Energy loss (XP) vs mean time in ERAM 13
Energy loss (XP) vs mean time in ERAM 9
Energy loss (XP) vs mean time in ERAM 2
Energy loss (XP) vs mean time in ERAM 26
Energy loss (XP) vs mean time in ERAM 17
Energy loss (XP) vs mean time in ERAM 23
Energy loss (XP) vs mean time in ERAM 29
Energy loss (XP) vs mean time in ERAM 1
Energy loss (XP) vs mean time in ERAM 10
Energy loss (XP) vs mean time in ERAM 11
Energy loss (XP) vs mean time in ERAM 3
dE/dx (ADC counts/cm)
mean time (ns)

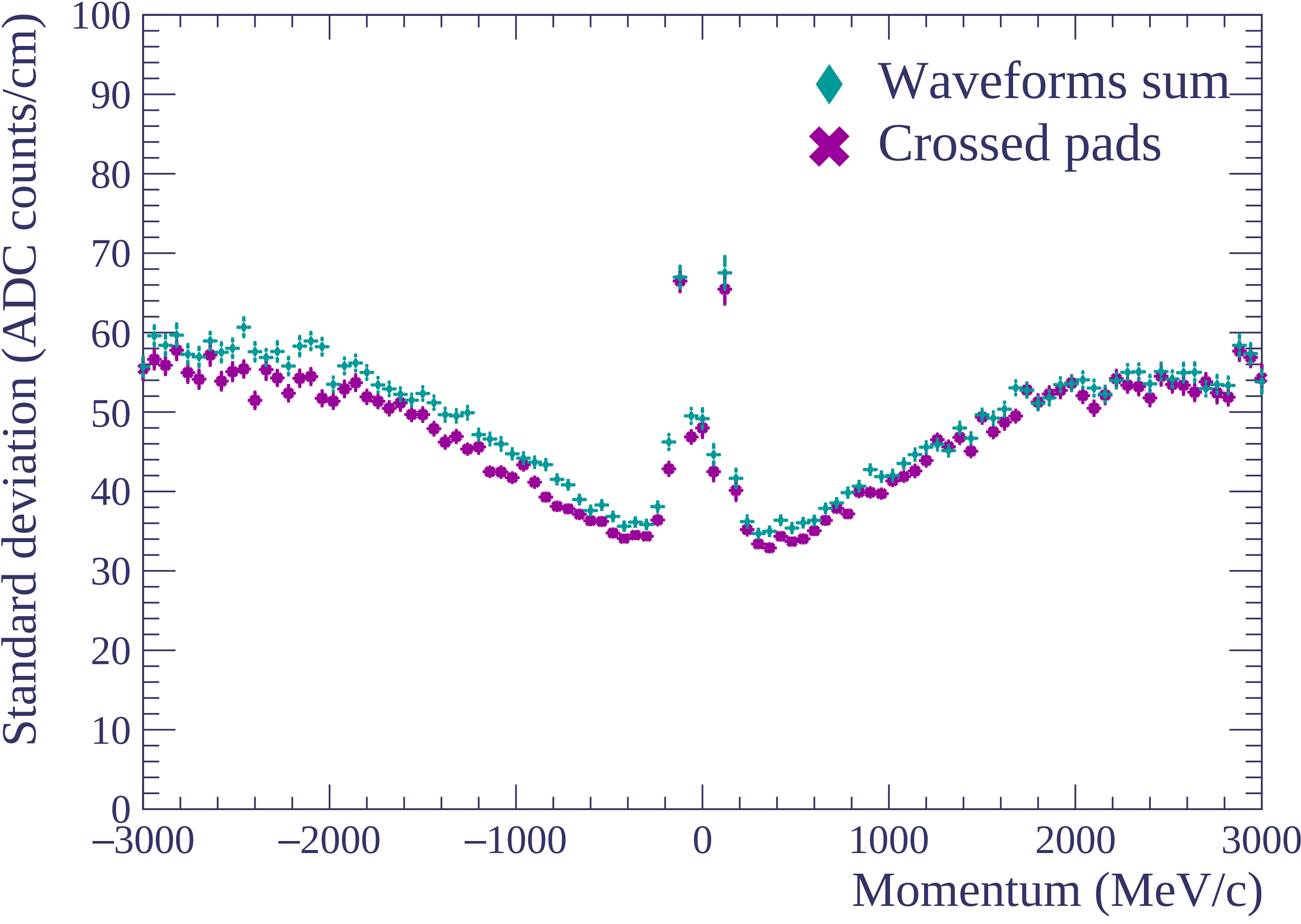

Energy loss (XP) vs mean time in ERAM 47
Energy loss (XP) vs mean time in ERAM 16
Energy loss (XP) vs mean time in ERAM 14
Energy loss (XP) vs mean time in ERAM 15
Energy loss (XP) vs mean time in ERAM 42
Energy loss (XP) vs mean time in ERAM 45
Energy loss (XP) vs mean time in ERAM 37
Energy loss (XP) vs mean time in ERAM 36
Energy loss (XP) vs mean time in ERAM 20
Energy loss (XP) vs mean time in ERAM 38
Energy loss (XP) vs mean time in ERAM 7
Energy loss (XP) vs mean time in ERAM 44
Energy loss (XP) vs mean time in ERAM 43
Energy loss (XP) vs mean time in ERAM 39
Energy loss (XP) vs mean time in ERAM 41
Energy loss (XP) vs mean time in ERAM 46
dE/dx (ADC counts/cm)
mean time (ns)

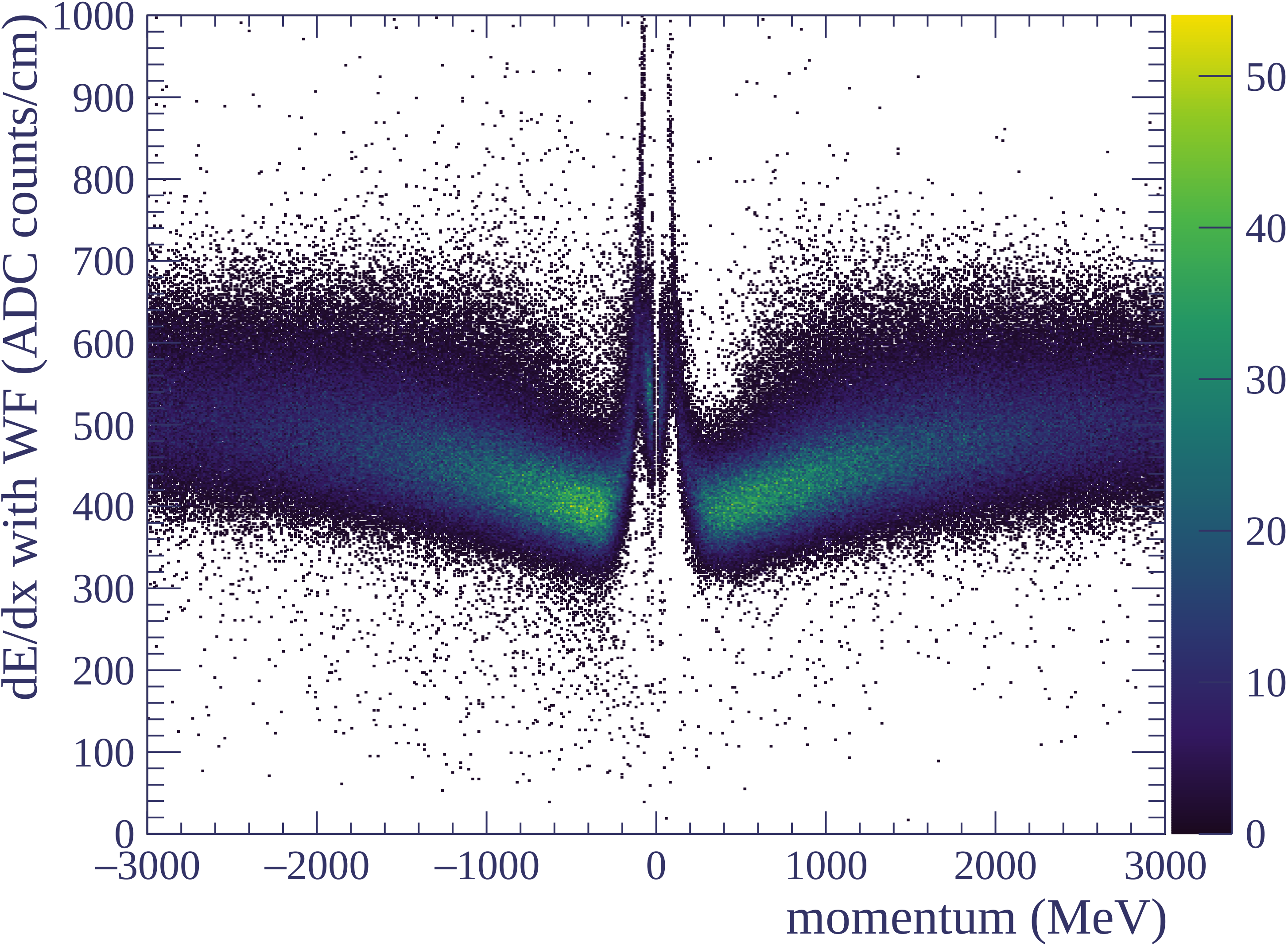


Resolution (%)
Momentum (MeV/c)
Waveforms sum
Crossed pads
0
2
4
6
8
10
12
14
16
18
20
−3000
−2000
−1000
0
1000
2000
3000

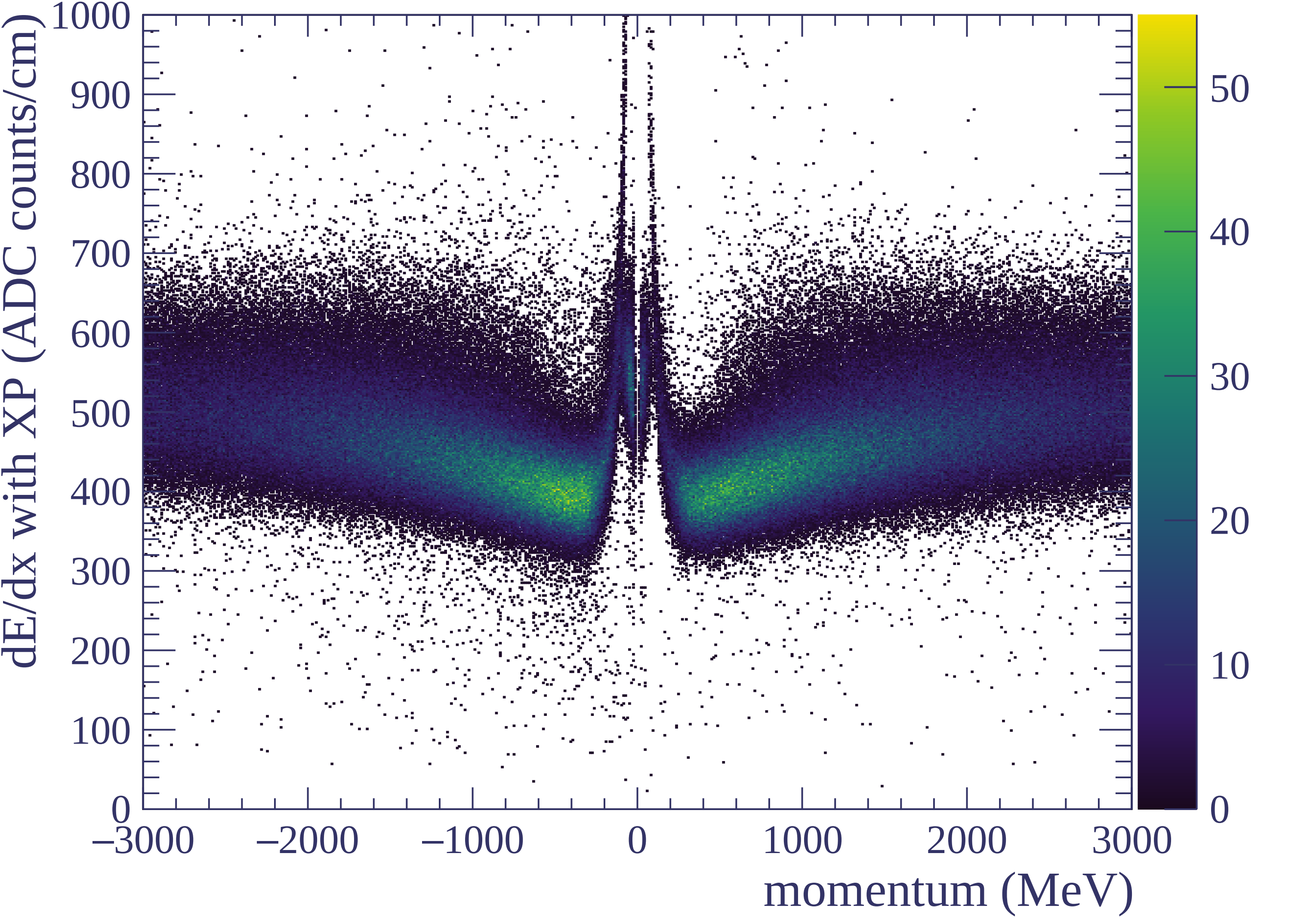


Waveforms sum
Crossed pads
Mean (ADC counts/cm)
Momentum (MeV/c)
800
700
600
500
400
300
200
−3000
−2000
−1000
0
1000
2000
3000

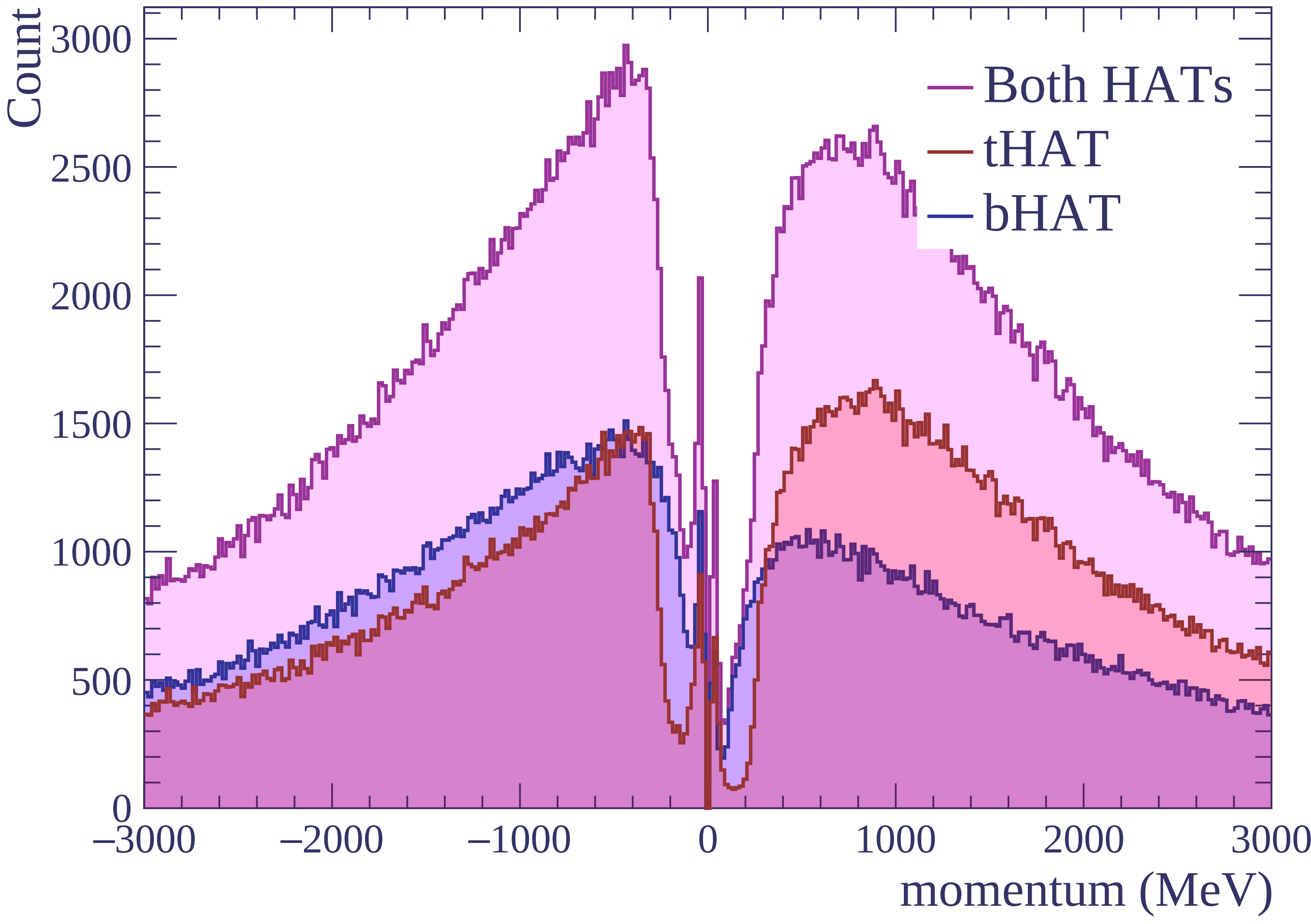

Waveforms sum
Crossed pads
Standard deviation (ADC counts/cm)
Momentum (MeV/c)
0
10
20
30
40
50
60
70
80
90
100
−3000
−2000
−1000
0
1000
2000
3000

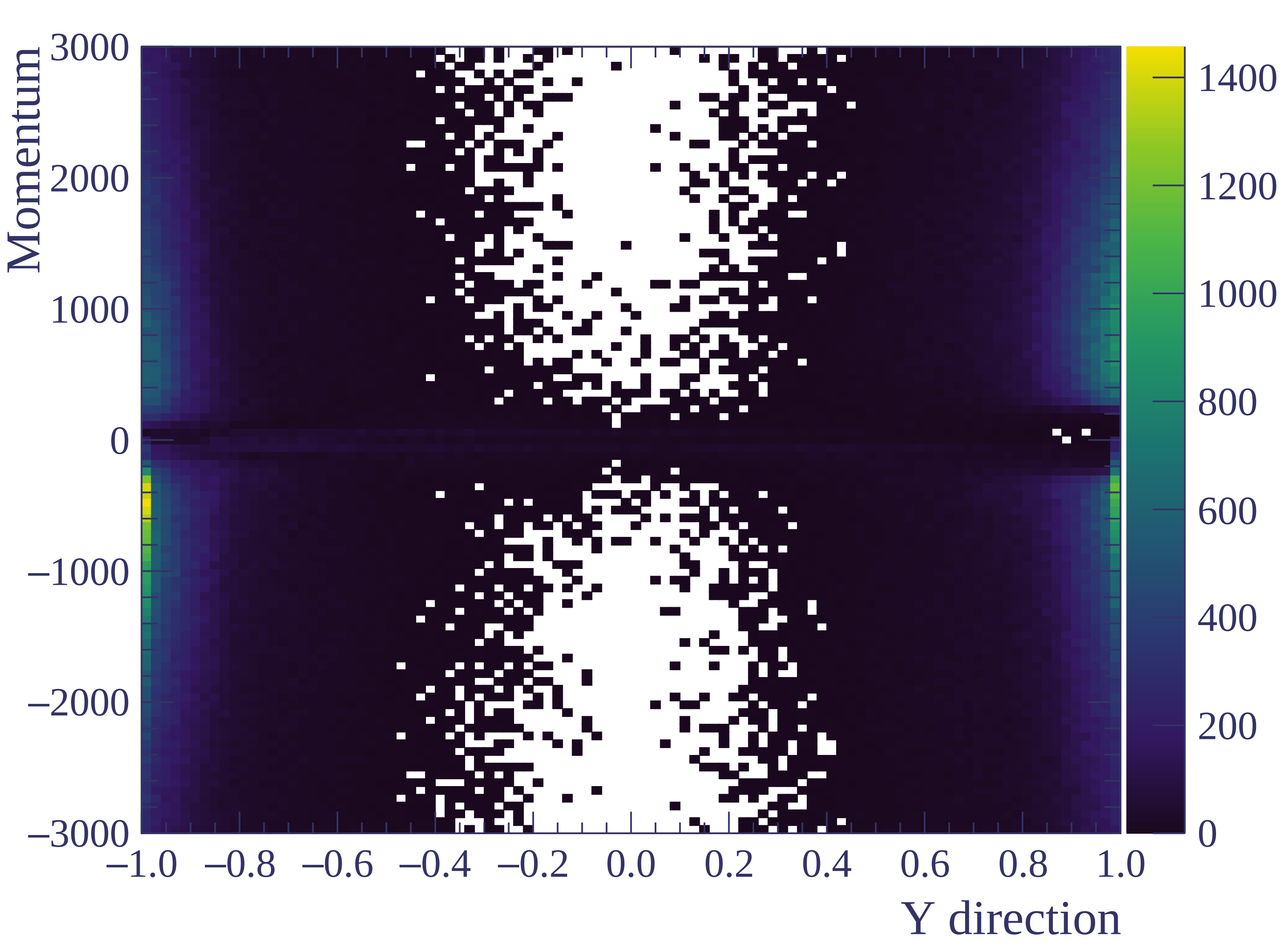


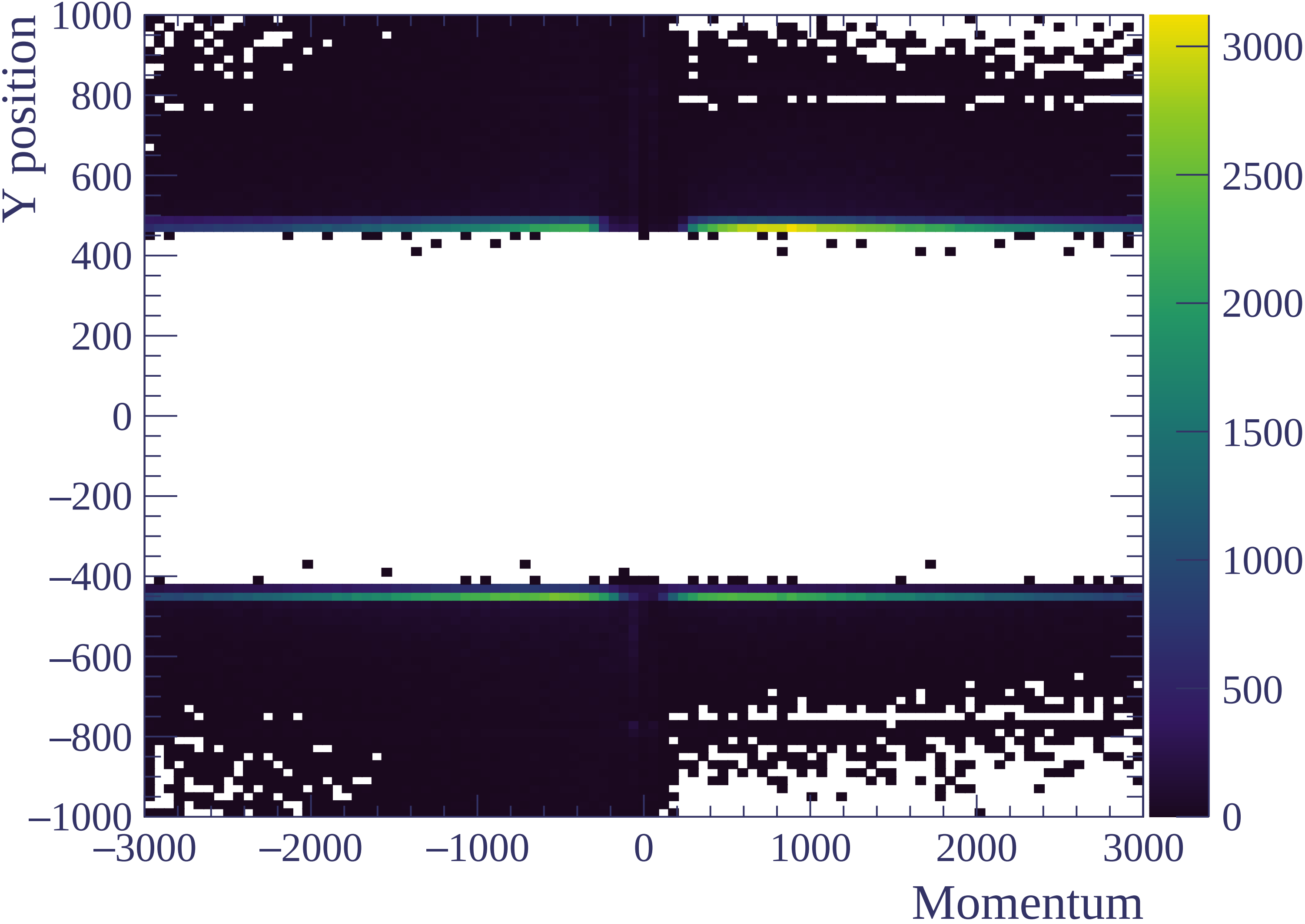
dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
50
40
30
20
10
0



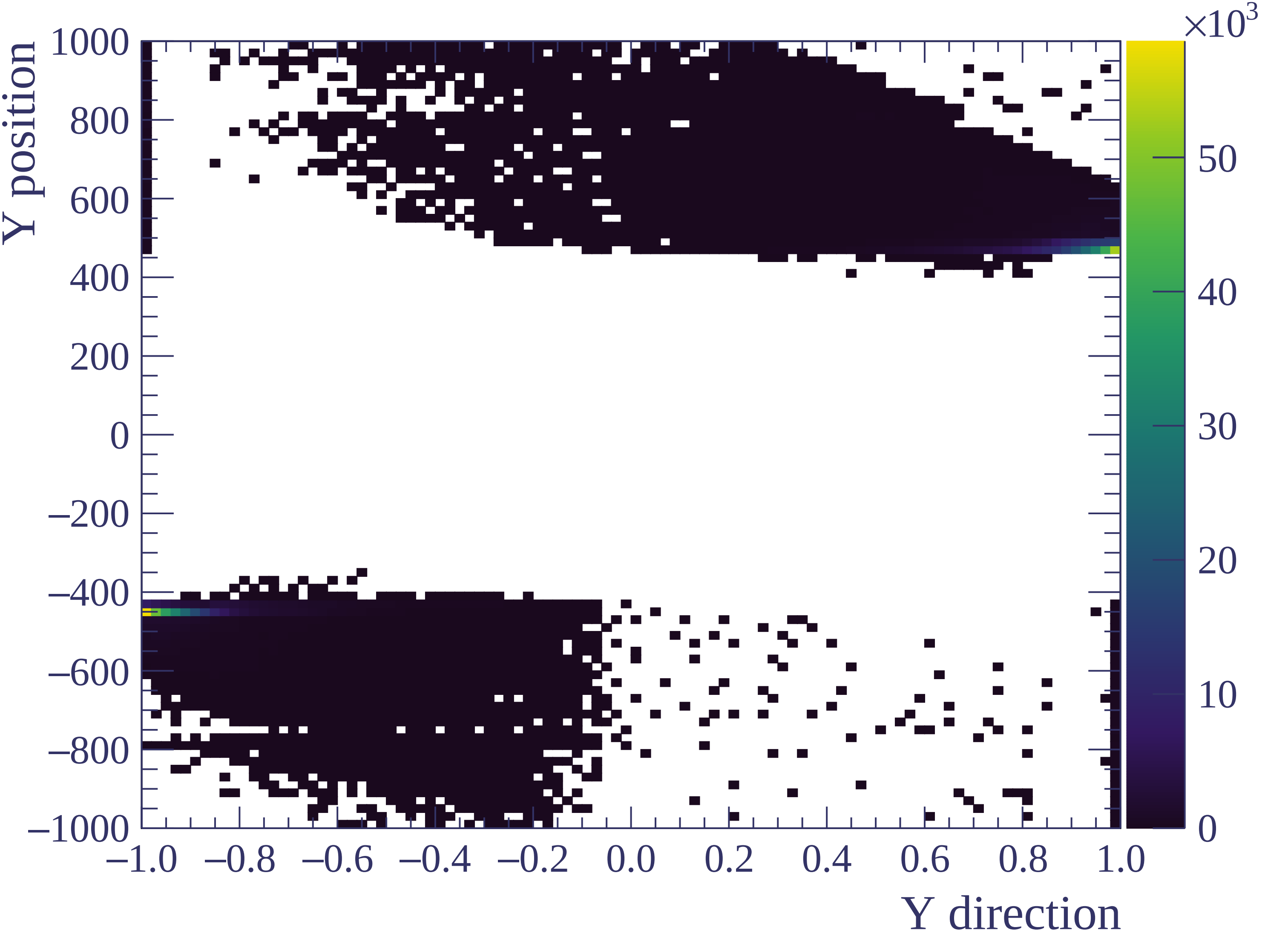
dE/dx with XP (ADC counts/cm)
momentum (MeV)
1000
900
800
700
600
500
400
300
200
100
0
−3000
−2000
−1000
0
1000
2000
3000
50
40
30
20
10
0


Count
momentum (MeV)
Both HATs
tHAT
bHAT
3000
2500
2000
1500
1000
500
0
−3000
−2000
−1000
0
1000
2000
3000

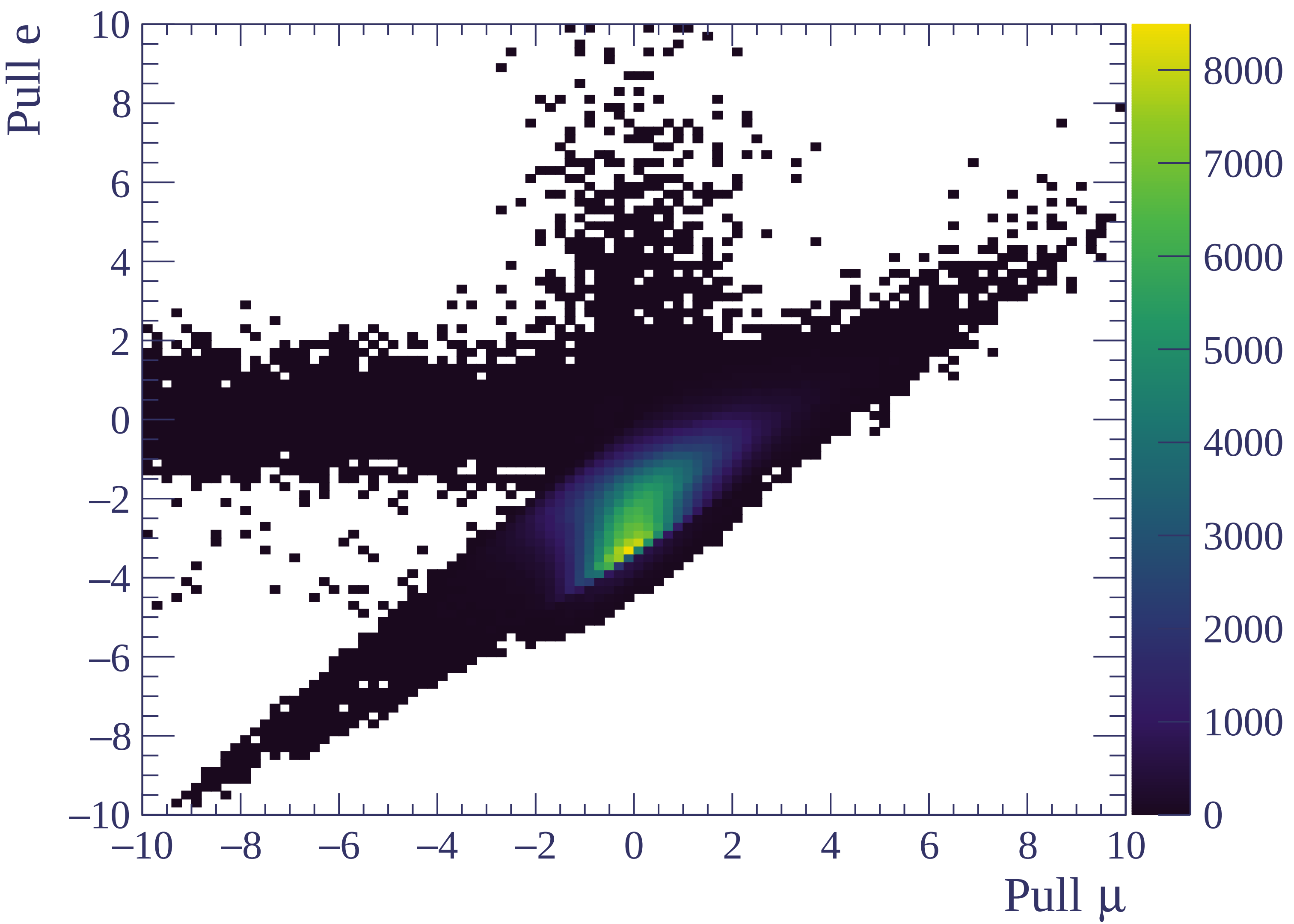

Momentum
3000
2000
1000
0
−1000
−2000
−3000
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y direction
1400
1200
1000
800
600
400
200
0

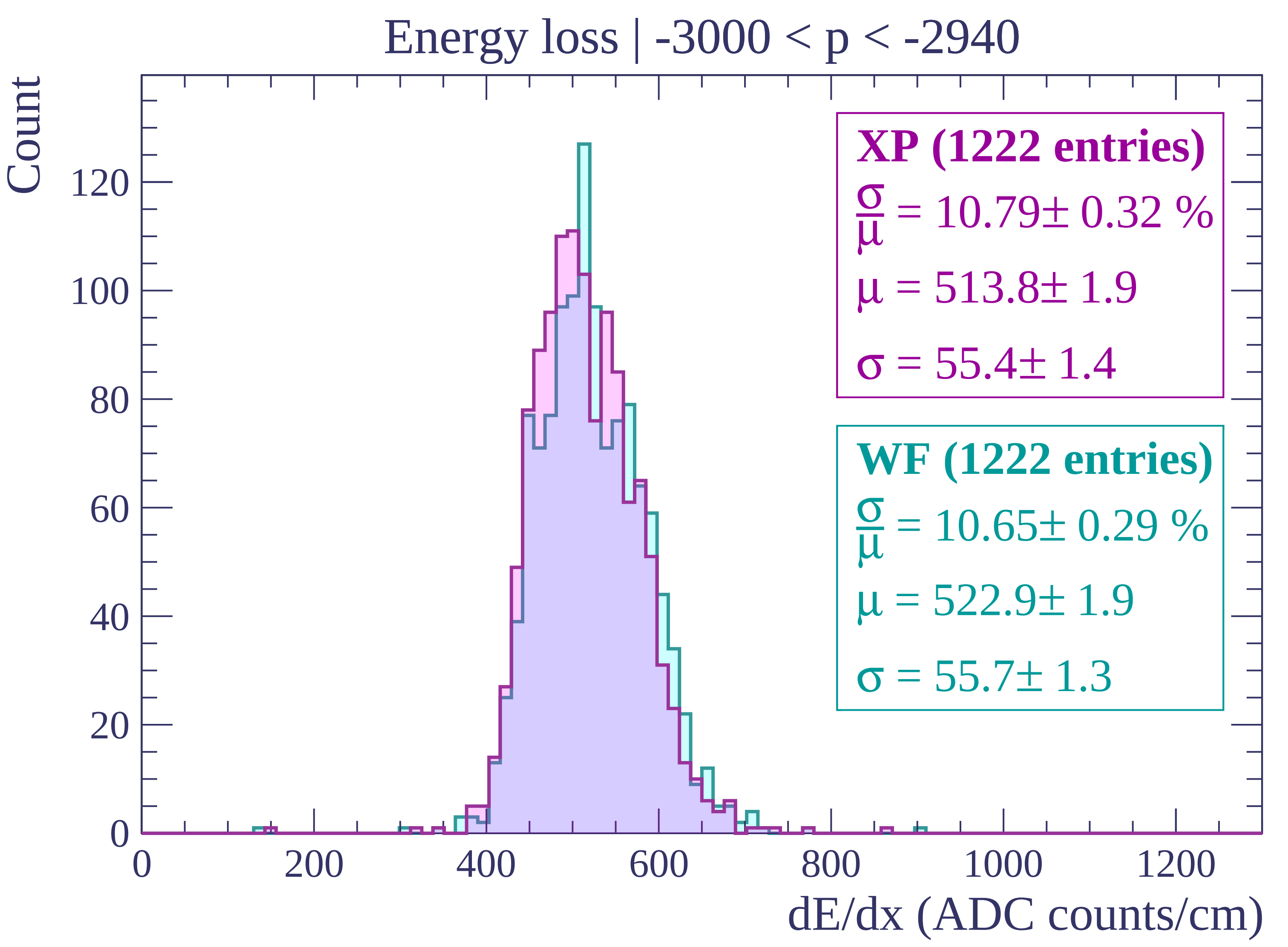


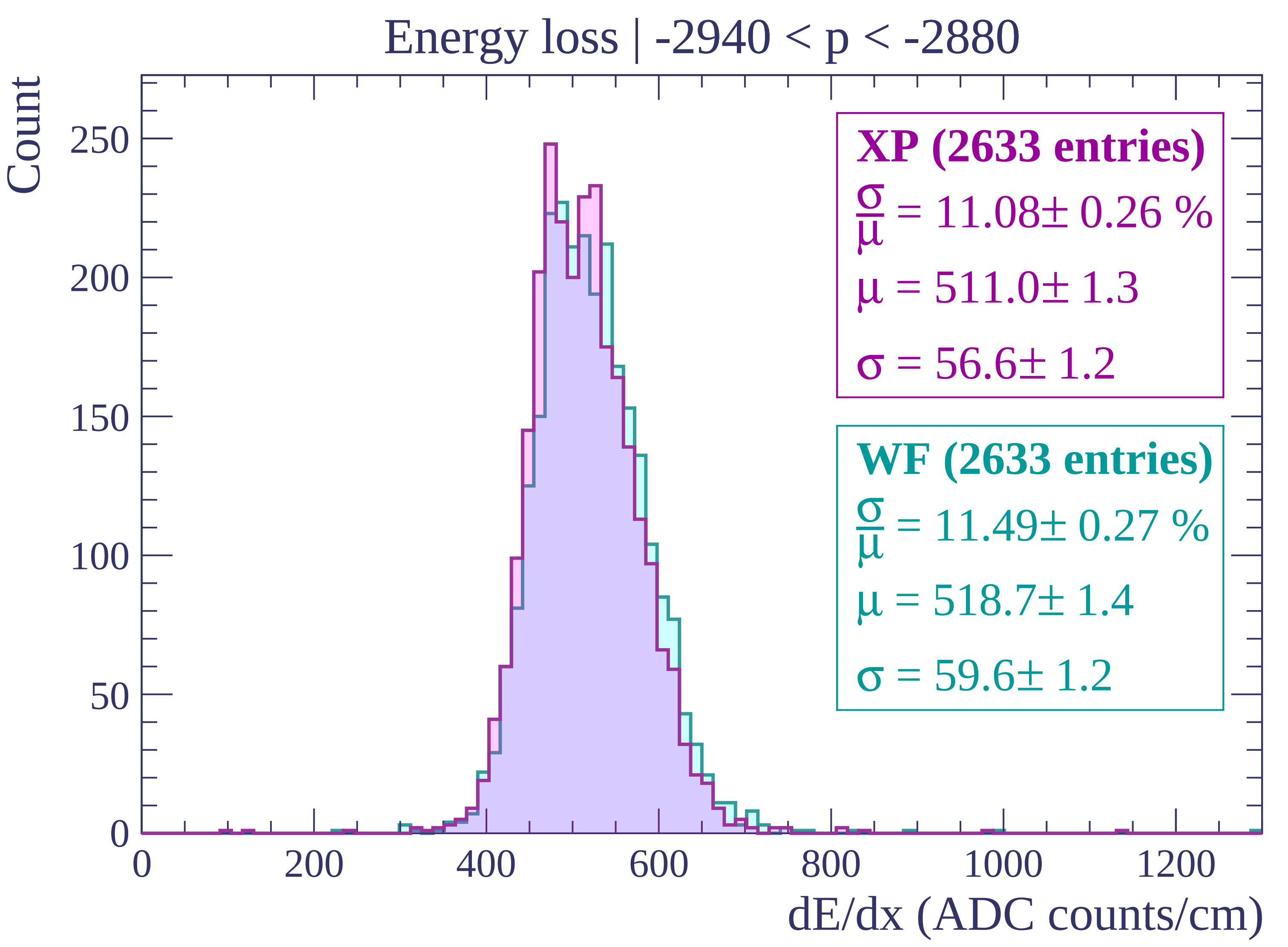
Y position
1000
800
600
400
200
0
−200
−400
−600
−800
−1000
−3000
−2000
−1000
0
1000
2000
3000
Momentum
3000
2500
2000
1500
1000
500
0



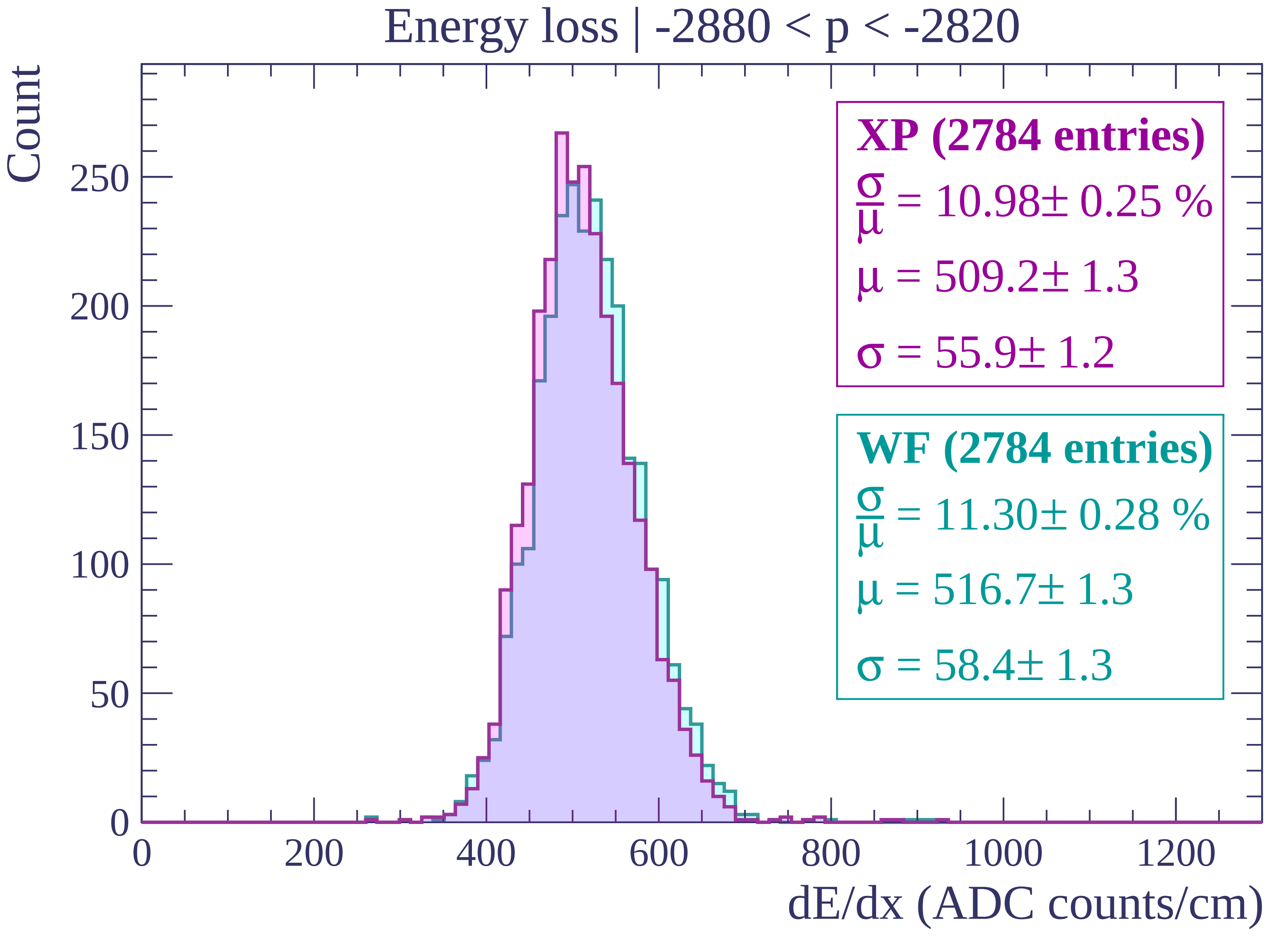
Y position
Y direction
1000
800
600
400
200
0
−200
−400
−600
−800
−1000
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
$\times10^3$
50
40
30
20
10
0


Pull e
Pull μ
10
8
6
4
2
0
−2
−4
−6
−8
−10
8000
7000
6000
5000
4000
3000
2000
1000
0

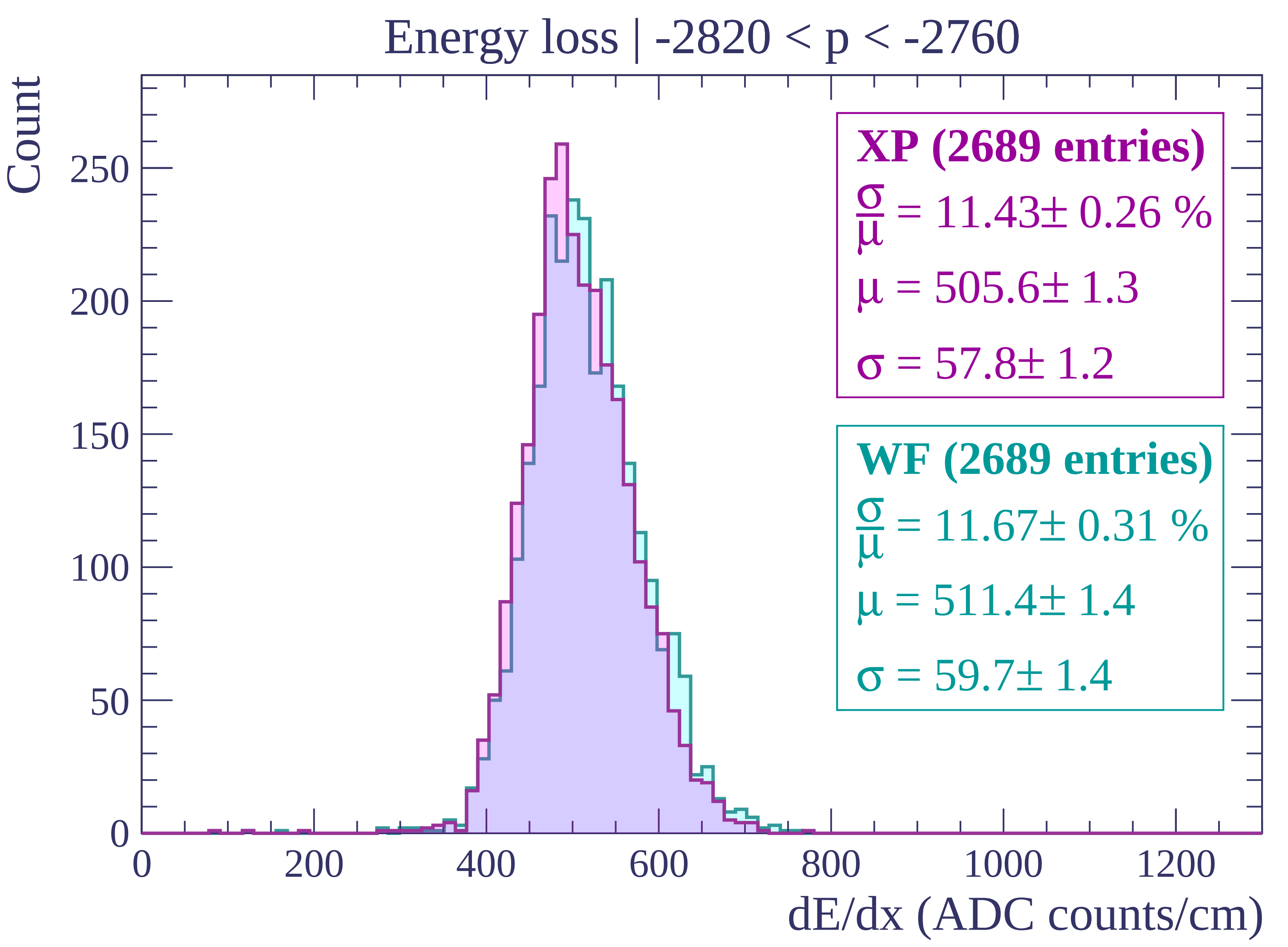

Energy loss | -3000 < p < -2940
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (1222 entries)
σ/μ = 10.79± 0.32 %
μ = 513.8± 1.9
σ = 55.4± 1.4
WF (1222 entries)
σ/μ = 10.65± 0.29 %
μ = 522.9± 1.9
σ = 55.7± 1.3



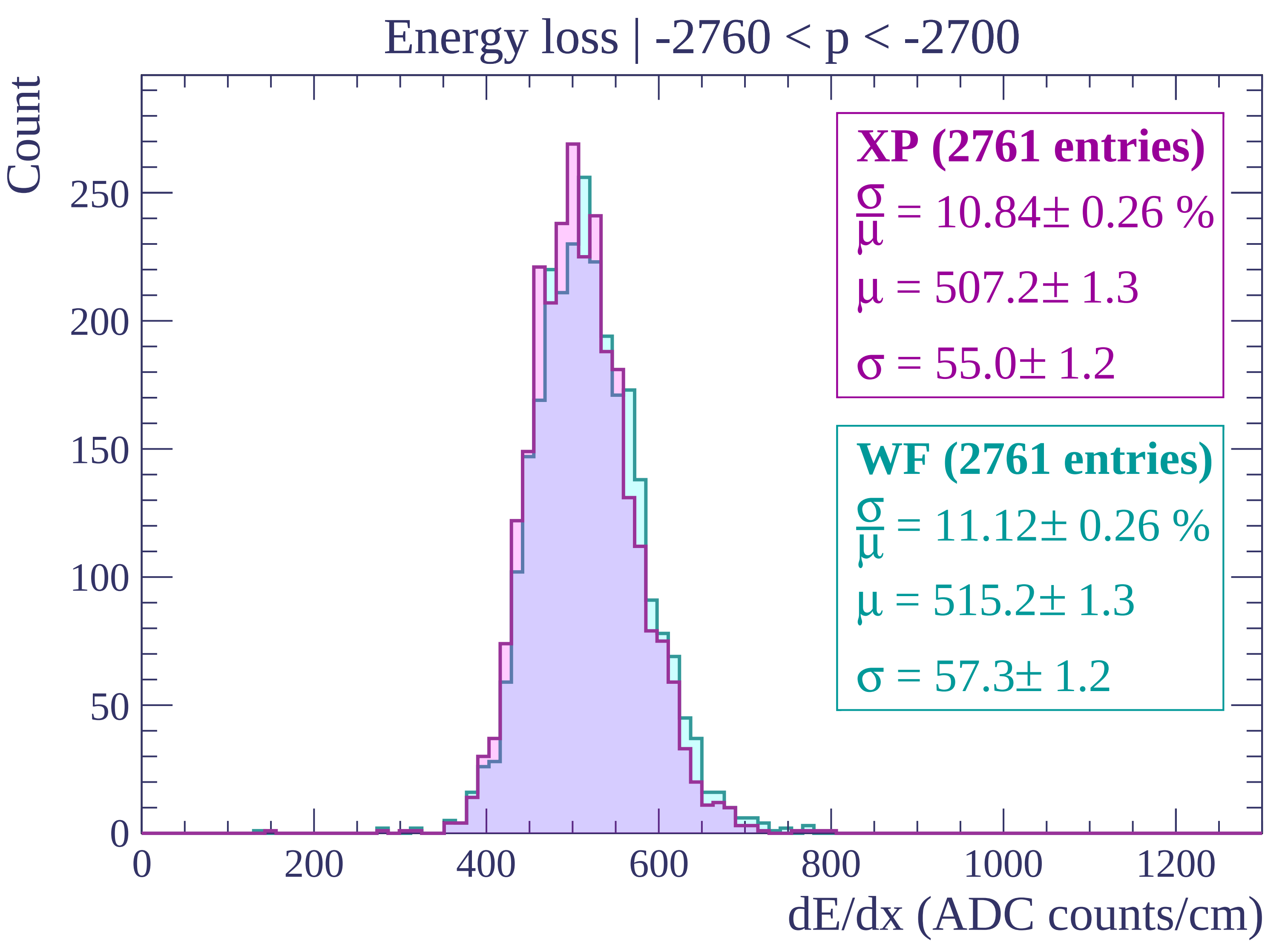
Energy loss | -2940 < p < -2880
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2633 entries)
σ/μ = 11.08± 0.26 %
μ = 511.0± 1.3
σ = 56.6± 1.2
WF (2633 entries)
σ/μ = 11.49± 0.27 %
μ = 518.7± 1.4
σ = 59.6± 1.2



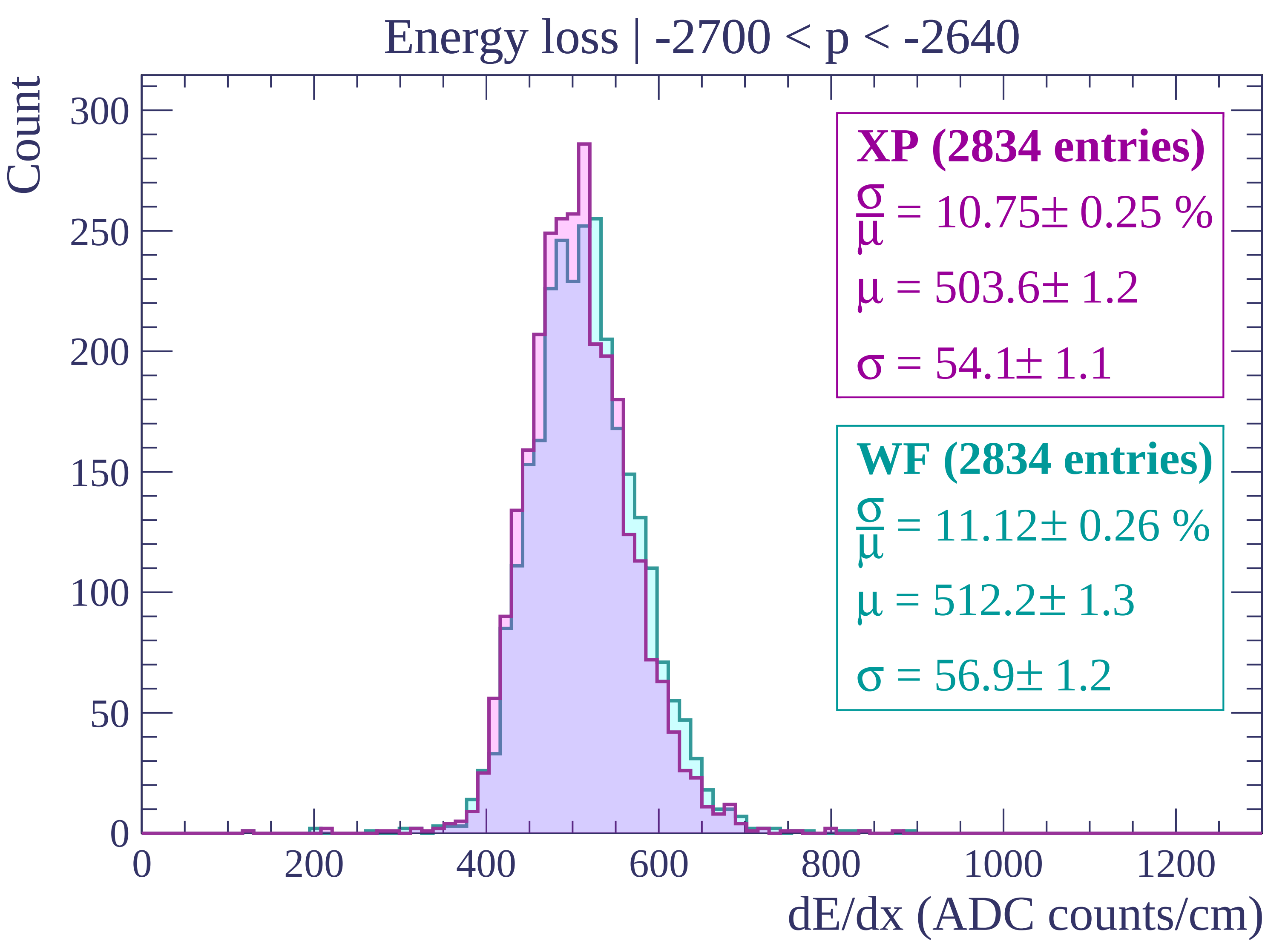
Energy loss | -2880 < p < -2820
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2784 entries)
$\frac{\sigma}{\mu}$ = 10.98± 0.25 %
μ = 509.2± 1.3
σ = 55.9± 1.2
WF (2784 entries)
$\frac{\sigma}{\mu}$ = 11.30± 0.28 %
μ = 516.7± 1.3
σ = 58.4± 1.3



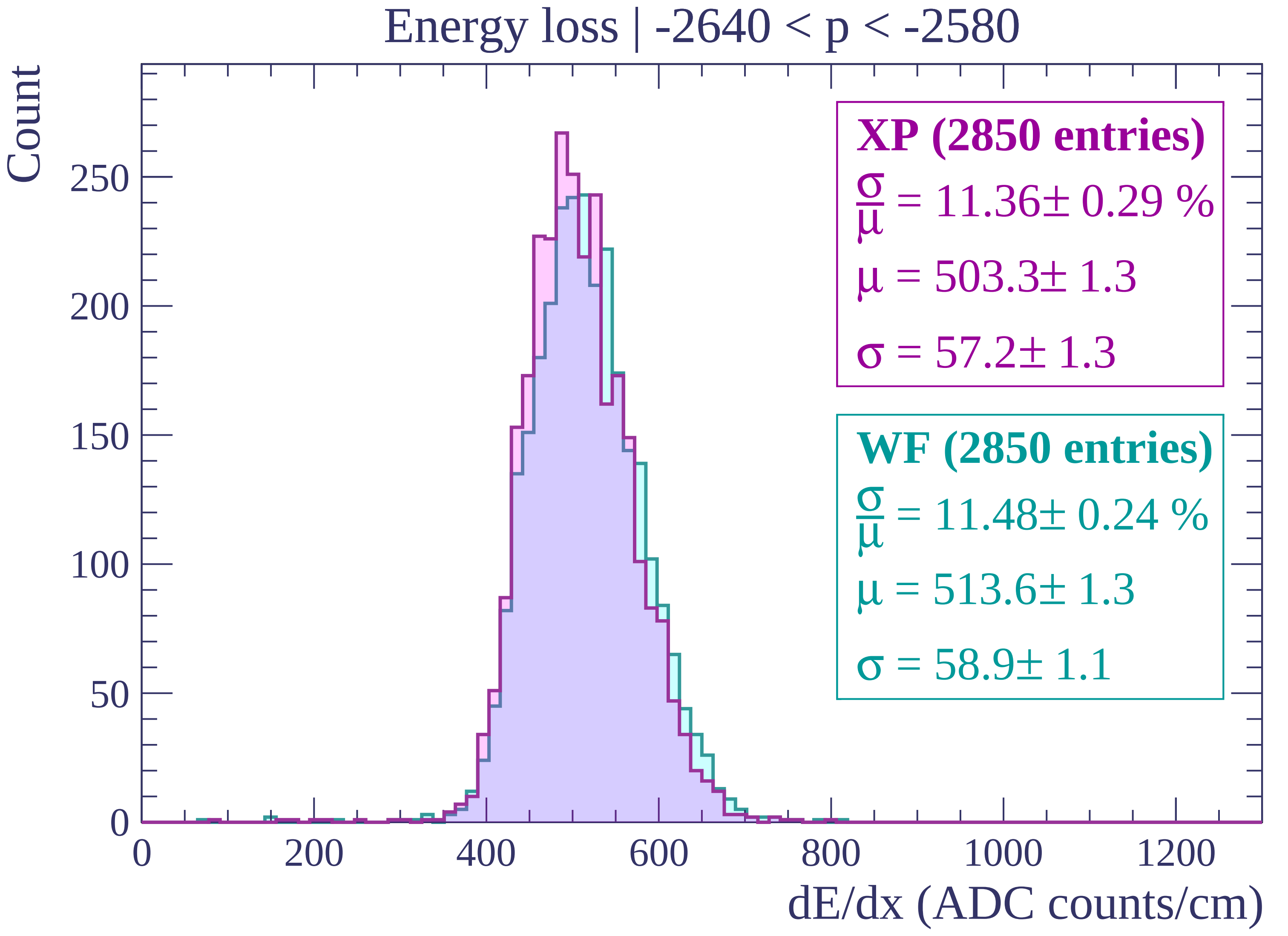
Energy loss | -2820 < p < -2760
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2689 entries)
σ/μ = 11.43± 0.26 %
μ = 505.6± 1.3
σ = 57.8± 1.2
WF (2689 entries)
σ/μ = 11.67± 0.31 %
μ = 511.4± 1.4
σ = 59.7± 1.4

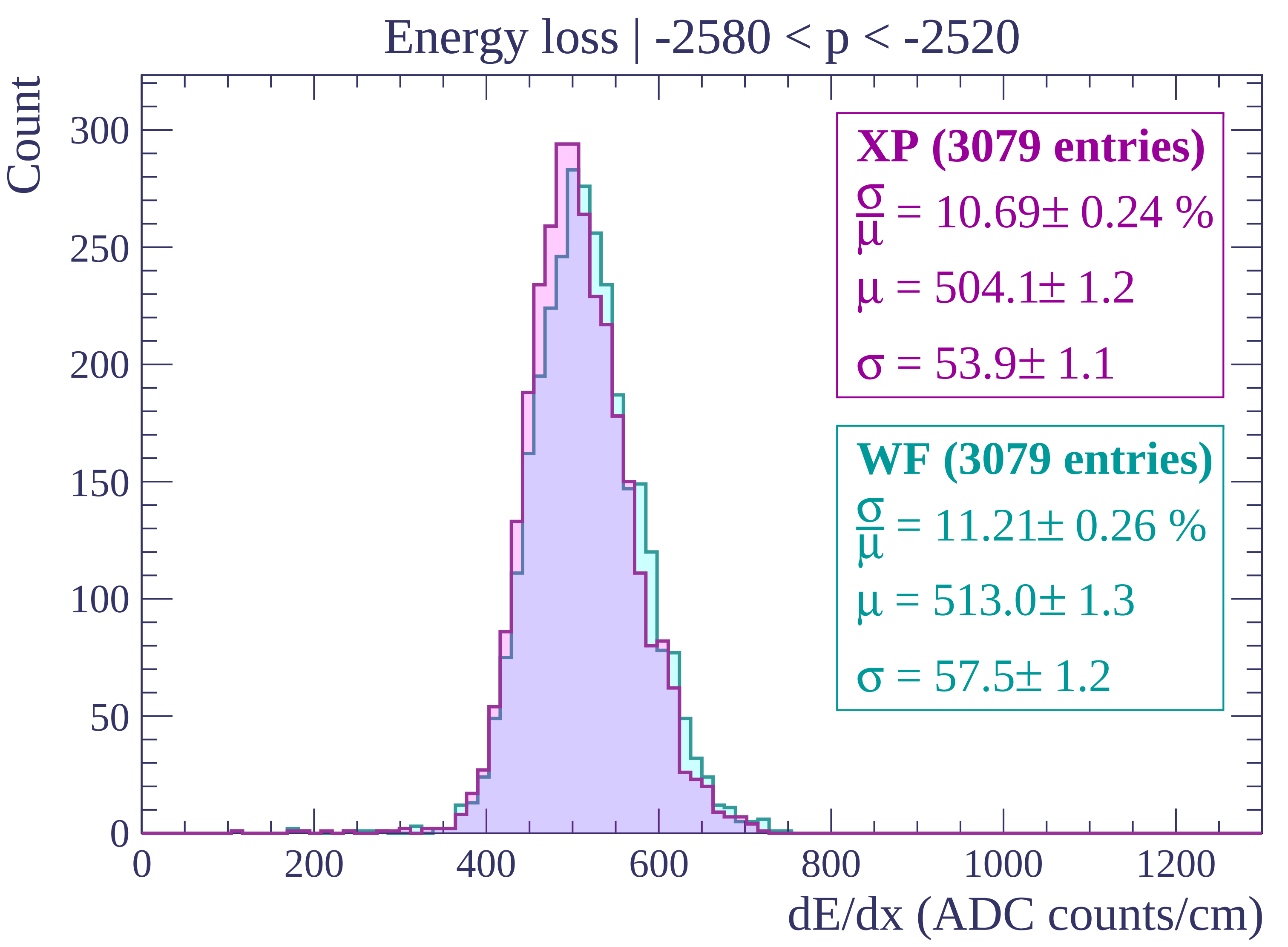


Energy loss | -2760 < p < -2700
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2761 entries)
σ/μ = 10.84± 0.26 %
μ = 507.2± 1.3
σ = 55.0± 1.2
WF (2761 entries)
σ/μ = 11.12± 0.26 %
μ = 515.2± 1.3
σ = 57.3± 1.2

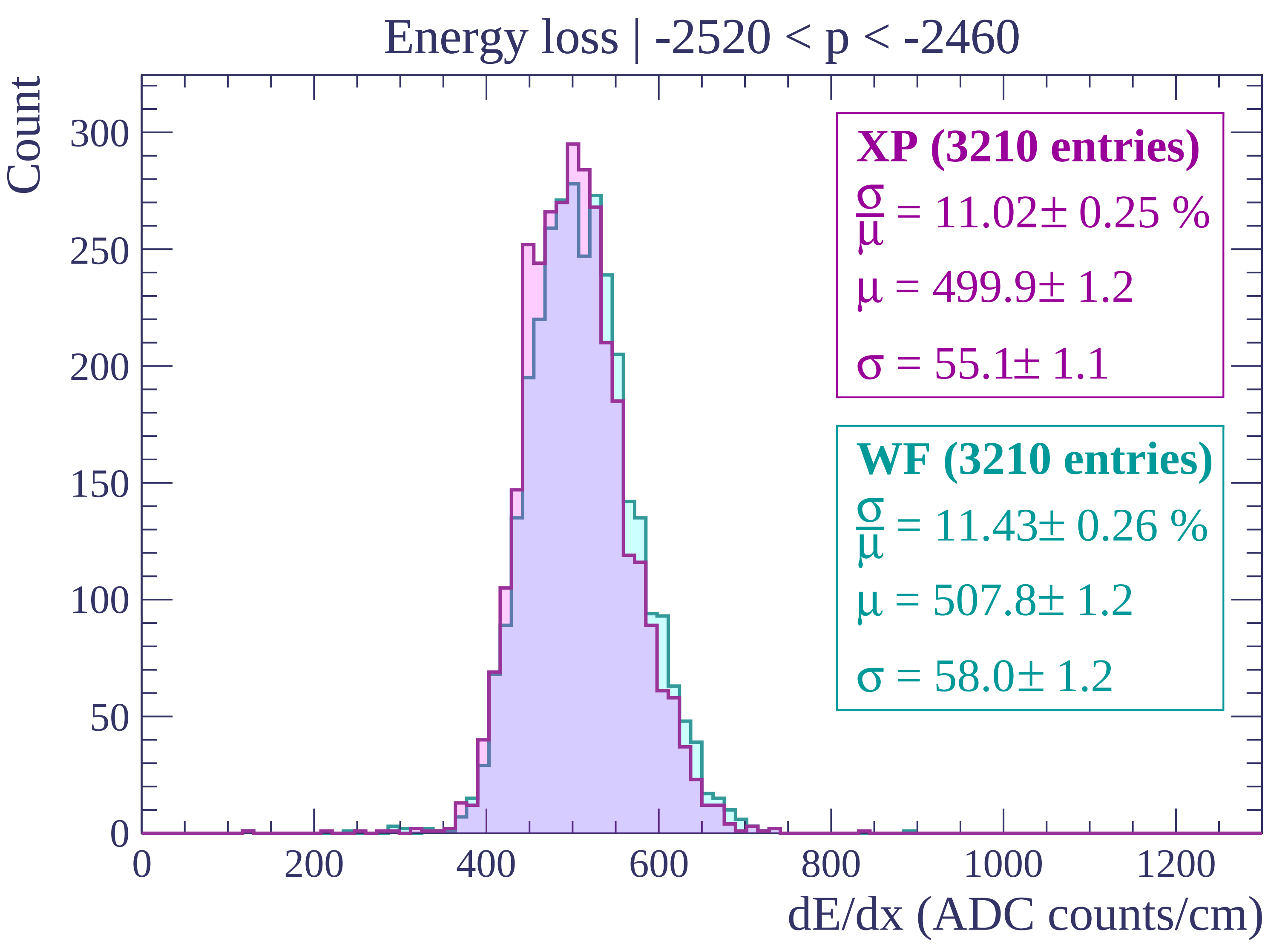


Energy loss | -2700 < p < -2640
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2834 entries)
σ/μ = 10.75± 0.25 %
μ = 503.6± 1.2
σ = 54.1± 1.1
WF (2834 entries)
σ/μ = 11.12± 0.26 %
μ = 512.2± 1.3
σ = 56.9± 1.2

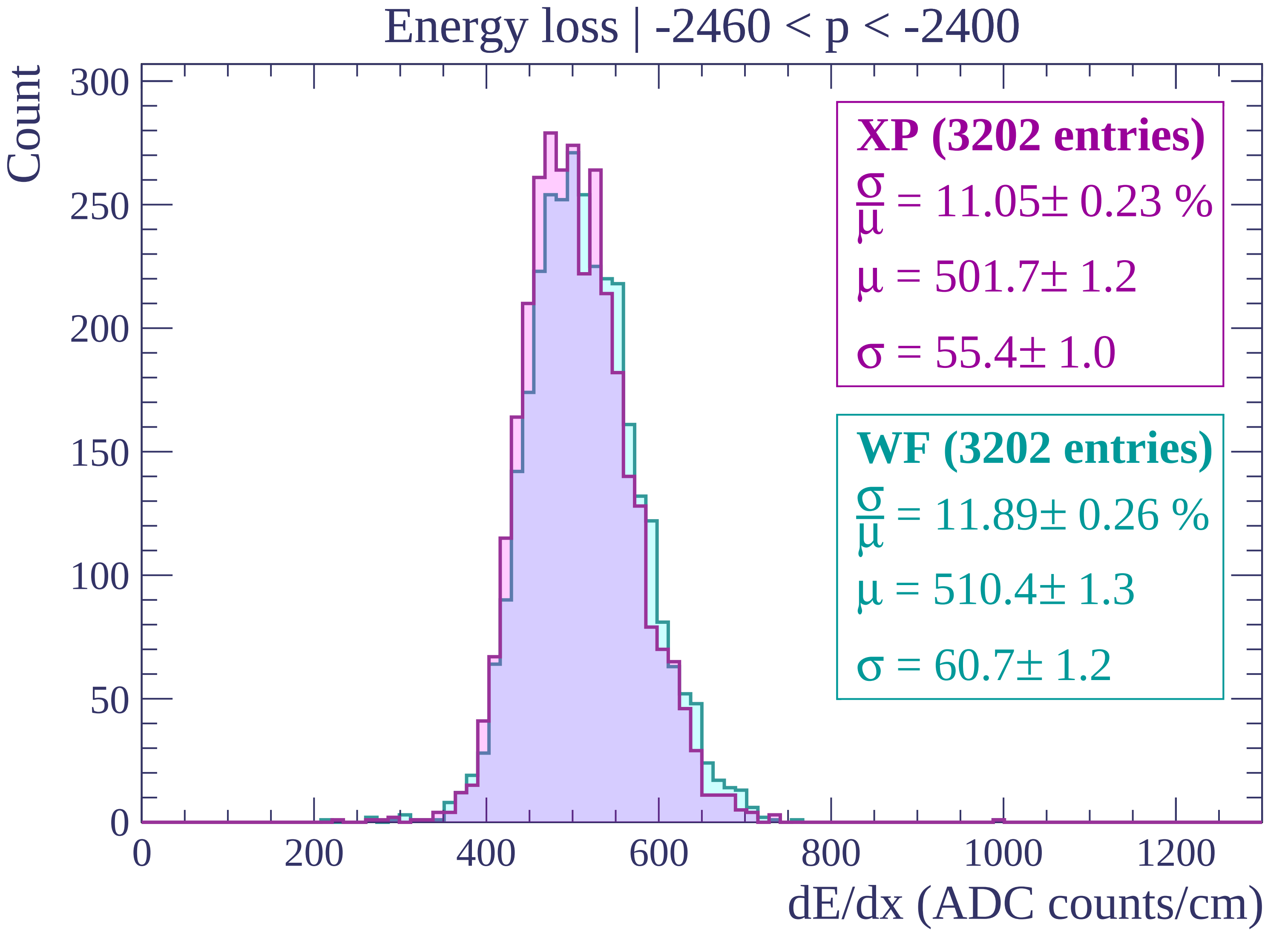

Energy loss | -2640 < p < -2580
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2850 entries)
σ/μ = 11.36± 0.29 %
μ = 503.3± 1.3
σ = 57.2± 1.3
WF (2850 entries)
σ/μ = 11.48± 0.24 %
μ = 513.6± 1.3
σ = 58.9± 1.1



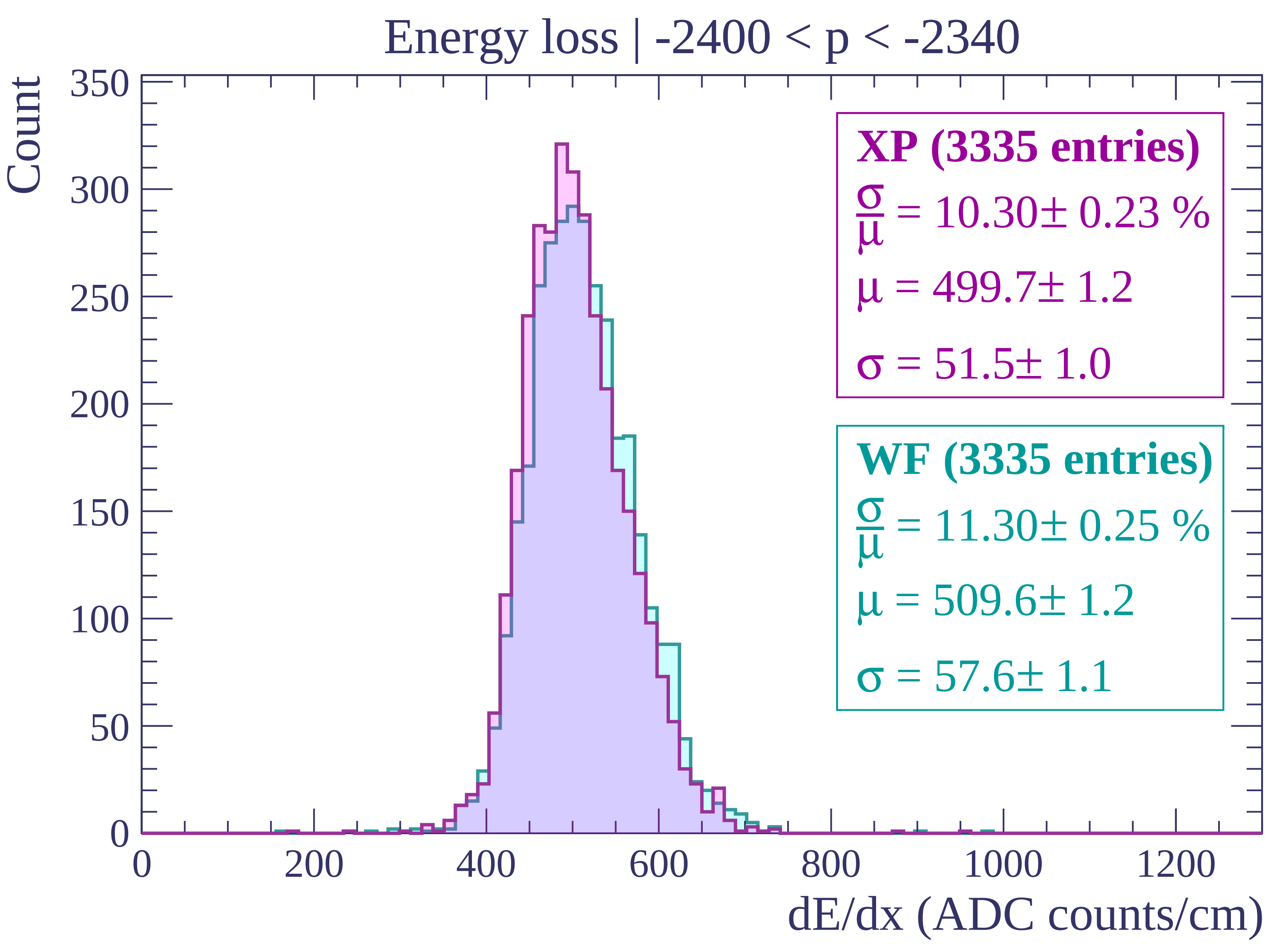
Energy loss | -2580 < p < -2520
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
XP (3079 entries)
σ/μ = 10.69± 0.24 %
μ = 504.1± 1.2
σ = 53.9± 1.1
WF (3079 entries)
σ/μ = 11.21± 0.26 %
μ = 513.0± 1.3
σ = 57.5± 1.2

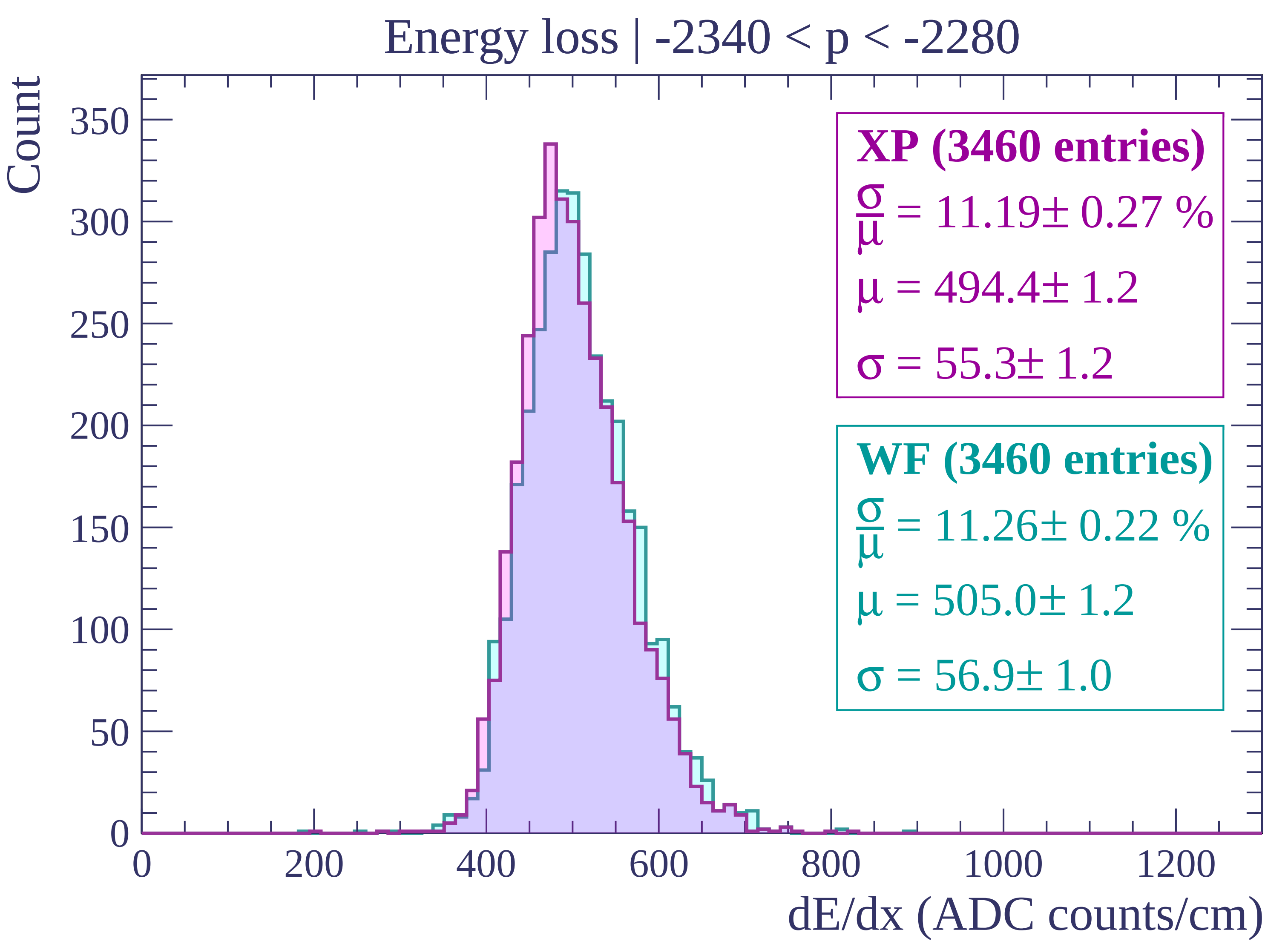


Energy loss | -2520 < p < -2460
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (3210 entries)
σ/μ = 11.02± 0.25 %
μ = 499.9± 1.2
σ = 55.1± 1.1
WF (3210 entries)
σ/μ = 11.43± 0.26 %
μ = 507.8± 1.2
σ = 58.0± 1.2

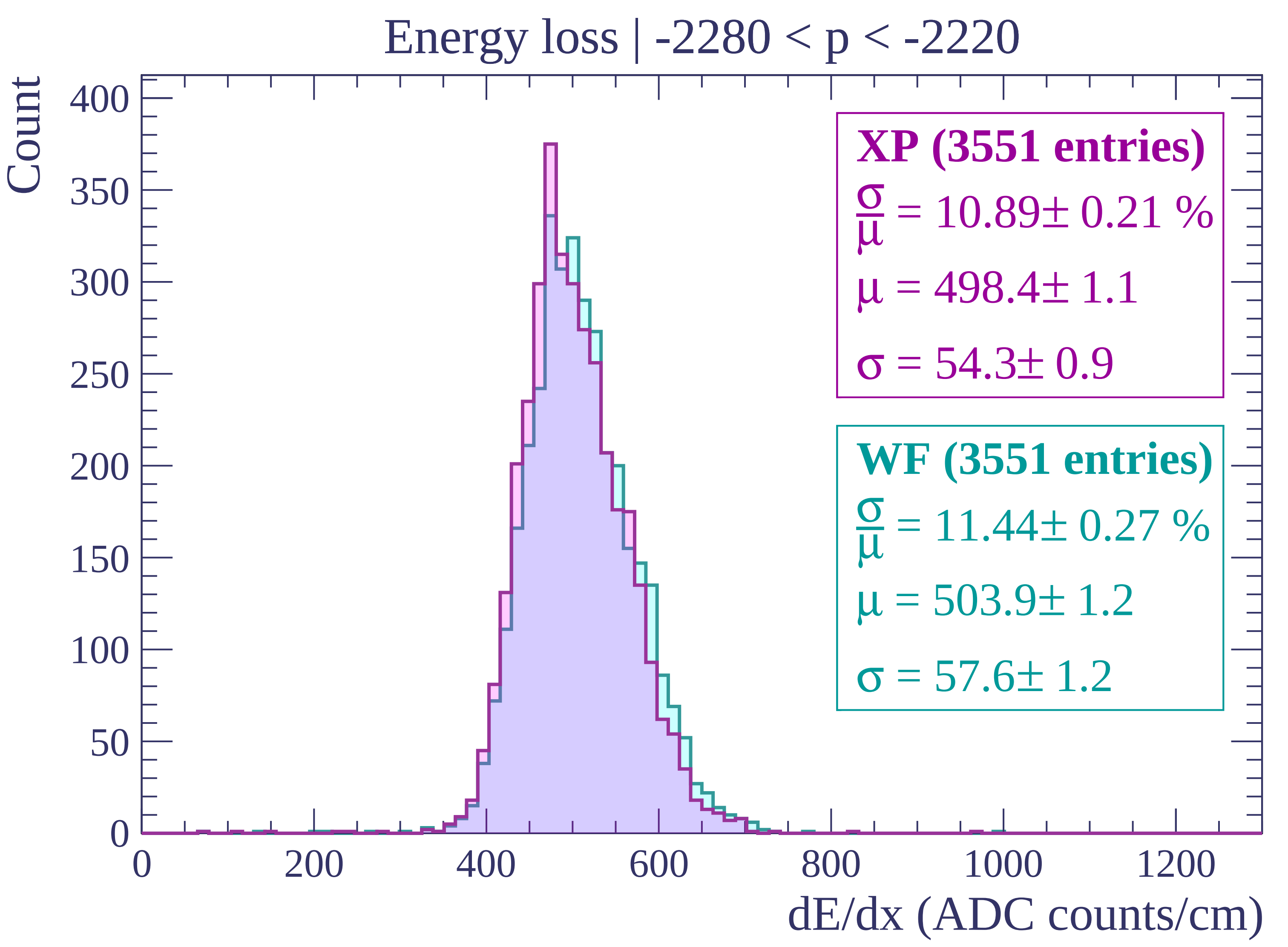


Energy loss | -2460 < p < -2400
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3202 entries)
σ/μ = 11.05± 0.23 %
μ = 501.7± 1.2
σ = 55.4± 1.0
WF (3202 entries)
σ/μ = 11.89± 0.26 %
μ = 510.4± 1.3
σ = 60.7± 1.2

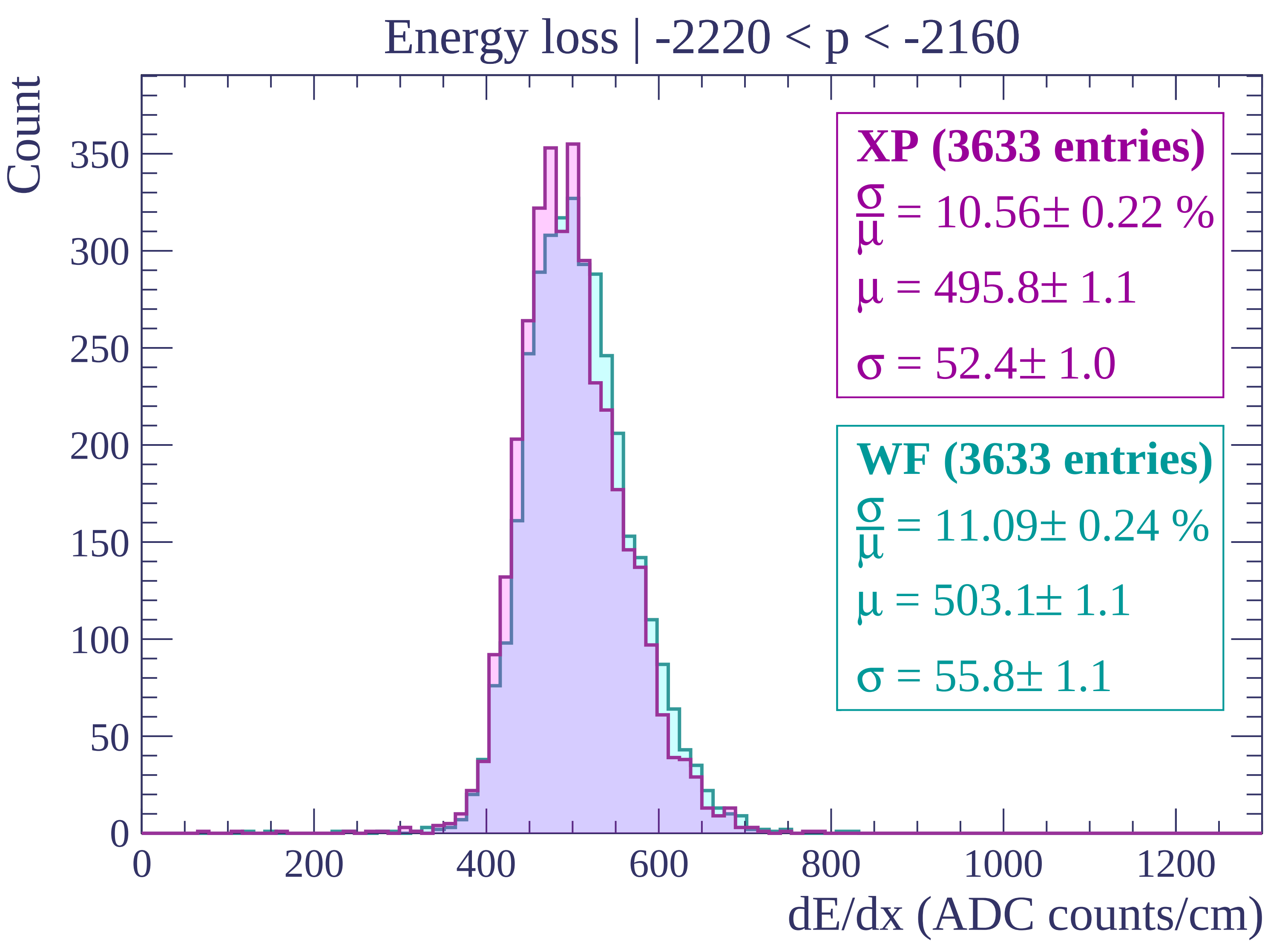

Energy loss | -2400 < p < -2340
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3335 entries)
σ/μ = 10.30± 0.23 %
μ = 499.7± 1.2
σ = 51.5± 1.0
WF (3335 entries)
σ/μ = 11.30± 0.25 %
μ = 509.6± 1.2
σ = 57.6± 1.1

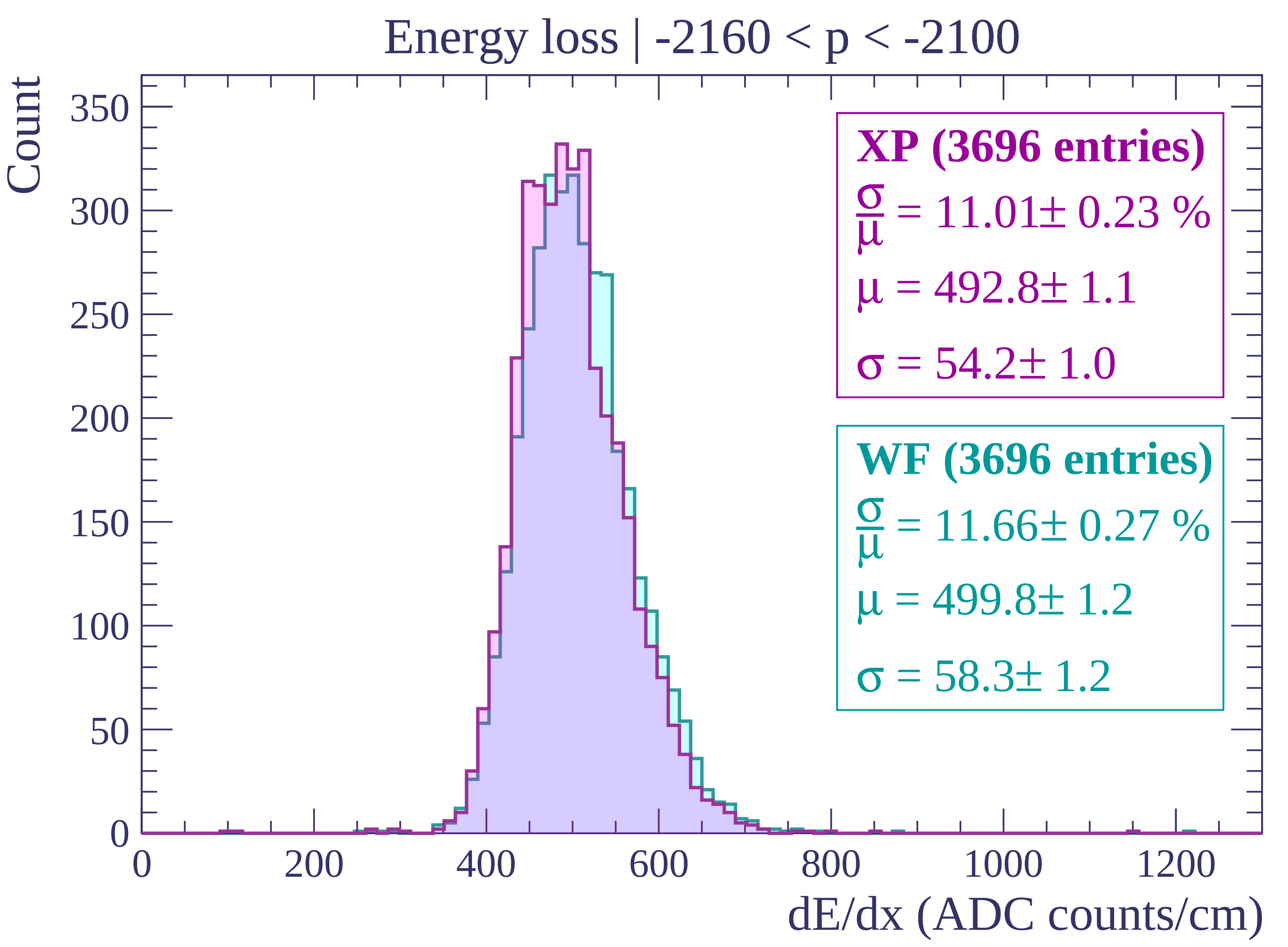


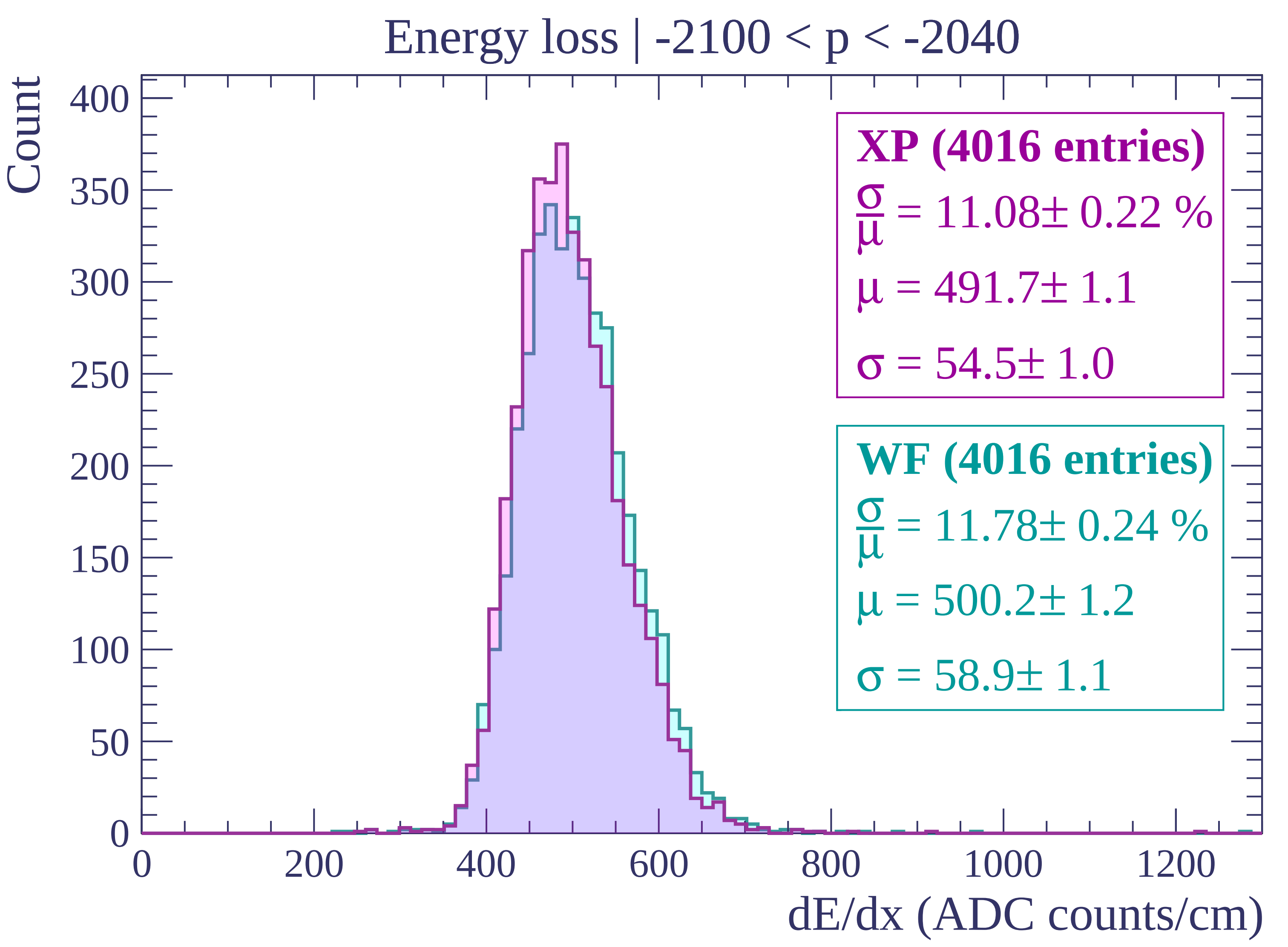
Energy loss | -2340 < p < -2280
Count
XP (3460 entries)
σ/μ = 11.19± 0.27 %
μ = 494.4± 1.2
σ = 55.3± 1.2
WF (3460 entries)
σ/μ = 11.26± 0.22 %
μ = 505.0± 1.2
σ = 56.9± 1.0
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200



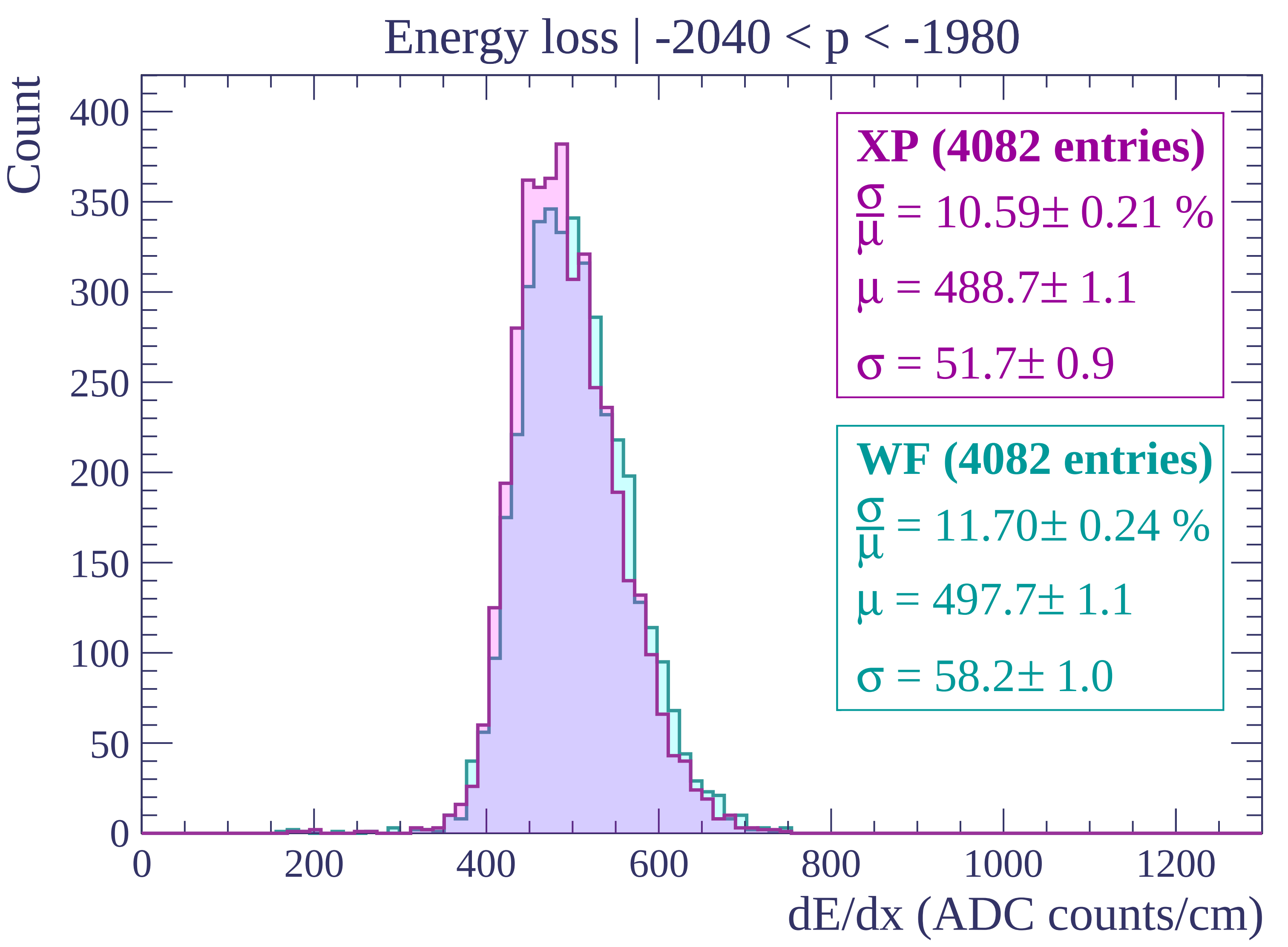
Energy loss | -2280 < p < -2220
Count
dE/dx (ADC counts/cm)
XP (3551 entries)
σ/μ = 10.89± 0.21 %
μ = 498.4± 1.1
σ = 54.3± 0.9
WF (3551 entries)
σ/μ = 11.44± 0.27 %
μ = 503.9± 1.2
σ = 57.6± 1.2

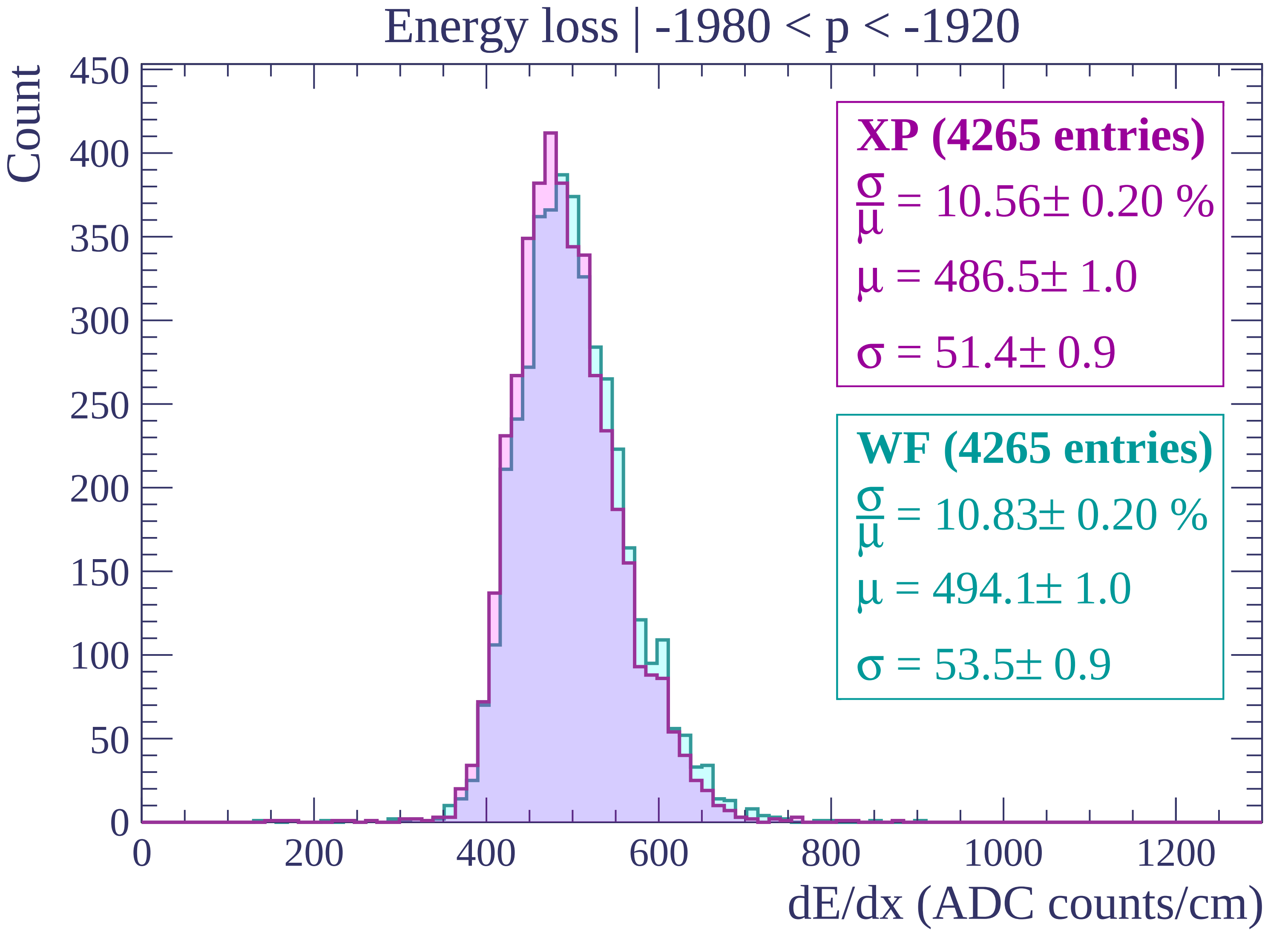

Energy loss | -2220 < p < -2160
Count
dE/dx (ADC counts/cm)
XP (3633 entries)
σ/μ = 10.56± 0.22 %
μ = 495.8± 1.1
σ = 52.4± 1.0
WF (3633 entries)
σ/μ = 11.09± 0.24 %
μ = 503.1± 1.1
σ = 55.8± 1.1


Energy loss | -2160 < p < -2100
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
XP (3696 entries)
σ/μ = 11.01± 0.23 %
μ = 492.8± 1.1
σ = 54.2± 1.0
WF (3696 entries)
σ/μ = 11.66± 0.27 %
μ = 499.8± 1.2
σ = 58.3± 1.2

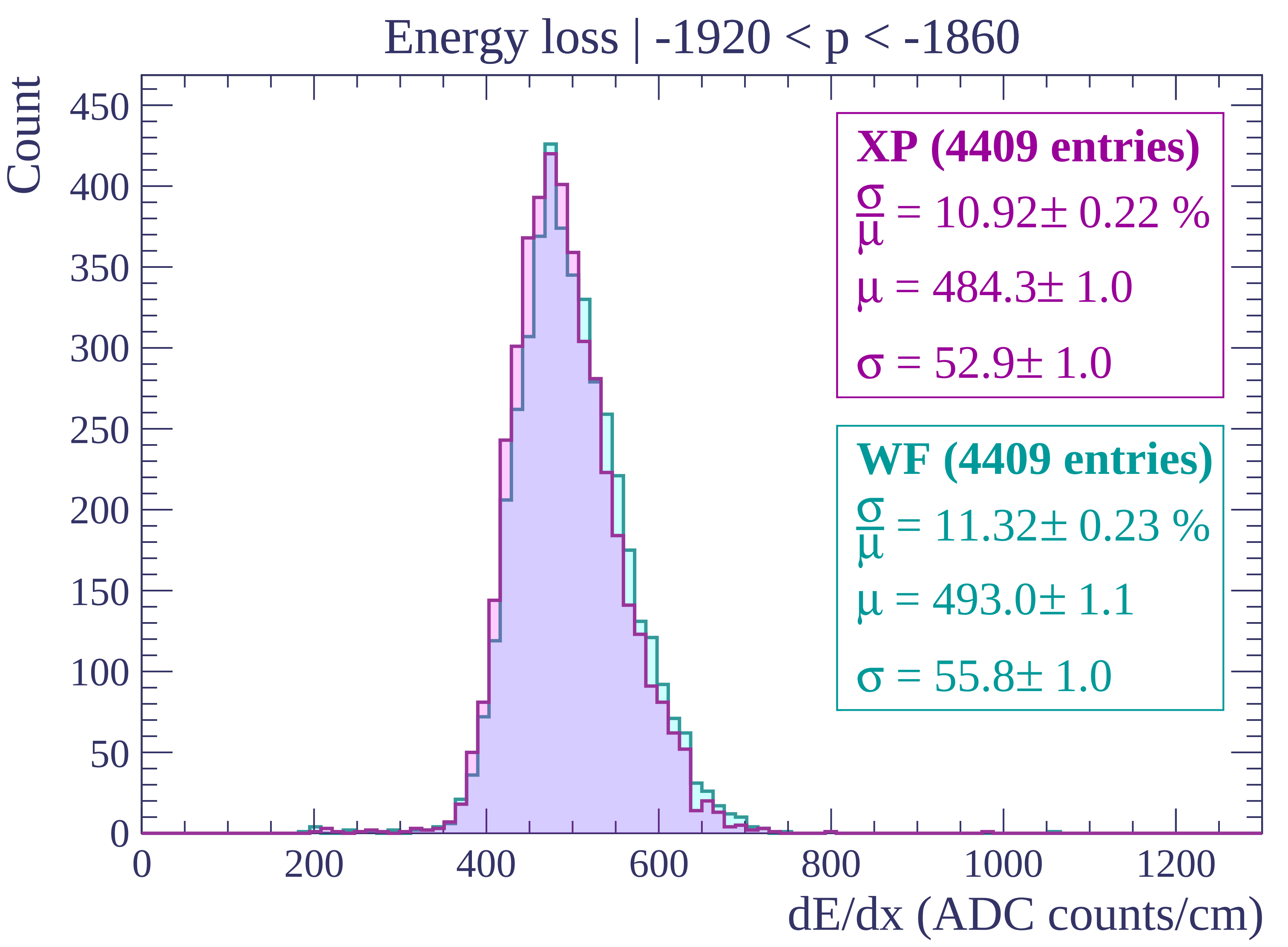


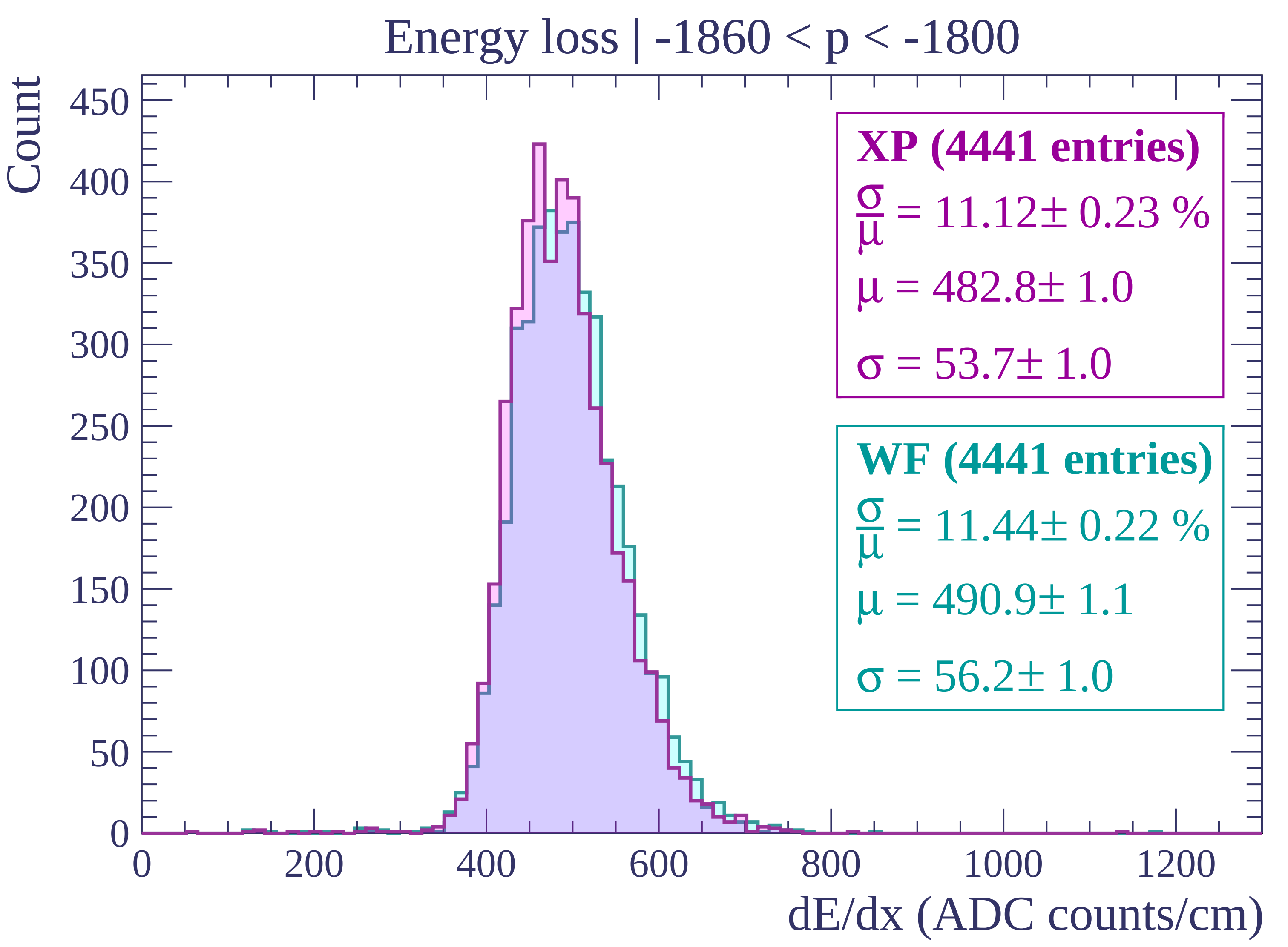
Energy loss | -2100 < p < -2040
Count
dE/dx (ADC counts/cm)
XP (4016 entries)
σ/μ = 11.08± 0.22 %
μ = 491.7± 1.1
σ = 54.5± 1.0
WF (4016 entries)
σ/μ = 11.78± 0.24 %
μ = 500.2± 1.2
σ = 58.9± 1.1

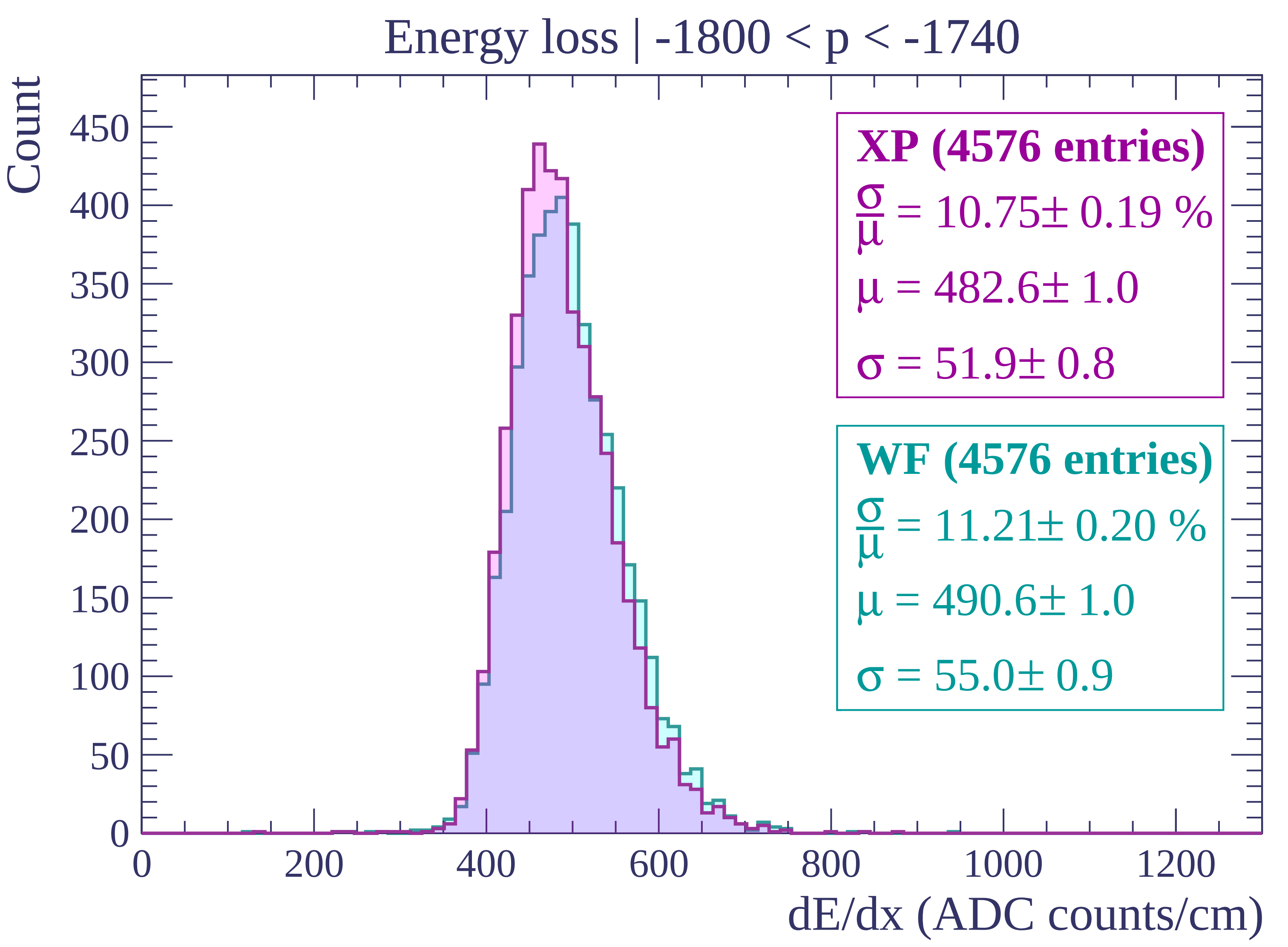

Energy loss | -2040 < p < -1980
Count
dE/dx (ADC counts/cm)
XP (4082 entries)
σ/μ = 10.59± 0.21 %
μ = 488.7± 1.1
σ = 51.7± 0.9
WF (4082 entries)
σ/μ = 11.70± 0.24 %
μ = 497.7± 1.1
σ = 58.2± 1.0
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200



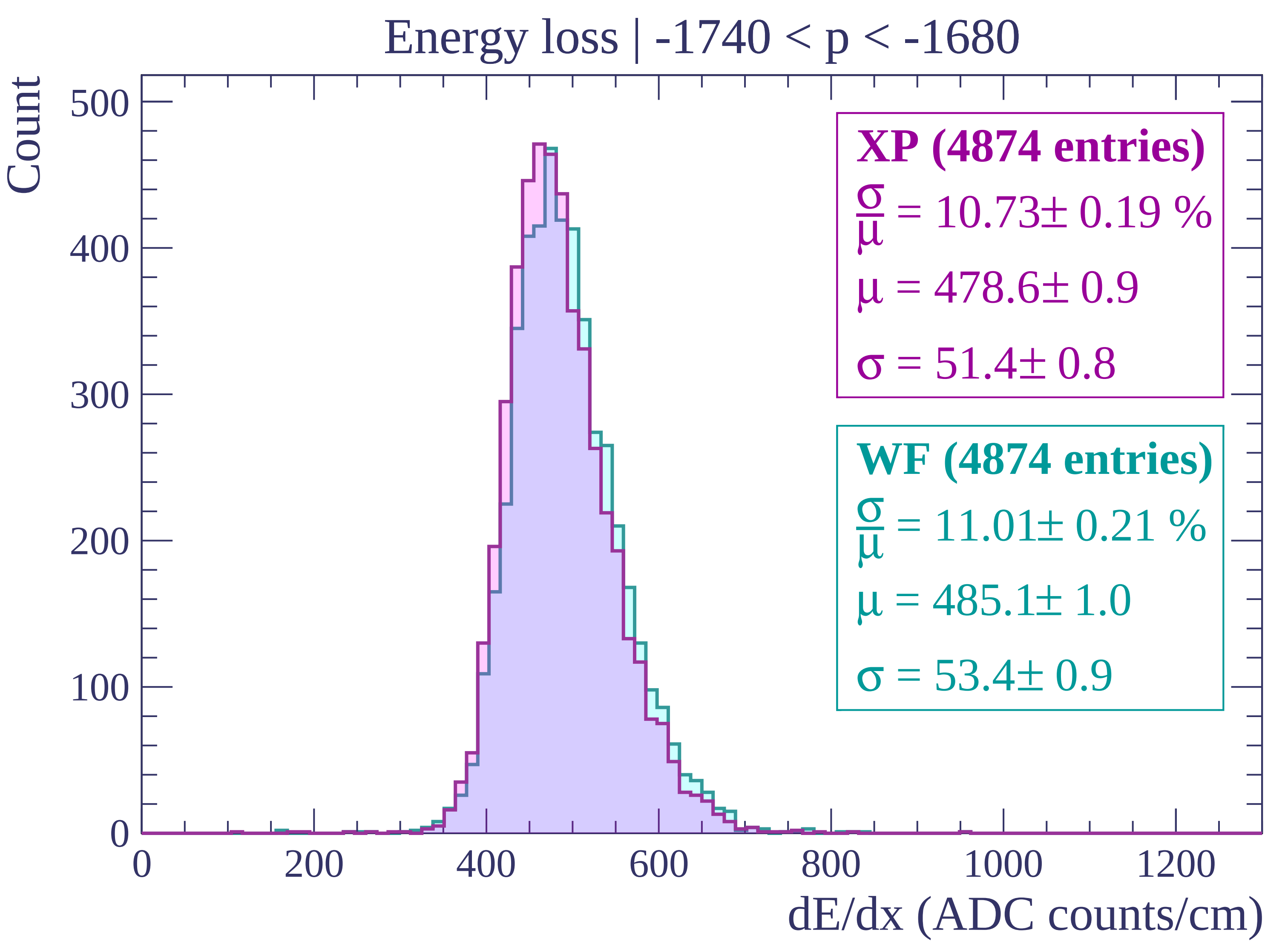
Energy loss | -1980 < p < -1920
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (4265 entries)
σ/μ = 10.56± 0.20 %
μ = 486.5± 1.0
σ = 51.4± 0.9
WF (4265 entries)
σ/μ = 10.83± 0.20 %
μ = 494.1± 1.0
σ = 53.5± 0.9


Energy loss | -1920 < p < -1860
Count
dE/dx (ADC counts/cm)
XP (4409 entries)
σ/μ = 10.92± 0.22 %
μ = 484.3± 1.0
σ = 52.9± 1.0
WF (4409 entries)
σ/μ = 11.32± 0.23 %
μ = 493.0± 1.1
σ = 55.8± 1.0
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200

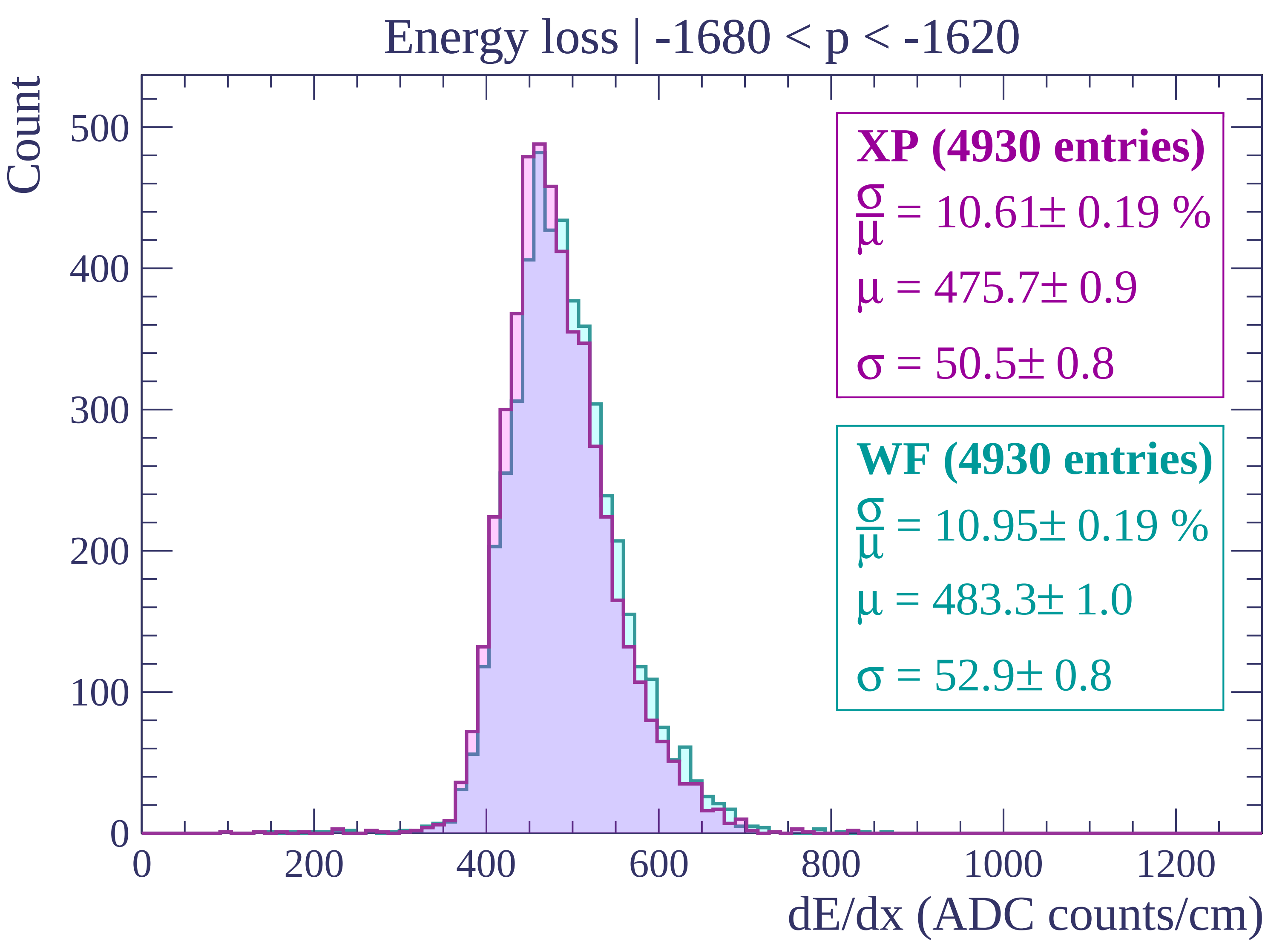

Energy loss | -1860 < p < -1800
Count
dE/dx (ADC counts/cm)
XP (4441 entries)
$\frac{\sigma}{\mu}$ = 11.12± 0.23 %
μ = 482.8± 1.0
σ = 53.7± 1.0
WF (4441 entries)
$\frac{\sigma}{\mu}$ = 11.44± 0.22 %
μ = 490.9± 1.1
σ = 56.2± 1.0

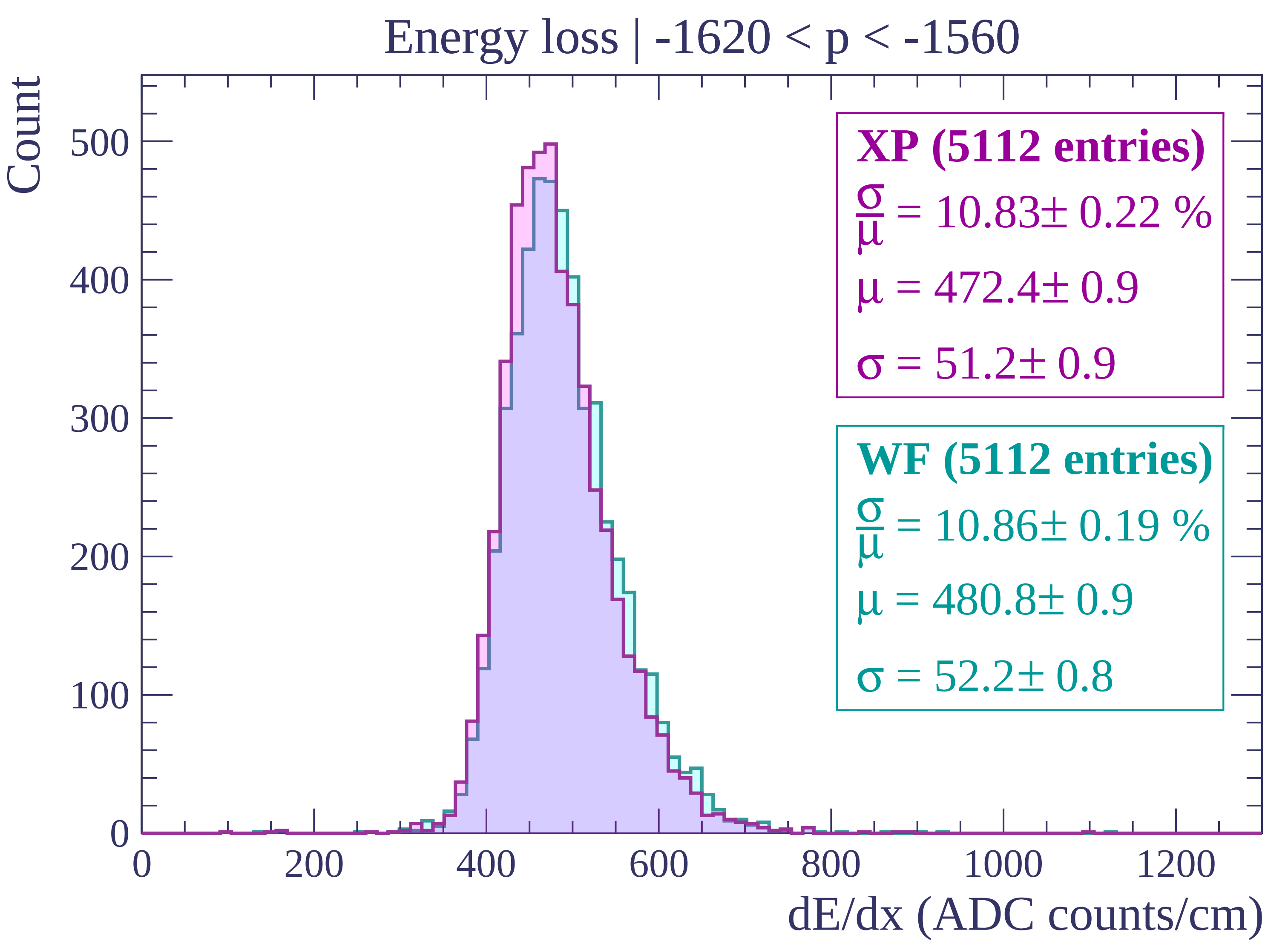


Energy loss | -1800 < p < -1740
Count
dE/dx (ADC counts/cm)
XP (4576 entries)
σ/μ = 10.75± 0.19 %
μ = 482.6± 1.0
σ = 51.9± 0.8
WF (4576 entries)
σ/μ = 11.21± 0.20 %
μ = 490.6± 1.0
σ = 55.0± 0.9
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200

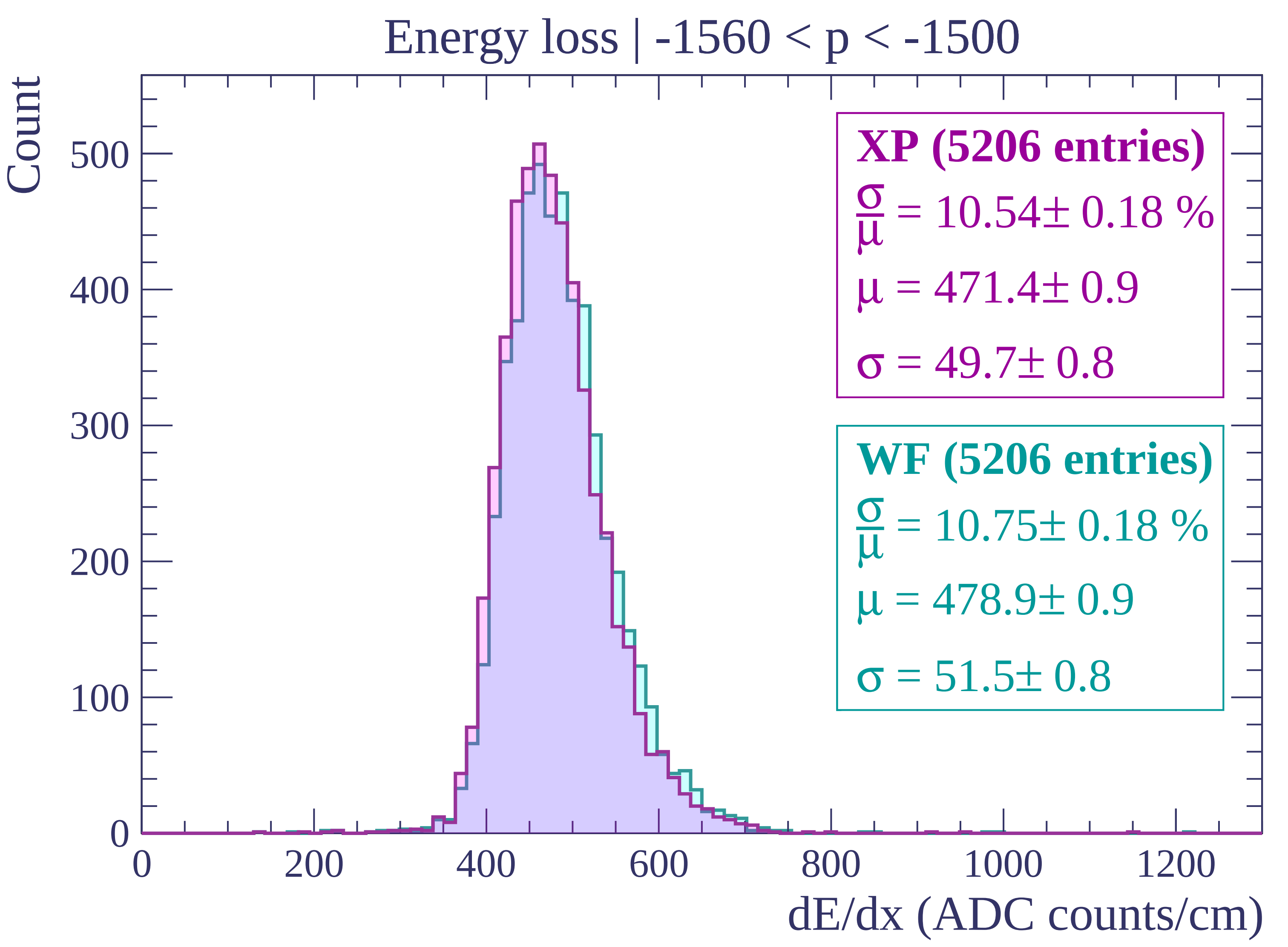


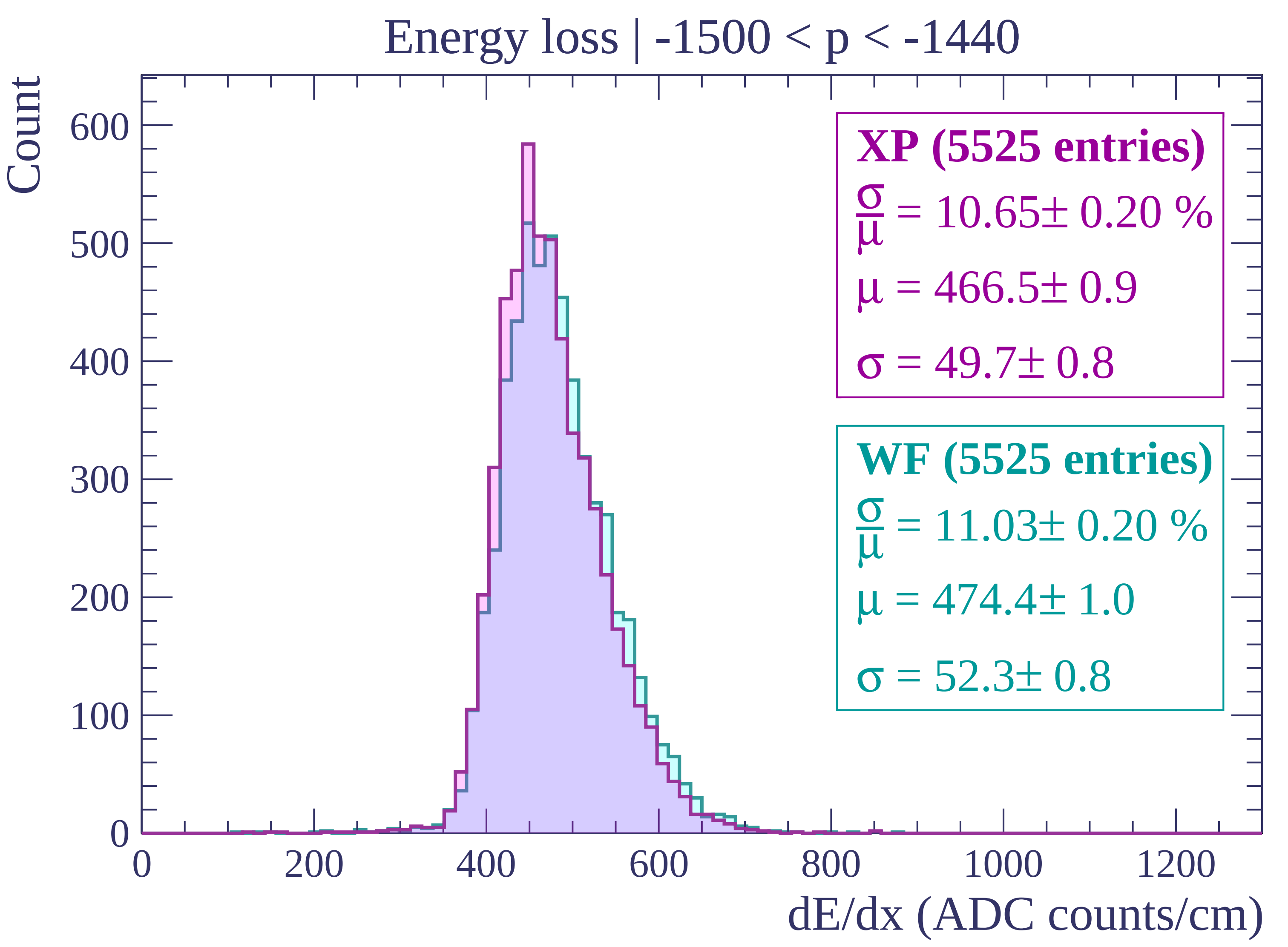
Energy loss | -1740 < p < -1680
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4874 entries)
σ/μ = 10.73± 0.19 %
μ = 478.6± 0.9
σ = 51.4± 0.8
WF (4874 entries)
σ/μ = 11.01± 0.21 %
μ = 485.1± 1.0
σ = 53.4± 0.9

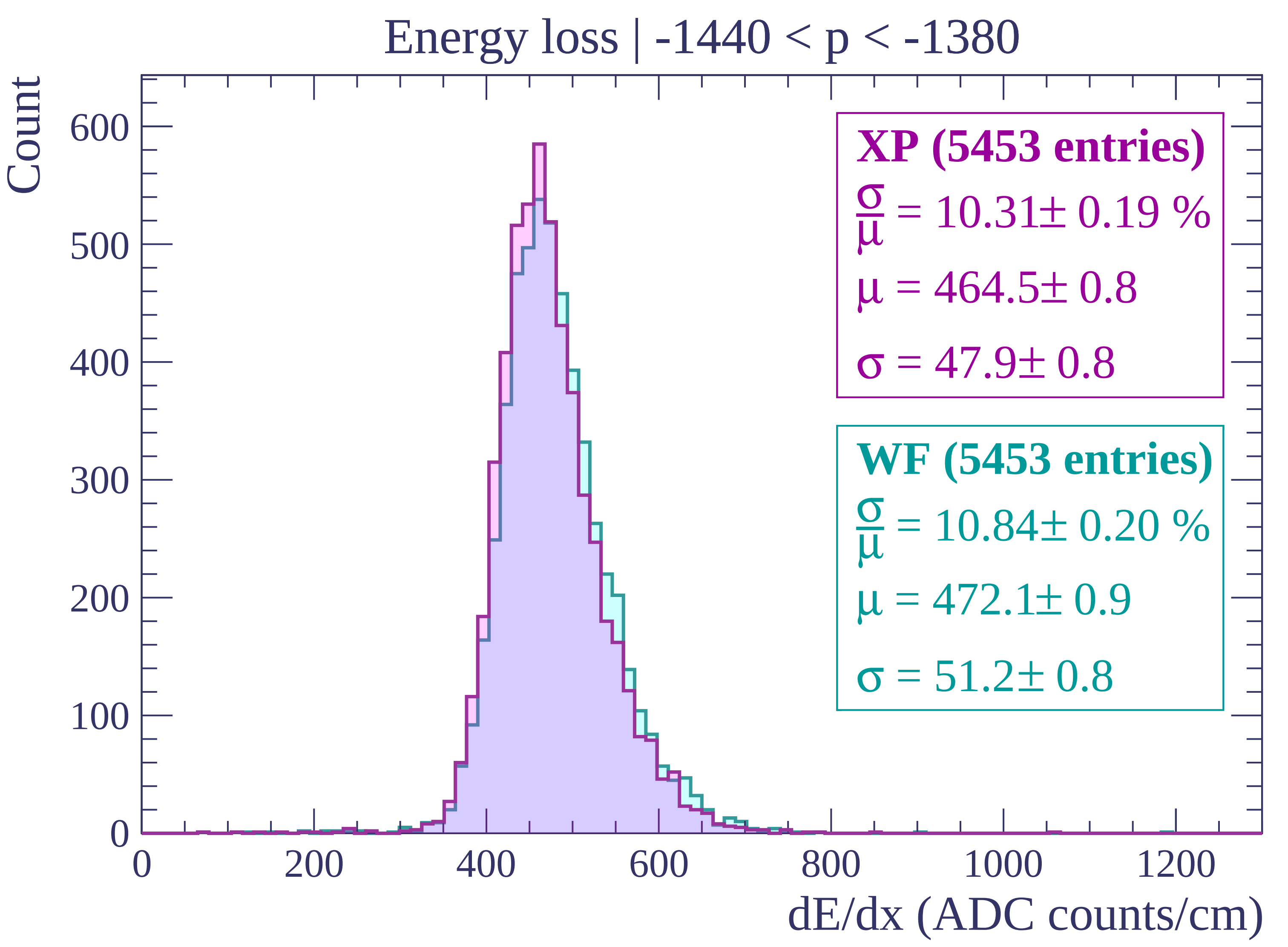

Energy loss | -1680 < p < -1620
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (4930 entries)
$\frac{\sigma}{\mu}$ = 10.61± 0.19 %
μ = 475.7± 0.9
σ = 50.5± 0.8
WF (4930 entries)
$\frac{\sigma}{\mu}$ = 10.95± 0.19 %
μ = 483.3± 1.0
σ = 52.9± 0.8

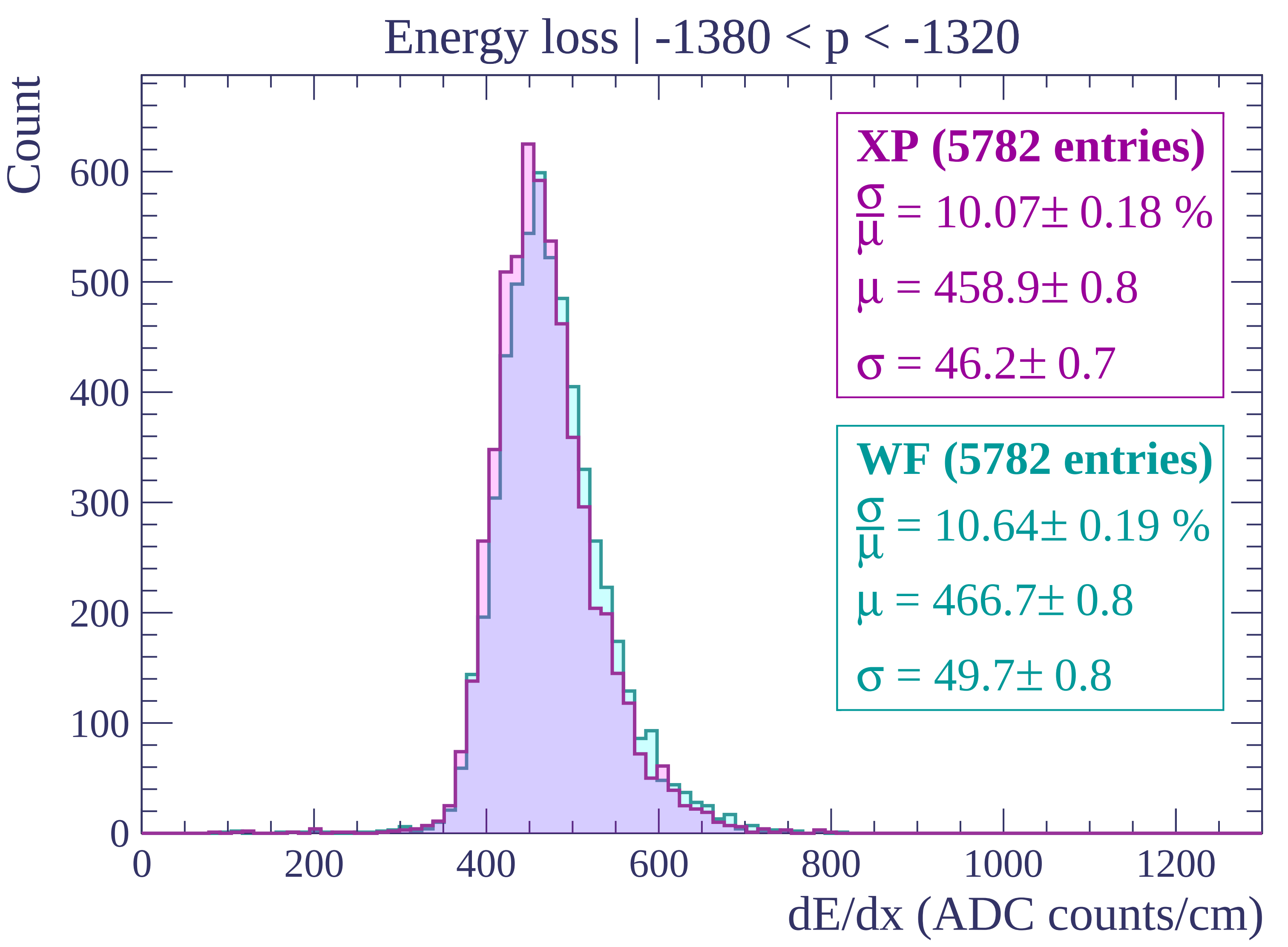


Energy loss | -1620 < p < -1560
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5112 entries)
$\frac{\sigma}{\mu}$ = 10.83± 0.22 %
μ = 472.4± 0.9
σ = 51.2± 0.9
WF (5112 entries)
$\frac{\sigma}{\mu}$ = 10.86± 0.19 %
μ = 480.8± 0.9
σ = 52.2± 0.8

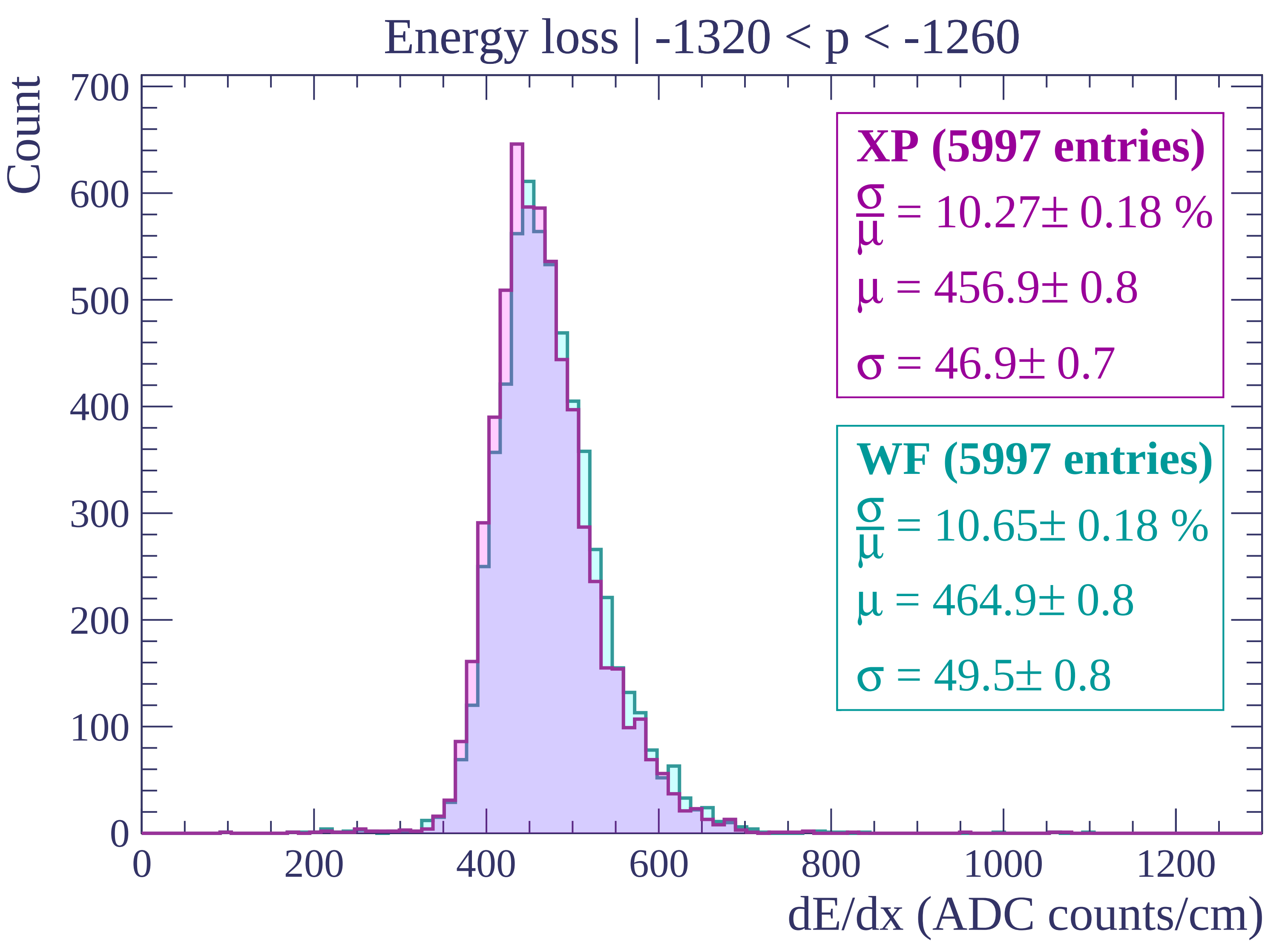

Energy loss | -1560 < p < -1500
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (5206 entries)
σ/μ = 10.54± 0.18 %
μ = 471.4± 0.9
σ = 49.7± 0.8
WF (5206 entries)
σ/μ = 10.75± 0.18 %
μ = 478.9± 0.9
σ = 51.5± 0.8


Energy loss | -1500 < p < -1440
Count
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5525 entries)
$\frac{\sigma}{\mu}$ = 10.65± 0.20 %
μ = 466.5± 0.9
σ = 49.7± 0.8
WF (5525 entries)
$\frac{\sigma}{\mu}$ = 11.03± 0.20 %
μ = 474.4± 1.0
σ = 52.3± 0.8

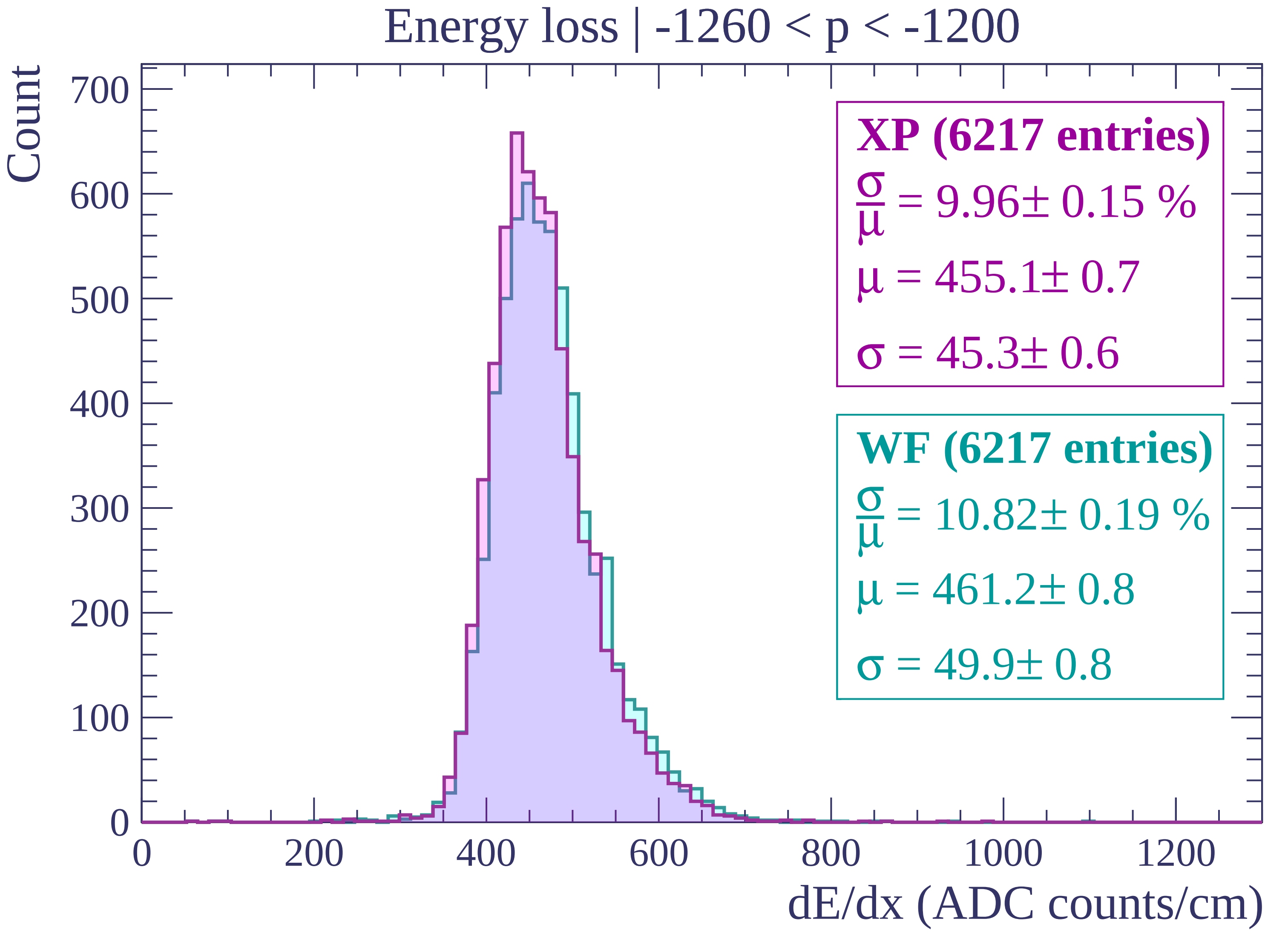

Energy loss | -1440 < p < -1380
Count
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5453 entries)
σ/μ = 10.31± 0.19 %
μ = 464.5± 0.8
σ = 47.9± 0.8
WF (5453 entries)
σ/μ = 10.84± 0.20 %
μ = 472.1± 0.9
σ = 51.2± 0.8



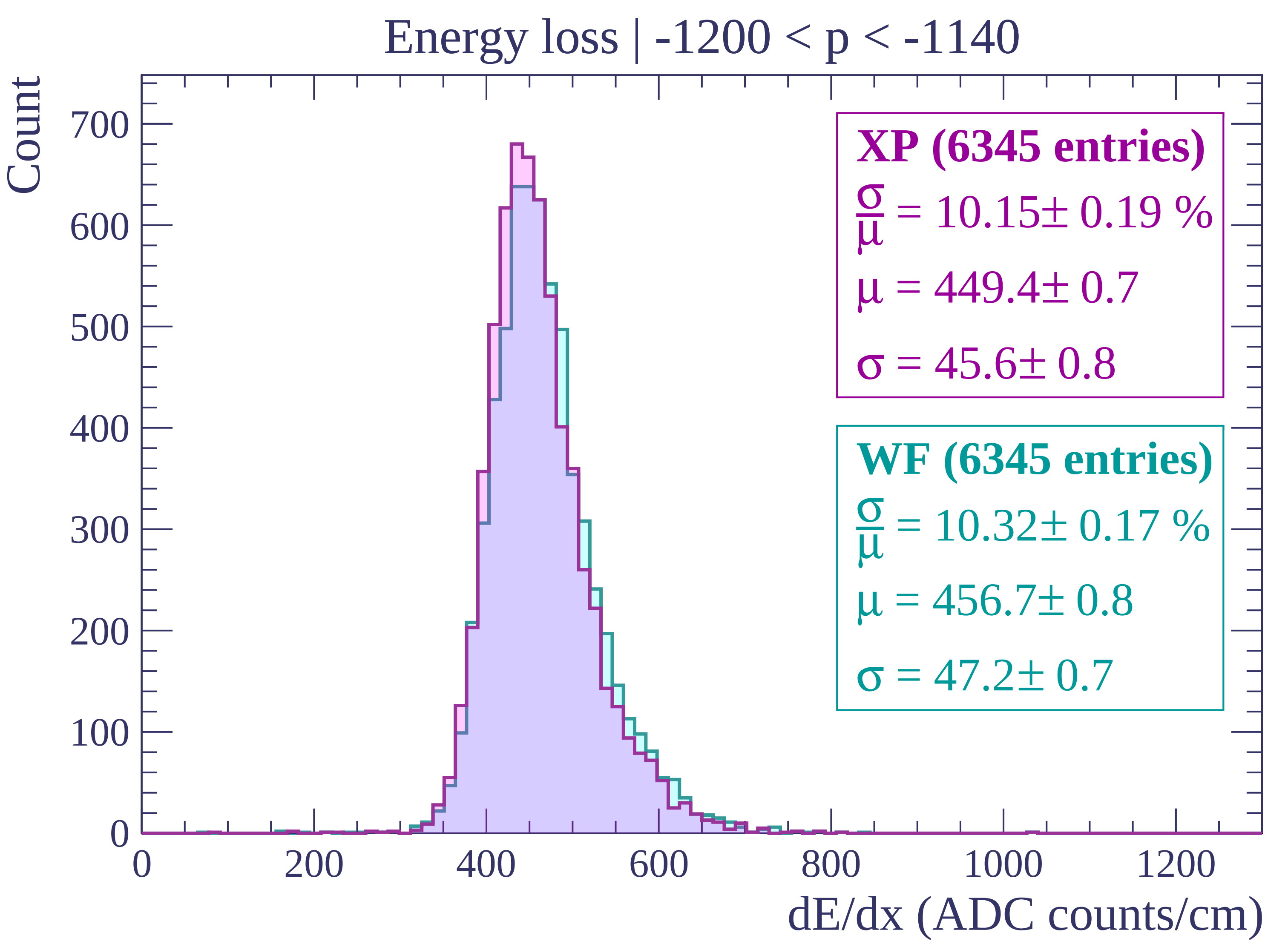
Energy loss | -1380 < p < -1320
Count
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5782 entries)
σ/μ = 10.07± 0.18 %
μ = 458.9± 0.8
σ = 46.2± 0.7
WF (5782 entries)
σ/μ = 10.64± 0.19 %
μ = 466.7± 0.8
σ = 49.7± 0.8



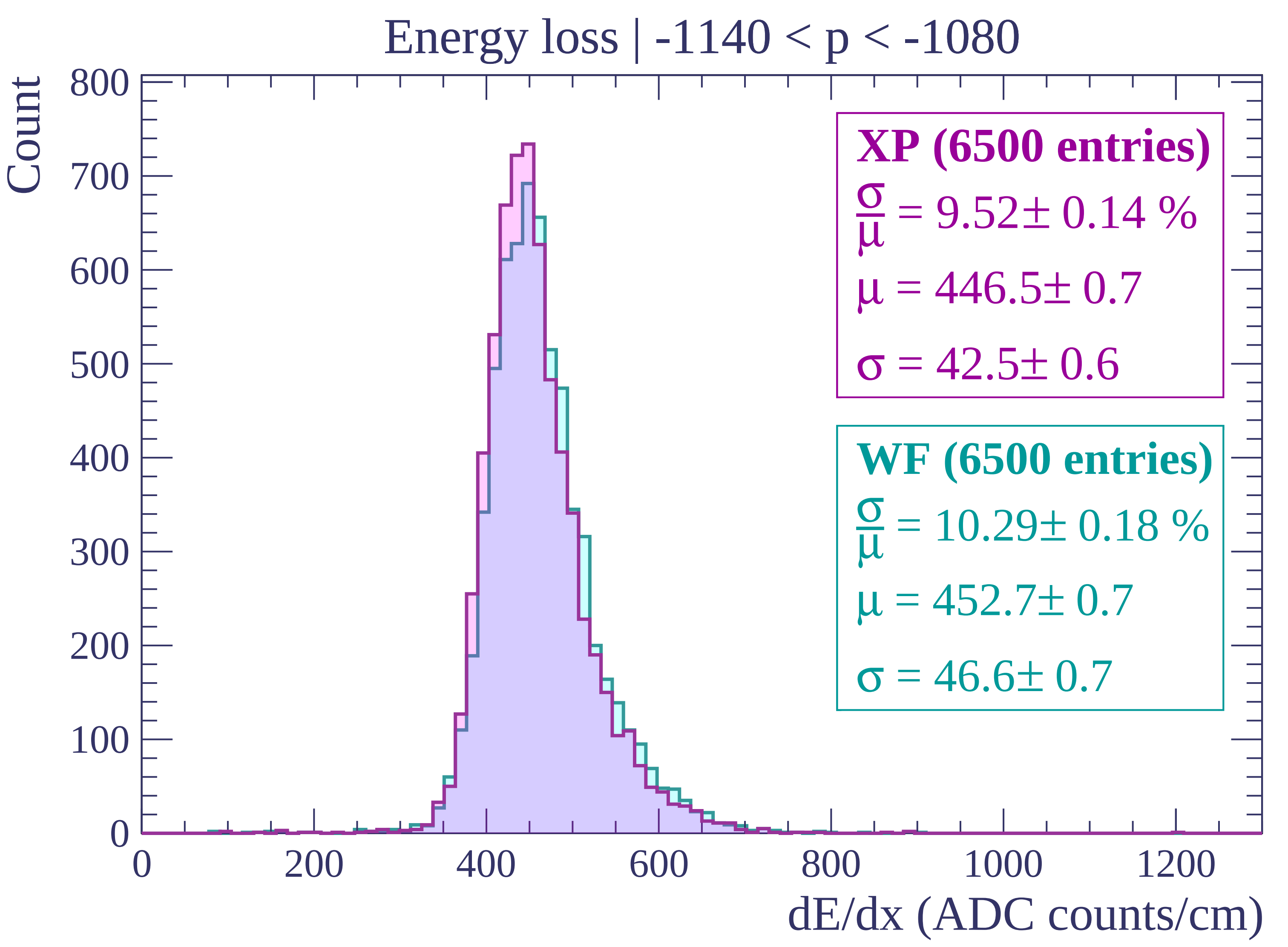
Energy loss | -1320 < p < -1260
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (5997 entries)
σ/μ = 10.27± 0.18 %
μ = 456.9± 0.8
σ = 46.9± 0.7
WF (5997 entries)
σ/μ = 10.65± 0.18 %
μ = 464.9± 0.8
σ = 49.5± 0.8

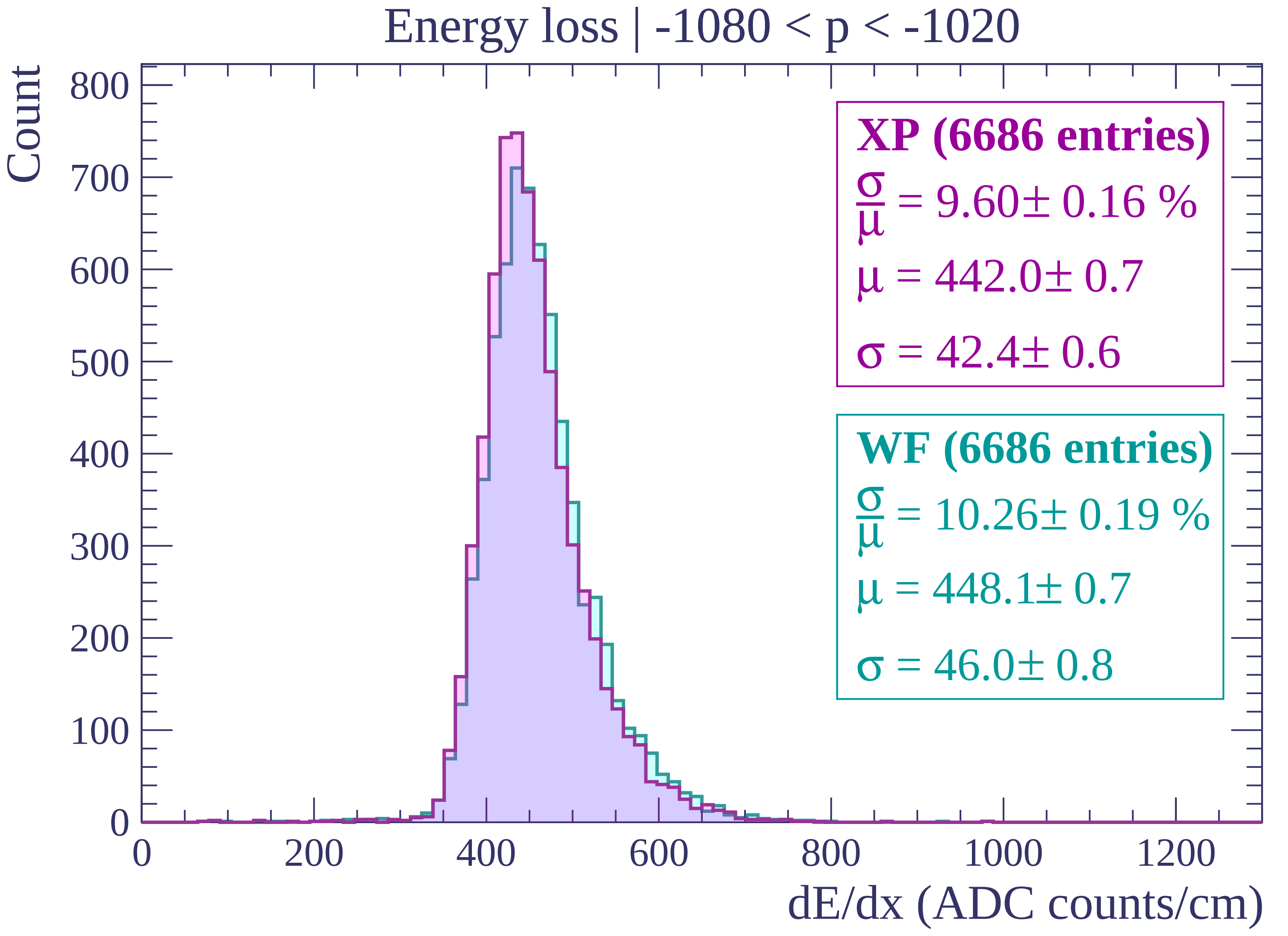


Energy loss | -1260 < p < -1200
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (6217 entries)
σ/μ = 9.96± 0.15 %
μ = 455.1± 0.7
σ = 45.3± 0.6
WF (6217 entries)
σ/μ = 10.82± 0.19 %
μ = 461.2± 0.8
σ = 49.9± 0.8

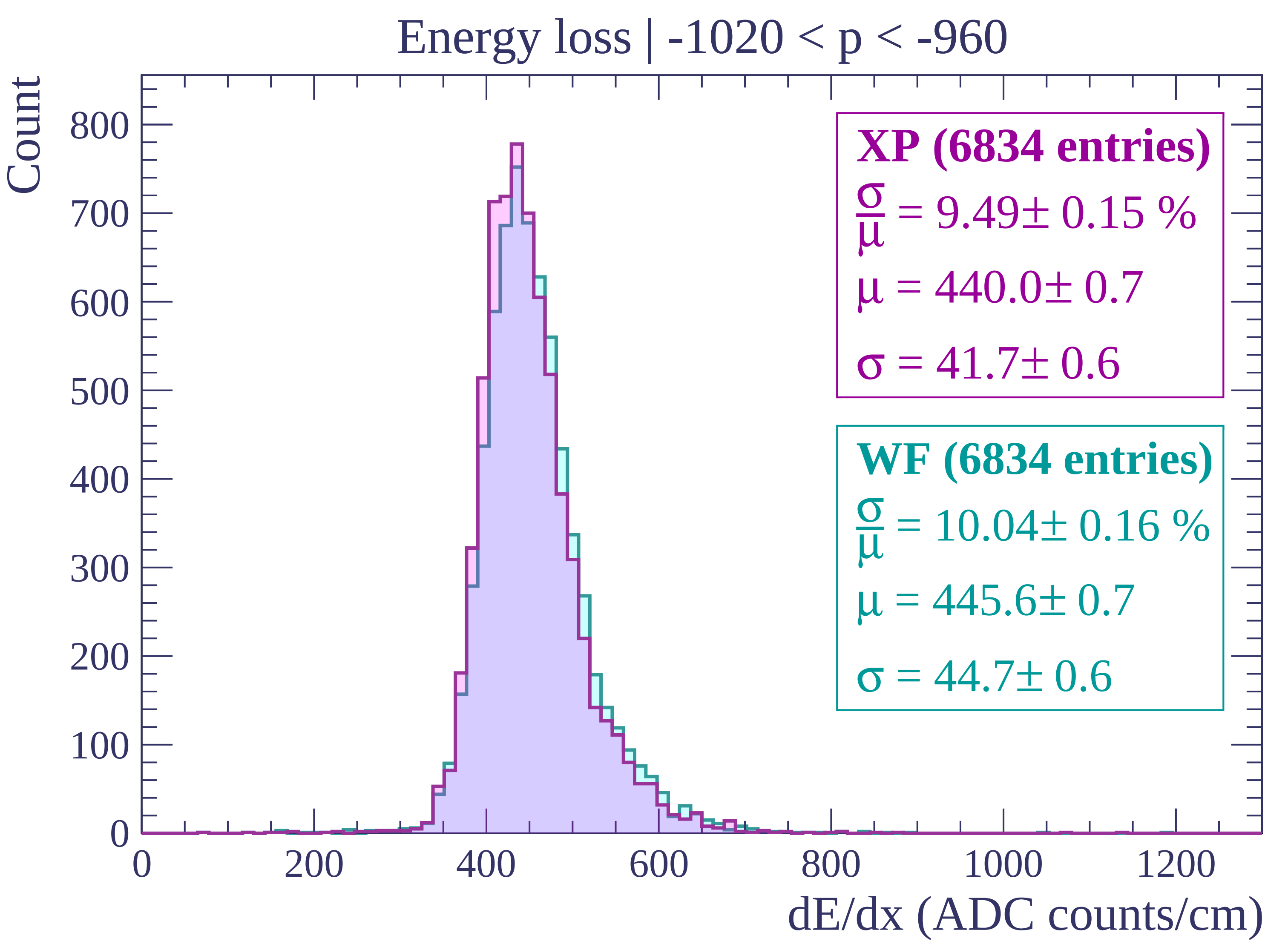


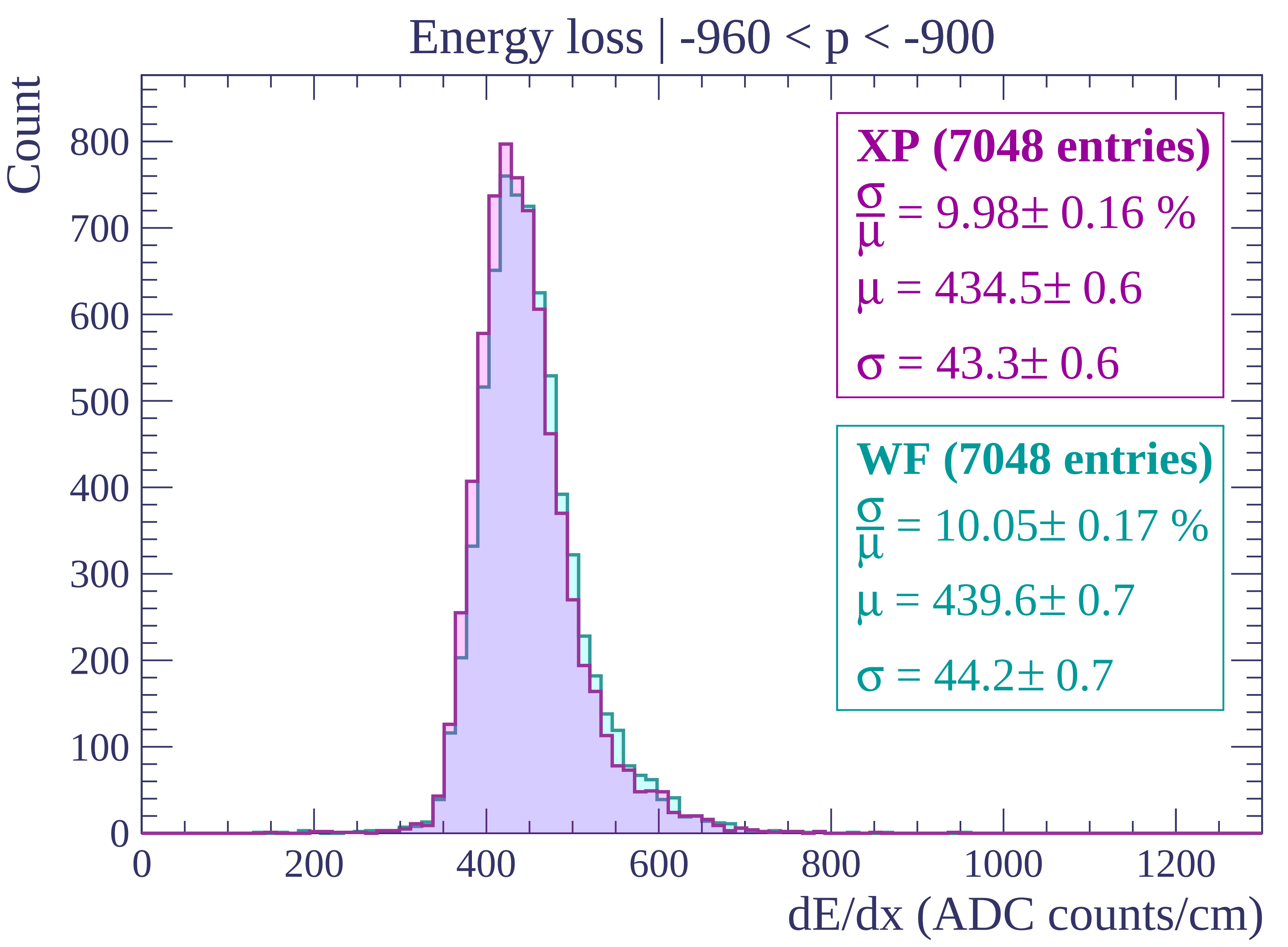
Energy loss | -1200 < p < -1140
Count
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6345 entries)
σ/μ = 10.15± 0.19 %
μ = 449.4± 0.7
σ = 45.6± 0.8
WF (6345 entries)
σ/μ = 10.32± 0.17 %
μ = 456.7± 0.8
σ = 47.2± 0.7


Energy loss | -1140 < p < -1080
Count
dE/dx (ADC counts/cm)
XP (6500 entries)
σ/μ = 9.52± 0.14 %
μ = 446.5± 0.7
σ = 42.5± 0.6
WF (6500 entries)
σ/μ = 10.29± 0.18 %
μ = 452.7± 0.7
σ = 46.6± 0.7
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200

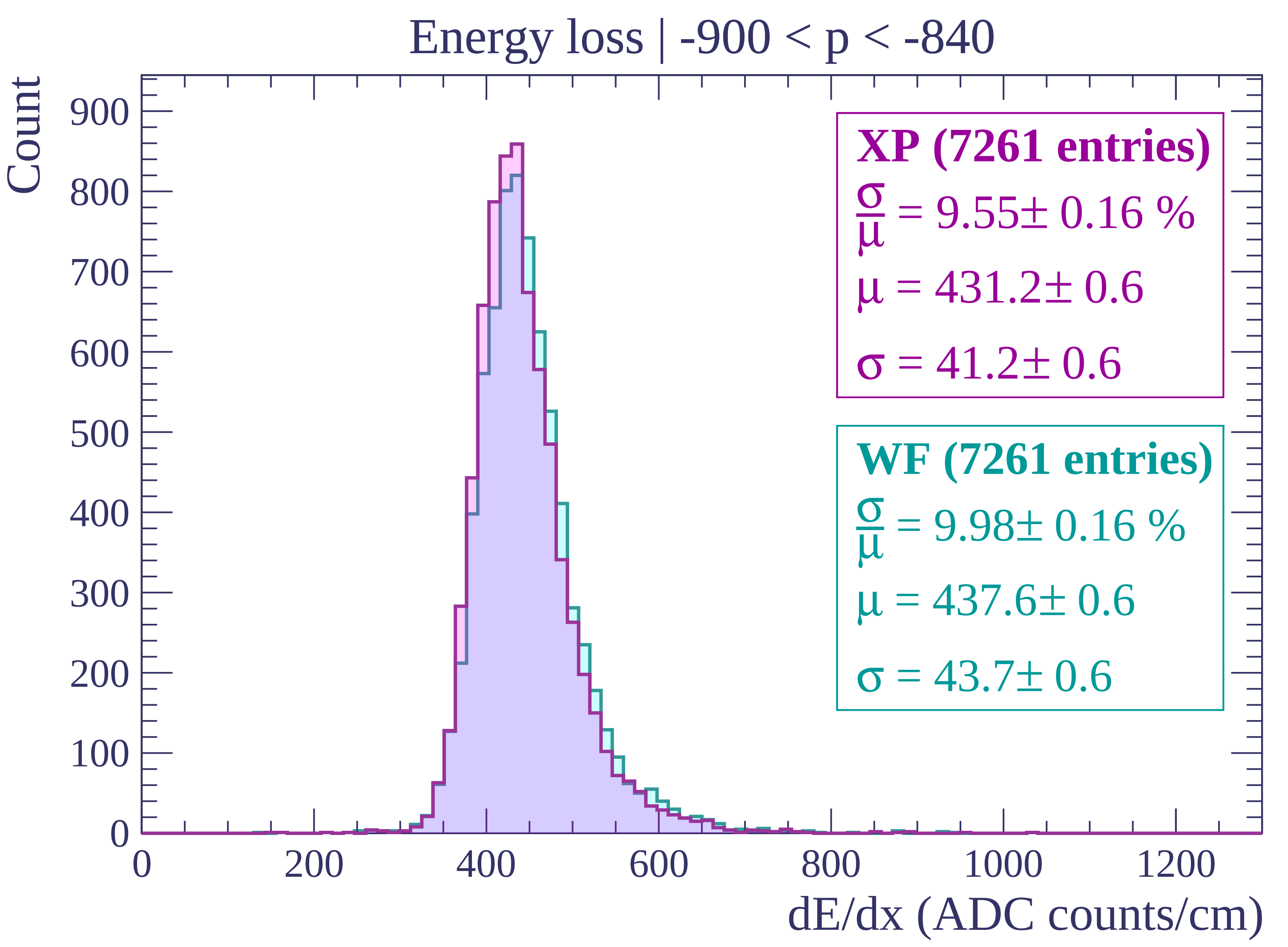


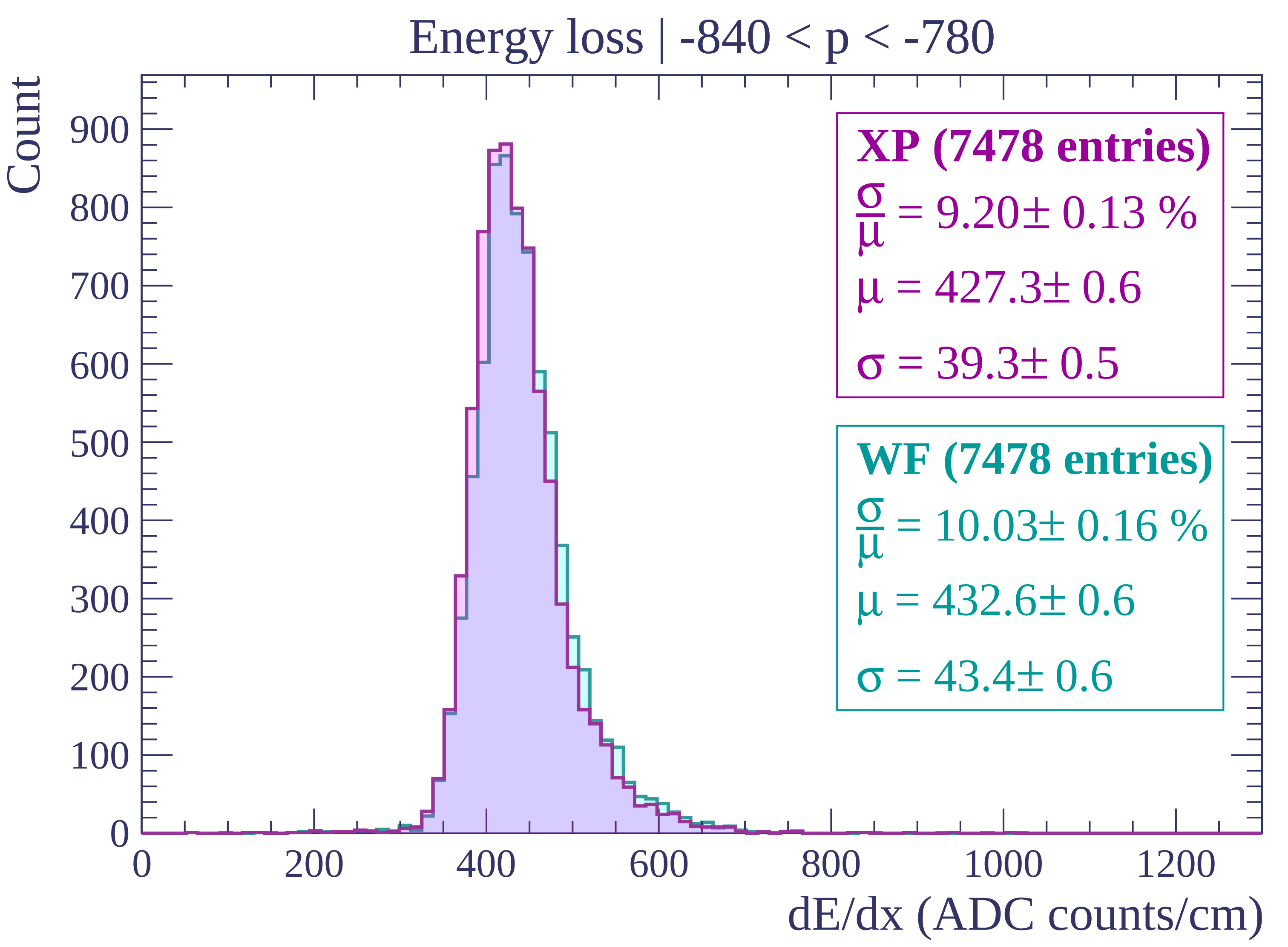
Energy loss | -1080 < p < -1020
Count
dE/dx (ADC counts/cm)
XP (6686 entries)
$\frac{\sigma}{\mu}$ = 9.60± 0.16 %
μ = 442.0± 0.7
σ = 42.4± 0.6
WF (6686 entries)
$\frac{\sigma}{\mu}$ = 10.26± 0.19 %
μ = 448.1± 0.7
σ = 46.0± 0.8
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200

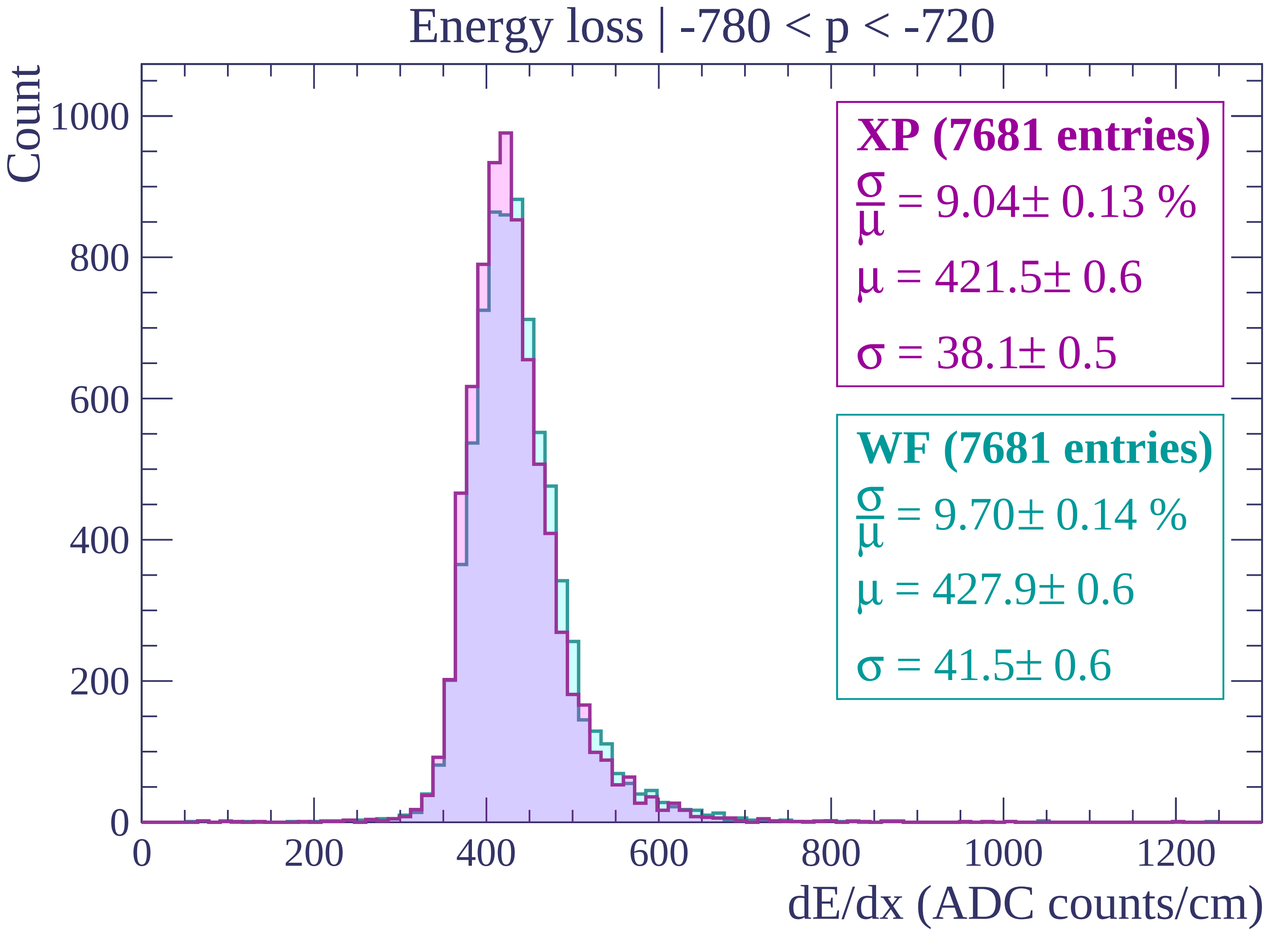

Energy loss | -1020 < p < -960
Count
dE/dx (ADC counts/cm)
XP (6834 entries)
σ/μ = 9.49± 0.15 %
μ = 440.0± 0.7
σ = 41.7± 0.6
WF (6834 entries)
σ/μ = 10.04± 0.16 %
μ = 445.6± 0.7
σ = 44.7± 0.6



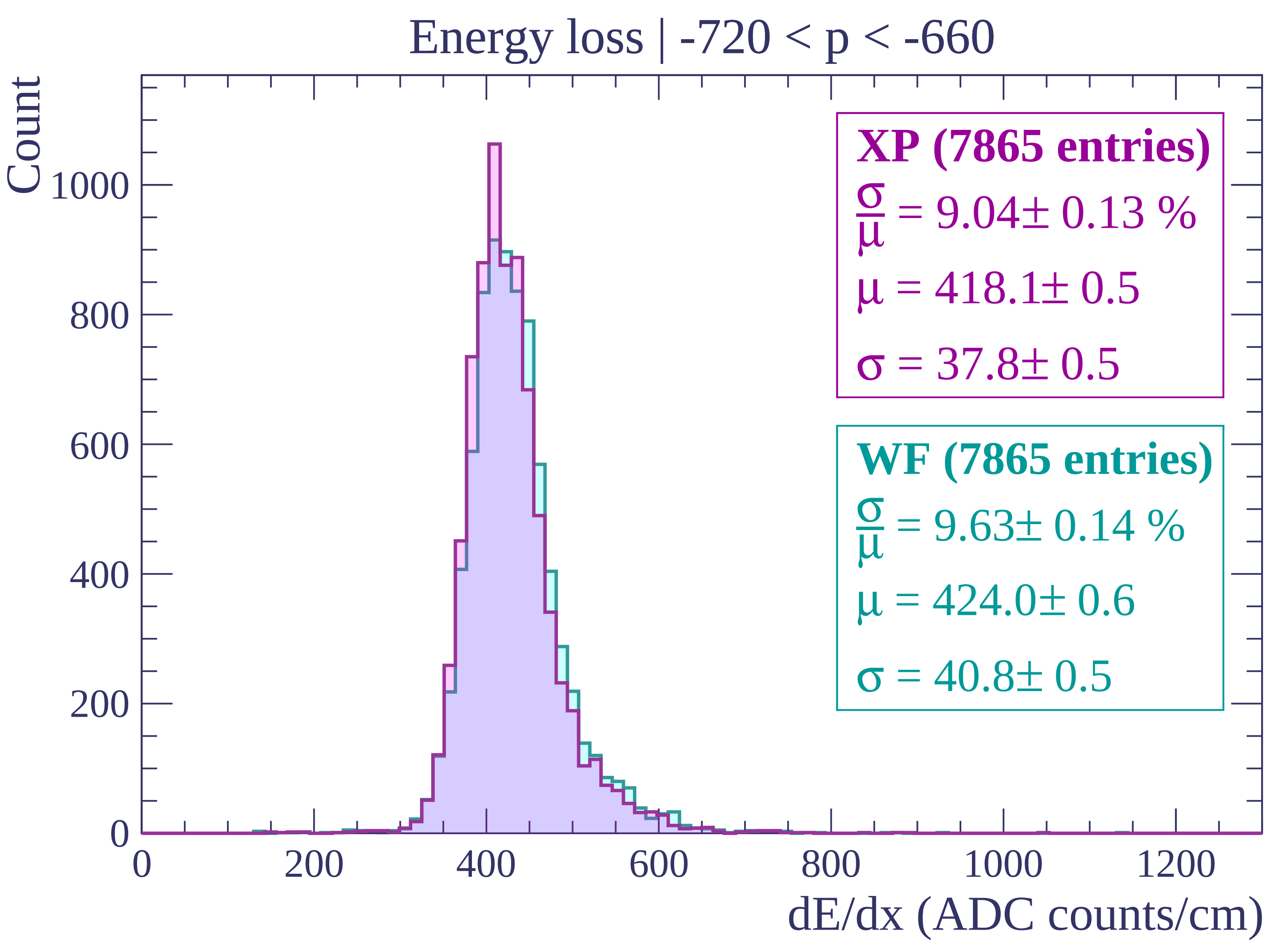
Energy loss | -960 < p < -900
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP (7048 entries)
σ/μ = 9.98± 0.16 %
μ = 434.5± 0.6
σ = 43.3± 0.6
WF (7048 entries)
σ/μ = 10.05± 0.17 %
μ = 439.6± 0.7
σ = 44.2± 0.7


Energy loss | -900 < p < -840
Count
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (7261 entries)
$\frac{\sigma}{\mu}$ = 9.55± 0.16 %
μ = 431.2± 0.6
σ = 41.2± 0.6
WF (7261 entries)
$\frac{\sigma}{\mu}$ = 9.98± 0.16 %
μ = 437.6± 0.6
σ = 43.7± 0.6

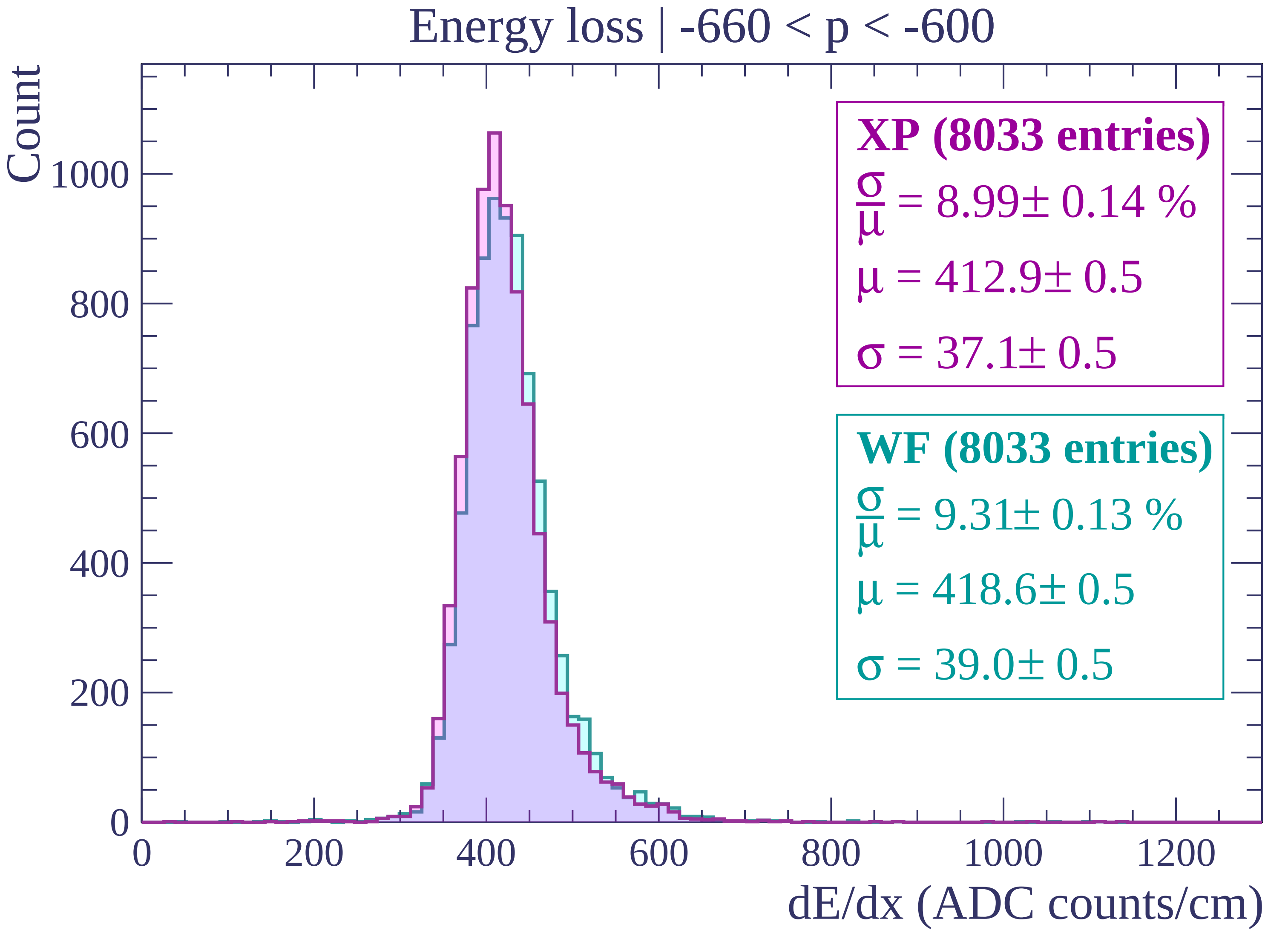

Energy loss | -840 < p < -780
Count
dE/dx (ADC counts/cm)
XP (7478 entries)
σ/μ = 9.20± 0.13 %
μ = 427.3± 0.6
σ = 39.3± 0.5
WF (7478 entries)
σ/μ = 10.03± 0.16 %
μ = 432.6± 0.6
σ = 43.4± 0.6
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200



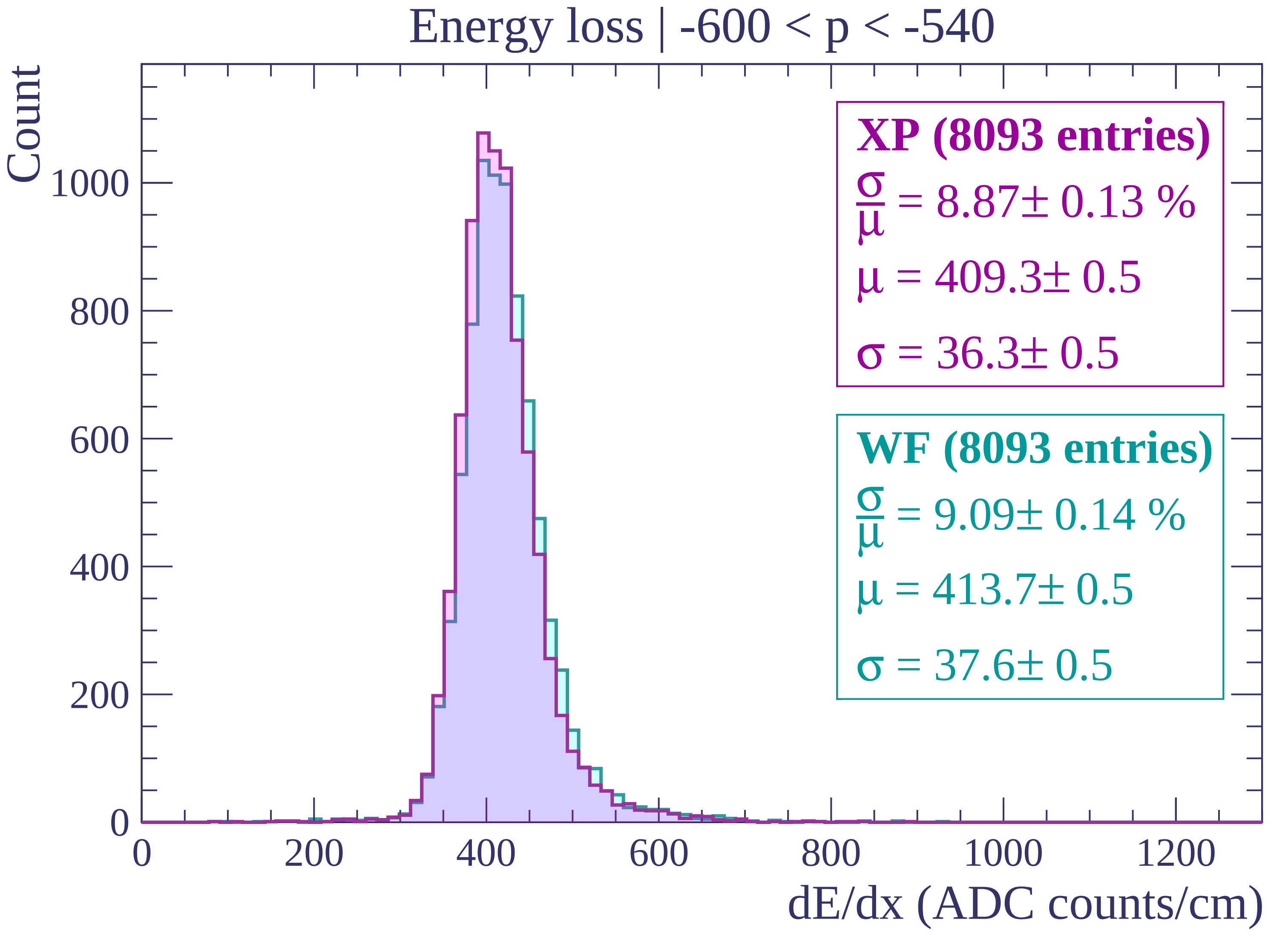
Energy loss | -780 < p < -720
Count
0
200
400
600
800
1000
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (7681 entries)
$\frac{\sigma}{\mu}$ = 9.04± 0.13 %
μ = 421.5± 0.6
σ = 38.1± 0.5
WF (7681 entries)
$\frac{\sigma}{\mu}$ = 9.70± 0.14 %
μ = 427.9± 0.6
σ = 41.5± 0.6



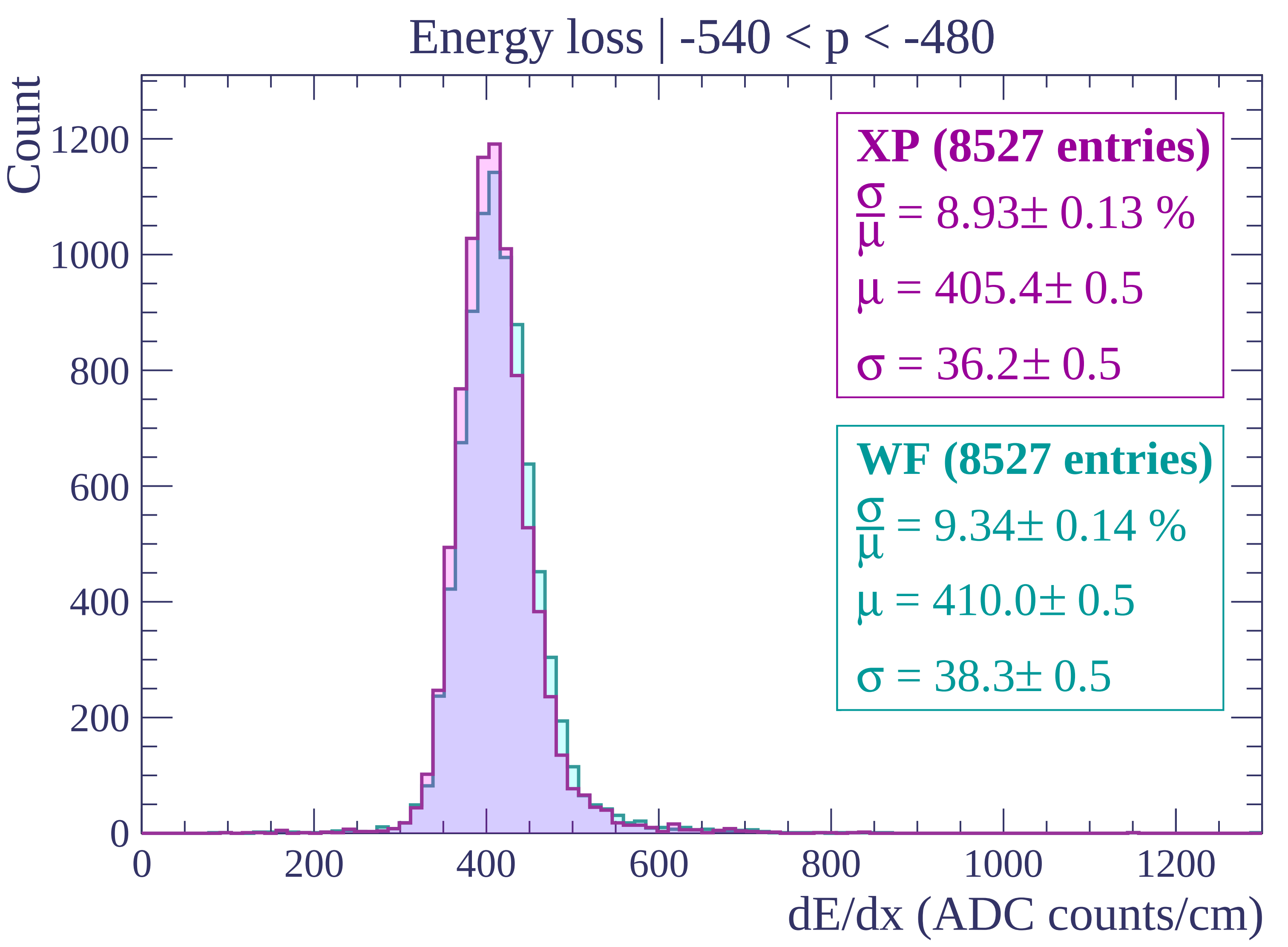
Energy loss | -720 < p < -660
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (7865 entries)
σ/μ = 9.04± 0.13 %
μ = 418.1± 0.5
σ = 37.8± 0.5
WF (7865 entries)
σ/μ = 9.63± 0.14 %
μ = 424.0± 0.6
σ = 40.8± 0.5



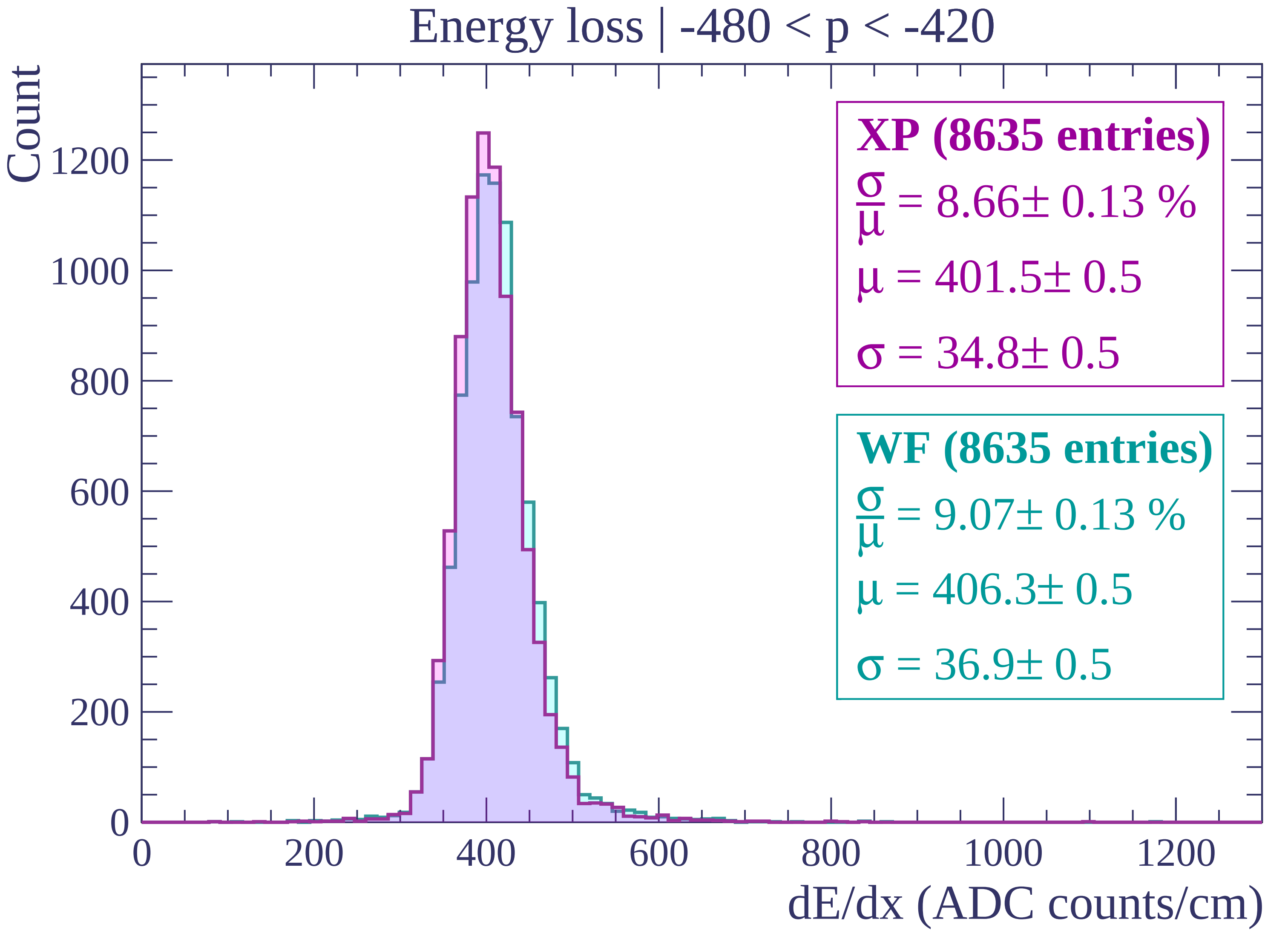
Energy loss | -660 < p < -600
Count
dE/dx (ADC counts/cm)
XP (8033 entries)
$\frac{\sigma}{\mu}$ = 8.99± 0.14 %
μ = 412.9± 0.5
σ = 37.1± 0.5
WF (8033 entries)
$\frac{\sigma}{\mu}$ = 9.31± 0.13 %
μ = 418.6± 0.5
σ = 39.0± 0.5

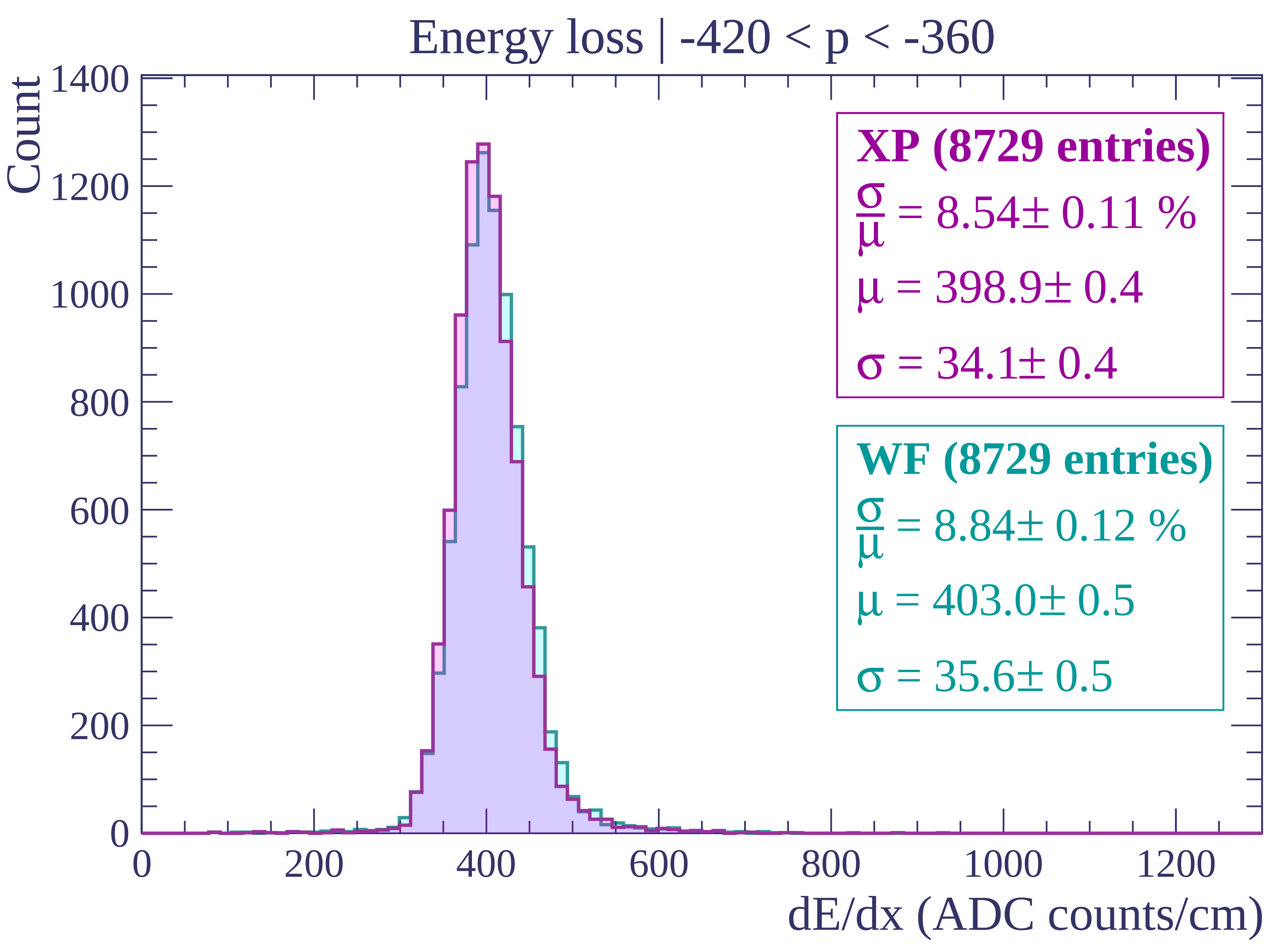


Energy loss | -600 < p < -540
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (8093 entries)
σ/μ = 8.87± 0.13 %
μ = 409.3± 0.5
σ = 36.3± 0.5
WF (8093 entries)
σ/μ = 9.09± 0.14 %
μ = 413.7± 0.5
σ = 37.6± 0.5

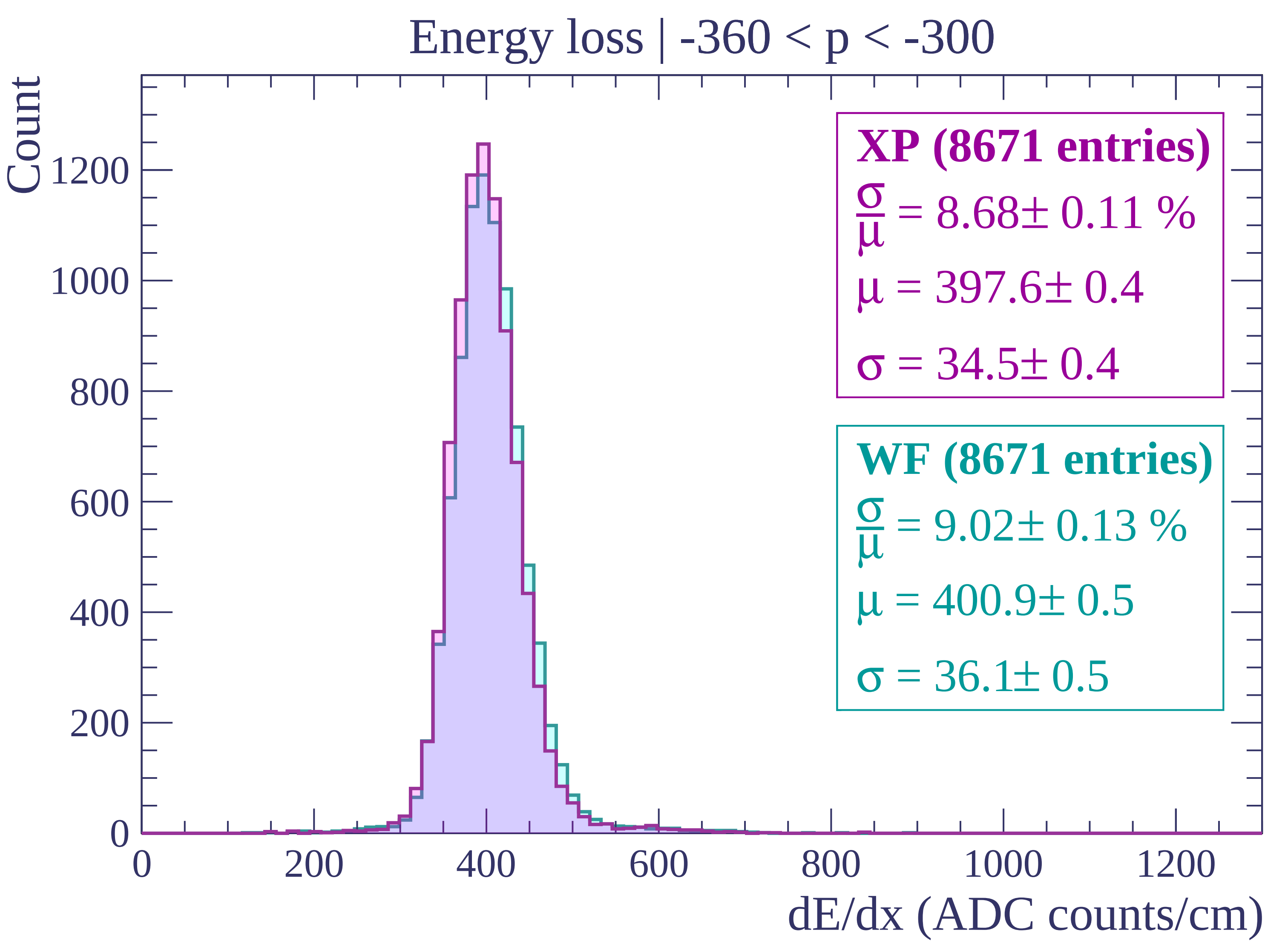


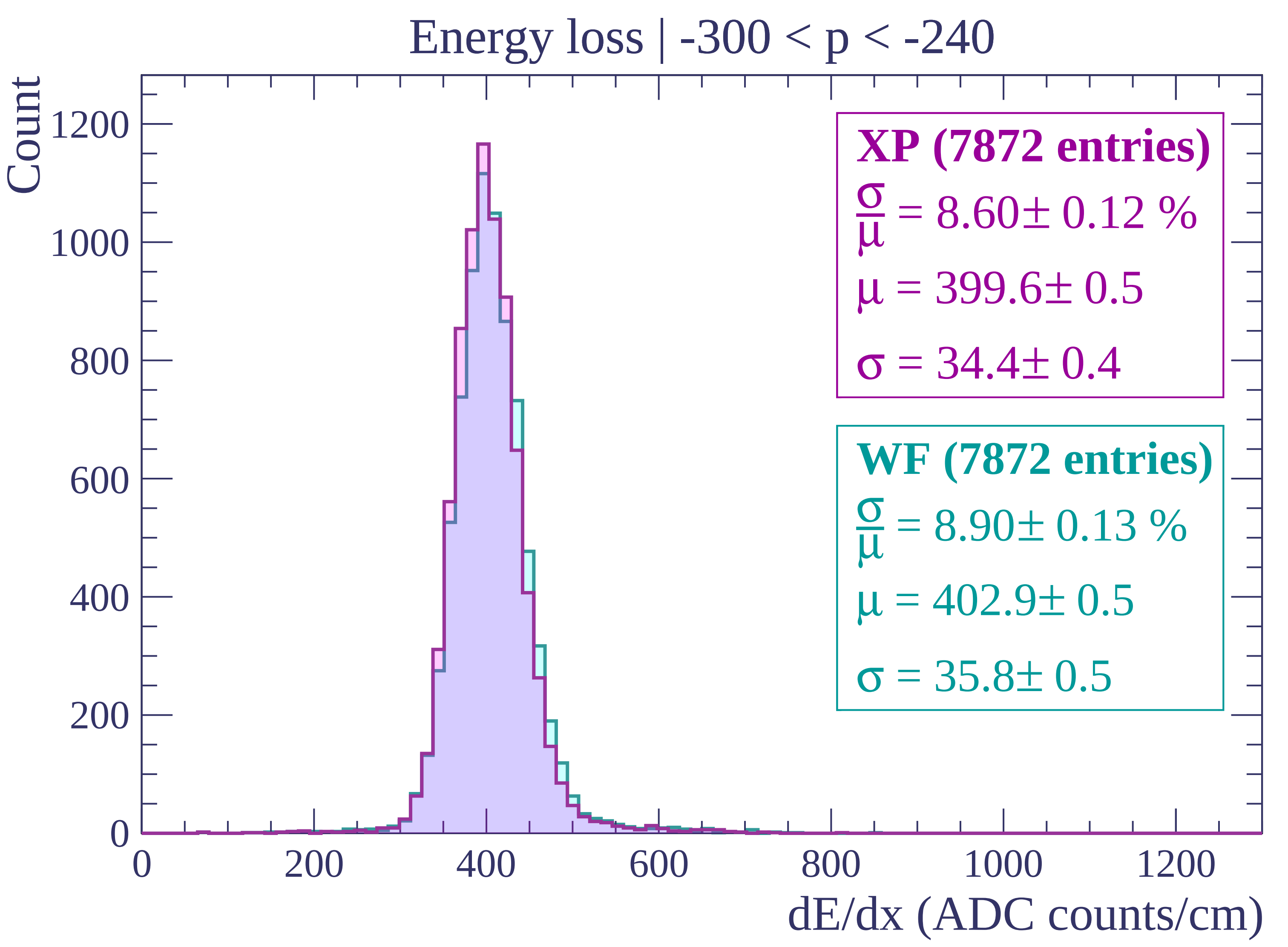
Energy loss | -540 < p < -480
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (8527 entries)
σ/μ = 8.93± 0.13 %
μ = 405.4± 0.5
σ = 36.2± 0.5
WF (8527 entries)
σ/μ = 9.34± 0.14 %
μ = 410.0± 0.5
σ = 38.3± 0.5


Energy loss | -480 < p < -420
Count
dE/dx (ADC counts/cm)
XP (8635 entries)
σ/μ = 8.66± 0.13 %
μ = 401.5± 0.5
σ = 34.8± 0.5
WF (8635 entries)
σ/μ = 9.07± 0.13 %
μ = 406.3± 0.5
σ = 36.9± 0.5
0
200
400
600
800
1000
1200

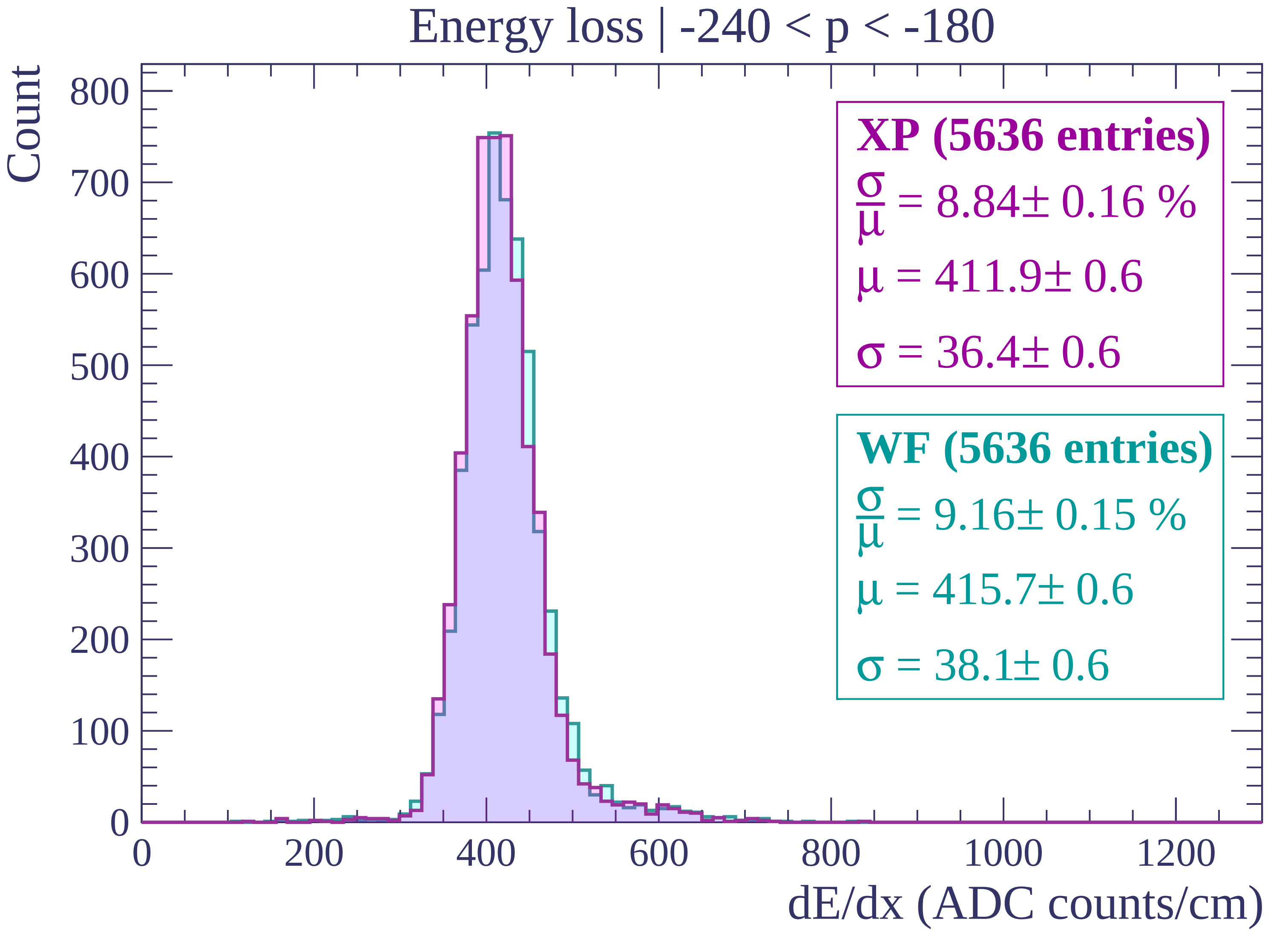


Energy loss | -420 < p < -360
Count
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (8729 entries)
$\frac{\sigma}{\mu}$ = 8.54± 0.11 %
μ = 398.9± 0.4
σ = 34.1± 0.4
WF (8729 entries)
$\frac{\sigma}{\mu}$ = 8.84± 0.12 %
μ = 403.0± 0.5
σ = 35.6± 0.5

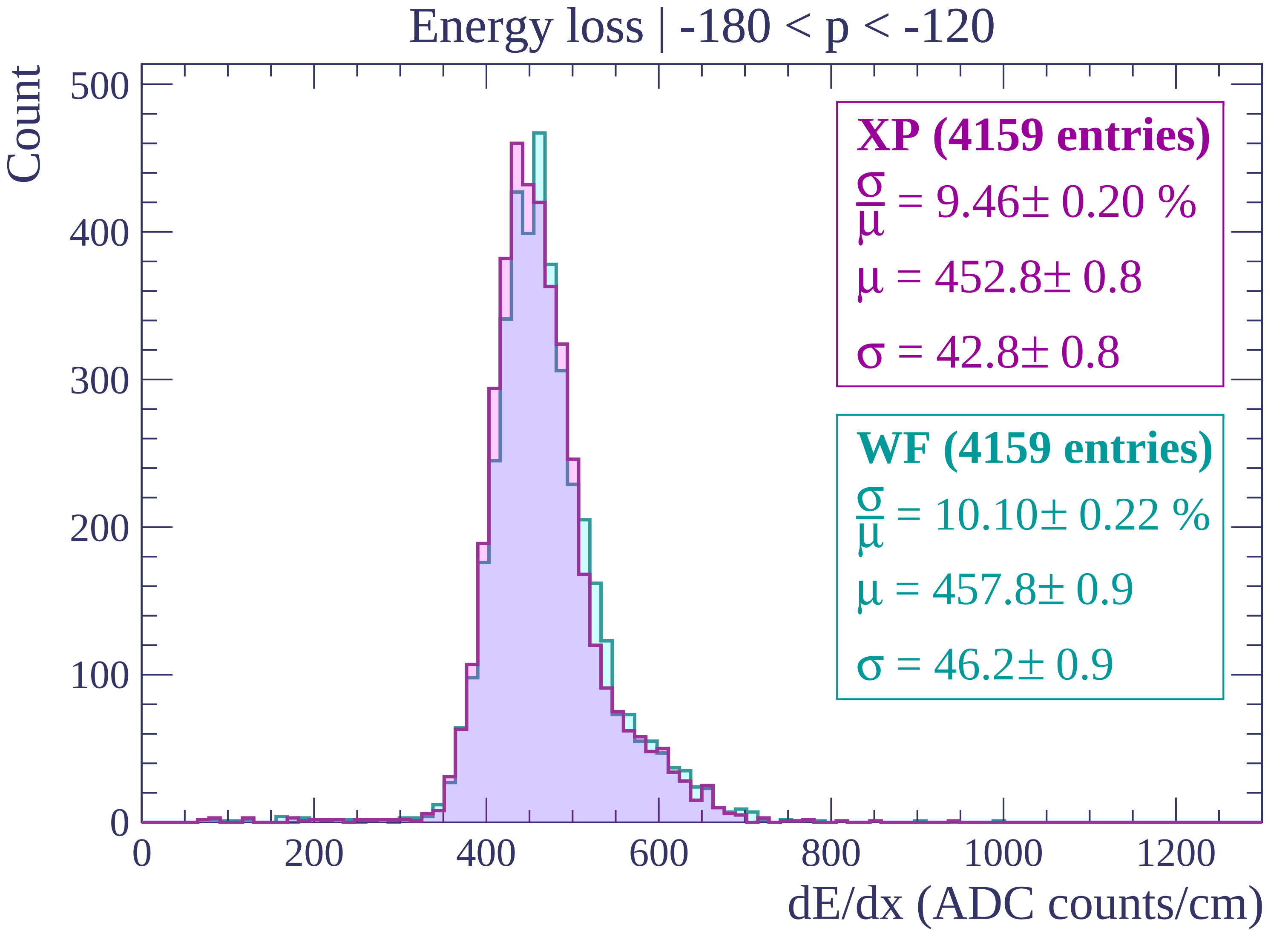

Energy loss | -360 < p < -300
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (8671 entries)
$\frac{\sigma}{\mu}$ = 8.68± 0.11 %
μ = 397.6± 0.4
σ = 34.5± 0.4
WF (8671 entries)
$\frac{\sigma}{\mu}$ = 9.02± 0.13 %
μ = 400.9± 0.5
σ = 36.1± 0.5

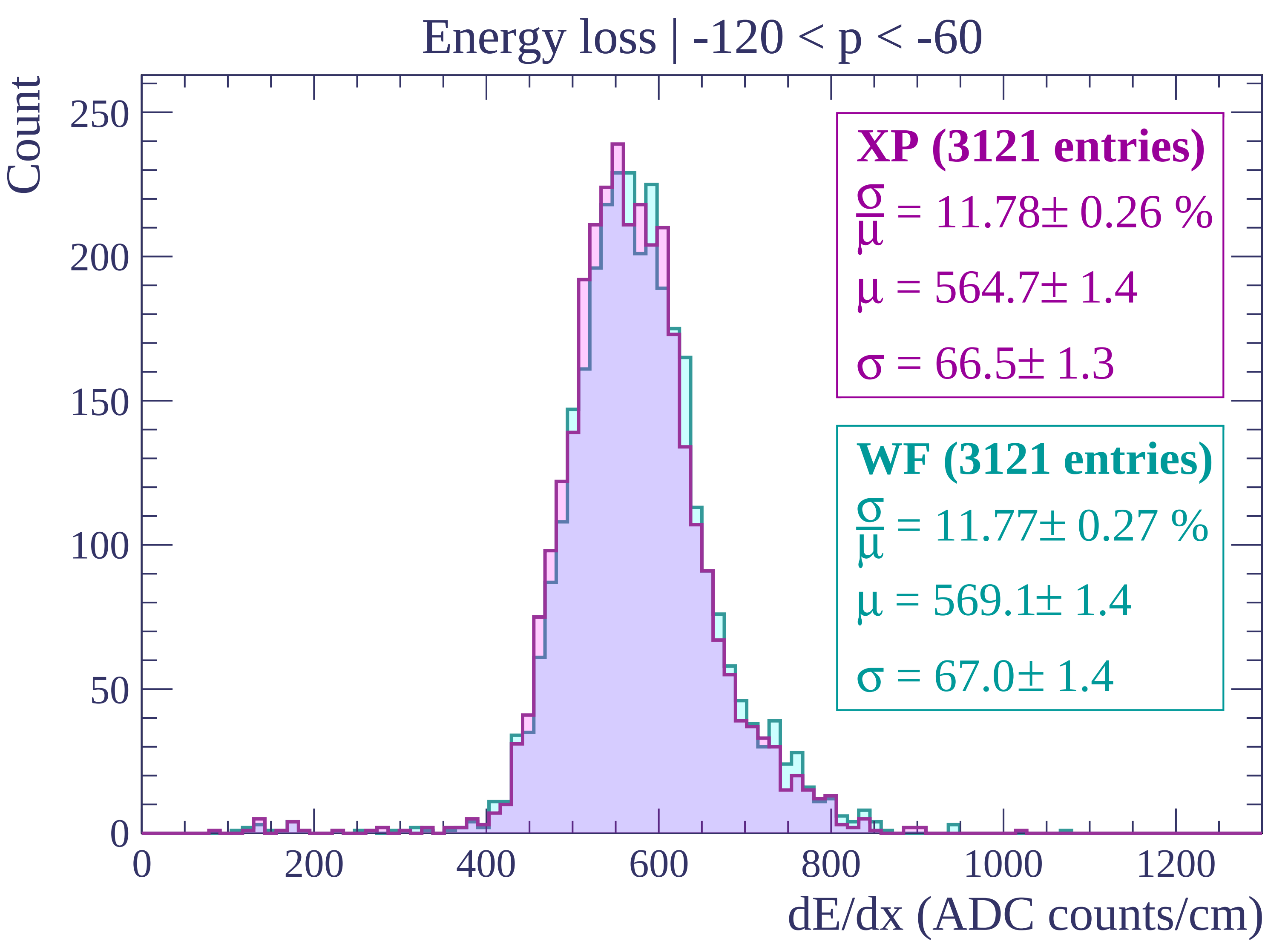


Energy loss | -300 < p < -240
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (7872 entries)
σ/μ = 8.60± 0.12 %
μ = 399.6± 0.5
σ = 34.4± 0.4
WF (7872 entries)
σ/μ = 8.90± 0.13 %
μ = 402.9± 0.5
σ = 35.8± 0.5

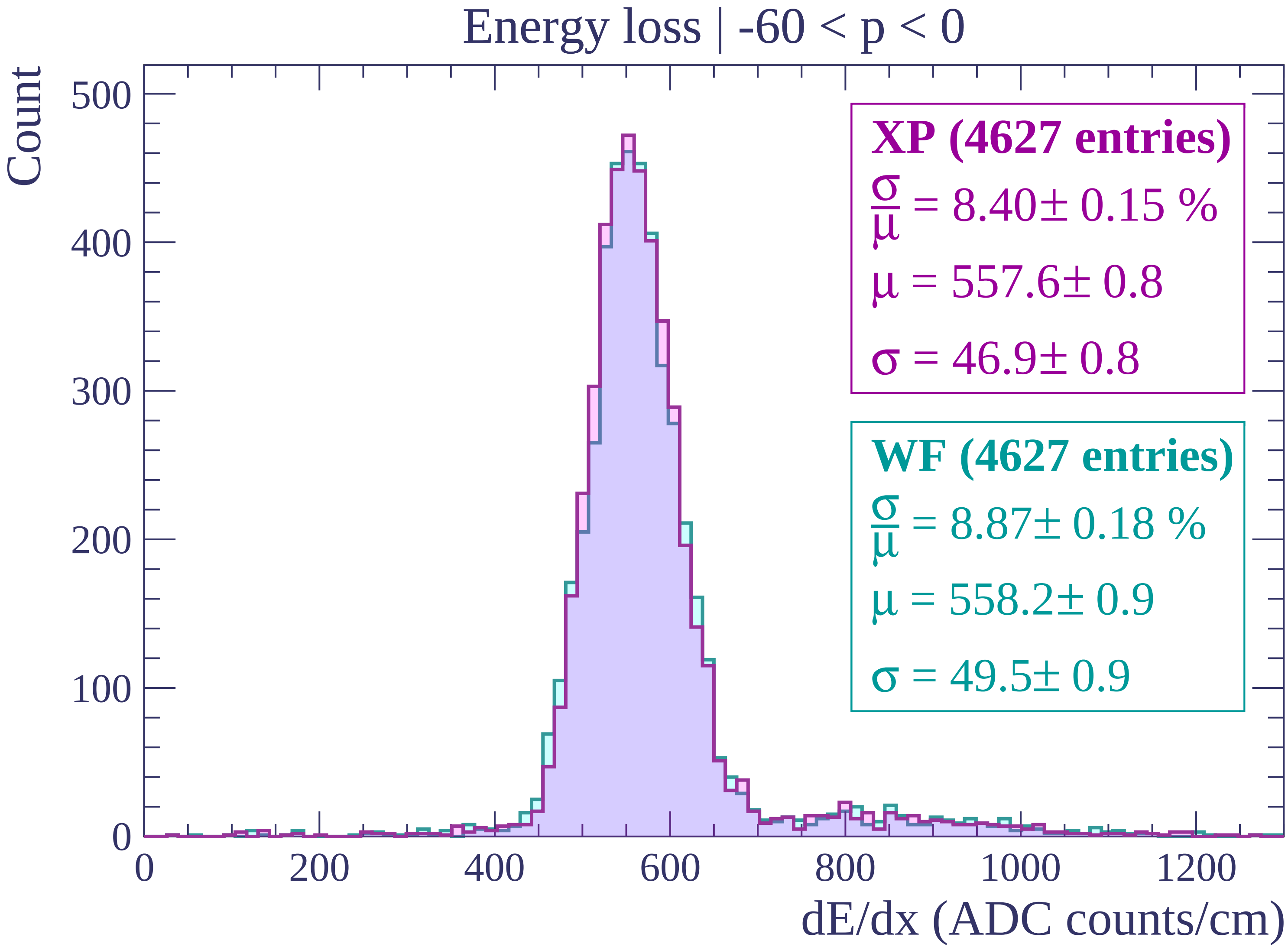


Energy loss | -240 < p < -180
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP (5636 entries)
σ/μ = 8.84± 0.16 %
μ = 411.9± 0.6
σ = 36.4± 0.6
WF (5636 entries)
σ/μ = 9.16± 0.15 %
μ = 415.7± 0.6
σ = 38.1± 0.6

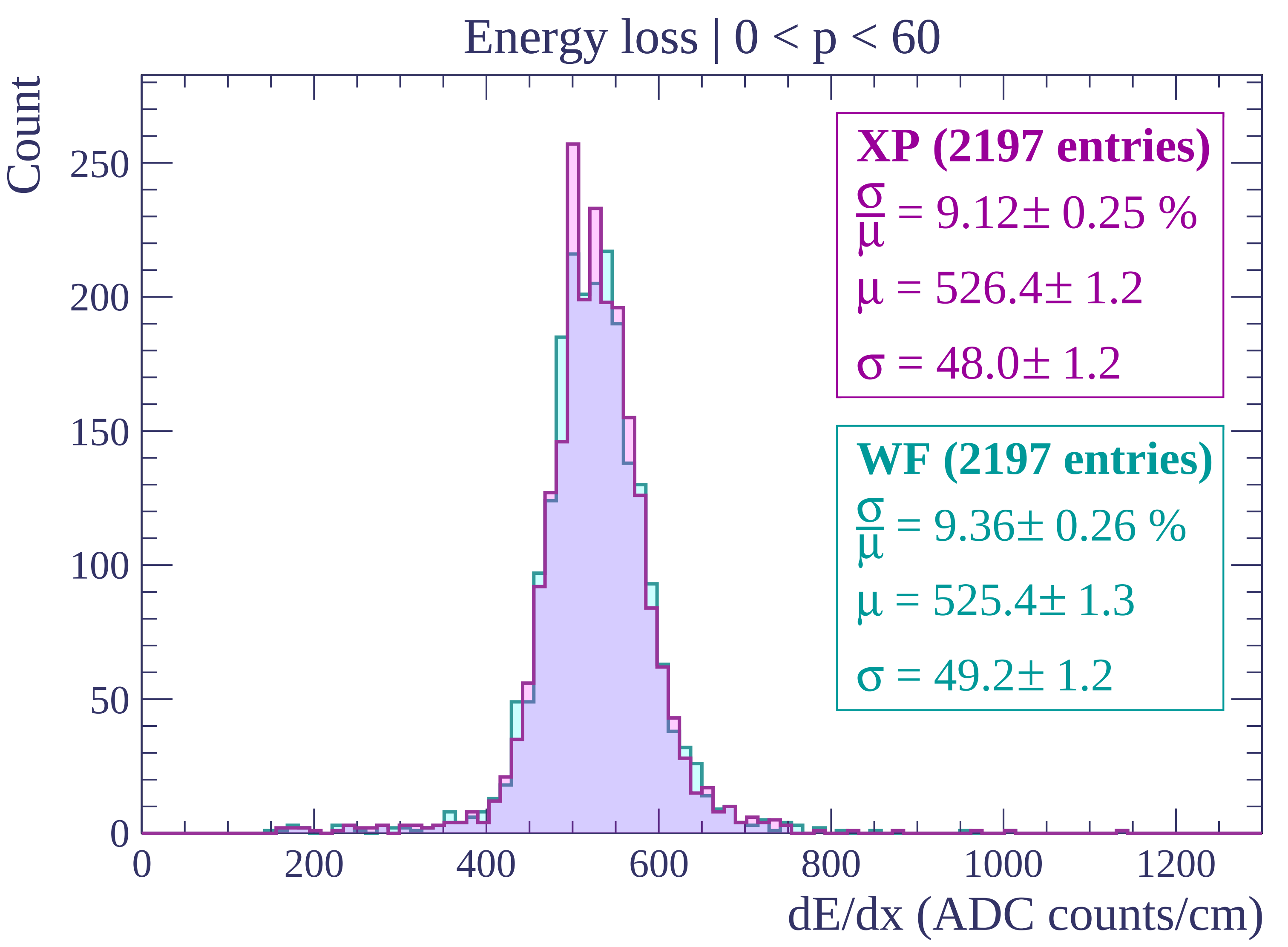

Energy loss | -180 < p < -120
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4159 entries)
$\frac{\sigma}{\mu}$ = 9.46± 0.20 %
μ = 452.8± 0.8
σ = 42.8± 0.8
WF (4159 entries)
$\frac{\sigma}{\mu}$ = 10.10± 0.22 %
μ = 457.8± 0.9
σ = 46.2± 0.9



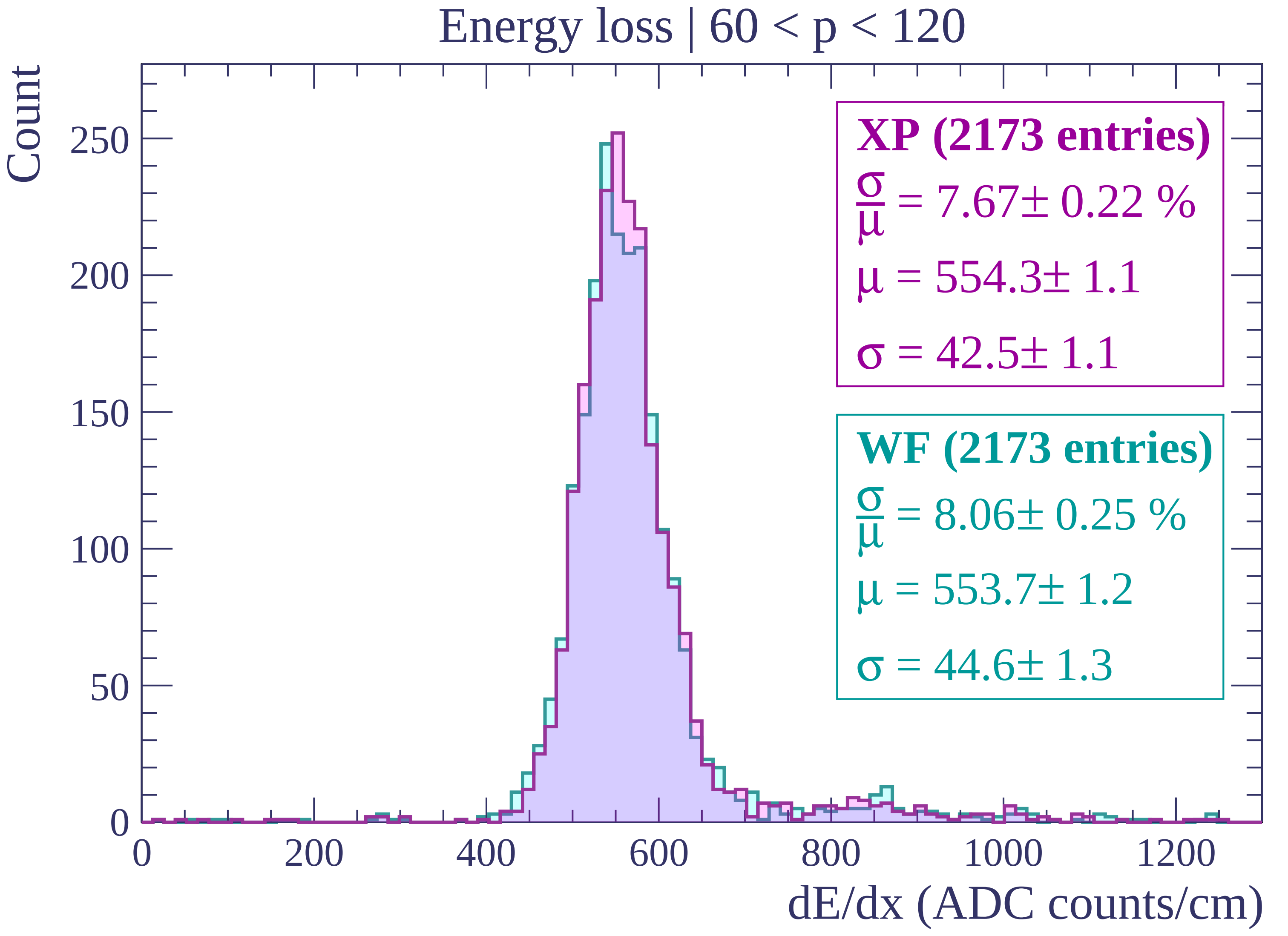
Energy loss | -120 < p < -60
Count
XP (3121 entries)
$\frac{\sigma}{\mu}$ = 11.78± 0.26 %
μ = 564.7± 1.4
σ = 66.5± 1.3
WF (3121 entries)
$\frac{\sigma}{\mu}$ = 11.77± 0.27 %
μ = 569.1± 1.4
σ = 67.0± 1.4
dE/dx (ADC counts/cm)



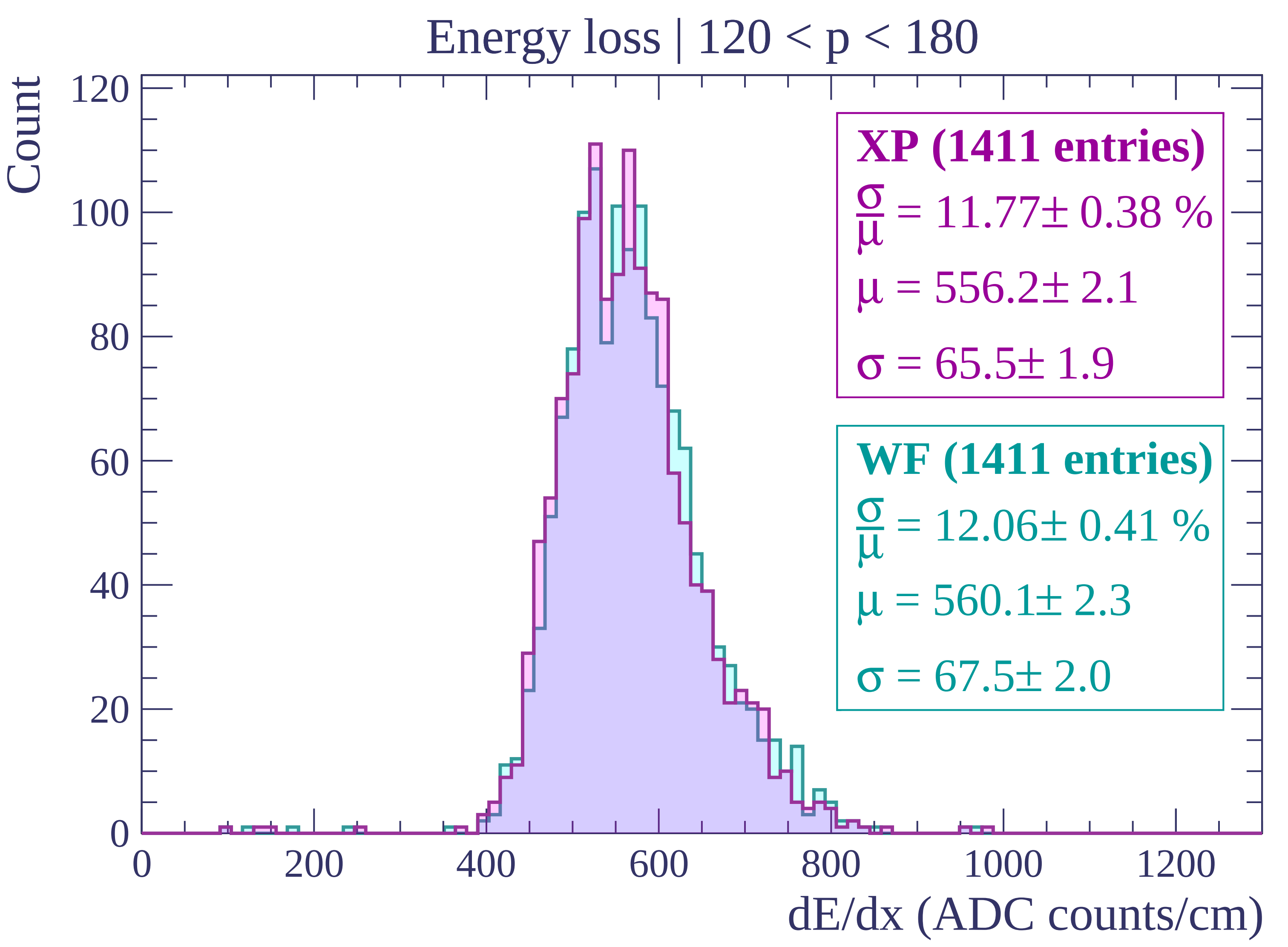
Energy loss | -60 < p < 0
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (4627 entries)
$\frac{\sigma}{\mu}$ = 8.40± 0.15 %
μ = 557.6± 0.8
σ = 46.9± 0.8
WF (4627 entries)
$\frac{\sigma}{\mu}$ = 8.87± 0.18 %
μ = 558.2± 0.9
σ = 49.5± 0.9



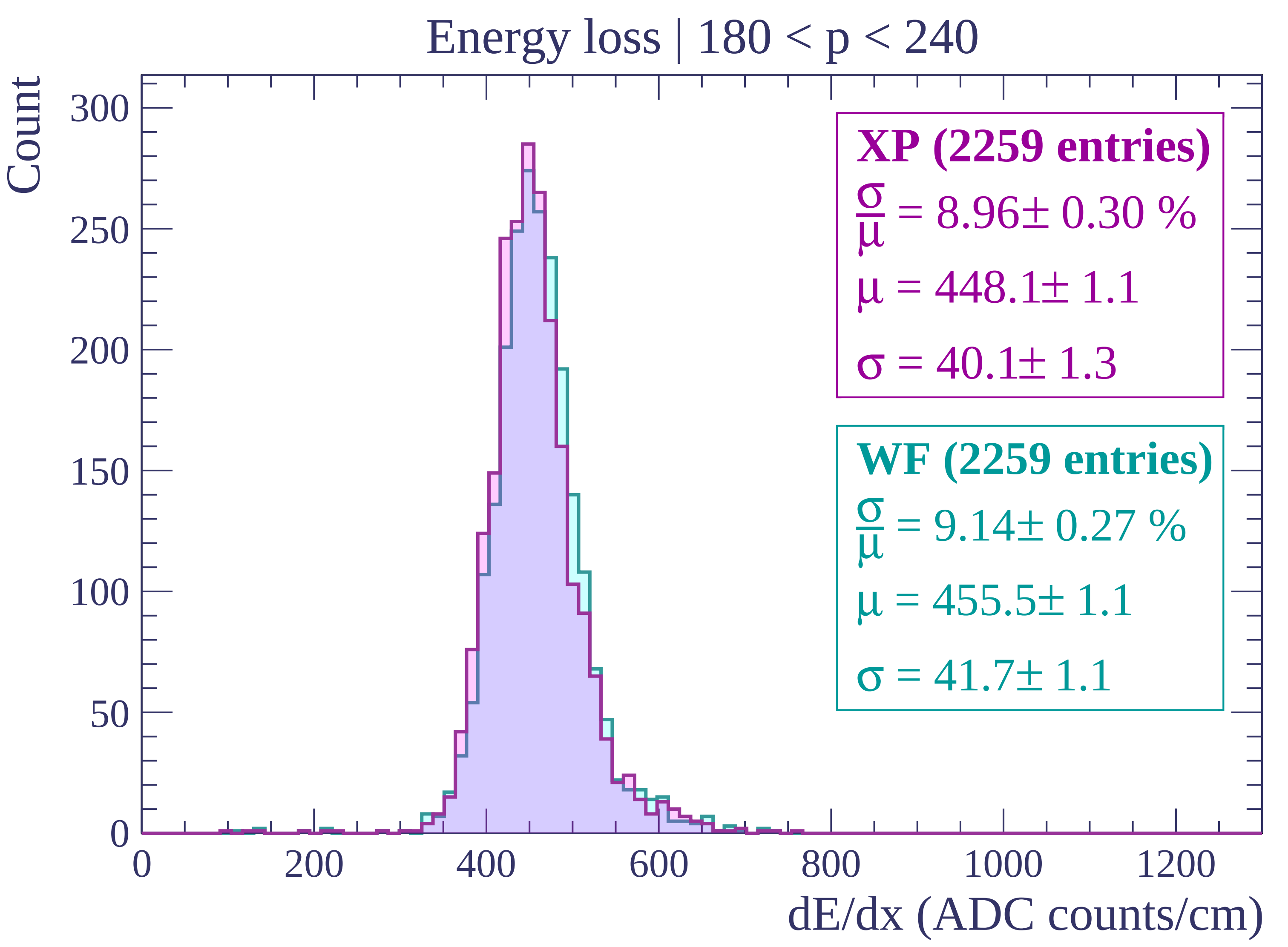
Energy loss | 0 < p < 60
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2197 entries)
σ/μ = 9.12± 0.25 %
μ = 526.4± 1.2
σ = 48.0± 1.2
WF (2197 entries)
σ/μ = 9.36± 0.26 %
μ = 525.4± 1.3
σ = 49.2± 1.2



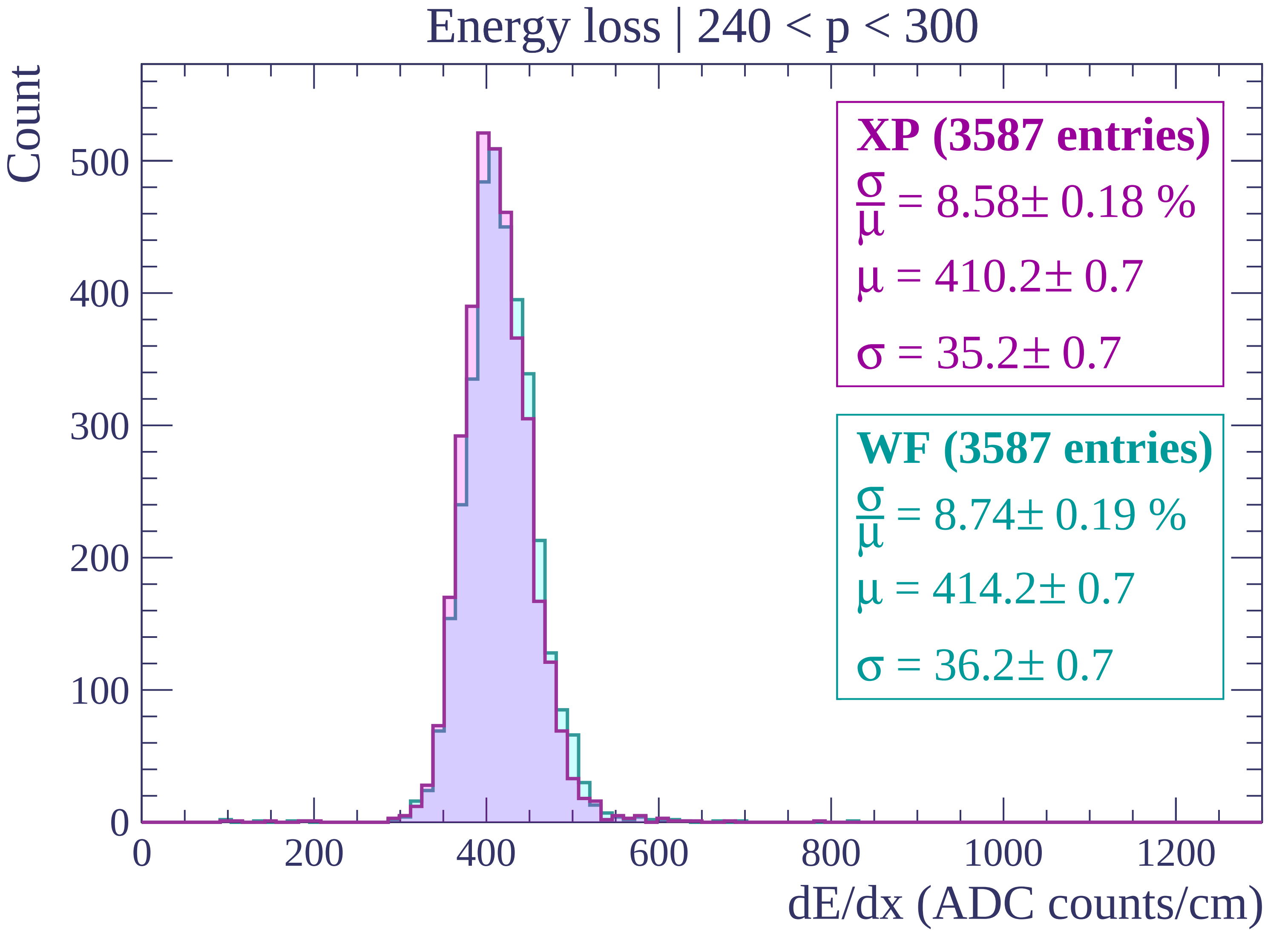
Energy loss | 60 < p < 120
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2173 entries)
σ/μ = 7.67± 0.22 %
μ = 554.3± 1.1
σ = 42.5± 1.1
WF (2173 entries)
σ/μ = 8.06± 0.25 %
μ = 553.7± 1.2
σ = 44.6± 1.3

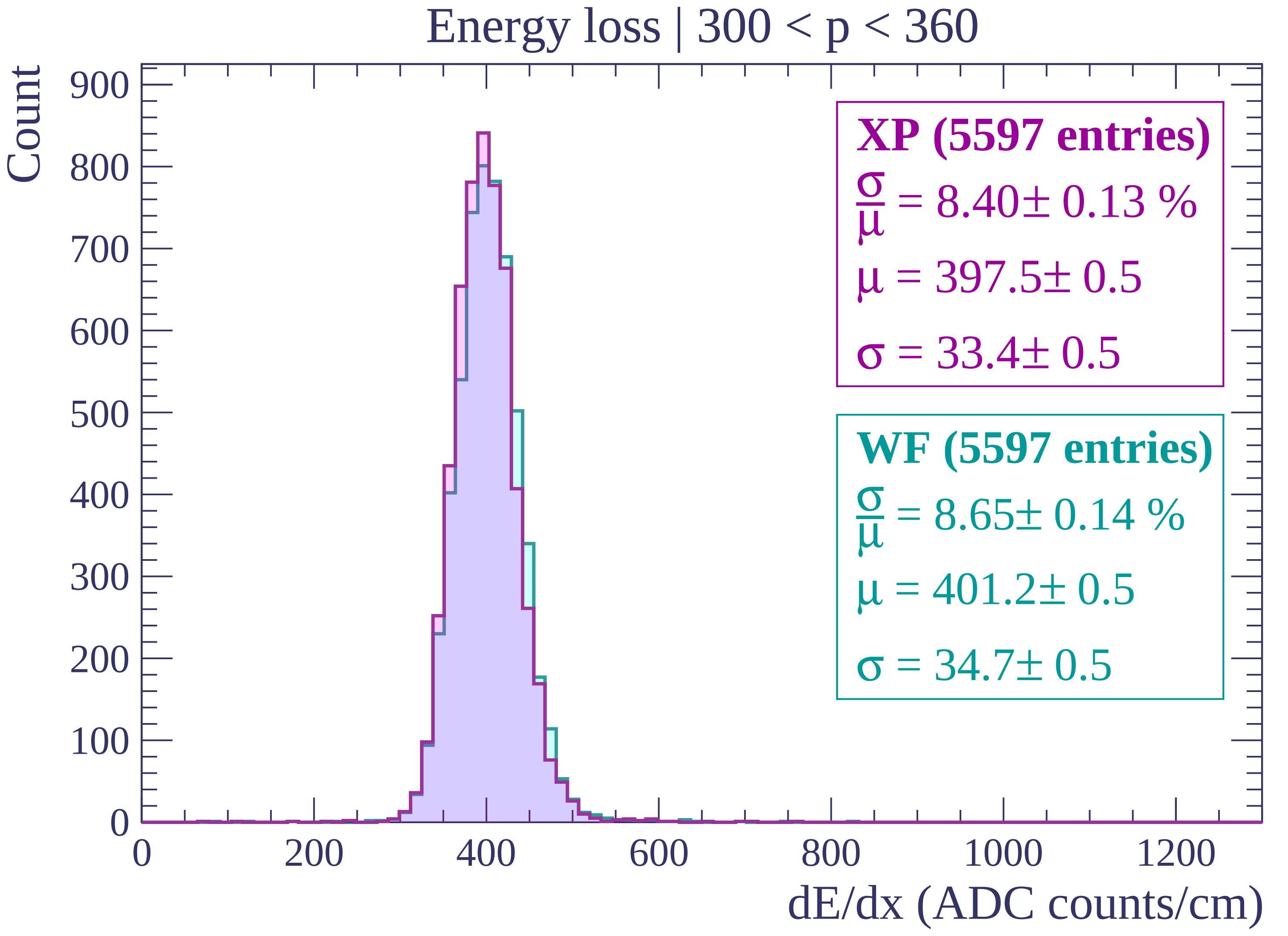


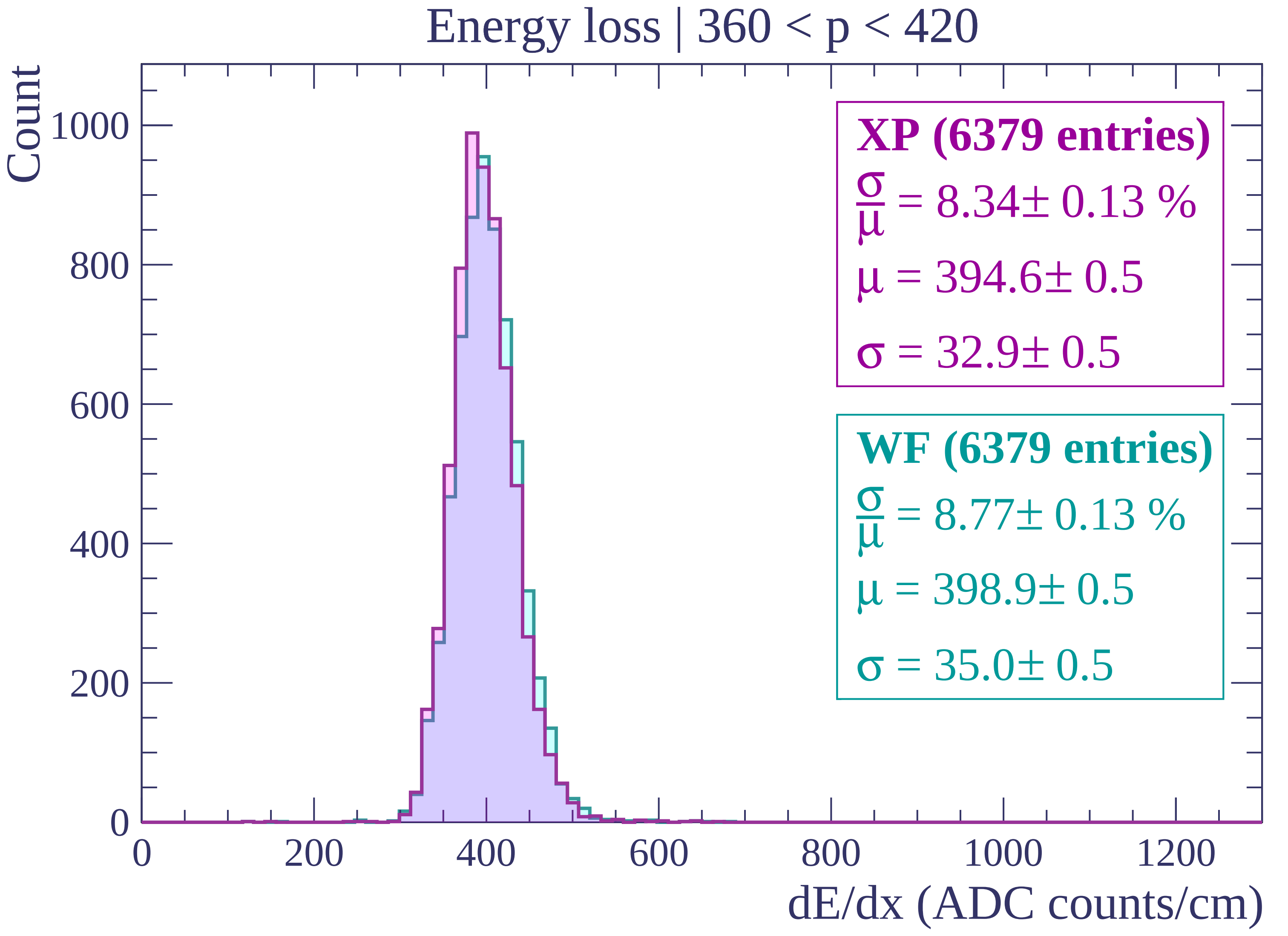
Energy loss | 120 < p < 180
Count
dE/dx (ADC counts/cm)
XP (1411 entries)
$\frac{\sigma}{\mu}$ = 11.77± 0.38 %
μ = 556.2± 2.1
σ = 65.5± 1.9
WF (1411 entries)
$\frac{\sigma}{\mu}$ = 12.06± 0.41 %
μ = 560.1± 2.3
σ = 67.5± 2.0
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200

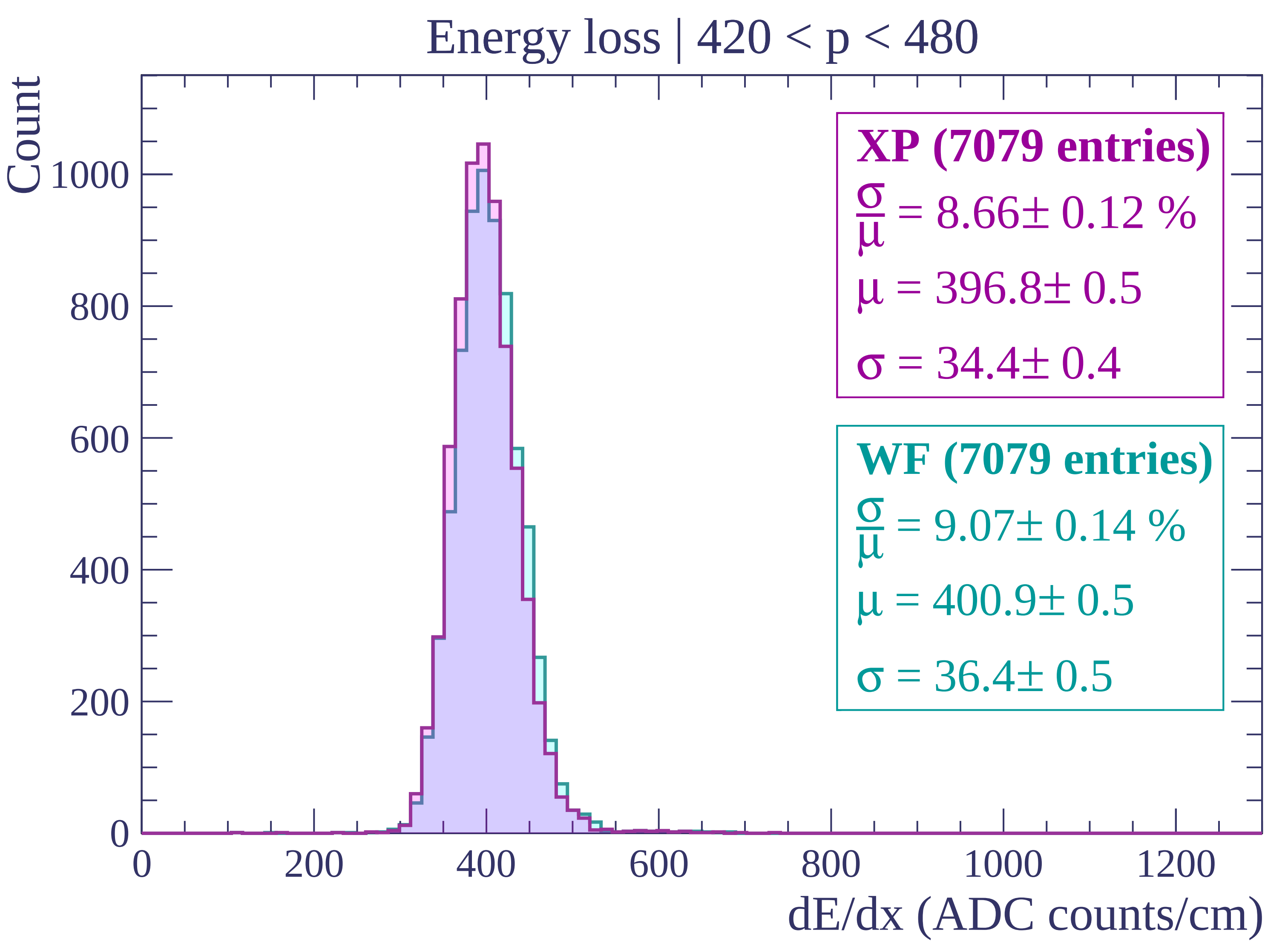


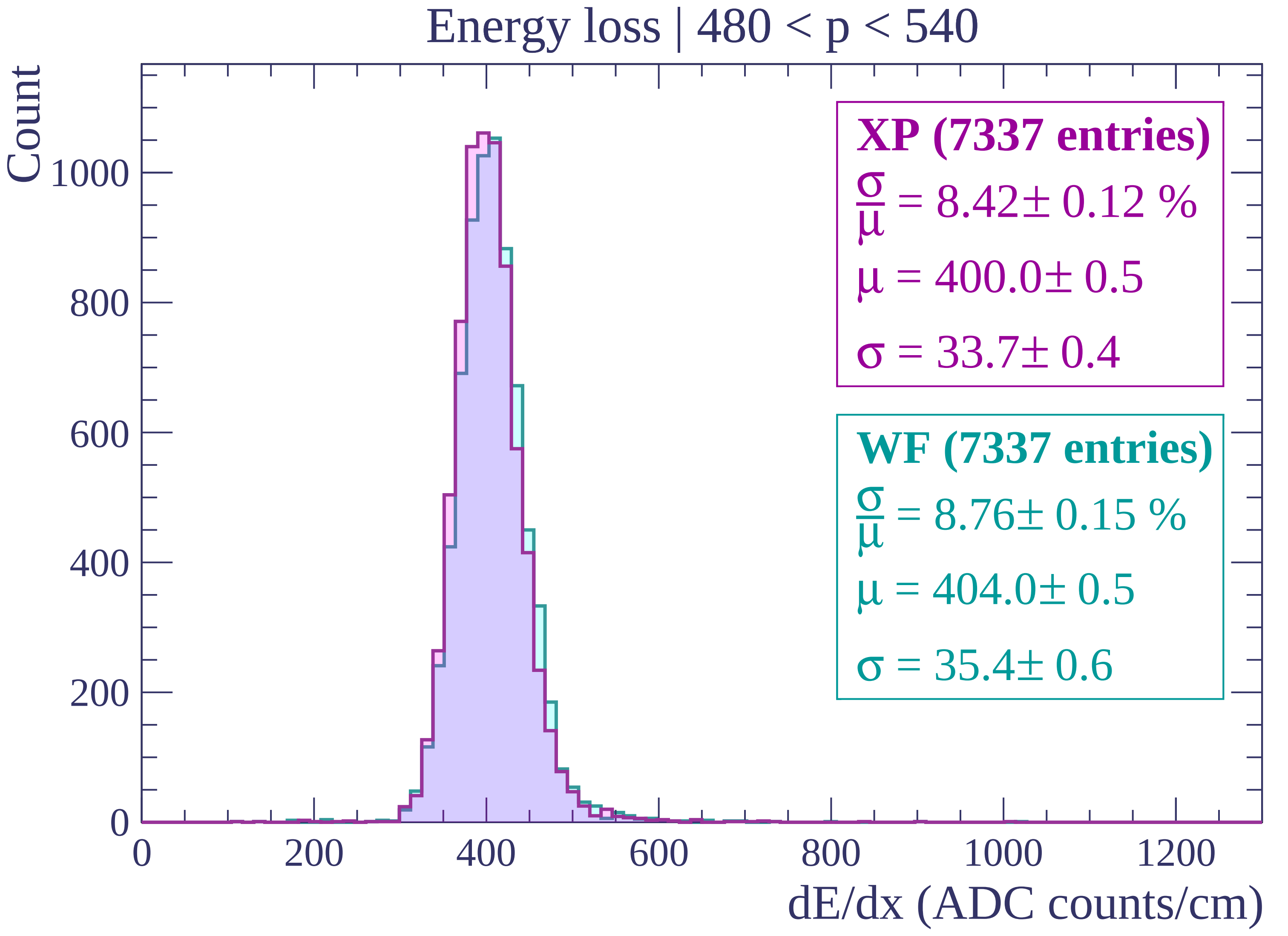
Energy loss | 180 < p < 240
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2259 entries)
σ/μ = 8.96± 0.30 %
μ = 448.1± 1.1
σ = 40.1± 1.3
WF (2259 entries)
σ/μ = 9.14± 0.27 %
μ = 455.5± 1.1
σ = 41.7± 1.1


Energy loss | 240 < p < 300
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (3587 entries)
σ/μ = 8.58± 0.18 %
μ = 410.2± 0.7
σ = 35.2± 0.7
WF (3587 entries)
σ/μ = 8.74± 0.19 %
μ = 414.2± 0.7
σ = 36.2± 0.7

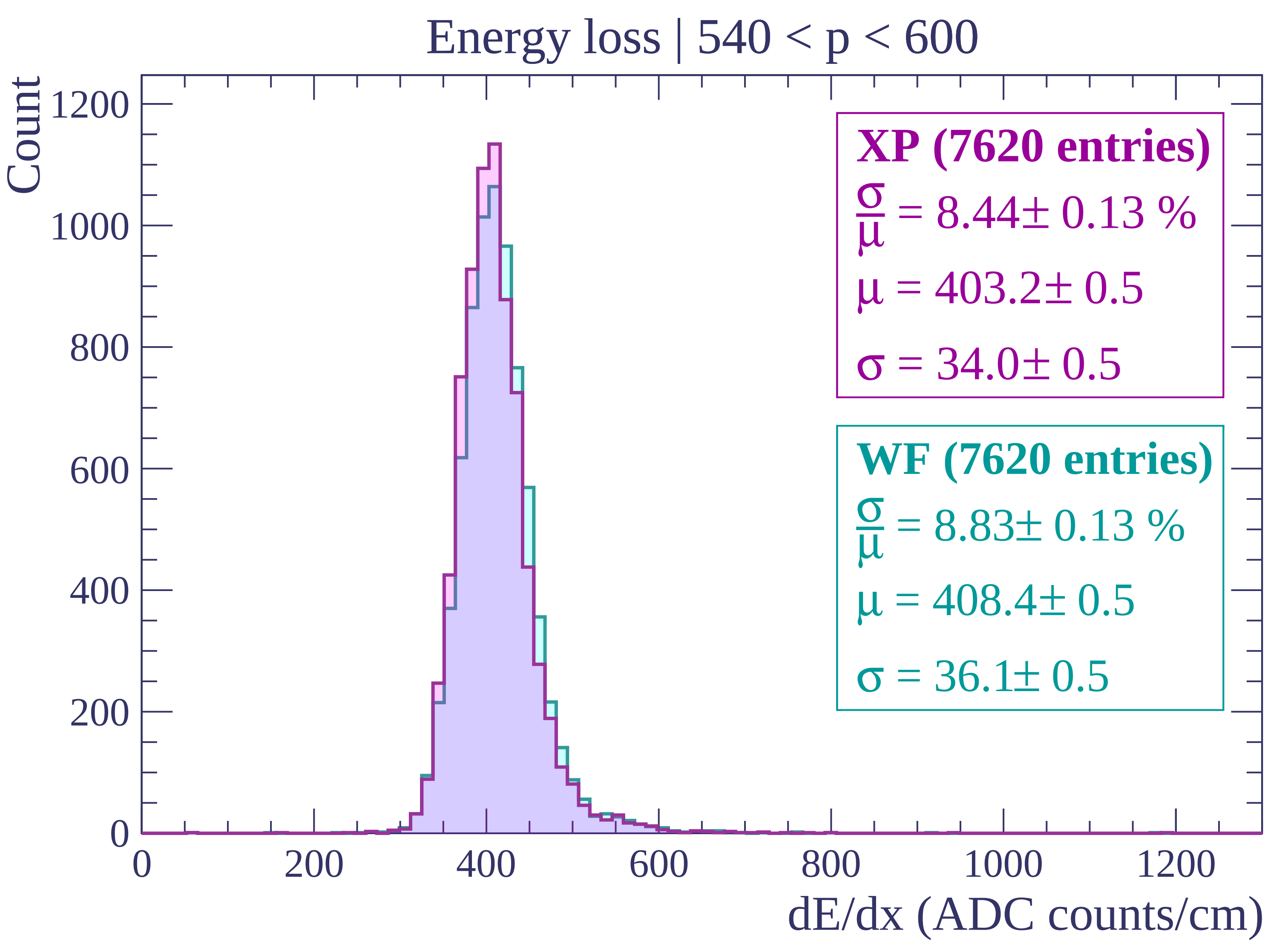

Energy loss | 300 < p < 360
Count
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5597 entries)
$\frac{\sigma}{\mu}$ = 8.40± 0.13 %
μ = 397.5± 0.5
σ = 33.4± 0.5
WF (5597 entries)
$\frac{\sigma}{\mu}$ = 8.65± 0.14 %
μ = 401.2± 0.5
σ = 34.7± 0.5


Energy loss | 360 < p < 420
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (6379 entries)
σ/μ = 8.34± 0.13 %
μ = 394.6± 0.5
σ = 32.9± 0.5
WF (6379 entries)
σ/μ = 8.77± 0.13 %
μ = 398.9± 0.5
σ = 35.0± 0.5

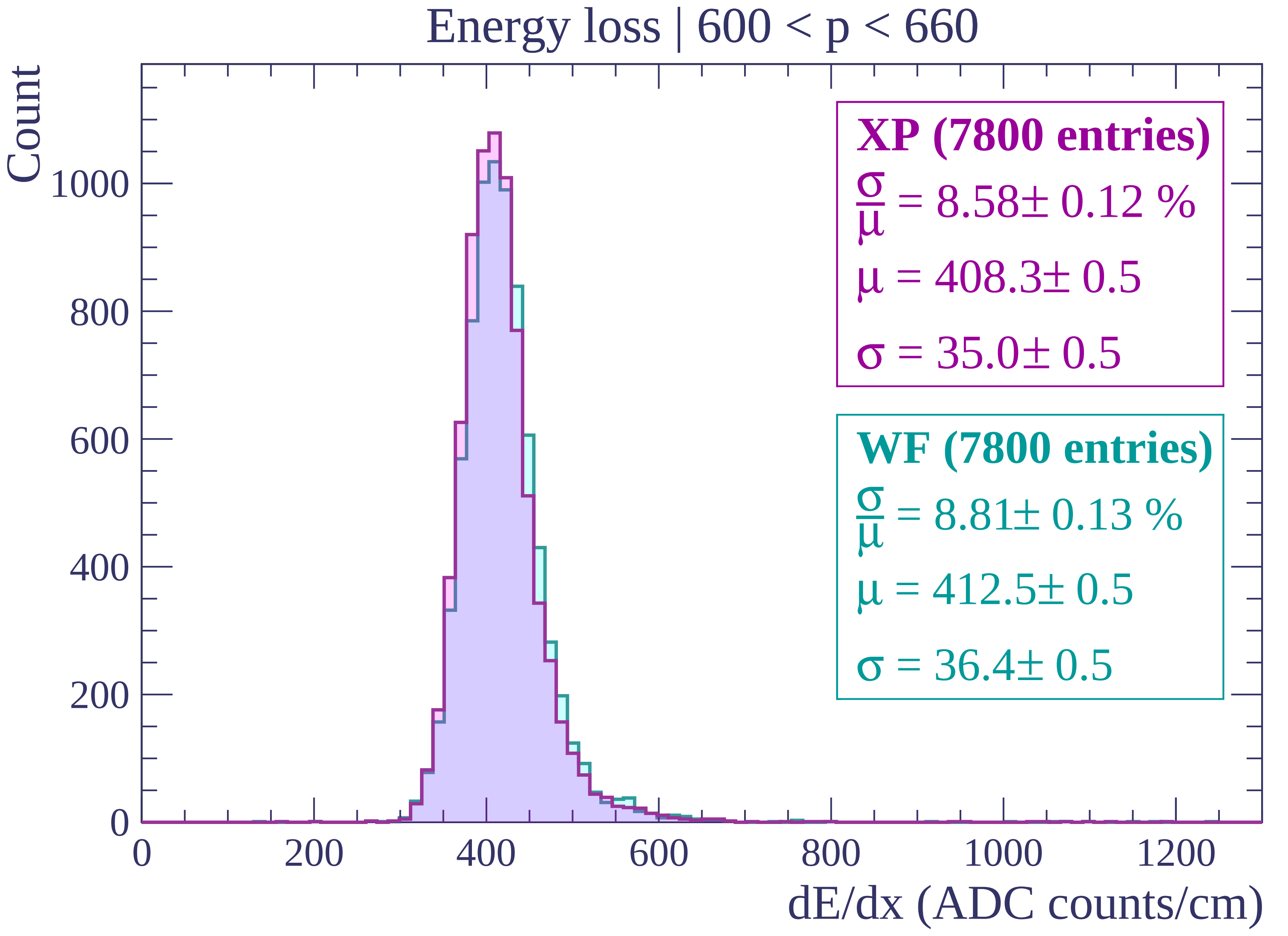

Energy loss | 420 < p < 480
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
0
200
400
600
800
1000
1200
XP (7079 entries)
σ/μ = 8.66± 0.12 %
μ = 396.8± 0.5
σ = 34.4± 0.4
WF (7079 entries)
σ/μ = 9.07± 0.14 %
μ = 400.9± 0.5
σ = 36.4± 0.5

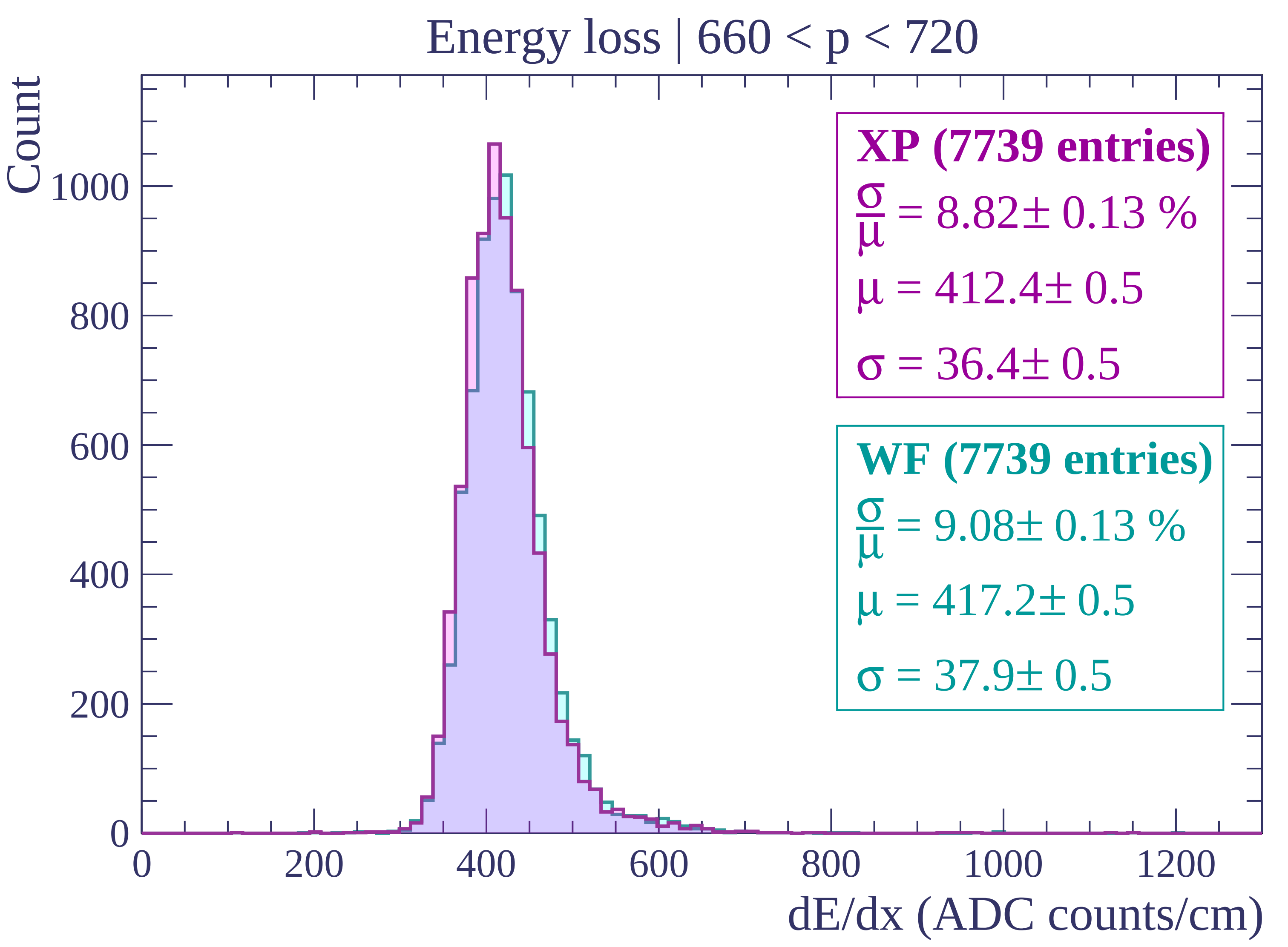


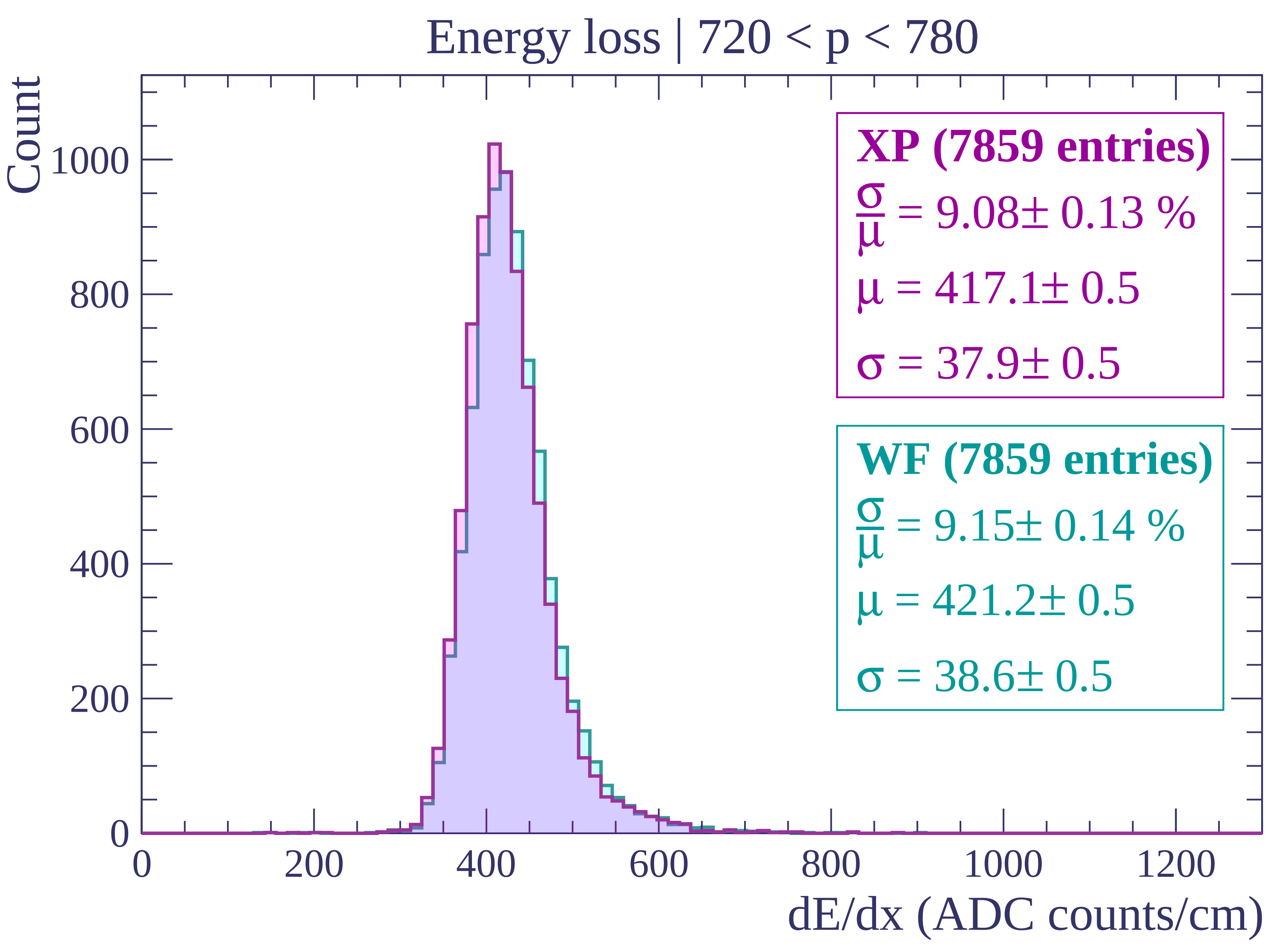
Energy loss | 480 < p < 540
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (7337 entries)
σ/μ = 8.42± 0.12 %
μ = 400.0± 0.5
σ = 33.7± 0.4
WF (7337 entries)
σ/μ = 8.76± 0.15 %
μ = 404.0± 0.5
σ = 35.4± 0.6

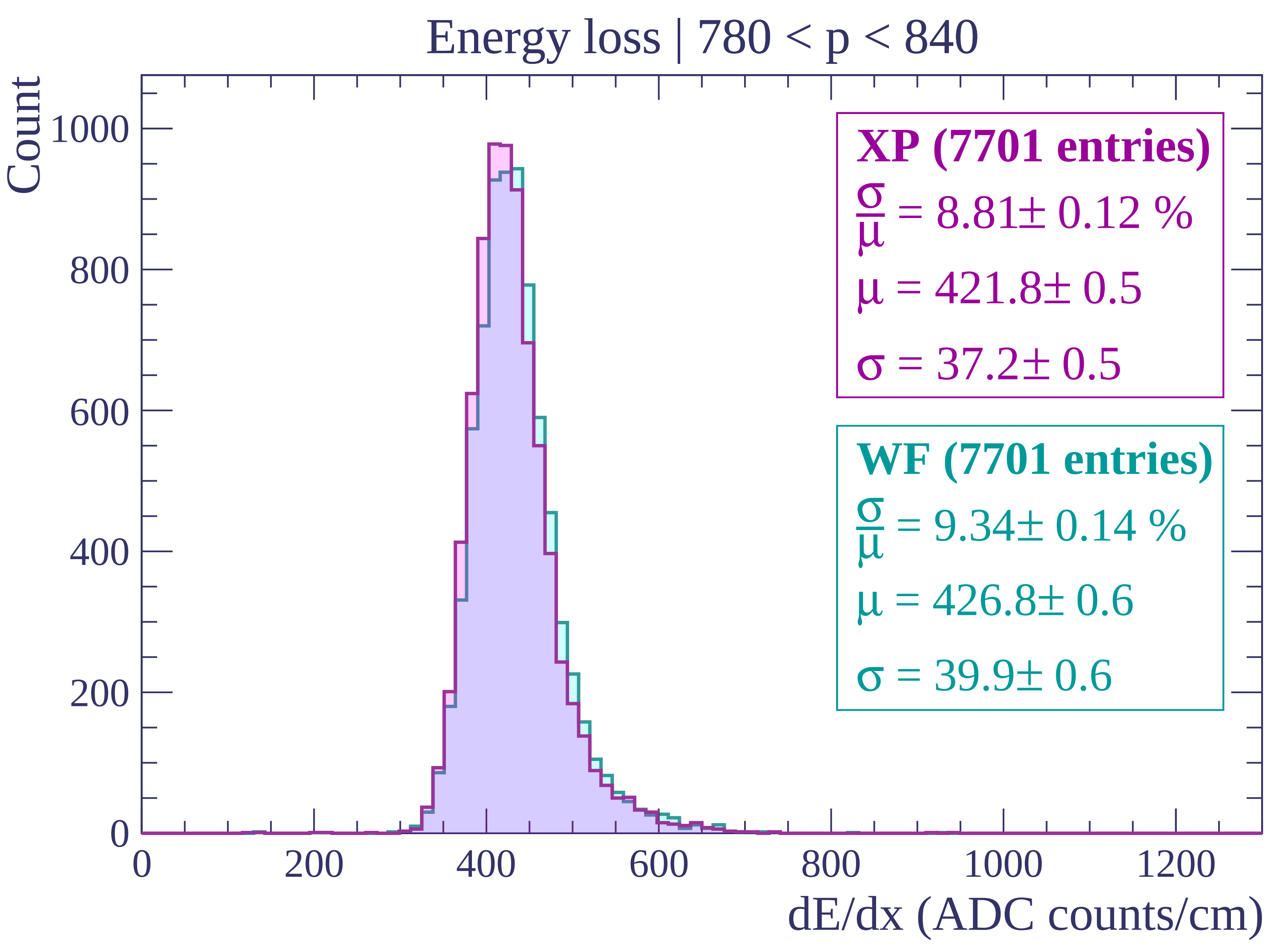


Energy loss | 540 < p < 600
Count
XP (7620 entries)
σ/μ = 8.44± 0.13 %
μ = 403.2± 0.5
σ = 34.0± 0.5
WF (7620 entries)
σ/μ = 8.83± 0.13 %
μ = 408.4± 0.5
σ = 36.1± 0.5
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

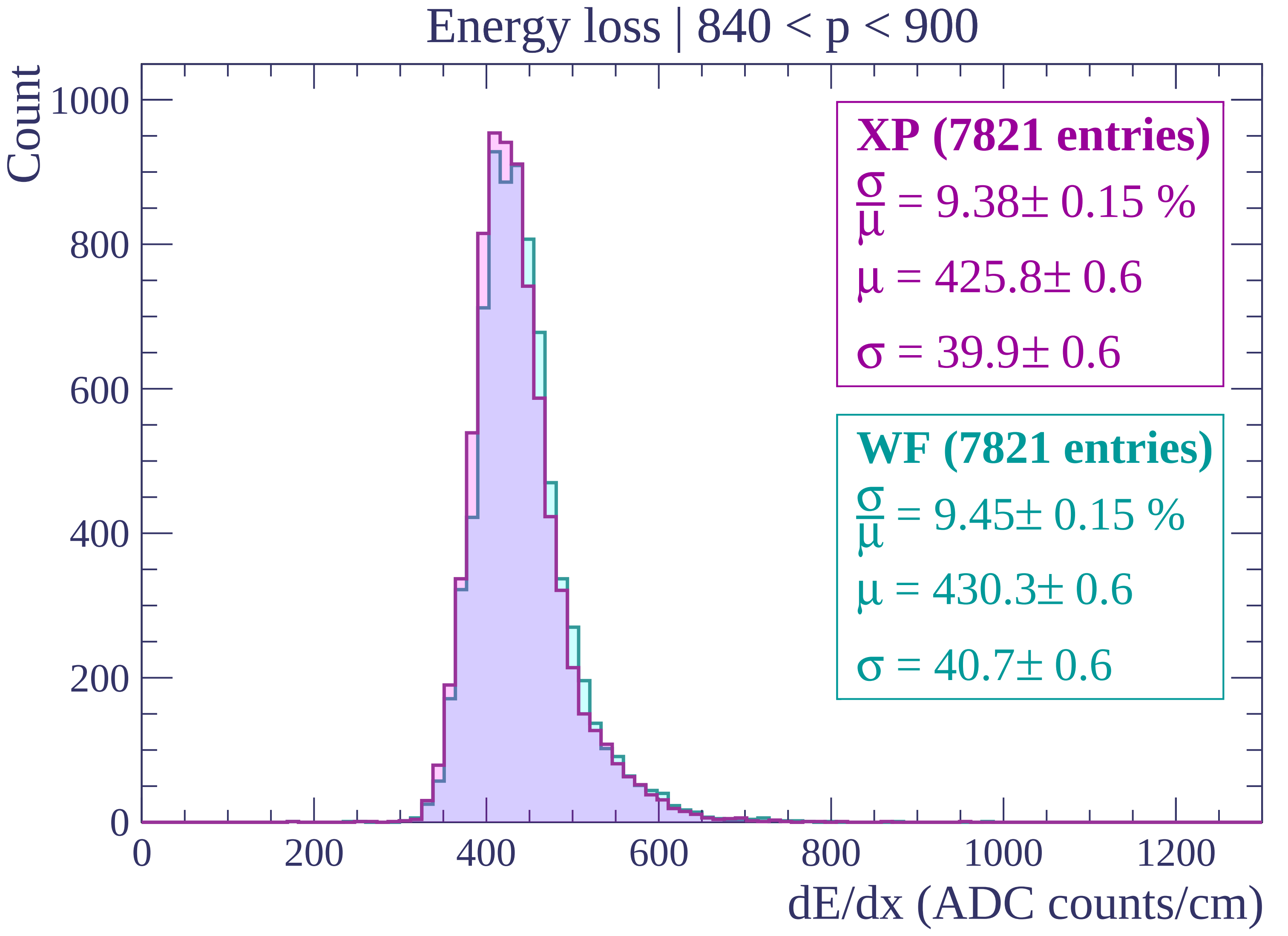

Energy loss | 600 < p < 660
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (7800 entries)
σ/μ = 8.58± 0.12 %
μ = 408.3± 0.5
σ = 35.0± 0.5
WF (7800 entries)
σ/μ = 8.81± 0.13 %
μ = 412.5± 0.5
σ = 36.4± 0.5


Energy loss | 660 < p < 720
Count
dE/dx (ADC counts/cm)
XP (7739 entries)
σ/μ = 8.82± 0.13 %
μ = 412.4± 0.5
σ = 36.4± 0.5
WF (7739 entries)
σ/μ = 9.08± 0.13 %
μ = 417.2± 0.5
σ = 37.9± 0.5
0
200
400
600
800
1000
1200

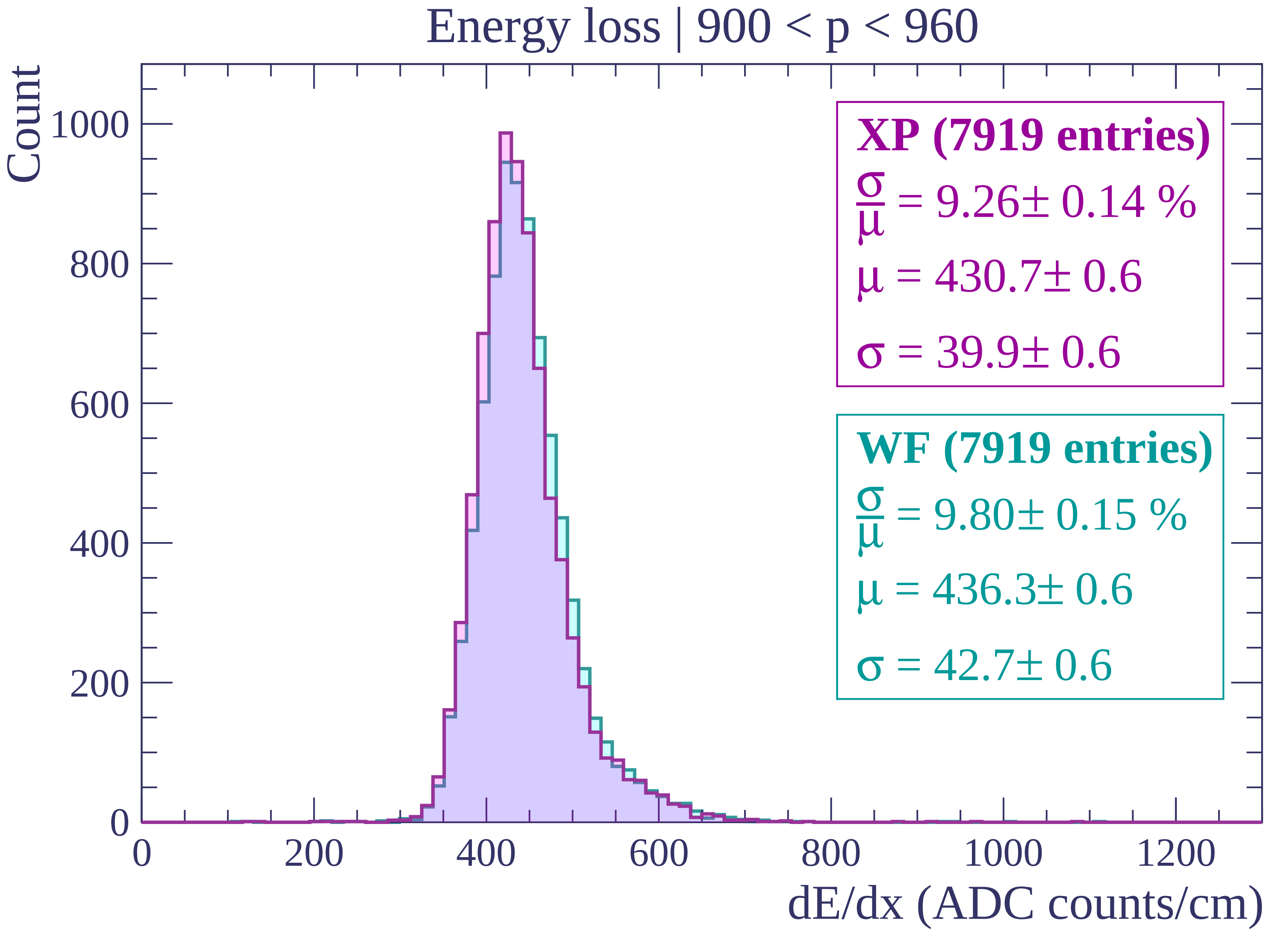

Energy loss | 720 < p < 780
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (7859 entries)
σ/μ = 9.08± 0.13 %
μ = 417.1± 0.5
σ = 37.9± 0.5
WF (7859 entries)
σ/μ = 9.15± 0.14 %
μ = 421.2± 0.5
σ = 38.6± 0.5



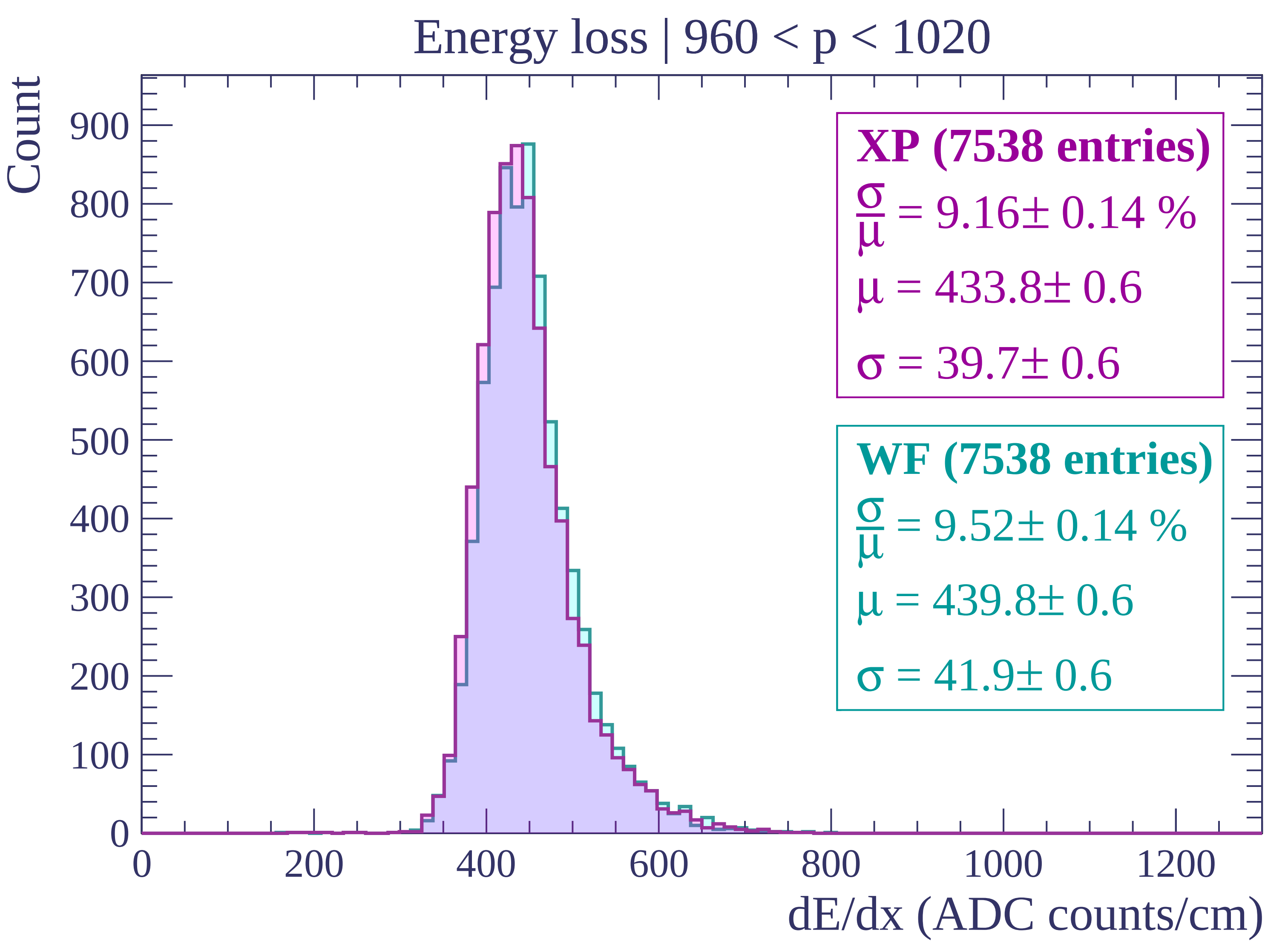
Energy loss | 780 < p < 840
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (7701 entries)
σ/μ = 8.81± 0.12 %
μ = 421.8± 0.5
σ = 37.2± 0.5
WF (7701 entries)
σ/μ = 9.34± 0.14 %
μ = 426.8± 0.6
σ = 39.9± 0.6



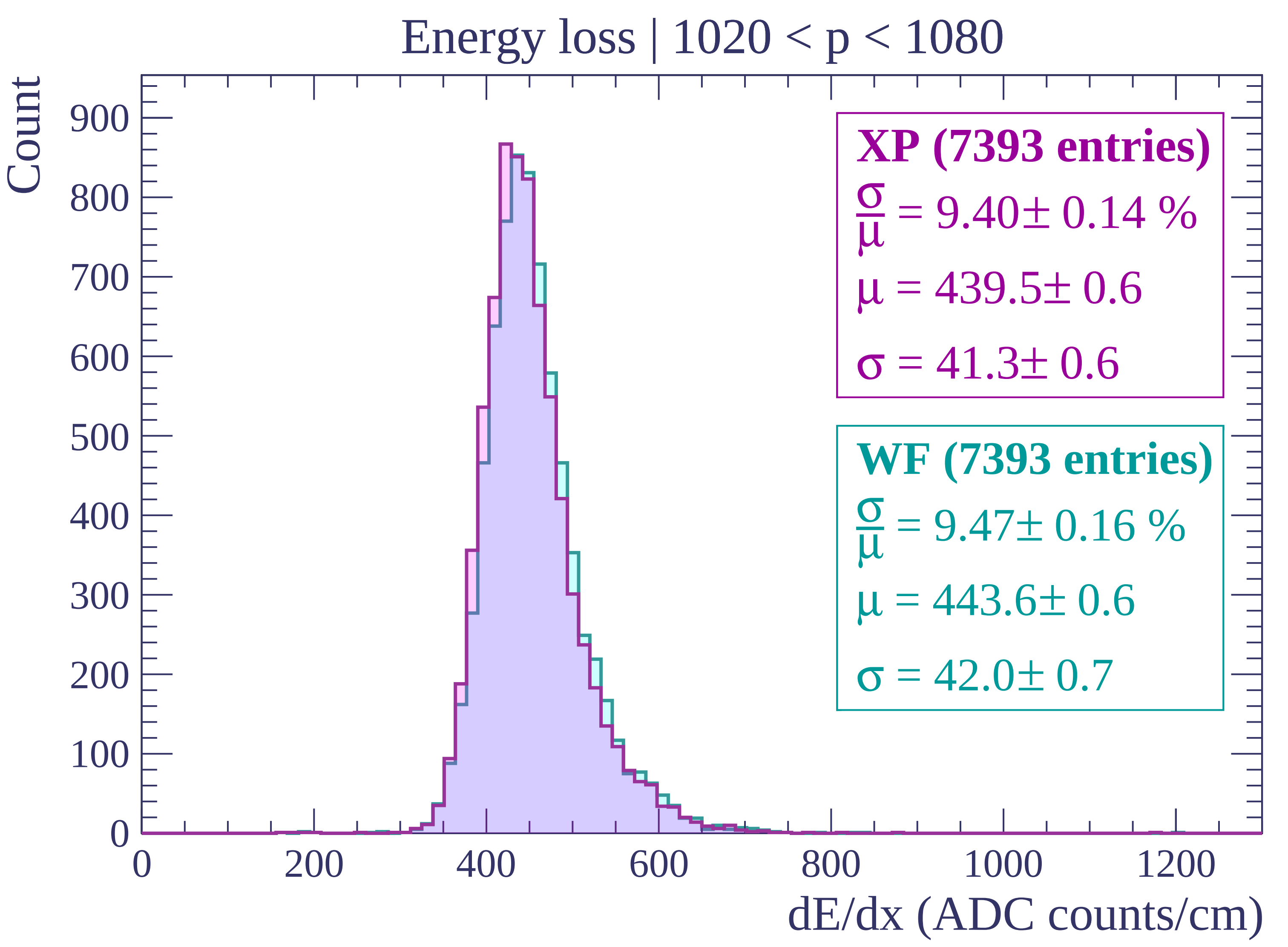
Energy loss | 840 < p < 900
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (7821 entries)
σ/μ = 9.38± 0.15 %
μ = 425.8± 0.6
σ = 39.9± 0.6
WF (7821 entries)
σ/μ = 9.45± 0.15 %
μ = 430.3± 0.6
σ = 40.7± 0.6



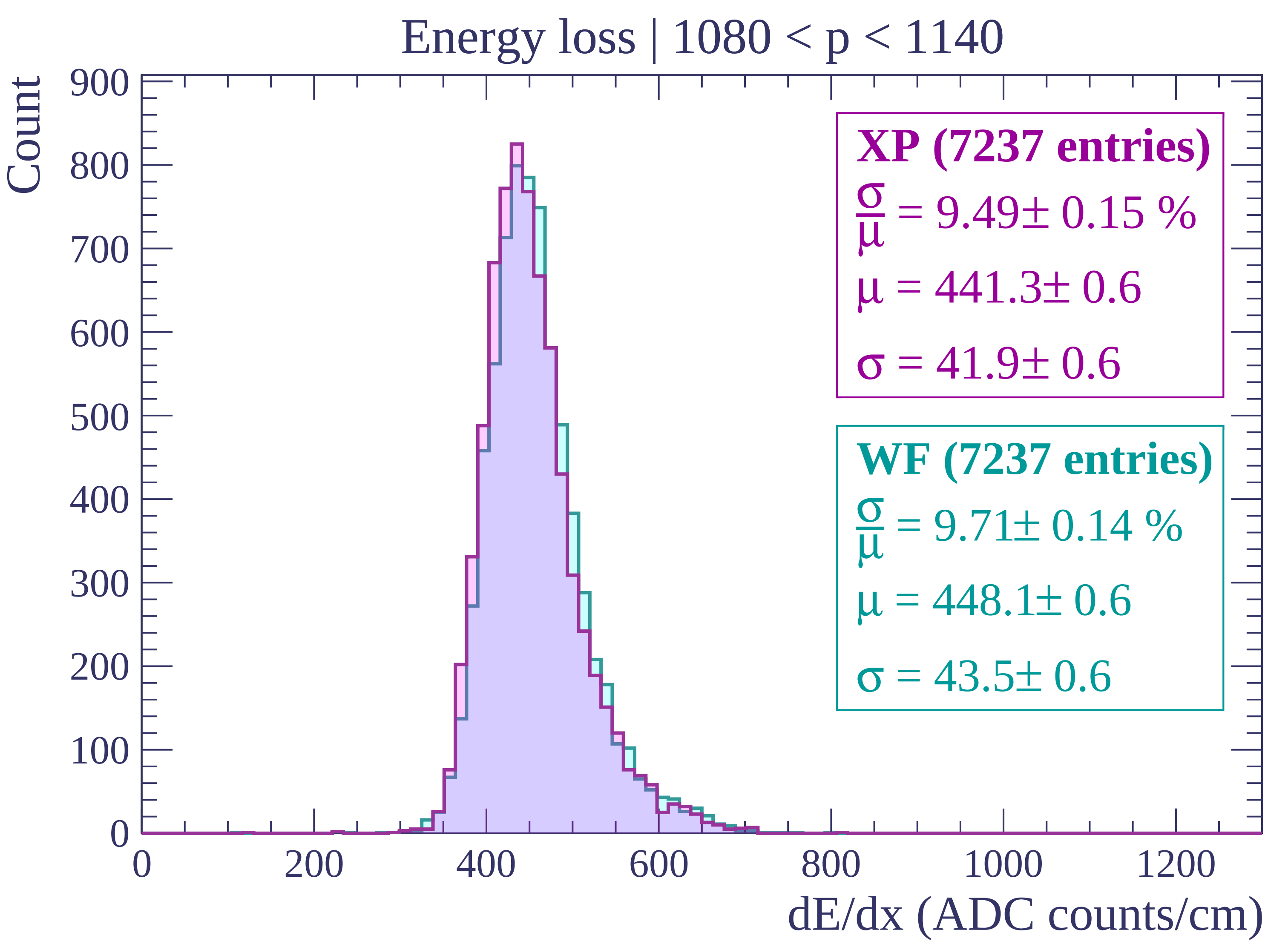
Energy loss | 900 < p < 960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (7919 entries)
σ/μ = 9.26± 0.14 %
μ = 430.7± 0.6
σ = 39.9± 0.6
WF (7919 entries)
σ/μ = 9.80± 0.15 %
μ = 436.3± 0.6
σ = 42.7± 0.6

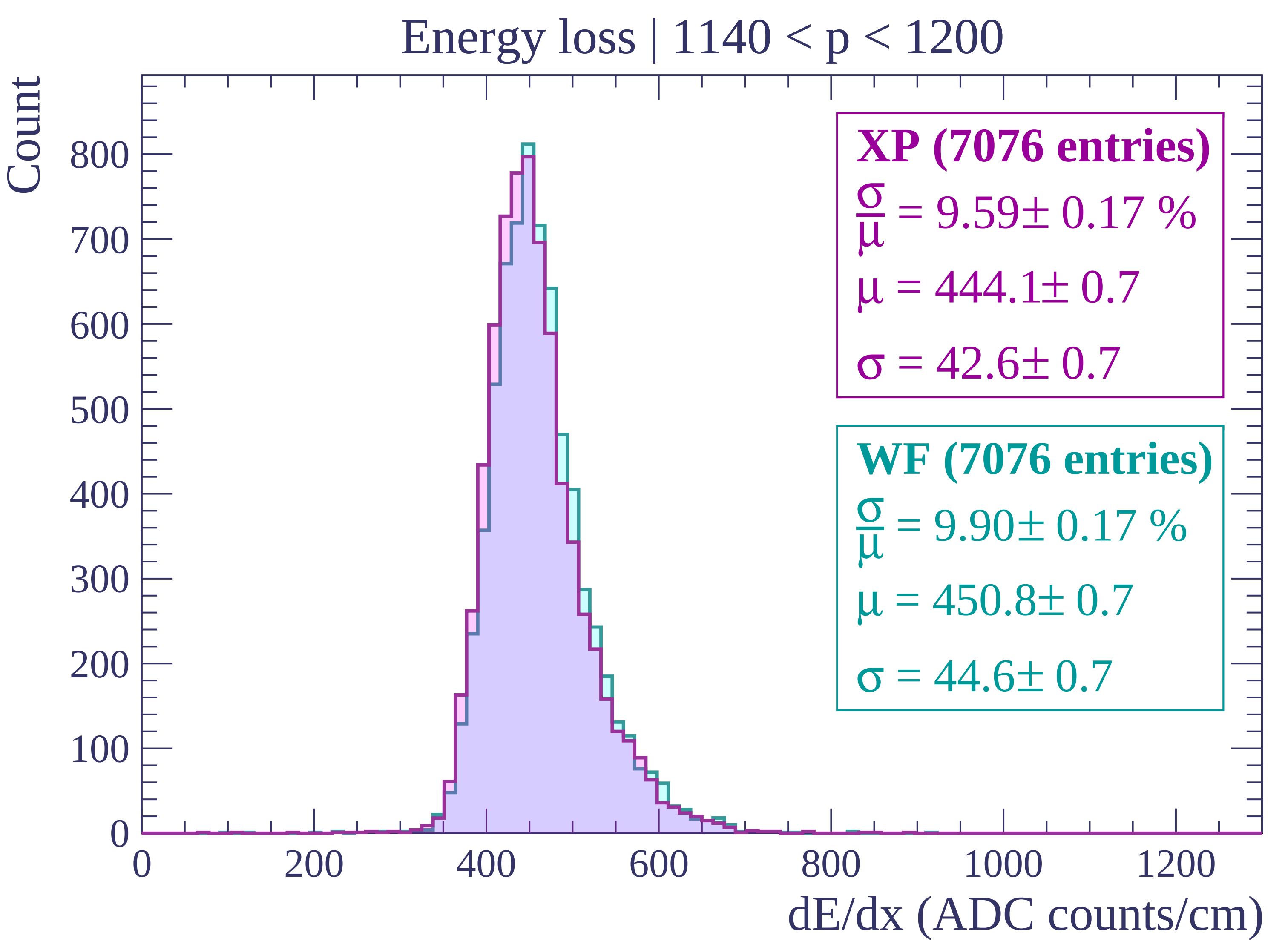


Energy loss | 960 < p < 1020
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP (7538 entries)
$\frac{\sigma}{\mu}$ = 9.16± 0.14 %
μ = 433.8± 0.6
σ = 39.7± 0.6
WF (7538 entries)
$\frac{\sigma}{\mu}$ = 9.52± 0.14 %
μ = 439.8± 0.6
σ = 41.9± 0.6

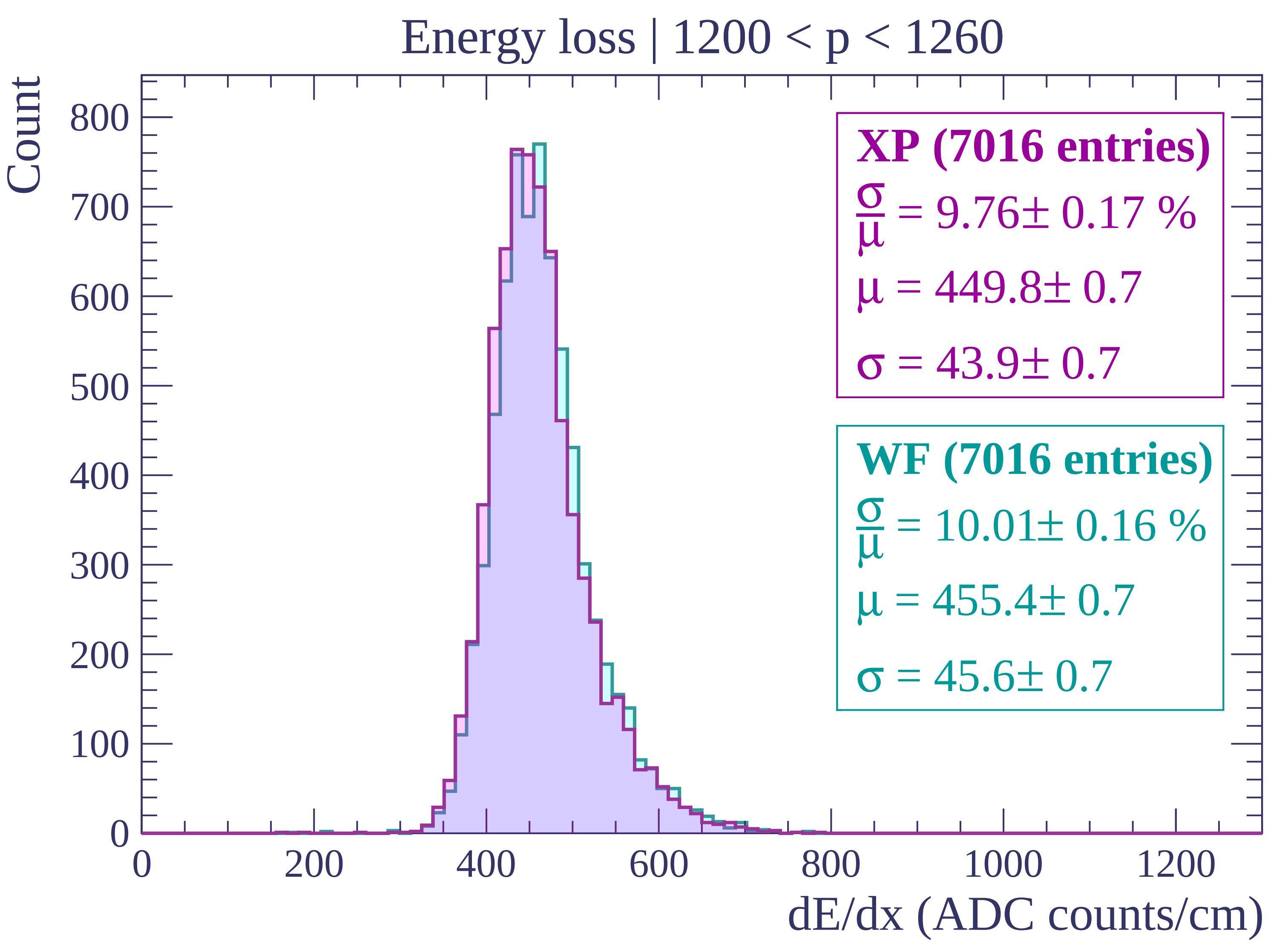


Energy loss | 1020 < p < 1080
Count
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (7393 entries)
σ/μ = 9.40± 0.14 %
μ = 439.5± 0.6
σ = 41.3± 0.6
WF (7393 entries)
σ/μ = 9.47± 0.16 %
μ = 443.6± 0.6
σ = 42.0± 0.7

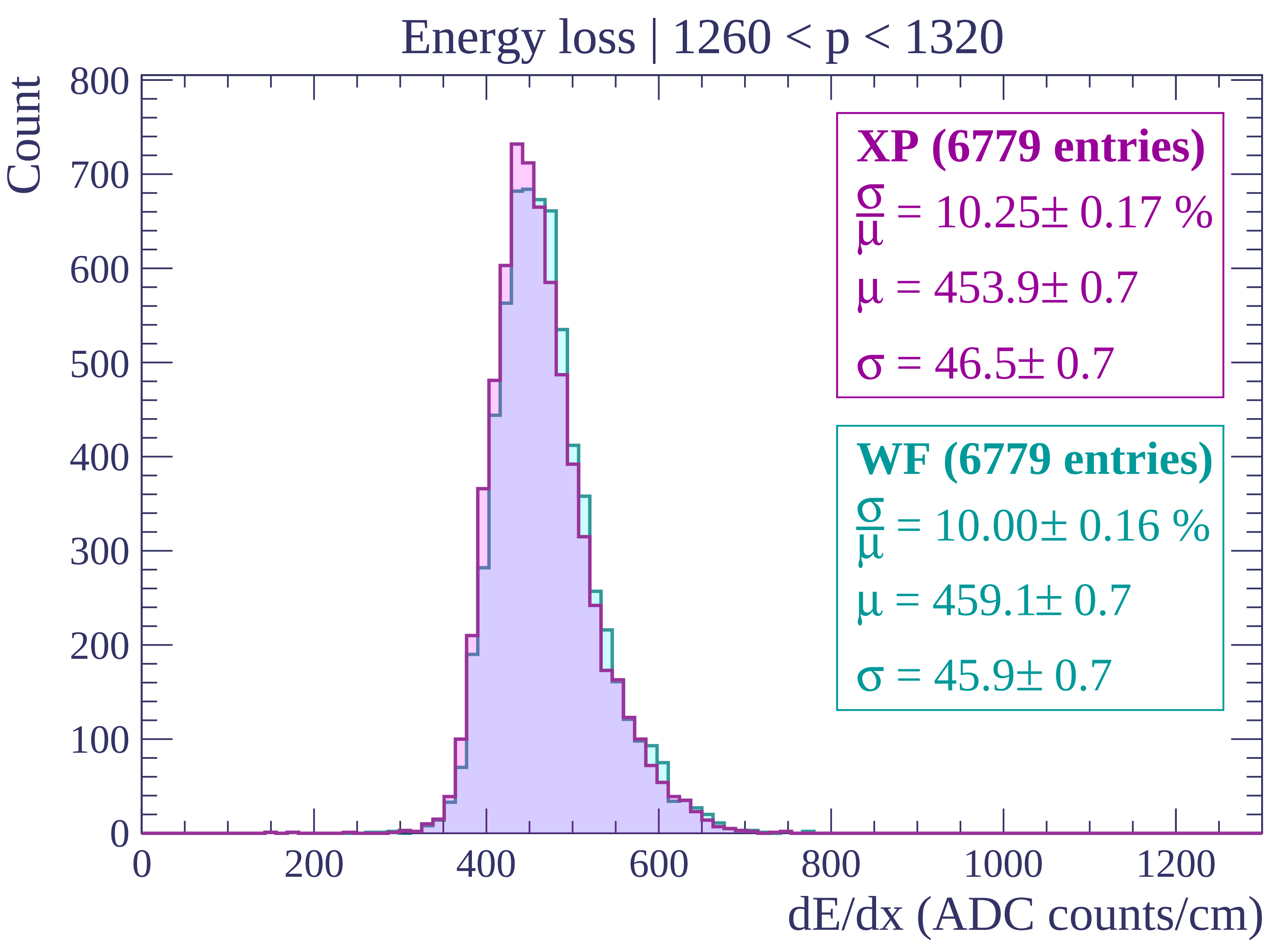

Energy loss | 1080 < p < 1140
Count
dE/dx (ADC counts/cm)
XP (7237 entries)
σ/μ = 9.49± 0.15 %
μ = 441.3± 0.6
σ = 41.9± 0.6
WF (7237 entries)
σ/μ = 9.71± 0.14 %
μ = 448.1± 0.6
σ = 43.5± 0.6

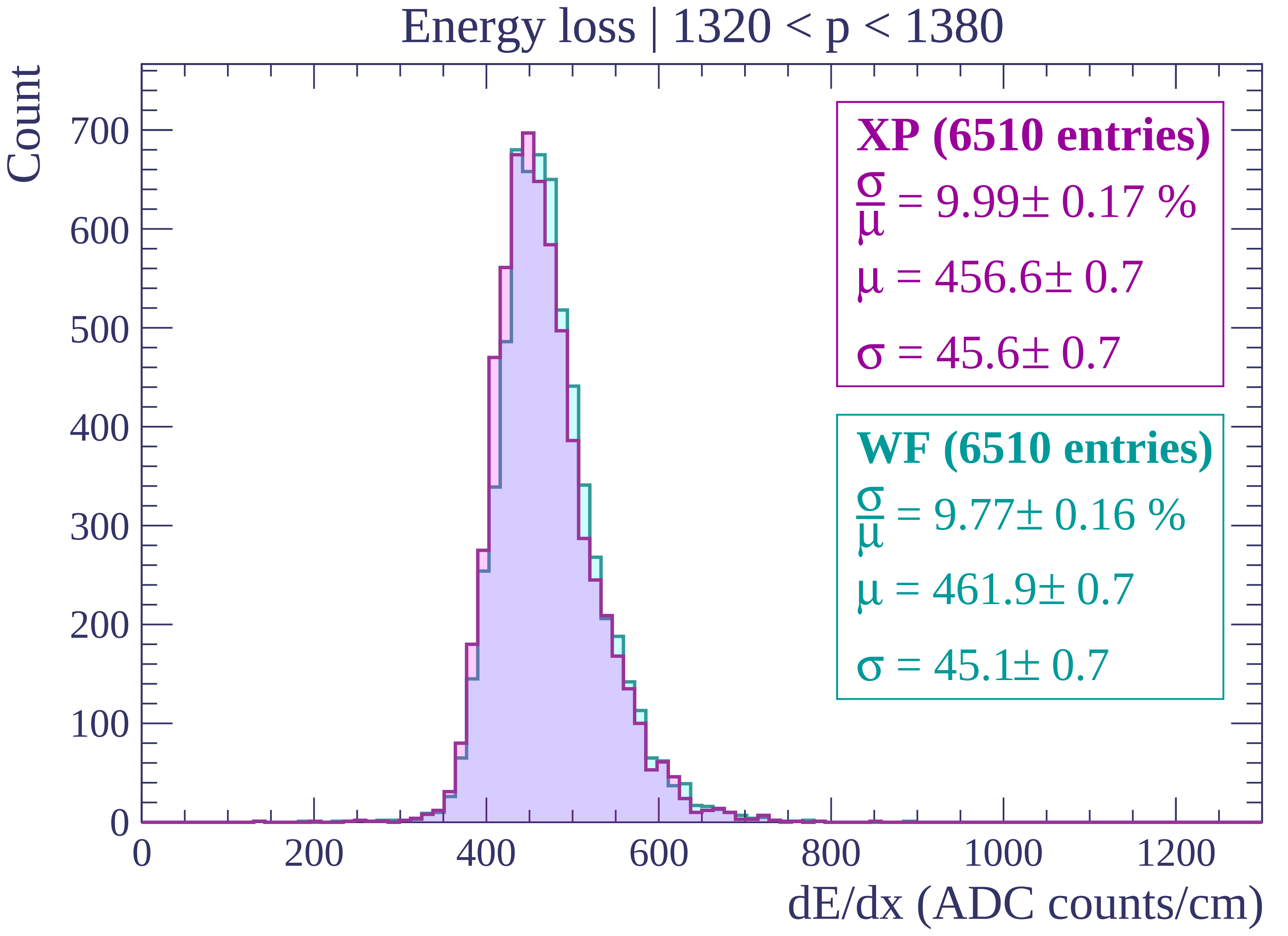


Energy loss | 1140 < p < 1200
Count
dE/dx (ADC counts/cm)
XP (7076 entries)
$\frac{\sigma}{\mu}$ = 9.59± 0.17 %
μ = 444.1± 0.7
σ = 42.6± 0.7
WF (7076 entries)
$\frac{\sigma}{\mu}$ = 9.90± 0.17 %
μ = 450.8± 0.7
σ = 44.6± 0.7
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200

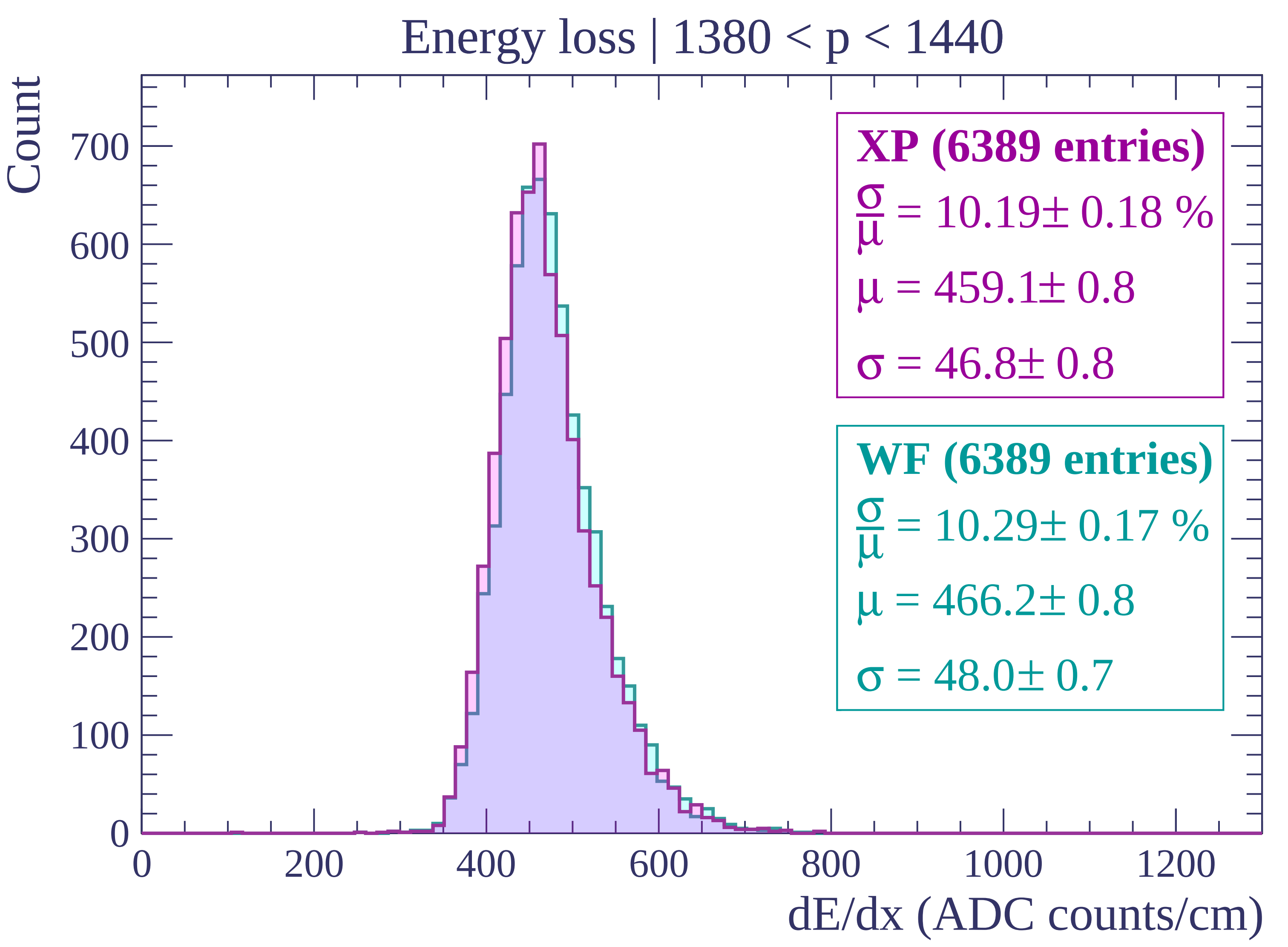


Energy loss | 1200 < p < 1260
Count
dE/dx (ADC counts/cm)
XP (7016 entries)
σ/μ = 9.76± 0.17 %
μ = 449.8± 0.7
σ = 43.9± 0.7
WF (7016 entries)
σ/μ = 10.01± 0.16 %
μ = 455.4± 0.7
σ = 45.6± 0.7
0
200
400
600
800
1000
1200
100
300
500
700

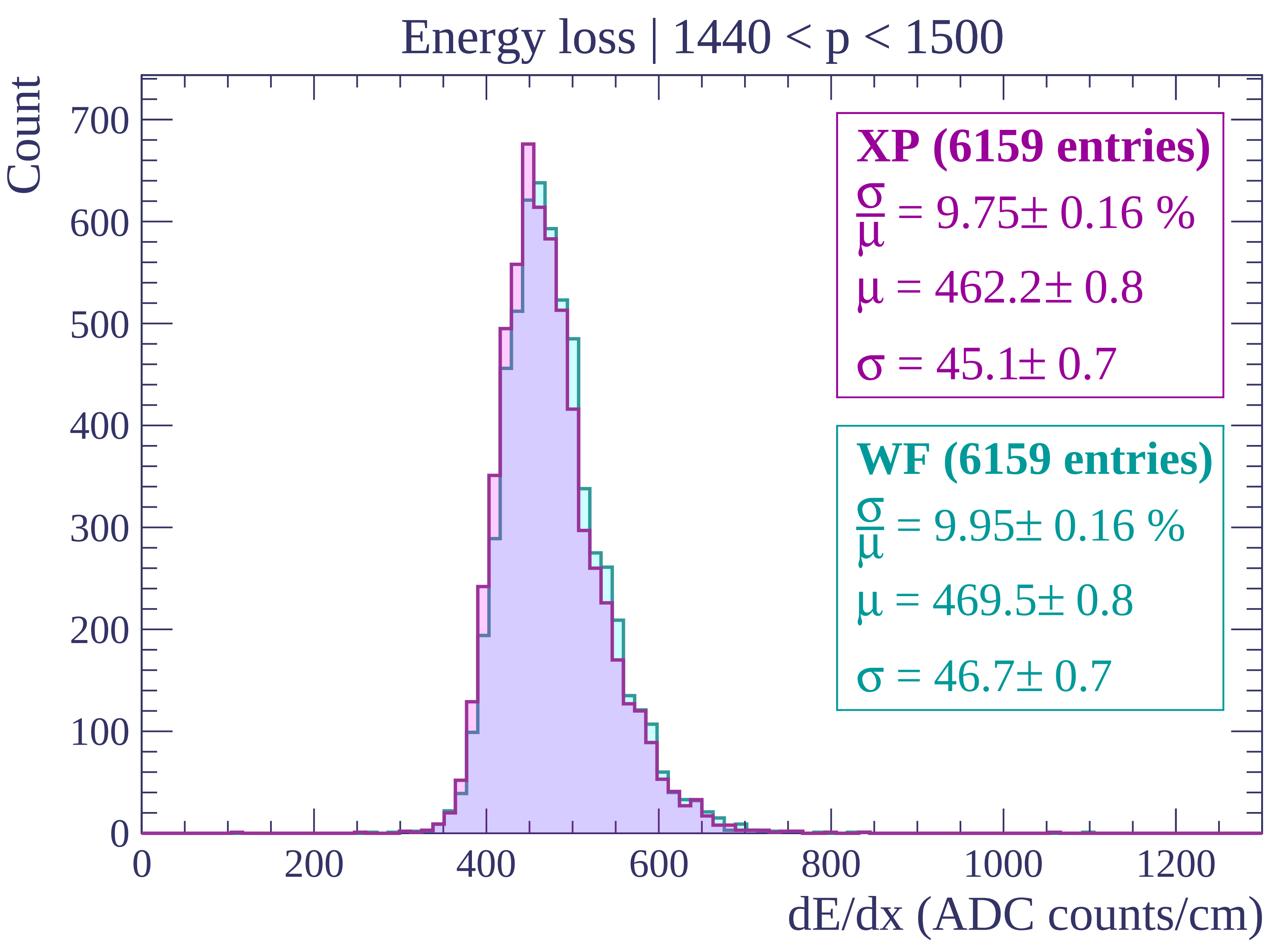

Energy loss | 1260 < p < 1320
Count
dE/dx (ADC counts/cm)
XP (6779 entries)
$\frac{\sigma}{\mu}$ = 10.25± 0.17 %
μ = 453.9± 0.7
σ = 46.5± 0.7
WF (6779 entries)
$\frac{\sigma}{\mu}$ = 10.00± 0.16 %
μ = 459.1± 0.7
σ = 45.9± 0.7
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200

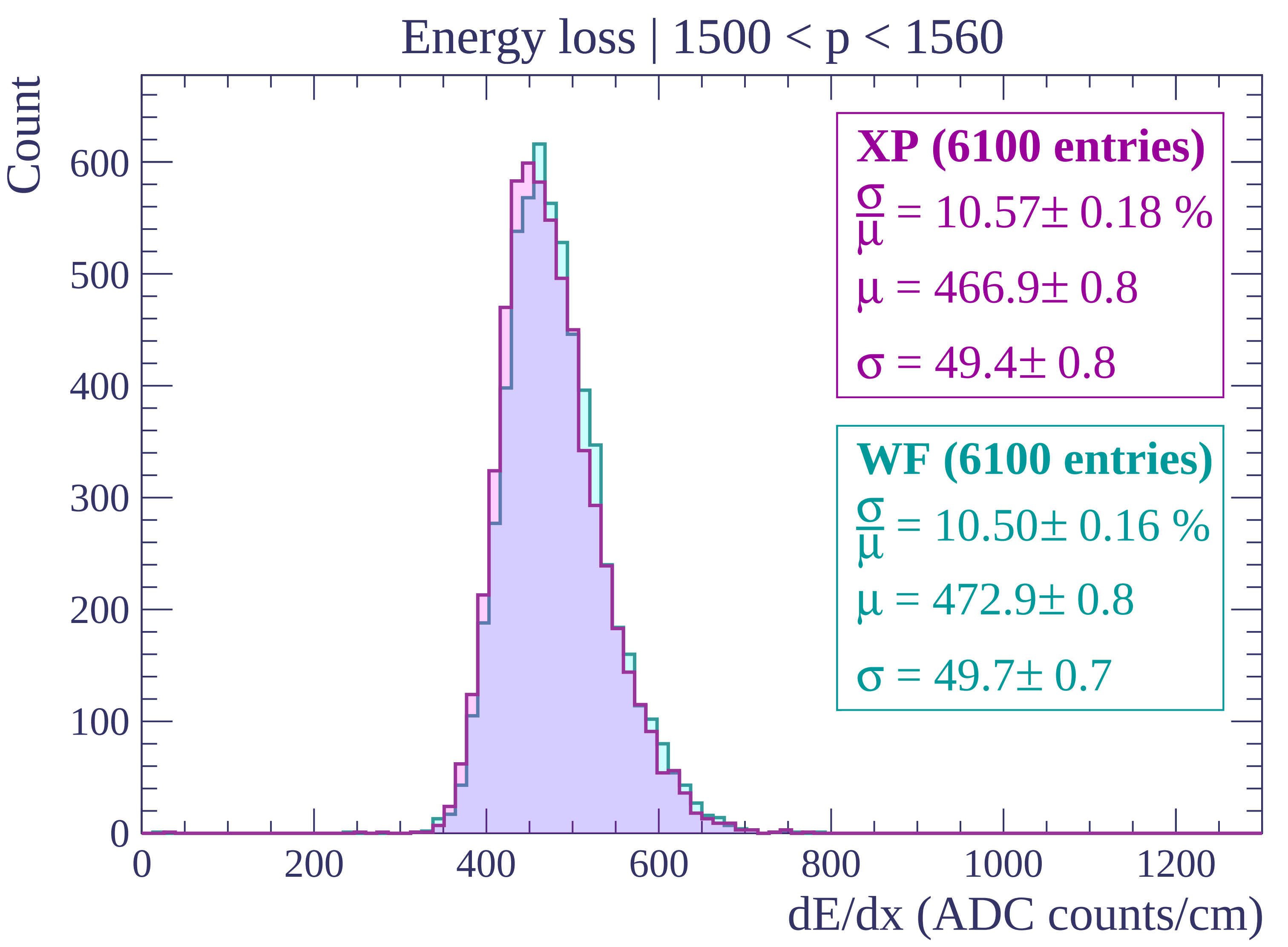


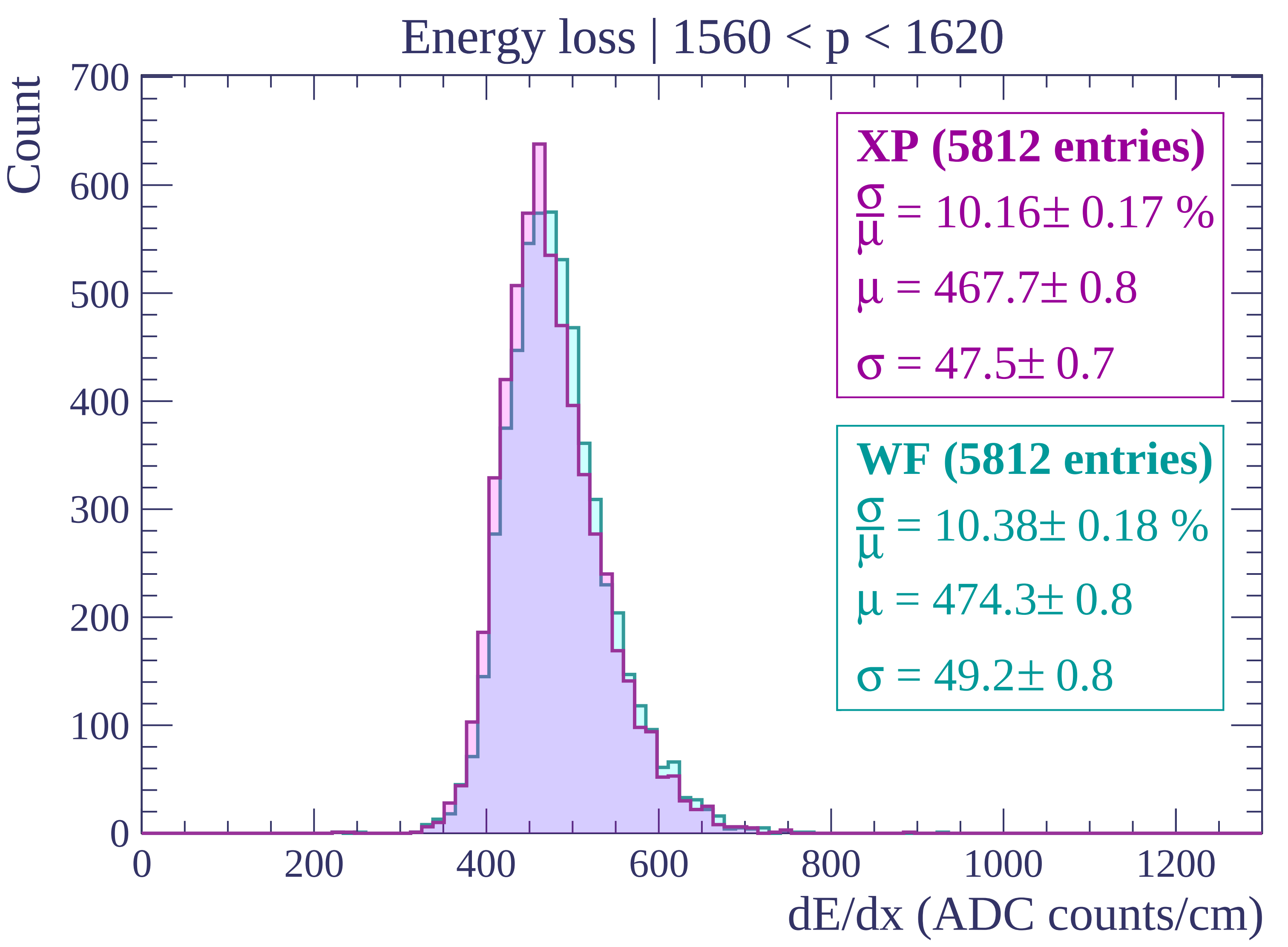
Energy loss | 1320 < p < 1380
Count
dE/dx (ADC counts/cm)
XP (6510 entries)
$\frac{\sigma}{\mu}$ = 9.99± 0.17 %
μ = 456.6± 0.7
σ = 45.6± 0.7
WF (6510 entries)
$\frac{\sigma}{\mu}$ = 9.77± 0.16 %
μ = 461.9± 0.7
σ = 45.1± 0.7

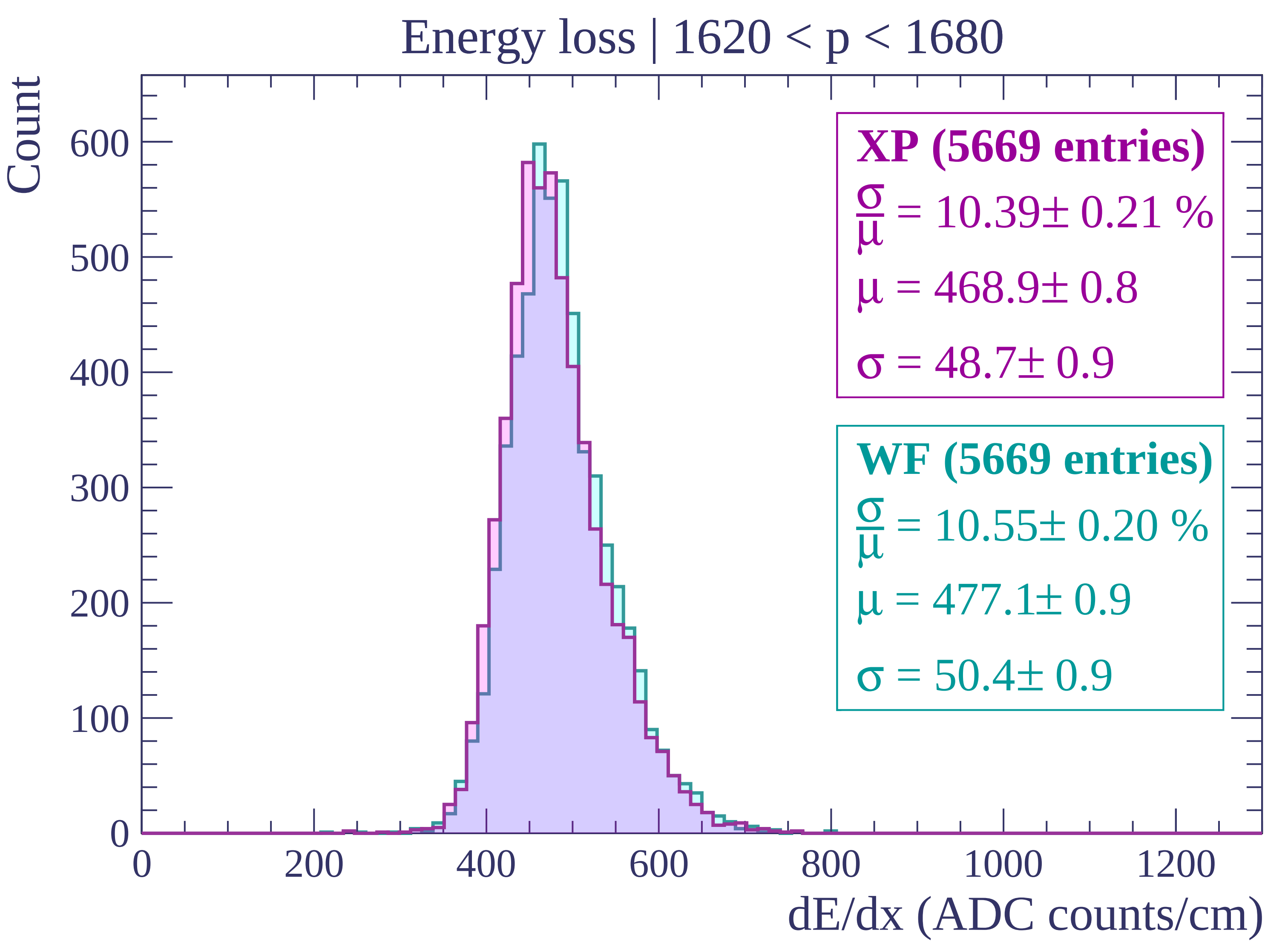


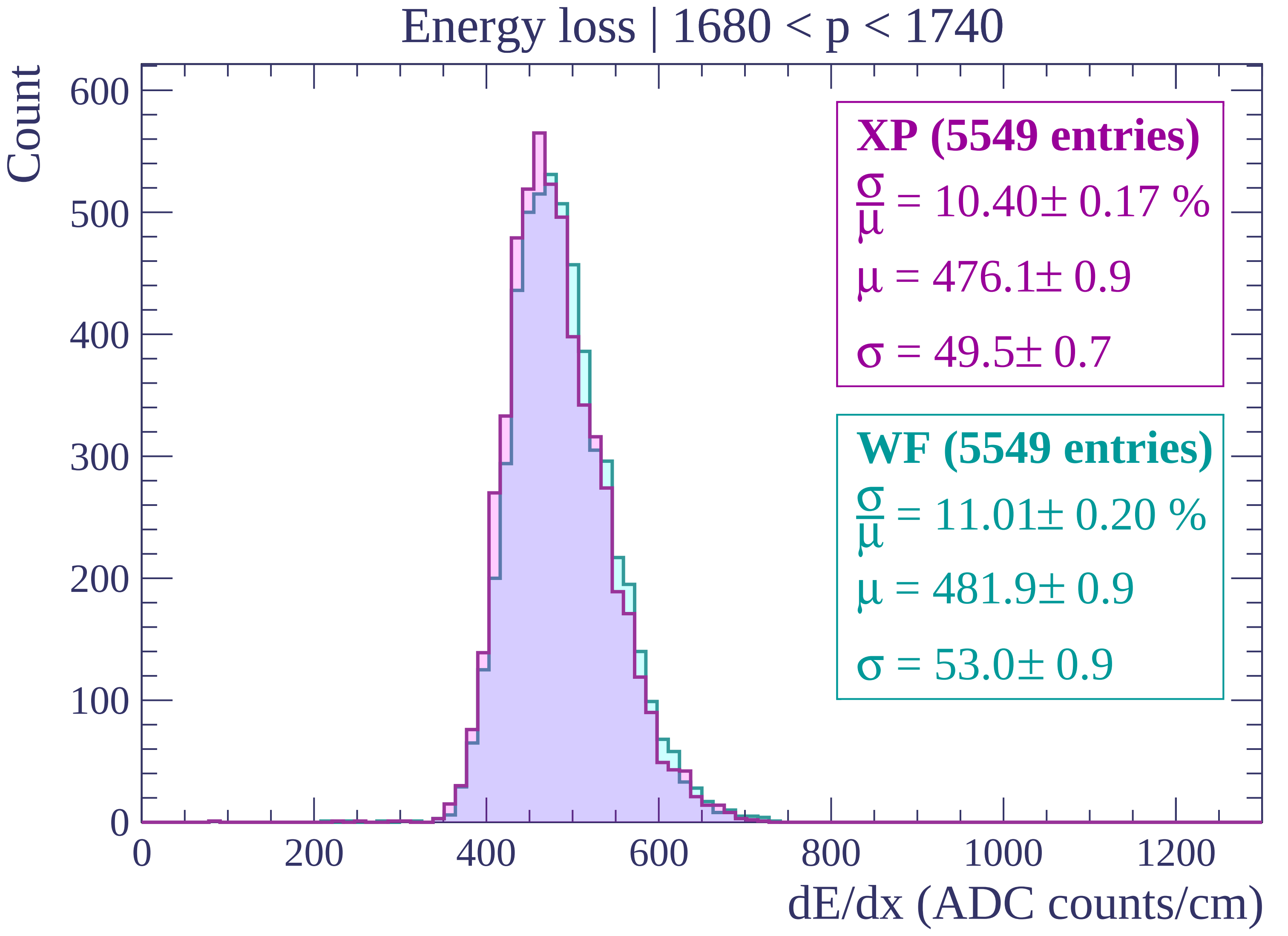
Energy loss | 1380 < p < 1440
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (6389 entries)
$\frac{\sigma}{\mu}$ = 10.19± 0.18 %
μ = 459.1± 0.8
σ = 46.8± 0.8
WF (6389 entries)
$\frac{\sigma}{\mu}$ = 10.29± 0.17 %
μ = 466.2± 0.8
σ = 48.0± 0.7


Energy loss | 1440 < p < 1500
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (6159 entries)
$\frac{\sigma}{\mu}$ = 9.75± 0.16 %
μ = 462.2± 0.8
σ = 45.1± 0.7
WF (6159 entries)
$\frac{\sigma}{\mu}$ = 9.95± 0.16 %
μ = 469.5± 0.8
σ = 46.7± 0.7

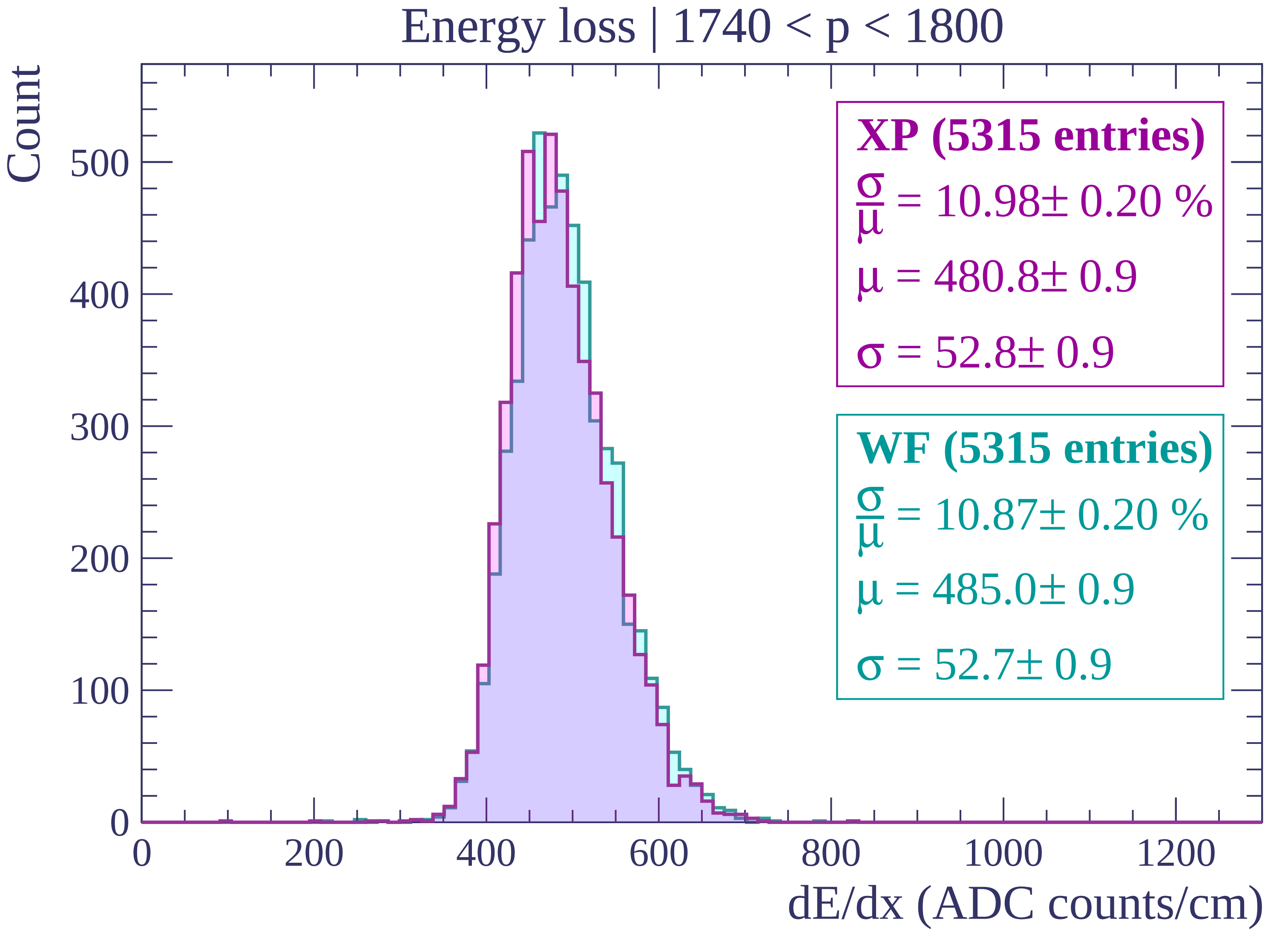

Energy loss | 1500 < p < 1560
Count
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6100 entries)
σ/μ = 10.57± 0.18 %
μ = 466.9± 0.8
σ = 49.4± 0.8
WF (6100 entries)
σ/μ = 10.50± 0.16 %
μ = 472.9± 0.8
σ = 49.7± 0.7


Energy loss | 1560 < p < 1620
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (5812 entries)
$\frac{\sigma}{\mu}$ = 10.16± 0.17 %
μ = 467.7± 0.8
σ = 47.5± 0.7
WF (5812 entries)
$\frac{\sigma}{\mu}$ = 10.38± 0.18 %
μ = 474.3± 0.8
σ = 49.2± 0.8

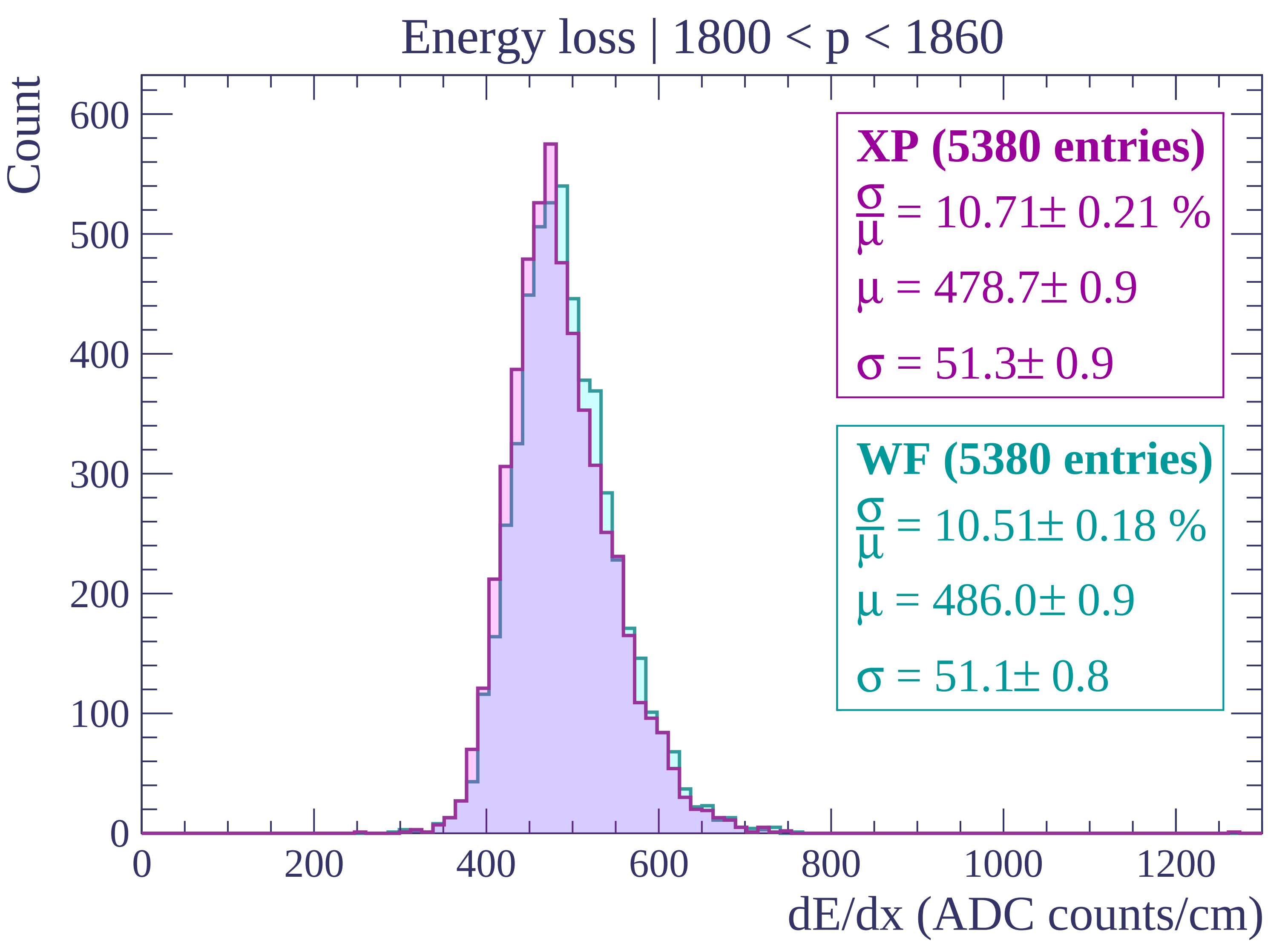

Energy loss | 1620 < p < 1680
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
XP (5669 entries)
$\frac{\sigma}{\mu}$ = 10.39± 0.21 %
μ = 468.9± 0.8
σ = 48.7± 0.9
WF (5669 entries)
$\frac{\sigma}{\mu}$ = 10.55± 0.20 %
μ = 477.1± 0.9
σ = 50.4± 0.9



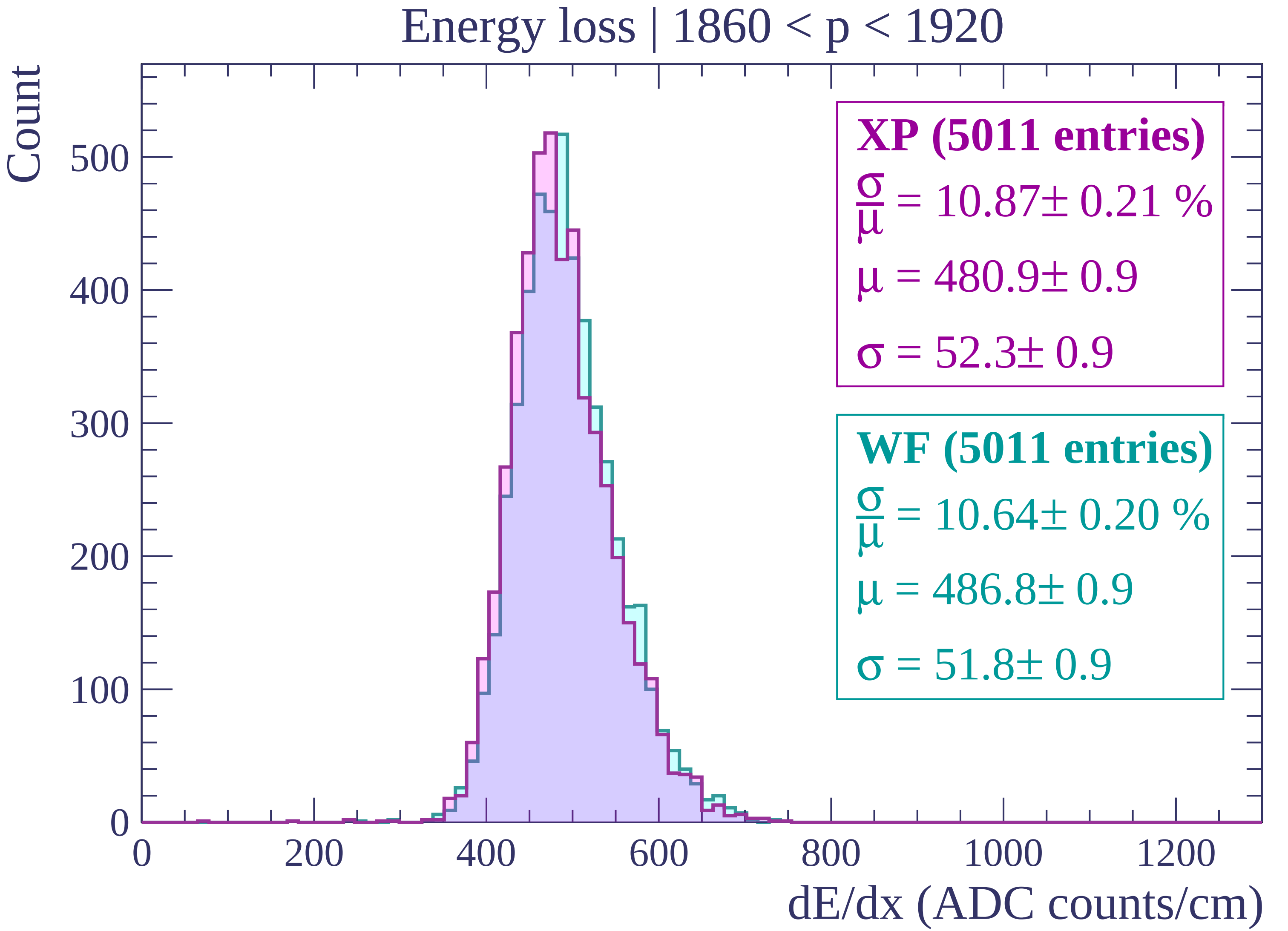
Energy loss | 1680 < p < 1740
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (5549 entries)
$\frac{\sigma}{\mu}$ = 10.40± 0.17 %
μ = 476.1± 0.9
σ = 49.5± 0.7
WF (5549 entries)
$\frac{\sigma}{\mu}$ = 11.01± 0.20 %
μ = 481.9± 0.9
σ = 53.0± 0.9



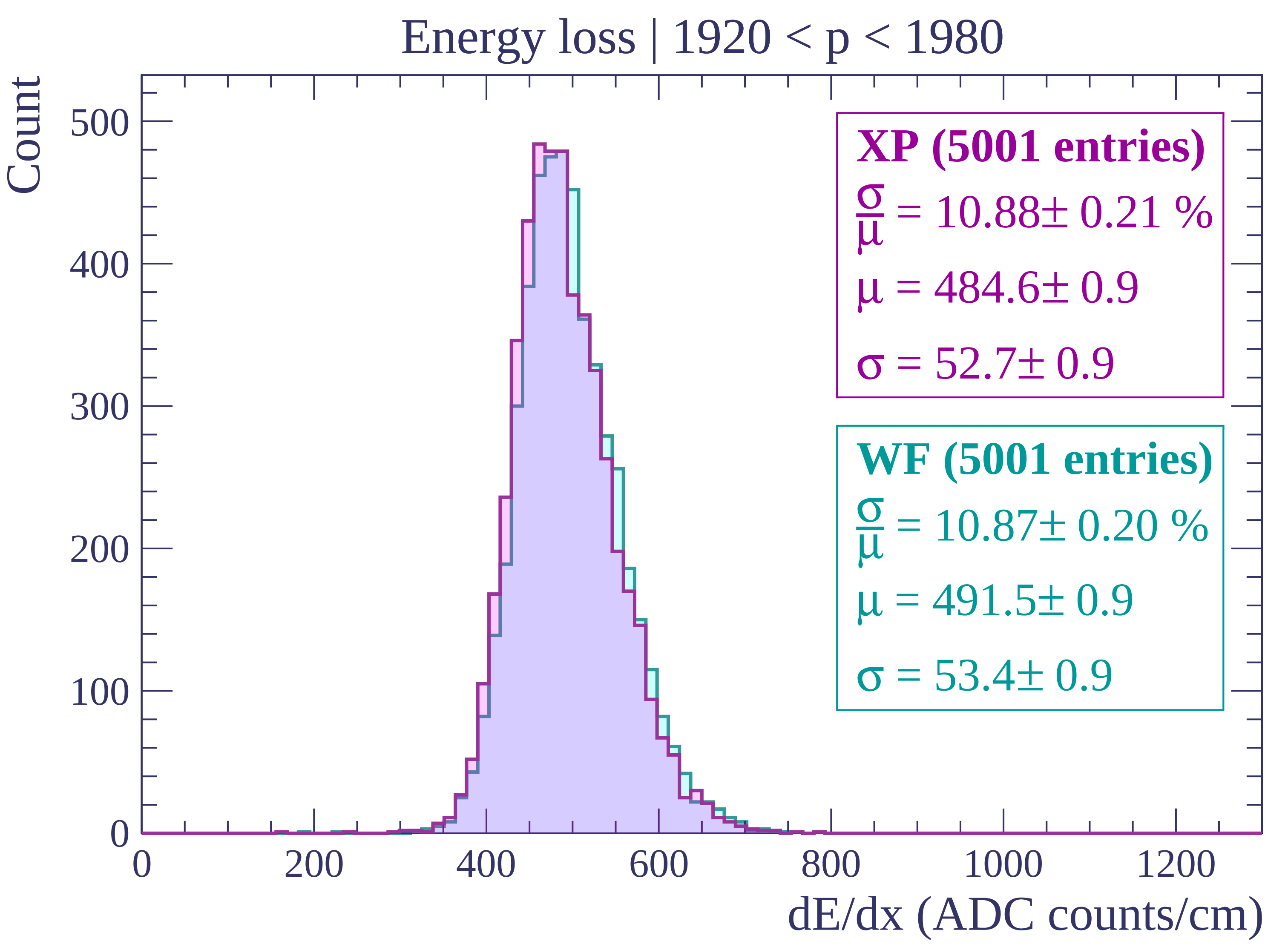
Energy loss | 1740 < p < 1800
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5315 entries)
σ/μ = 10.98± 0.20 %
μ = 480.8± 0.9
σ = 52.8± 0.9
WF (5315 entries)
σ/μ = 10.87± 0.20 %
μ = 485.0± 0.9
σ = 52.7± 0.9



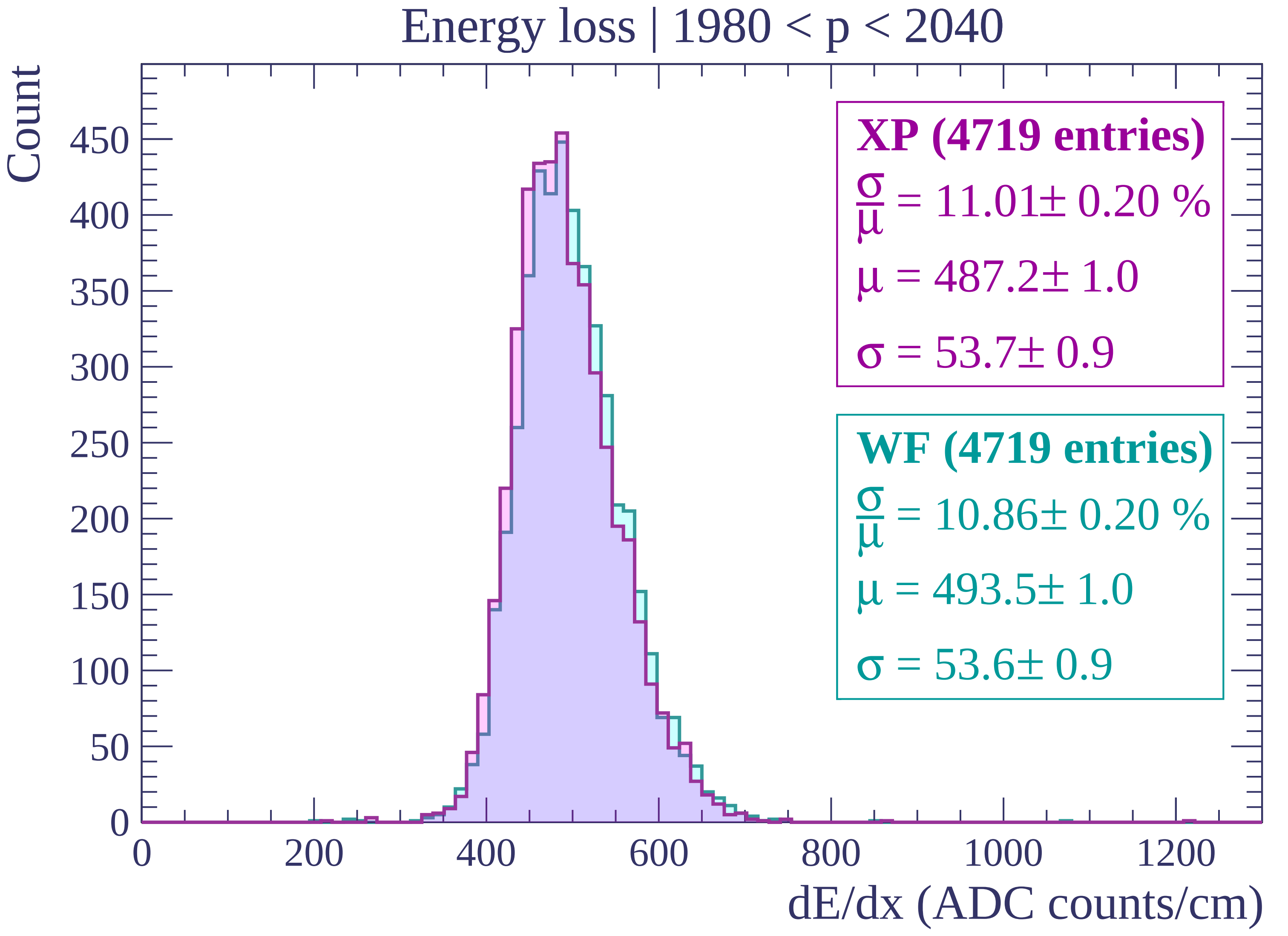
Energy loss | 1800 < p < 1860
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
XP (5380 entries)
σ/μ = 10.71± 0.21 %
μ = 478.7± 0.9
σ = 51.3± 0.9
WF (5380 entries)
σ/μ = 10.51± 0.18 %
μ = 486.0± 0.9
σ = 51.1± 0.8

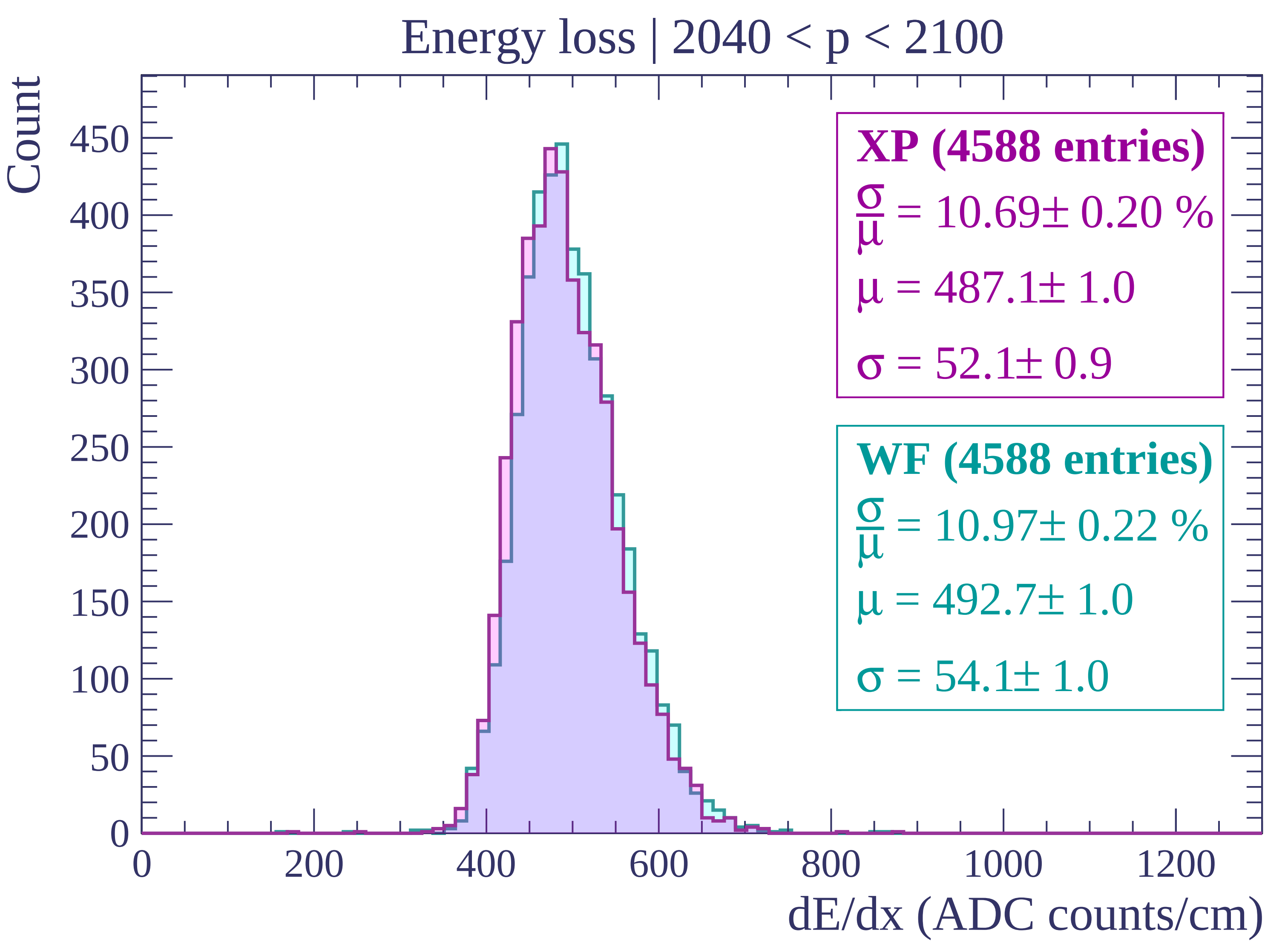


Energy loss | 1860 < p < 1920
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (5011 entries)
$\frac{\sigma}{\mu}$ = 10.87± 0.21 %
μ = 480.9± 0.9
σ = 52.3± 0.9
WF (5011 entries)
$\frac{\sigma}{\mu}$ = 10.64± 0.20 %
μ = 486.8± 0.9
σ = 51.8± 0.9

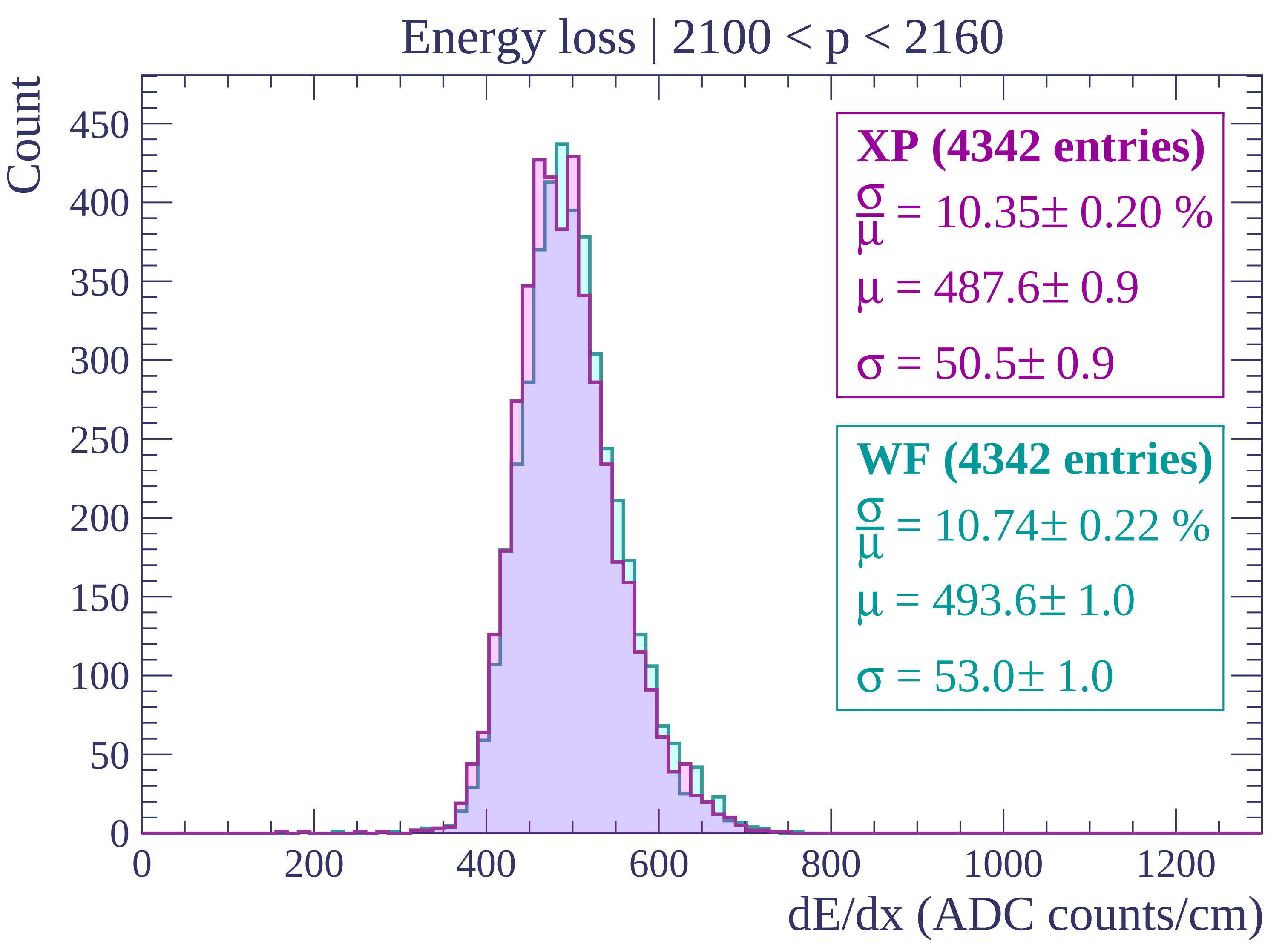


Energy loss | 1920 < p < 1980
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (5001 entries)
σ/μ = 10.88± 0.21 %
μ = 484.6± 0.9
σ = 52.7± 0.9
WF (5001 entries)
σ/μ = 10.87± 0.20 %
μ = 491.5± 0.9
σ = 53.4± 0.9

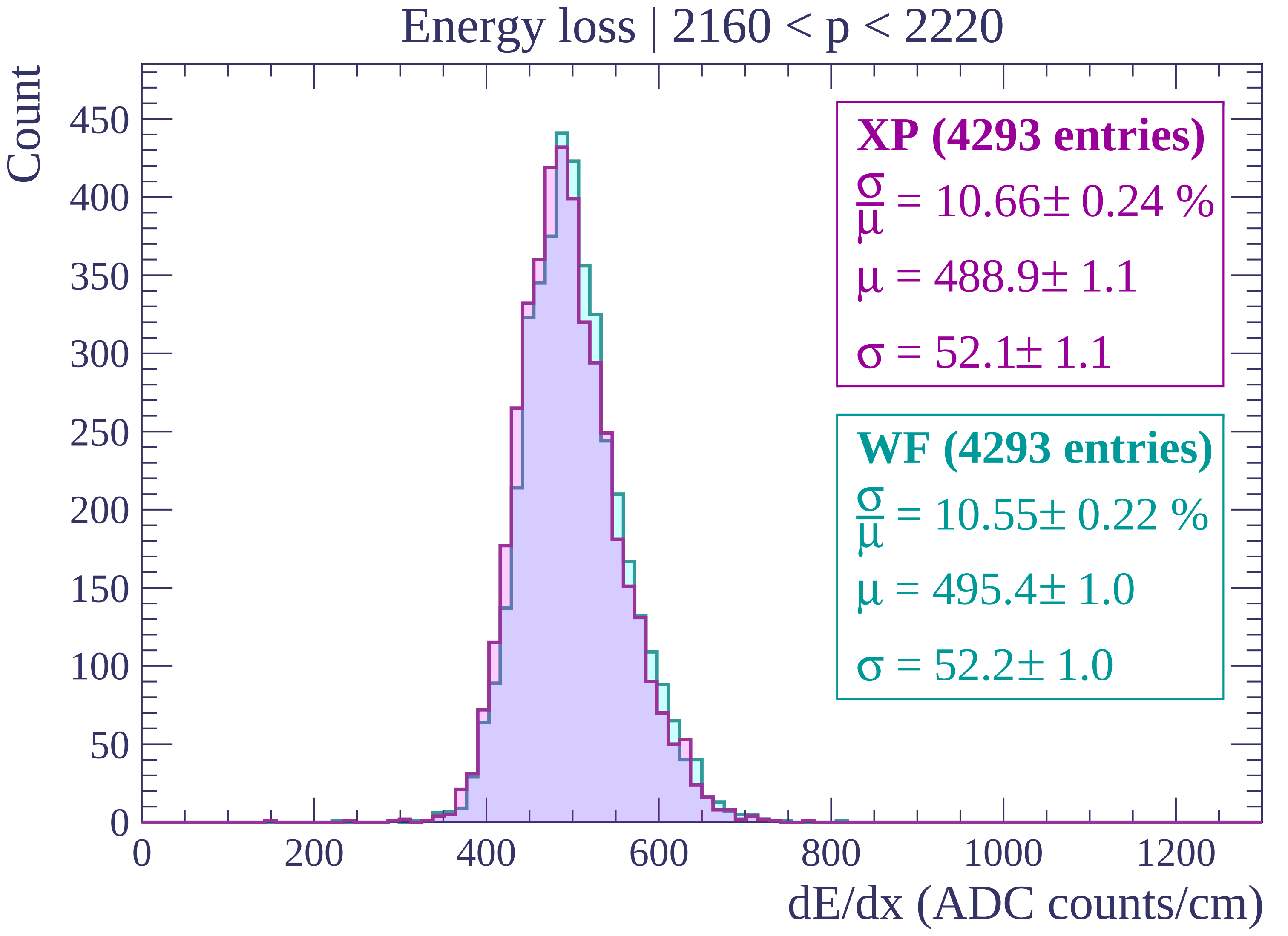

Energy loss | 1980 < p < 2040
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (4719 entries)
σ/μ = 11.01± 0.20 %
μ = 487.2± 1.0
σ = 53.7± 0.9
WF (4719 entries)
σ/μ = 10.86± 0.20 %
μ = 493.5± 1.0
σ = 53.6± 0.9



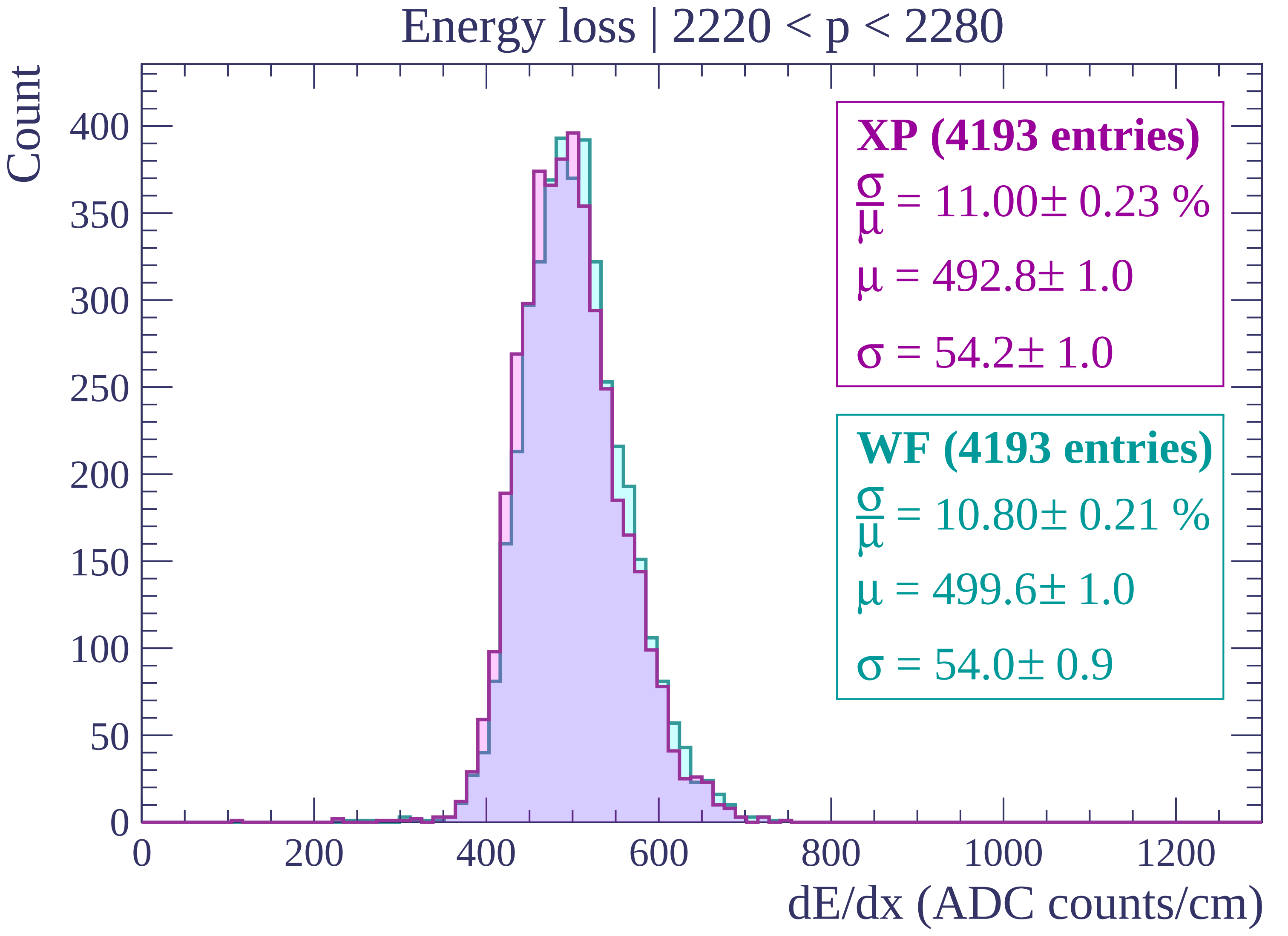
Energy loss | 2040 < p < 2100
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (4588 entries)
σ/μ = 10.69± 0.20 %
μ = 487.1± 1.0
σ = 52.1± 0.9
WF (4588 entries)
σ/μ = 10.97± 0.22 %
μ = 492.7± 1.0
σ = 54.1± 1.0

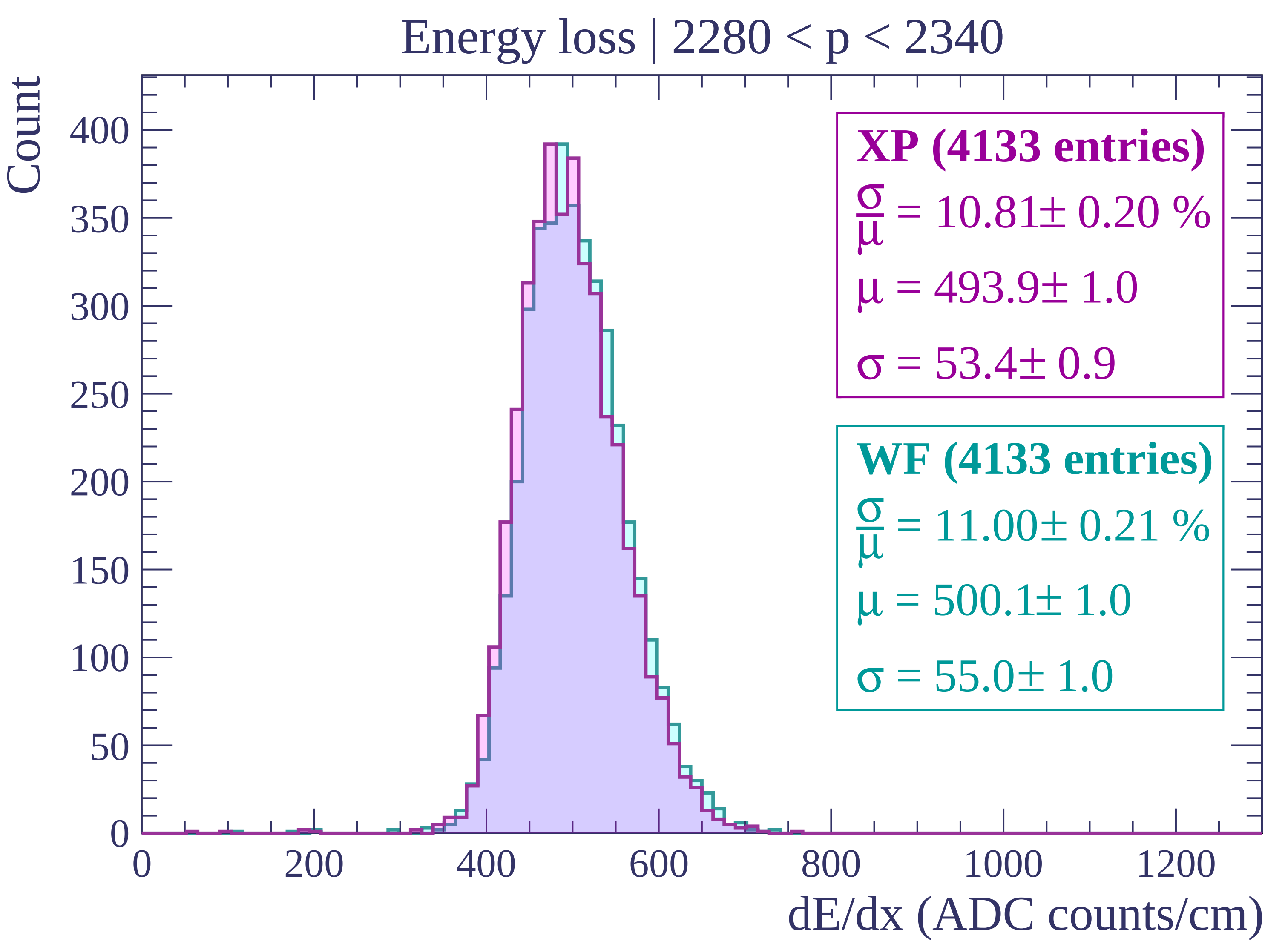


Energy loss | 2100 < p < 2160
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (4342 entries)
σ/μ = 10.35± 0.20 %
μ = 487.6± 0.9
σ = 50.5± 0.9
WF (4342 entries)
σ/μ = 10.74± 0.22 %
μ = 493.6± 1.0
σ = 53.0± 1.0

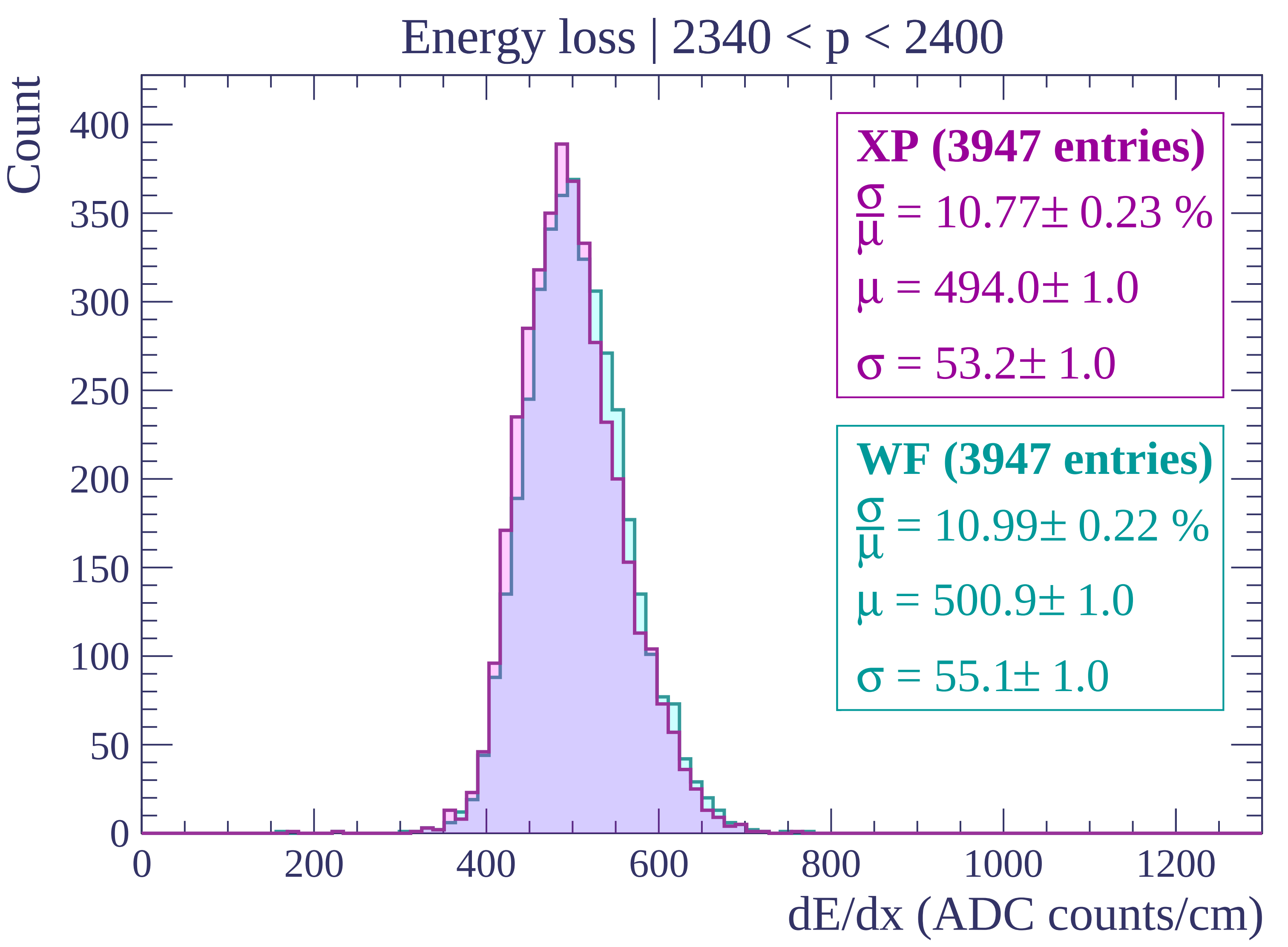


Energy loss | 2160 < p < 2220
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (4293 entries)
σ/μ = 10.66± 0.24 %
μ = 488.9± 1.1
σ = 52.1± 1.1
WF (4293 entries)
σ/μ = 10.55± 0.22 %
μ = 495.4± 1.0
σ = 52.2± 1.0

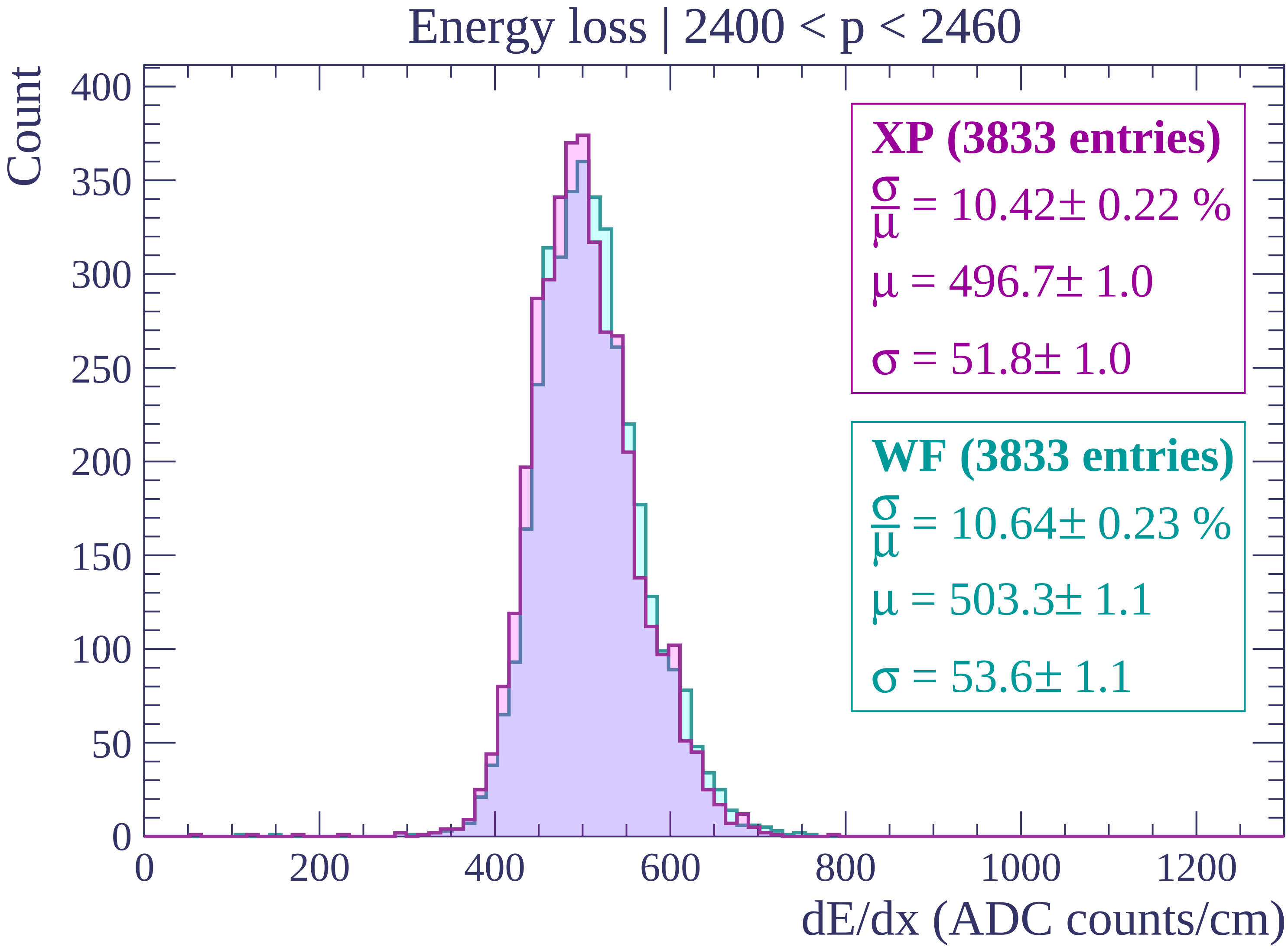

Energy loss | 2220 < p < 2280
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
400
XP (4193 entries)
σ/μ = 11.00± 0.23 %
μ = 492.8± 1.0
σ = 54.2± 1.0
WF (4193 entries)
σ/μ = 10.80± 0.21 %
μ = 499.6± 1.0
σ = 54.0± 0.9

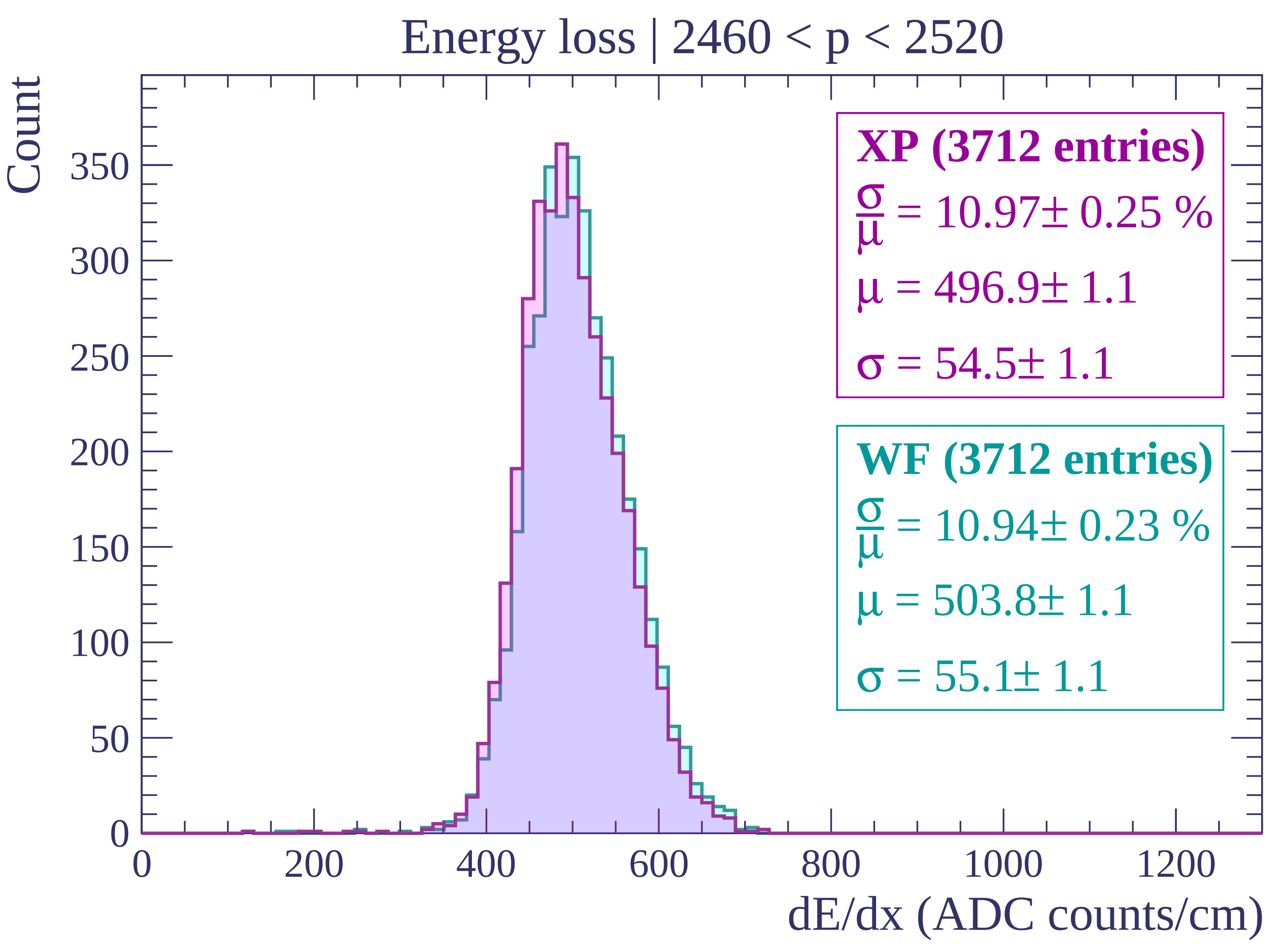


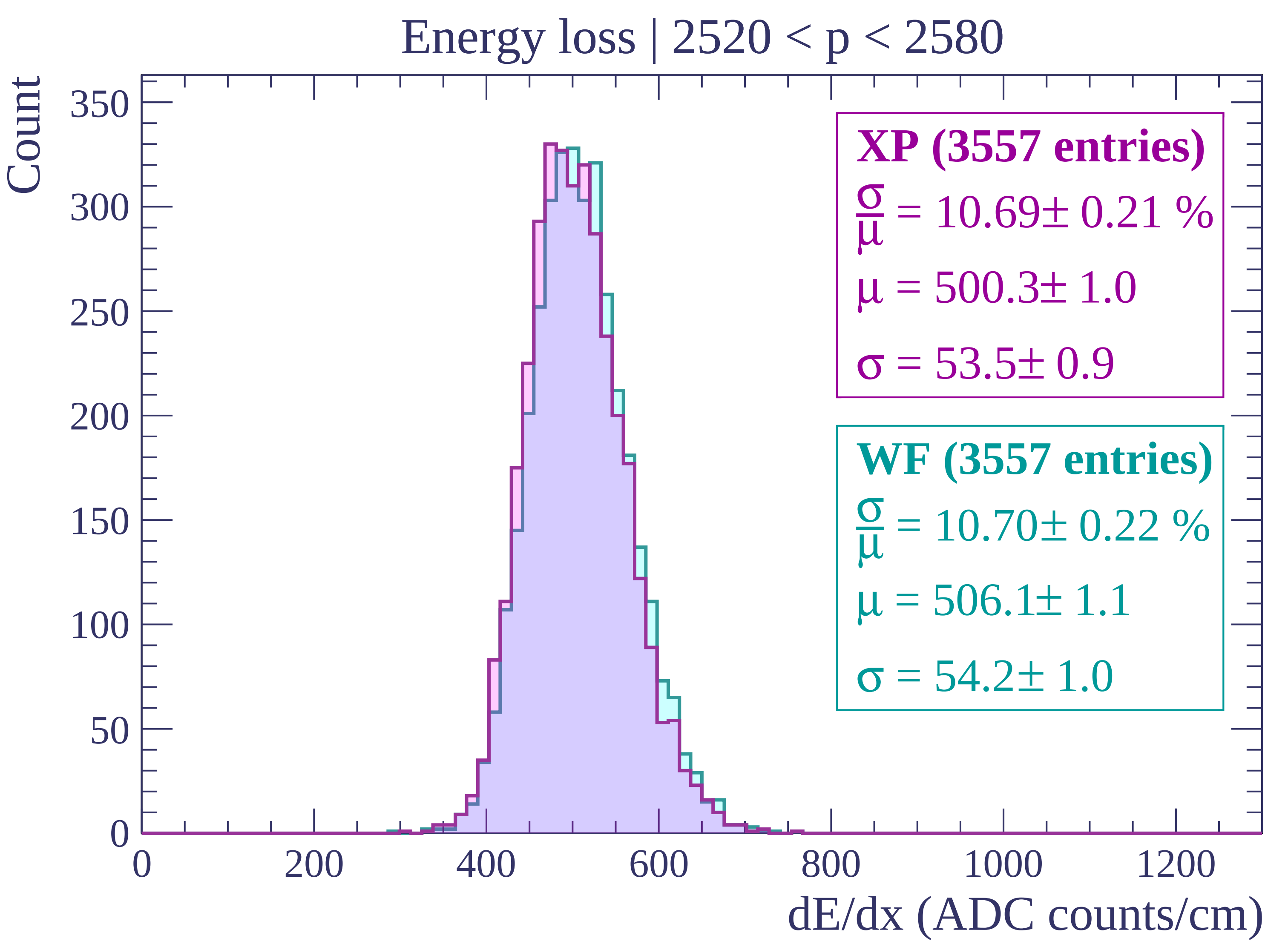
Energy loss | 2280 < p < 2340
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
400
XP (4133 entries)
σ/μ = 10.81± 0.20 %
μ = 493.9± 1.0
σ = 53.4± 0.9
WF (4133 entries)
σ/μ = 11.00± 0.21 %
μ = 500.1± 1.0
σ = 55.0± 1.0

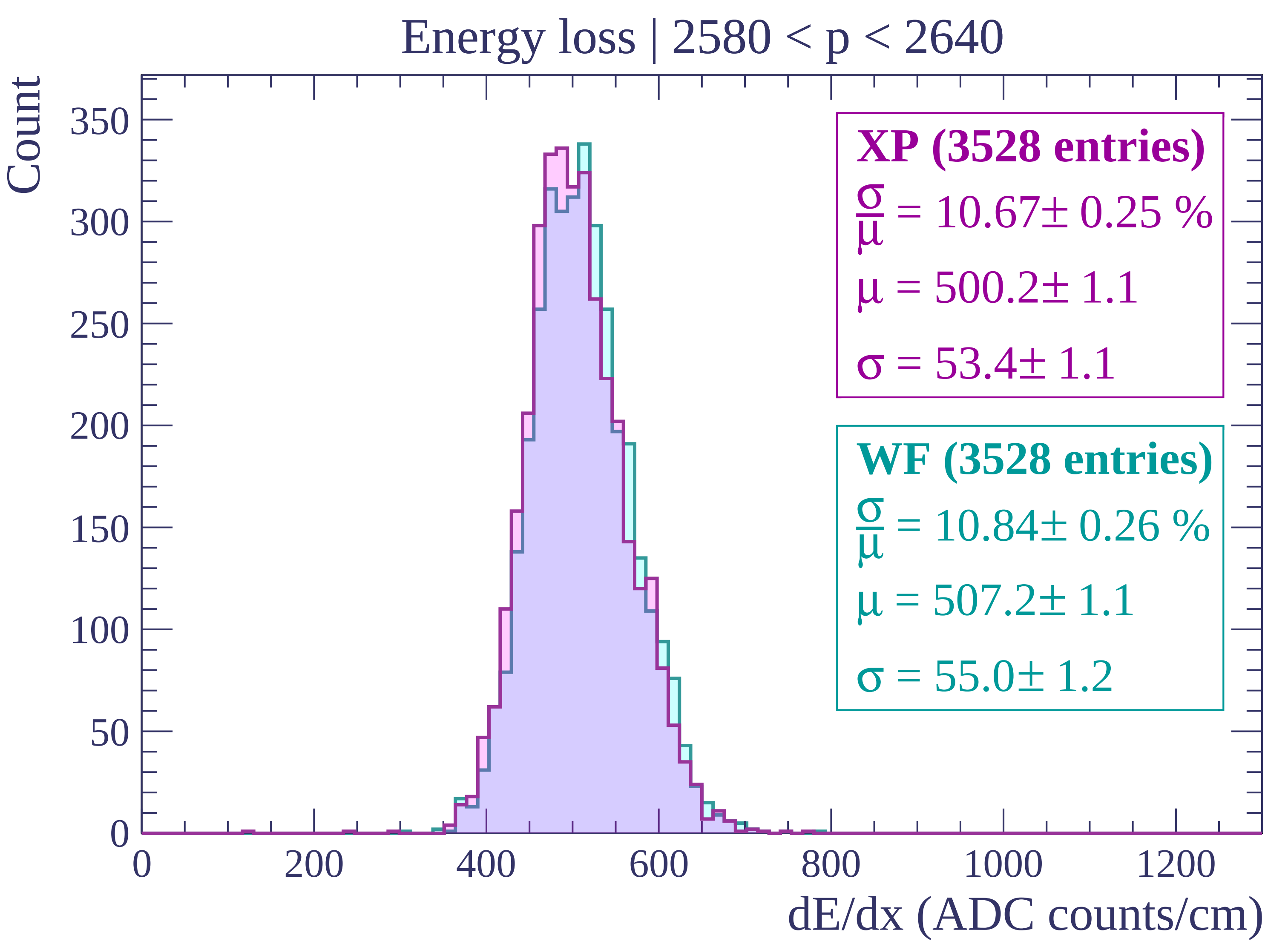


Energy loss | 2340 < p < 2400
Count
dE/dx (ADC counts/cm)
XP (3947 entries)
$\frac{\sigma}{\mu}$ = 10.77± 0.23 %
μ = 494.0± 1.0
σ = 53.2± 1.0
WF (3947 entries)
$\frac{\sigma}{\mu}$ = 10.99± 0.22 %
μ = 500.9± 1.0
σ = 55.1± 1.0

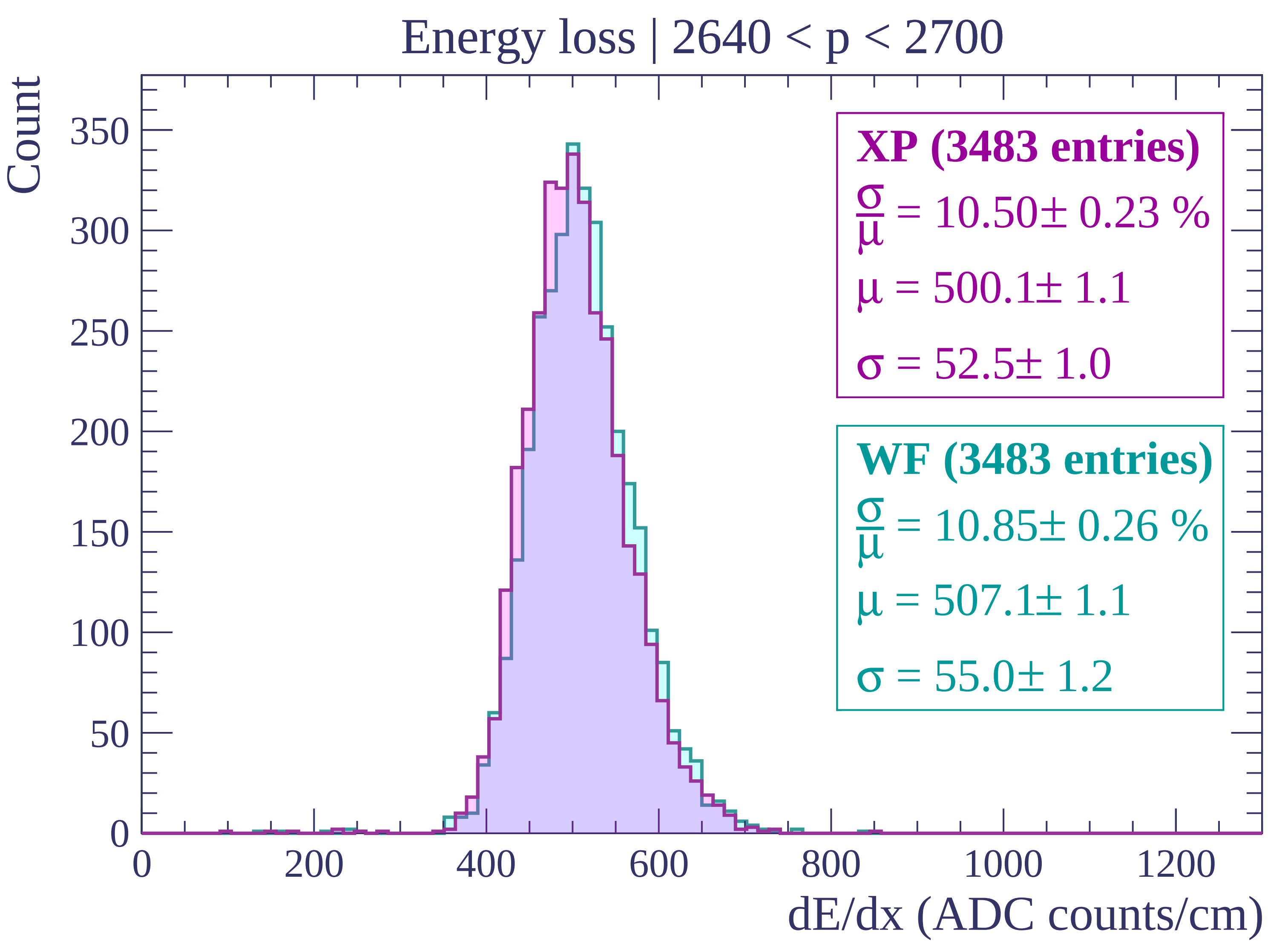

Energy loss | 2400 < p < 2460
Count
dE/dx (ADC counts/cm)
XP (3833 entries)
$\frac{\sigma}{\mu}$ = 10.42± 0.22 %
μ = 496.7± 1.0
σ = 51.8± 1.0
WF (3833 entries)
$\frac{\sigma}{\mu}$ = 10.64± 0.23 %
μ = 503.3± 1.1
σ = 53.6± 1.1
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200


Energy loss | 2460 < p < 2520
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3712 entries)
σ/μ = 10.97± 0.25 %
μ = 496.9± 1.1
σ = 54.5± 1.1
WF (3712 entries)
σ/μ = 10.94± 0.23 %
μ = 503.8± 1.1
σ = 55.1± 1.1

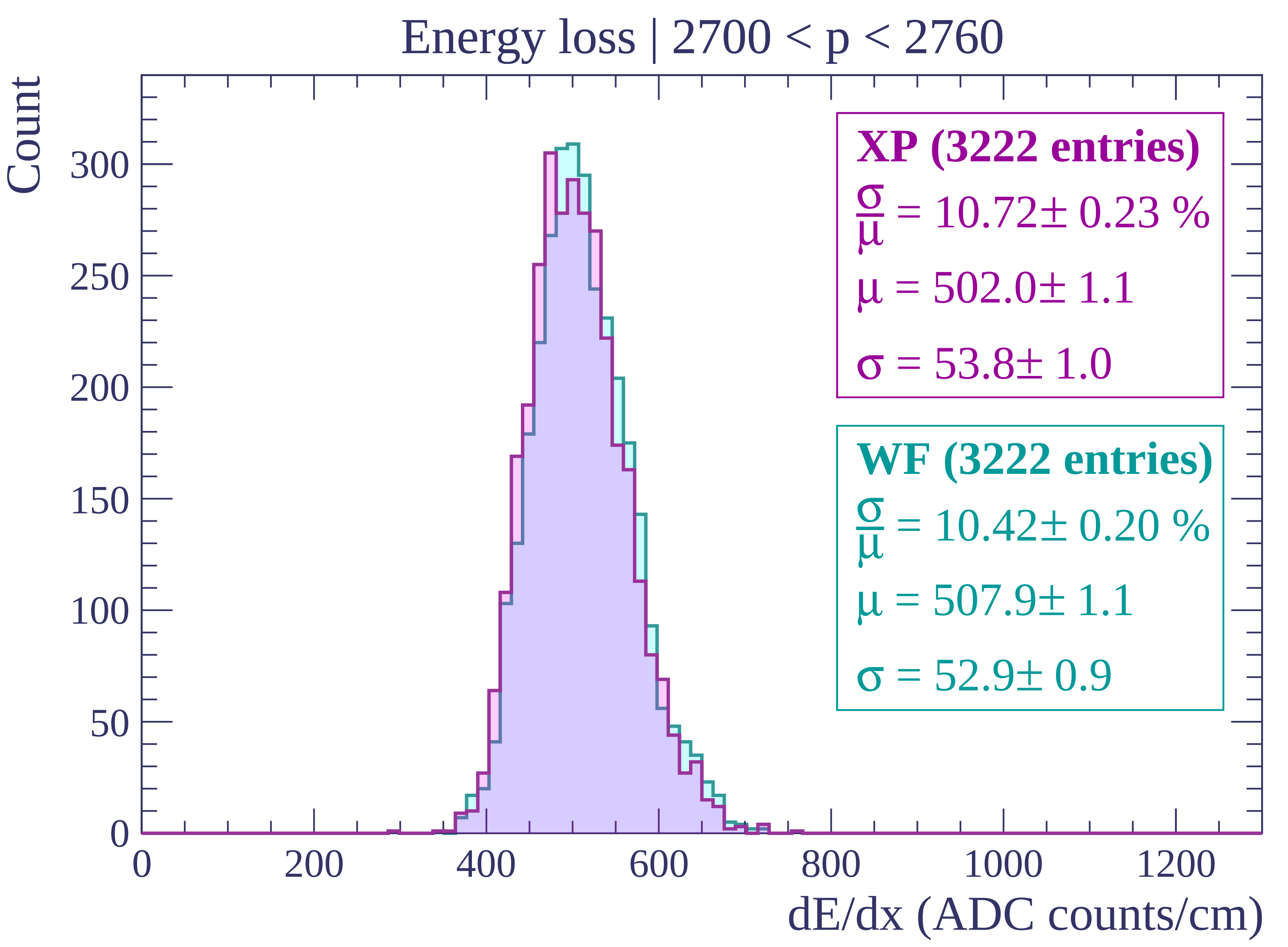

Energy loss | 2520 < p < 2580
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3557 entries)
$\frac{\sigma}{\mu}$ = 10.69± 0.21 %
μ = 500.3± 1.0
σ = 53.5± 0.9
WF (3557 entries)
$\frac{\sigma}{\mu}$ = 10.70± 0.22 %
μ = 506.1± 1.1
σ = 54.2± 1.0



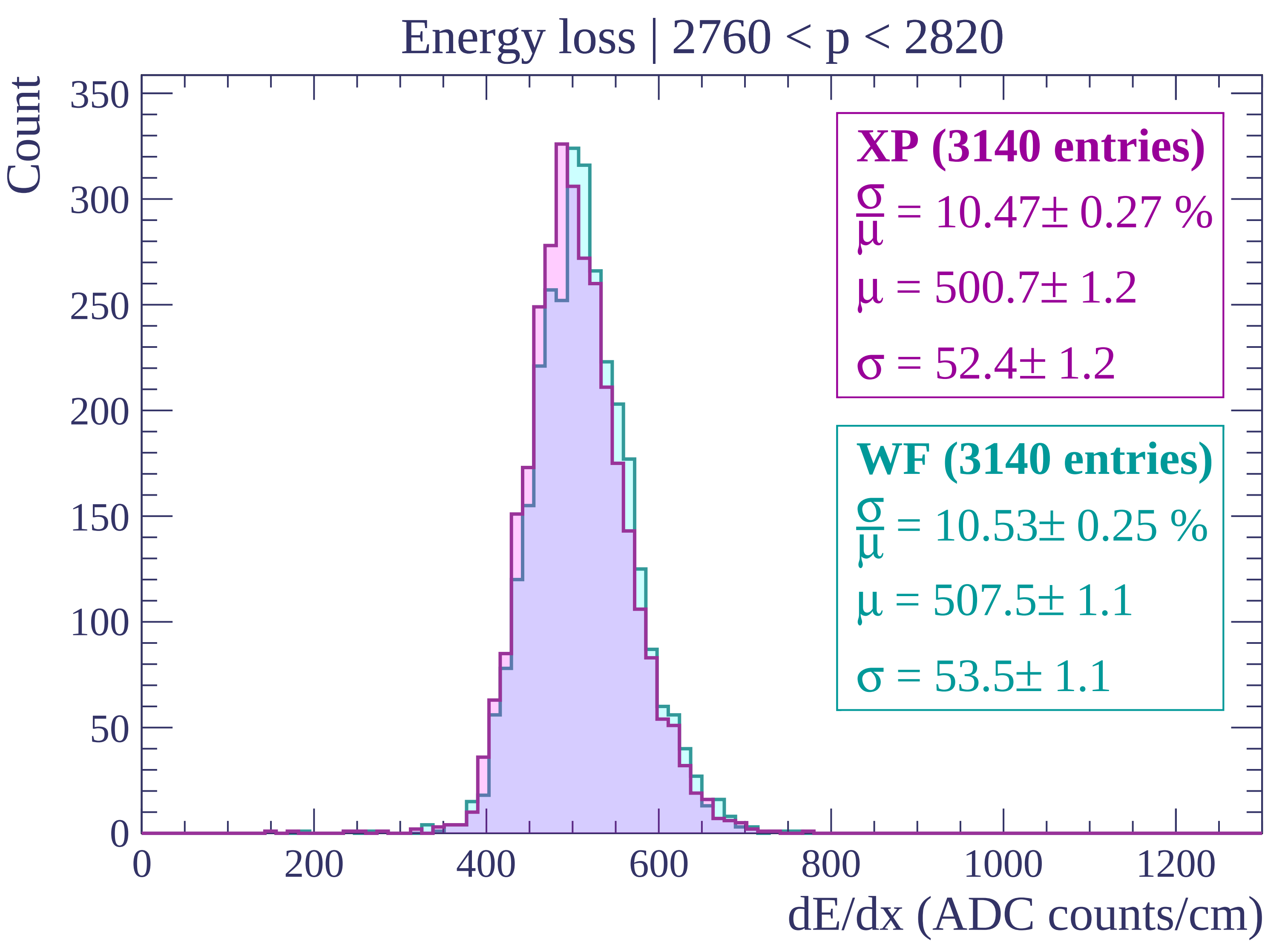
Energy loss | 2580 < p < 2640
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3528 entries)
σ/μ = 10.67± 0.25 %
μ = 500.2± 1.1
σ = 53.4± 1.1
WF (3528 entries)
σ/μ = 10.84± 0.26 %
μ = 507.2± 1.1
σ = 55.0± 1.2



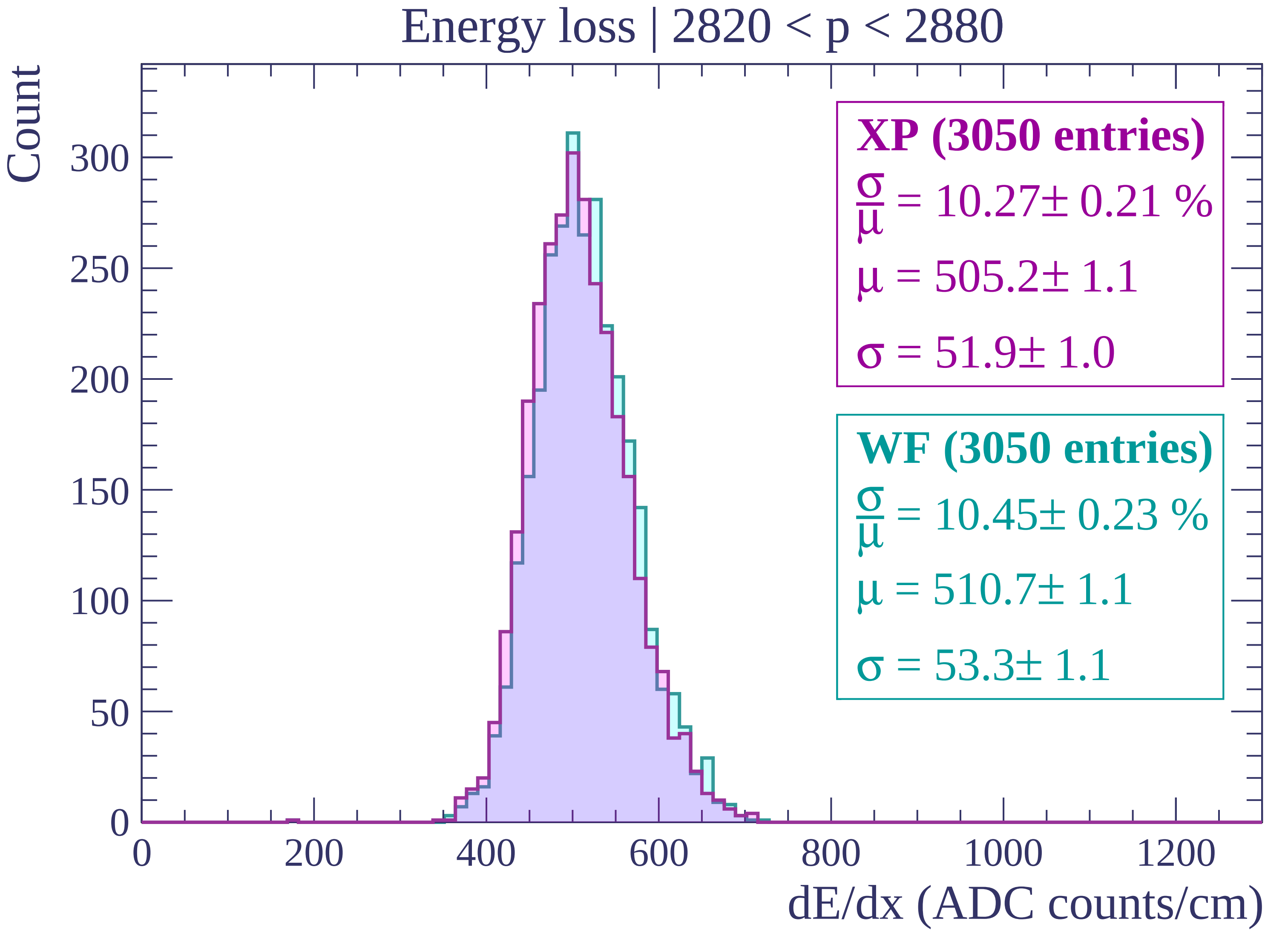
Energy loss | 2640 < p < 2700
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3483 entries)
σ/μ = 10.50± 0.23 %
μ = 500.1± 1.1
σ = 52.5± 1.0
WF (3483 entries)
σ/μ = 10.85± 0.26 %
μ = 507.1± 1.1
σ = 55.0± 1.2



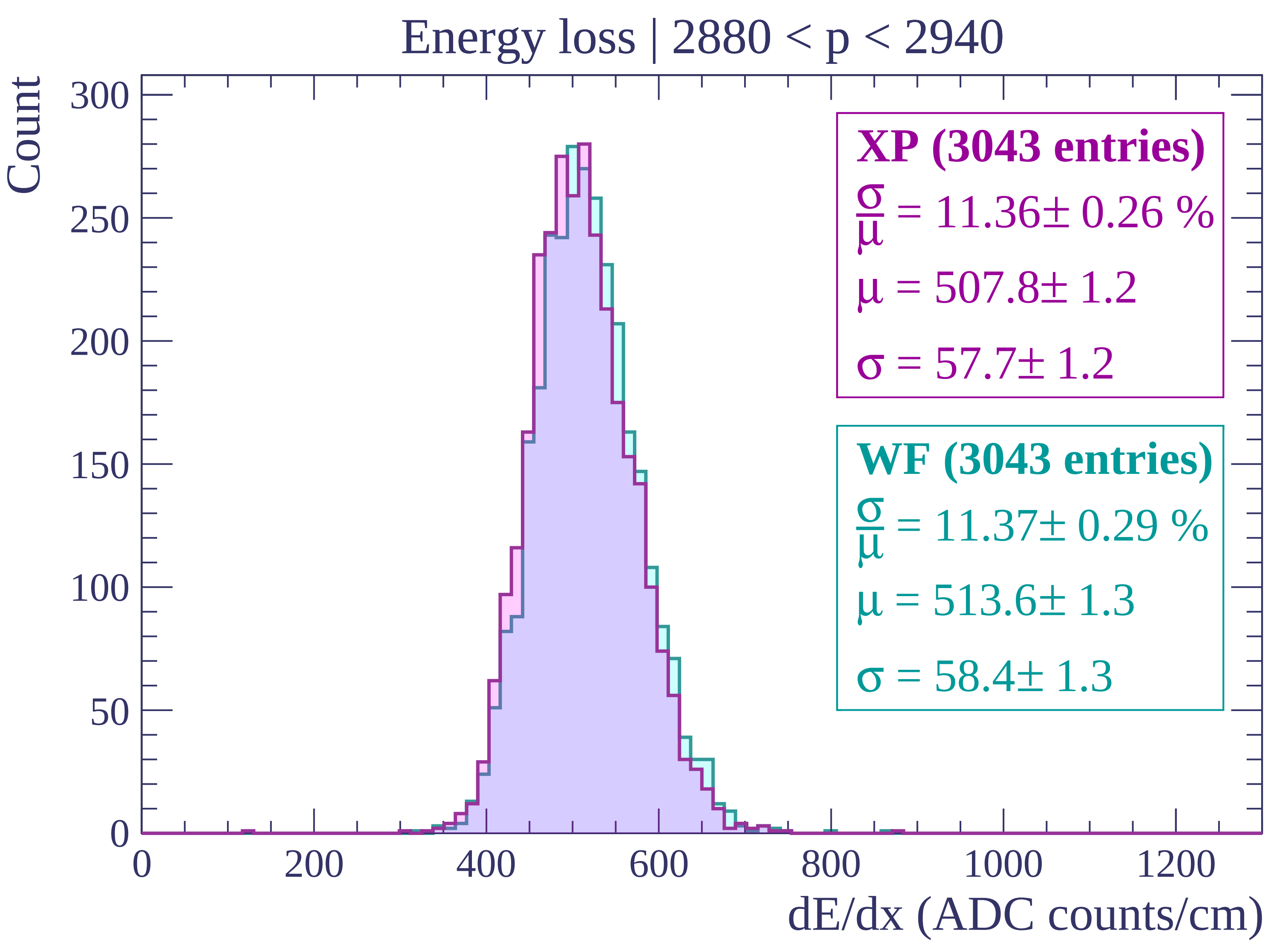
Energy loss | 2700 < p < 2760
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (3222 entries)
σ/μ = 10.72± 0.23 %
μ = 502.0± 1.1
σ = 53.8± 1.0
WF (3222 entries)
σ/μ = 10.42± 0.20 %
μ = 507.9± 1.1
σ = 52.9± 0.9

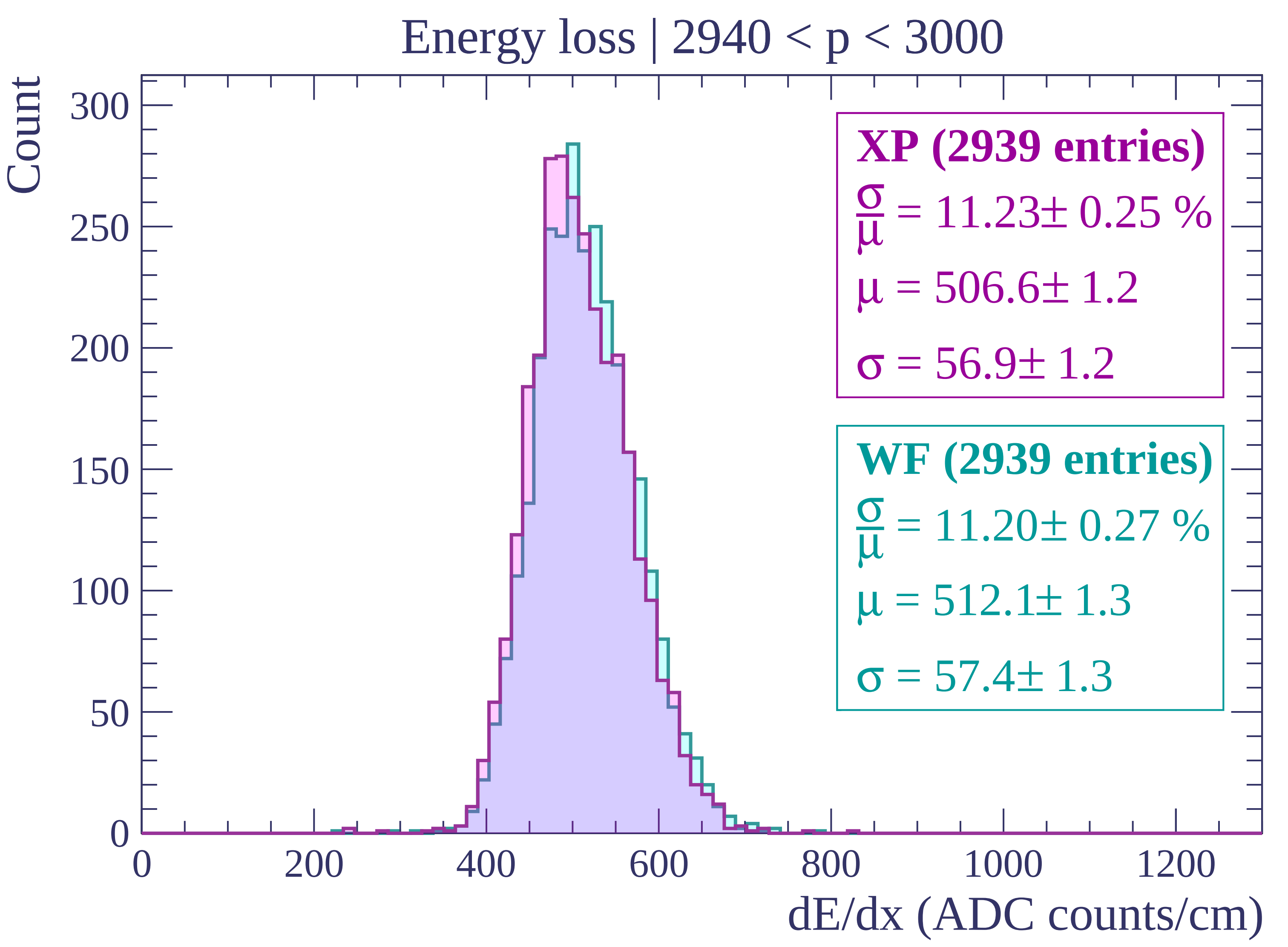


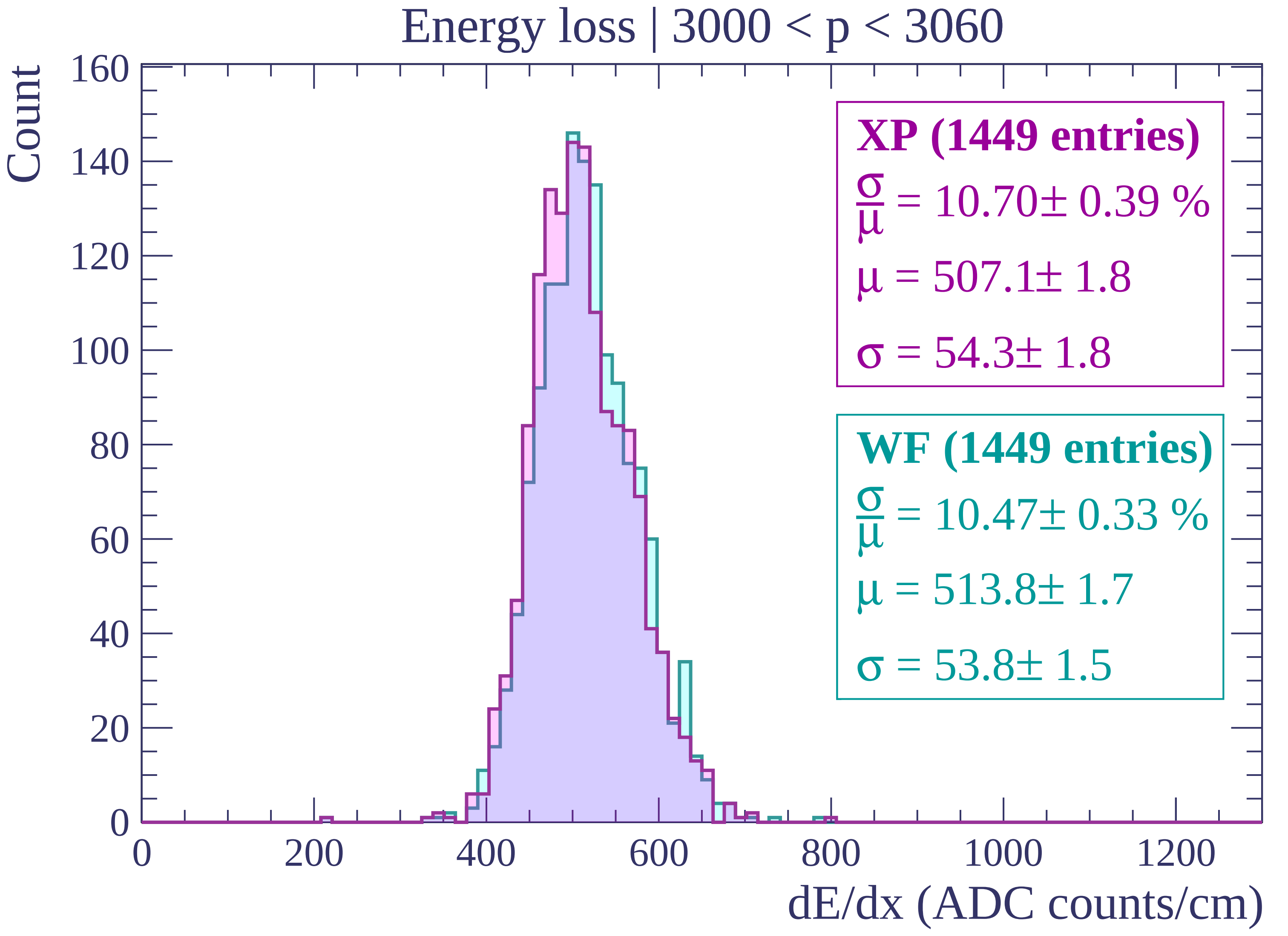
Energy loss | 2760 < p < 2820
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3140 entries)
σ/μ = 10.47± 0.27 %
μ = 500.7± 1.2
σ = 52.4± 1.2
WF (3140 entries)
σ/μ = 10.53± 0.25 %
μ = 507.5± 1.1
σ = 53.5± 1.1

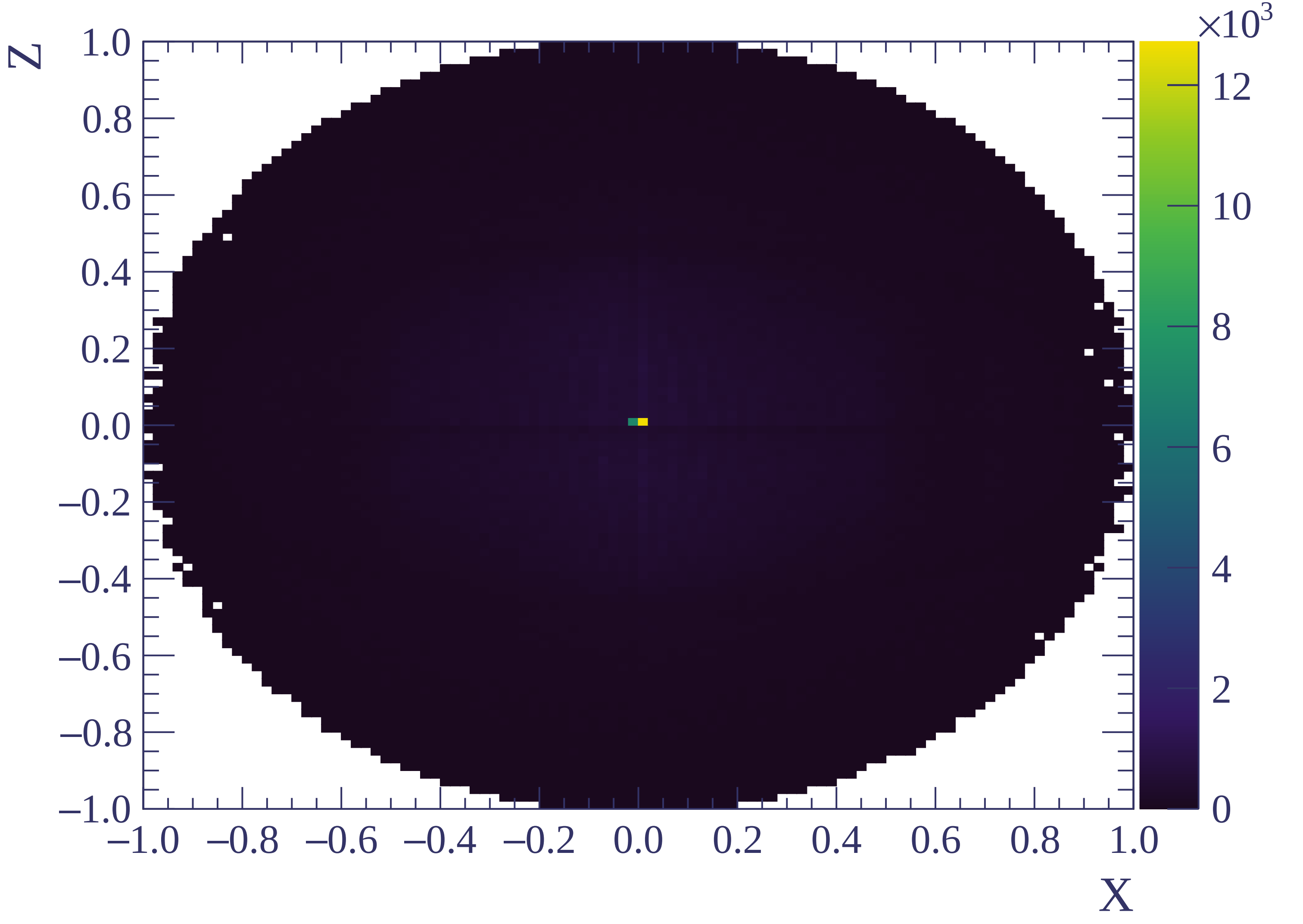


Energy loss | 2820 < p < 2880
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3050 entries)
σ/μ = 10.27± 0.21 %
μ = 505.2± 1.1
σ = 51.9± 1.0
WF (3050 entries)
σ/μ = 10.45± 0.23 %
μ = 510.7± 1.1
σ = 53.3± 1.1

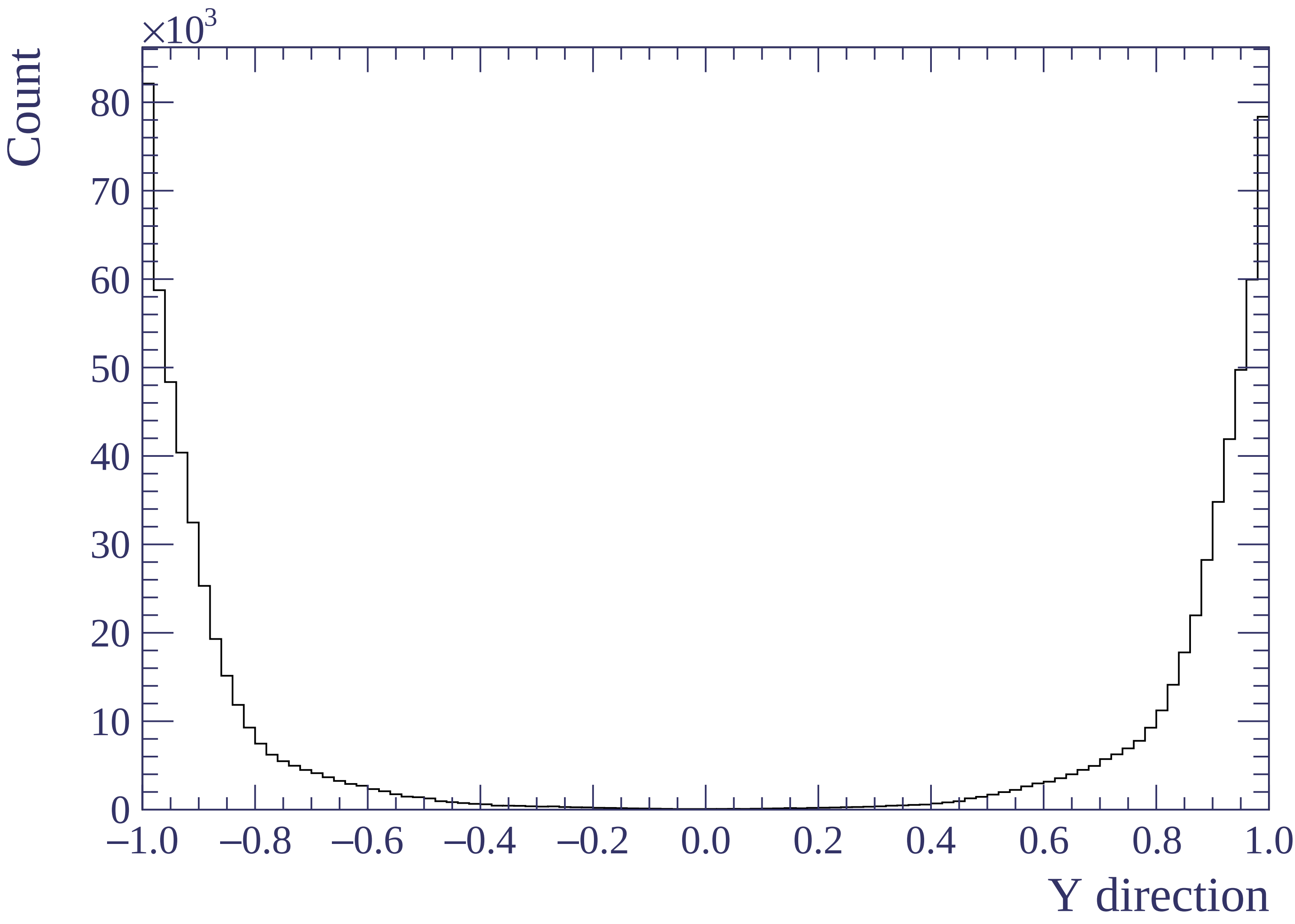

Energy loss | 2880 < p < 2940
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (3043 entries)
σ/μ = 11.36± 0.26 %
μ = 507.8± 1.2
σ = 57.7± 1.2
WF (3043 entries)
σ/μ = 11.37± 0.29 %
μ = 513.6± 1.3
σ = 58.4± 1.3


Energy loss | 2940 < p < 3000
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2939 entries)
σ/μ = 11.23± 0.25 %
μ = 506.6± 1.2
σ = 56.9± 1.2
WF (2939 entries)
σ/μ = 11.20± 0.27 %
μ = 512.1± 1.3
σ = 57.4± 1.3

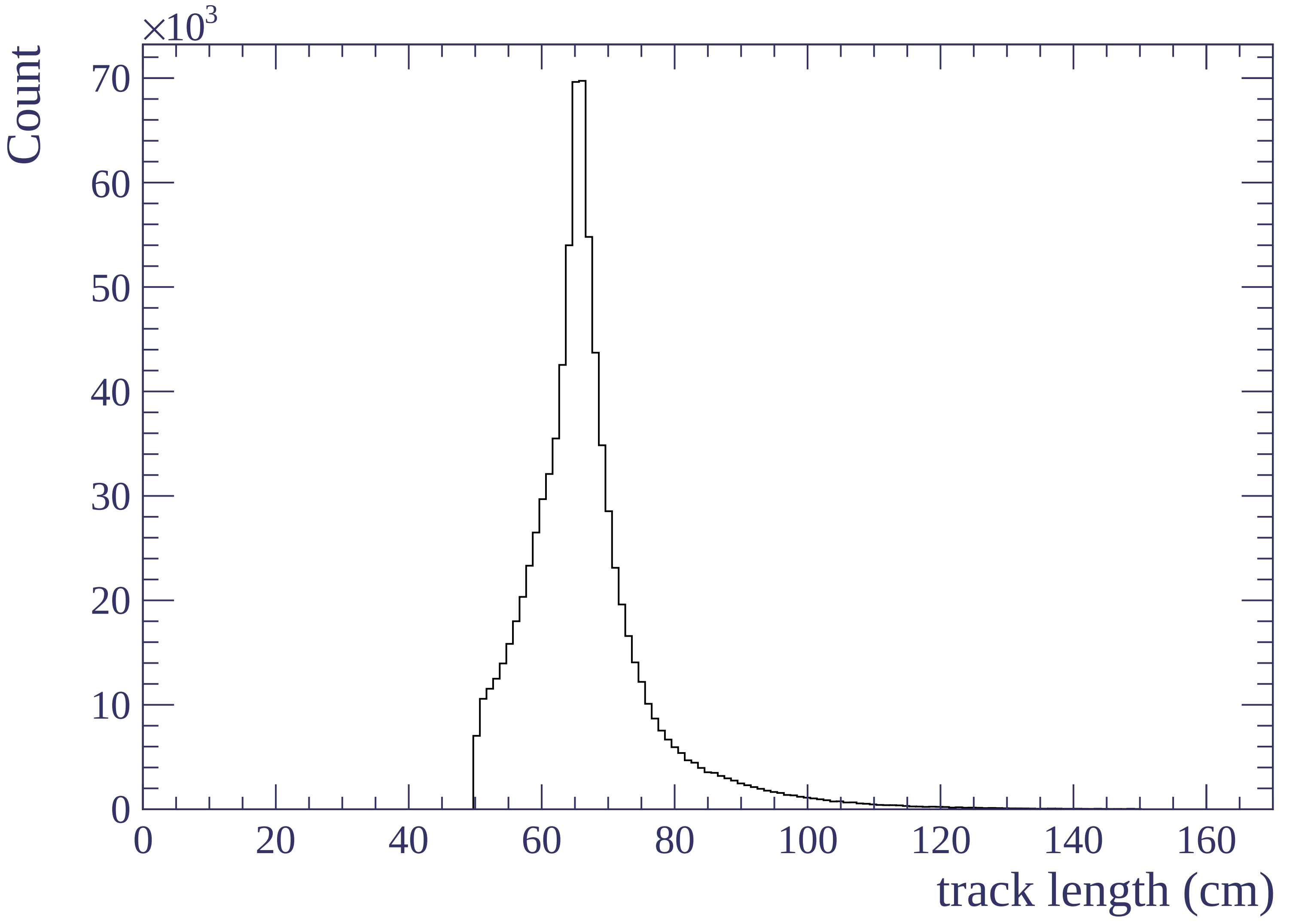

Energy loss | 3000 < p < 3060
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP (1449 entries)
σ/μ = 10.70± 0.39 %
μ = 507.1± 1.8
σ = 54.3± 1.8
WF (1449 entries)
σ/μ = 10.47± 0.33 %
μ = 513.8± 1.7
σ = 53.8± 1.5

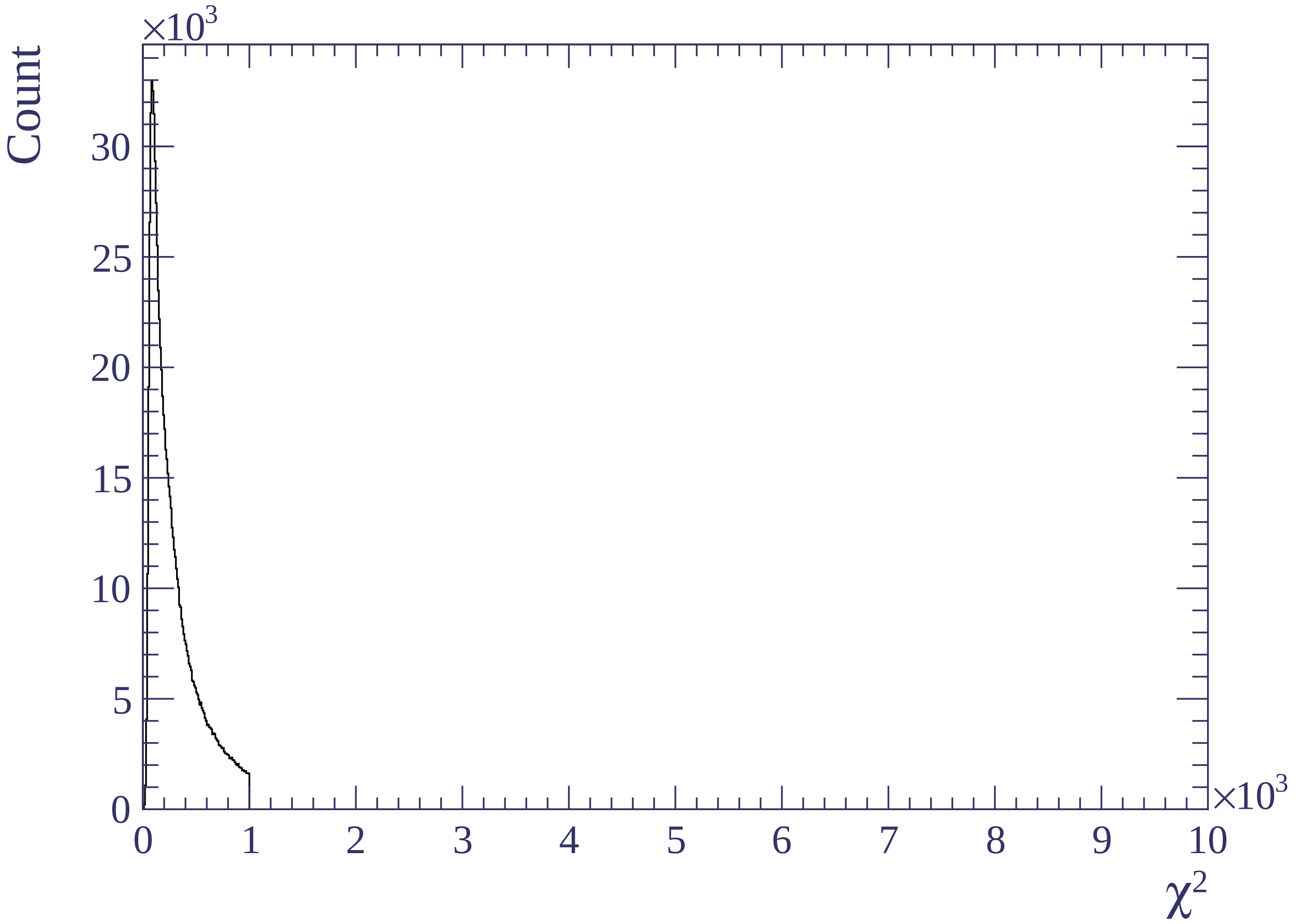


Z
X
×10^3
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
12
10
8
6
4
2
0

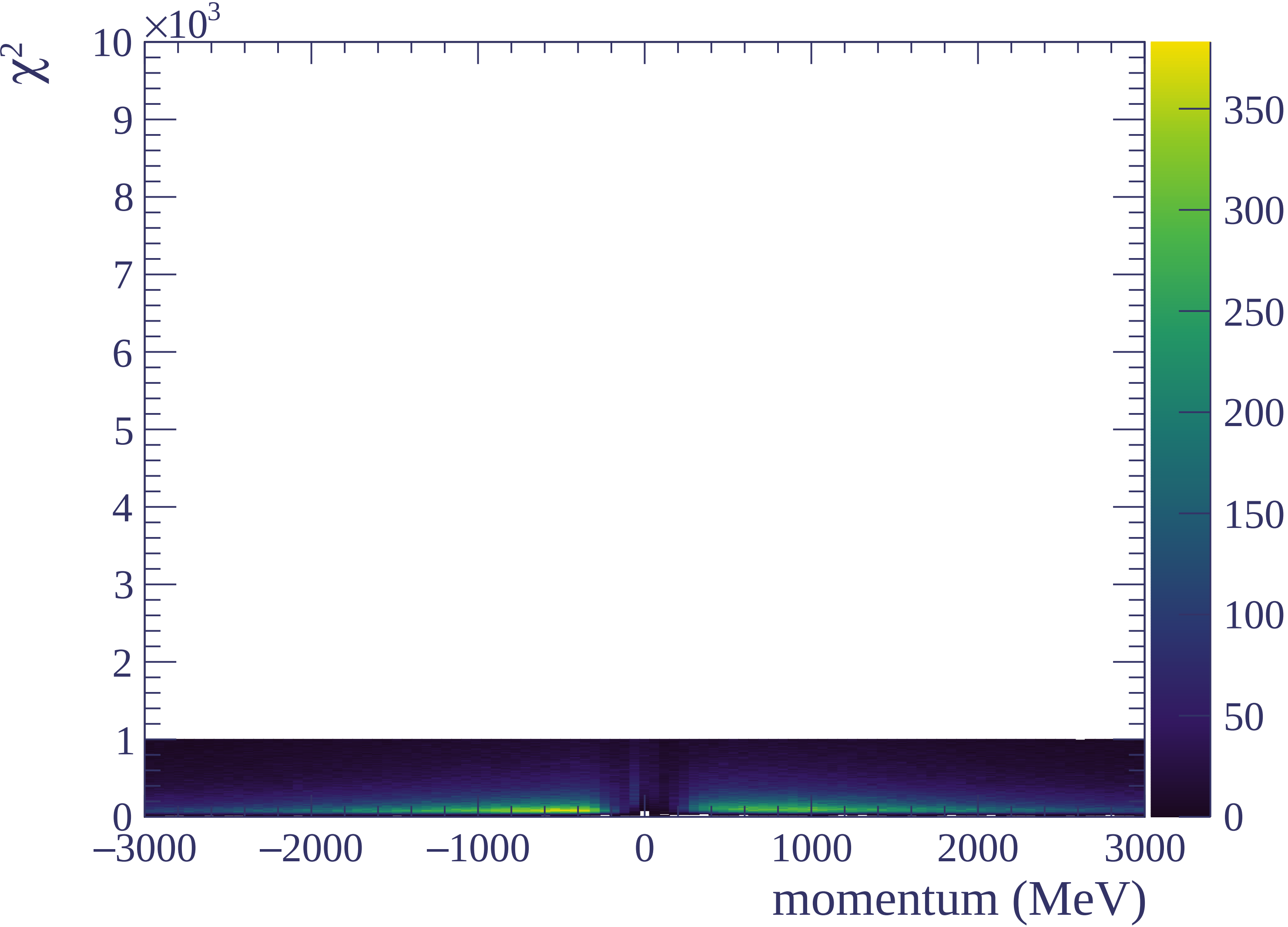


×10^3
Count
80
70
60
50
40
30
20
10
0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y direction

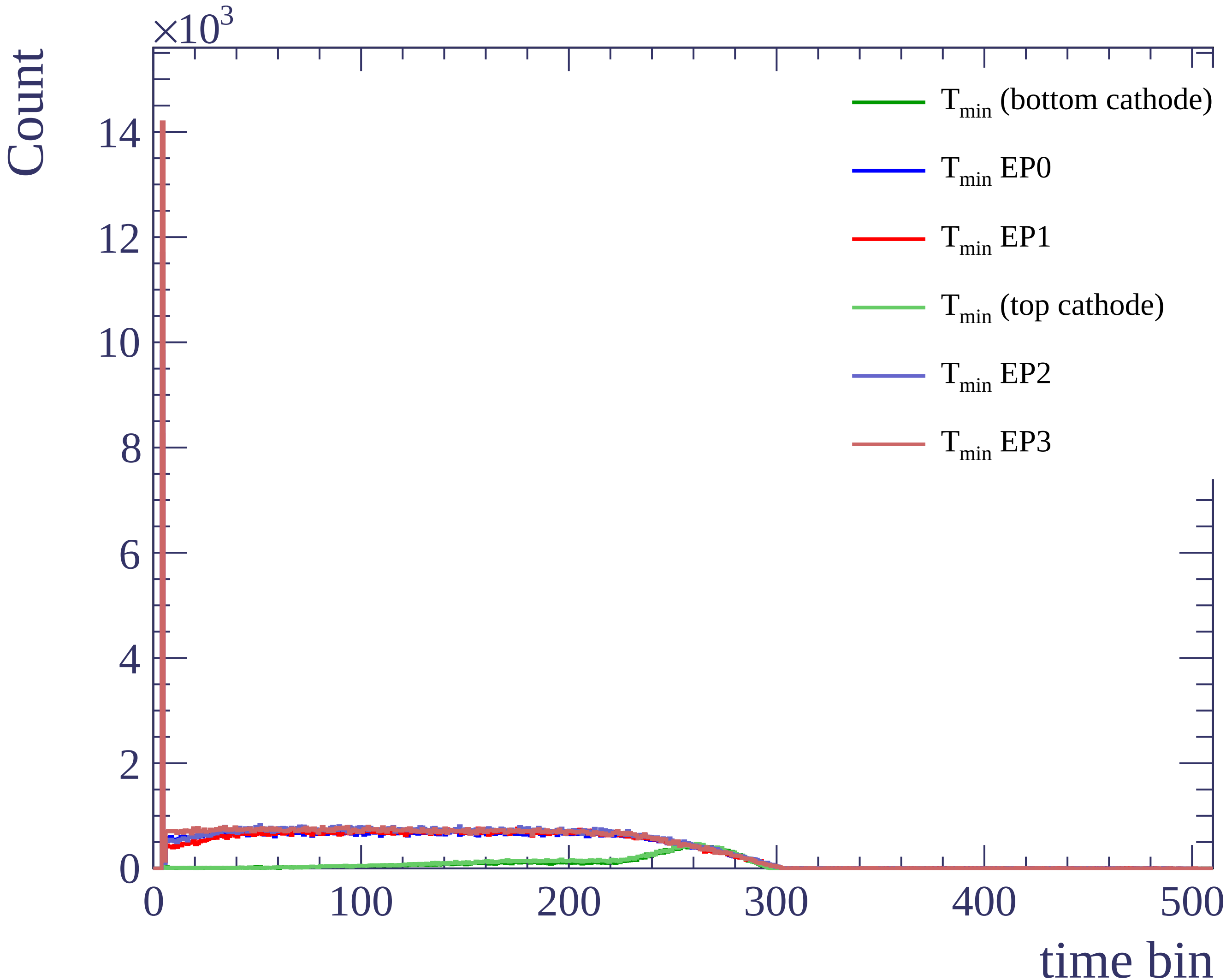

×10^3
Count
0
10
20
30
40
50
60
70
0
20
40
60
80
100
120
140
160
track length (cm)



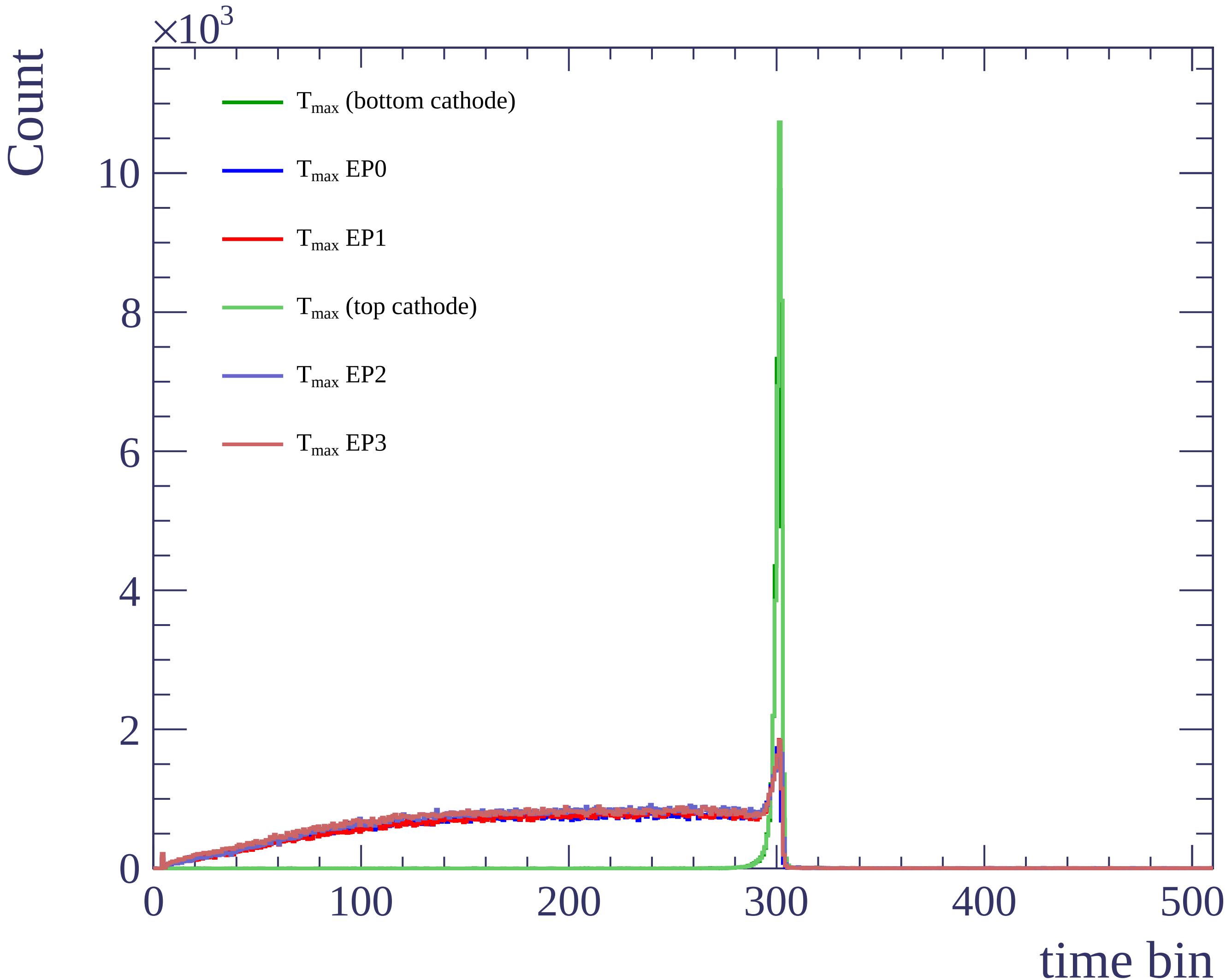
Count
×10^3
30
25
20
15
10
5
0
0
1
2
3
4
5
6
7
8
9
10
×10^3
χ^2



$\chi^2$
$\times 10^3$
10
9
8
7
6
5
4
3
2
1
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
350
300
250
200
150
100
50
0



$\times 10^3$
Count
time bin
0
2
4
6
8
10
12
14
0
100
200
300
400
500
$T_{min}$ (bottom cathode)
$T_{min}$ EP0
$T_{min}$ EP1
$T_{min}$ (top cathode)
$T_{min}$ EP2
$T_{min}$ EP3



×10^3
Count
T_max (bottom cathode)
T_max EP0
T_max EP1
T_max (top cathode)
T_max EP2
T_max EP3
0
2
4
6
8
10
0
100
200
300
400
500
time bin